

Interdisciplinary Public Finance, Business and Economics Studies
Volume II

Adil Akinci (ed.)

Interdisciplinary Public Finance, Business and Economics Studies

Volume II



PETER LANG

**Bibliographic Information published by the
Deutsche Nationalbibliothek**

The Deutsche Nationalbibliothek lists this publication in the Deutsche Nationalbibliografie; detailed bibliographic data is available online at <http://dnb.d-nb.de>.

Library of Congress Cataloging-in-Publication Data

A CIP catalog record for this book has been applied for at the
Library of Congress.

Printed by CPI books GmbH, Leck

ISBN 978-3-631-79727-3 (Print)
E-ISBN 978-3-631-80293-9 (E-PDF)
E-ISBN 978-3-631-80294-6 (EPUB)
E-ISBN 978-3-631-80295-3 (MOBI)
DOI 10.3726/b16175

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Internationaler Verlag der Wissenschaften
Berlin 2019

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This publication has been peer reviewed.

www.peterlang.com

Foreword

The book entitled *Interdisciplinary Public Finance, Economy and Business Studies – Volume II* aims to reveal the current and historical developments experienced in the field of economics and administrative sciences in the world and in Turkey, to produce solutions and to make a general evaluation by bringing together the studies of researchers and academicians working in the field of economics and administrative sciences.

In his chapter, Leba analyzes and evaluates the status of agricultural sector in income tax in Turkey. In his chapter, Sengur analyzes the effects of income and other factors on access to technology by logistic analysis method. In his chapter, Guler evaluates the coordination between monetary and fiscal policies within the context of financial discipline and price stability. In their chapter, Ozturk and Demir examine the effects of FED and ECB monetary policy decisions on the economies of developing countries. In his chapter, Kasa examines the relationship between R&D expenditures, high technology export and economic growth in OECD countries. In his chapter, Celebi examines the relationship between political stability, fixed capital, employment, and economic growth in MENA countries through panel cointegration analysis method. In their chapter, Mete and Usta evaluate the blue economy issues within the Turkish economy. In his chapter, Durmaz performs an analysis based on historical documents about the place of Edirne's Industry in economy during the first years of the republic. In his chapter, Kilci performs an econometric analysis on debt sustainability in Brazil, Turkey, India, Indonesia and South Africa by Fourier analysis method. In their chapter, Turk and Ak Bingul examine after EU enlargement development or deterioration. In his chapter, Celik makes evaluations about the credits and credit analysis in the banking sector. In his chapter, Tuzunturk evaluates the essentials and processes of sample survey study. In his chapter, Kanli evaluates the proposal of neighborhood parking cooperatives within the framework of parking problem in cities. Yalçın examines the strategic cost management process in businesses. In her chapter, Dombekci Ozcelik examines the liquidity management performance of 20 international airlines between the years 2011 and 2017. In his chapter, Karahan examines significant audit issues in the independent audit report. In his chapter, Cevik conducts a field research regarding the determination of financial literacy. In his chapter, Calık evaluates strategic management accounting practices in terms of marketing. In his chapter, Zeytinoglu examines the effects of some aspects of corporate governance on

financial performance with the data of 52 companies in BIST 100 index by panel data analysis method. In their chapter, Kayakıran, Dogan and Kilic conduct an application study related to accounting ethics. In their chapter, Senkayas and Can conduct an application related to the awareness of industry 4.0. In his chapter, Kahya makes an evaluation on the role and importance of innovative human resources capital in businesses. In his chapter, Aslan conducts a field study on the mediator effect of job stress in the effect of leader-member interaction on intention to leave work. In his chapter, Soylemez examines the conceptual framework of digital marketing. In their chapter, Kahraman and Duger make an evaluation on the role of human resources management, leadership and marketing in the employer brand. In his chapter, Er makes an evaluation on the evolution of social commerce. In their chapter, Cakir and Dogantan conduct a field study related to local tourism entrepreneurship.

Associate Professor Adil Akinci

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Reyhan Leba

Evaluation of Agricultural Sector in Turkey Within Income Tax¹

1 Introduction

As countries develop, the share of agricultural sector in domestic income has begun to reduce along with the increasing industrialization. According to the data of 2017, in Turkey the share of agriculture in domestic income is 6,1 %, and it is still in a considerable amount. However, the share of agricultural sector in employment is 19,4 % and this is a very high rate. Today, one out of every five people works in agricultural sector. Therefore, it can be thought that there is a serious tax potential in agricultural sector.

Income tax is a type of tax collected on the incomes of real persons. In Turkish Income Tax System, income is defined in 1st article of Income Tax Law (ITL) numbered 193. Accordingly, income is the net amount of incomes and revenues that a real person obtains in a calendar year. The incomes and revenues that will be subject to annual income tax consist of the income items in 2nd Article of ITL. These related items are business incomes, agricultural incomes, wages, self-employment incomes, real property incomes, movable assets incomes and other incomes and earnings. Income tax is a tax based on declaration. The unitary structure of Turkish Income Tax System necessitates the declaration and taxation of agricultural incomes based on real taxation by gathering in annual declaration along with the incomes obtained from the seven income items mentioned above. However, stoppage at source was adopted as the main principle in taxation of agricultural incomes in Turkey. But stoppage at source is against the essence of unitary income taxation system.

The role of income tax in taxation of agricultural sector is emphasized first and then the taxation of agricultural sector in Turkey within income tax is analyzed and evaluated in this study, which aims to analyze and evaluate the taxation of agricultural sector within income tax.

1 This study was presented as a summary notification at the 2nd International Annual Meeting of Socioeconomy Society held in Amsterdam-Holland on 28th–29 October 2016.

2 The Role of Income Tax in Taxation of Agricultural Sector

In taxation of agricultural sector, there are three types of taxation. These are taxation of agricultural incomes, taxation of goods and services that the people in agricultural sector consume and taxation of agricultural land (Görgün, 1973; Lewis, 1976).

Taxation of agricultural sector is inevitable especially in the countries in which agricultural sector occupies a big place in the economy and the taxes to be collected also need to have some special properties. Therefore, a taxation system to be established for agricultural sector should provide the necessary resources for public services and industrialization and also contribute to solve the problems in the traditional structure in agricultural sector. The first objective, that is, funding, depends on conditions such as the ultimate use of agricultural land, the use of current technology and increasing the productivity per labour. Second objective of the taxes in agricultural sector is to change the traditional structure of agriculture. Here, the traditional structure refers to the following two points. First one being ownership structure of agricultural land and second one the diverse action and reaction relationship between market mechanisms and agricultural sector. These factors, on the one hand, prevent potential agricultural production, and, on the other hand, prevent the economic and political development of the system. Therefore, these related problems in traditional structures should be solved as soon as possible (Önder, 2012).

As long as agricultural sector is not charged with tax obligation in order to realize the objectives above, it will take a longer time to solve the problems in traditional structure and the necessary resources will not develop automatically in agricultural sector. Because when this sector, which has not yet been paid for, or has a very weak relationship with the market, is left alone, it develops quite slowly. The resources created with this way will be used by various bodies ineffectively in pieces. However, it is necessary to create value surpluses in agricultural sector as well as to collect these surpluses in small, integrated pieces, that is, without broken pieces. Therefore, on the one hand, finance of public services on the other hand, creating resources for industrialization necessitate the application of direct taxation in agricultural sector (Önder, 2012).

Moreover, it is thought that income tax, one of the direct taxes, is the most ideal and modern method in terms of justice of taxation and taxation technique in the taxation of agricultural incomes (Saygılıoğlu, 1977). However, applying a tax based on real income in agricultural sector has some troubles even in developed countries. First of all, recognition of the activities in agricultural

sector is not easy. This derives from the characteristics of agricultural activities. For instance, some inputs used in agriculture such as fertilizers will not only increase the fertility of the soil in short term but also in the long term as well. In this case, the problem of whether the fertilization costs will be considered as operating costs or investment costs makes it difficult to calculate the annual net income. Another problem is that small family businesses are widespread in agricultural sector, and they do not keep the necessary business accounts for application of tax based on real income. That being said, the fact that majority of the public, especially in less developed countries, is illiterate also affects the registry and account-keeping activities (Görgün, 1973; Yuluğ, 1968; Yücelik, 1971). Moreover, if the barter economy, exchange is common in agricultural sector, in an economy and monetary economy does not work, there will even be no tax base because the incomes do not convert into actual money. When the agricultural sector in these countries is in question to be included in the scope of income tax, most of the businesses in agricultural sector are subject to taxation upon lump sum income, not the real incomes. However, the troubles in tax application may lead to great injustice in the taxes based on lump sum income method. In order to avoid this situation, there is a usual tendency to keep tax exemptions and reductions too broad. The fact that the structure of lump sum income taxes is suitable for tax evasion and this type of taxes cannot monitor the changes in real incomes closely reduces the efficiency of these taxes (Görgün, 1973).

However, the principle of universality besides the principle of justice in taxation also necessitates the taxation of agricultural incomes within income tax (Feyzioğlu, 1954). In addition, the fact that income tax cannot cover the agricultural sector facilitates the income items to be concealed in other sectors, and this is the main factor for low income tax efficiency. Therefore, it will be appropriate to tax the agricultural sector – despite the troubles it has – within the income tax system (Yaşa, 1965).

In fact, income taxation has many various beneficial aspects. Through this system, taxation is attributed to the concept of income which is the best sign of taxation capacity. As long as applied well, we can obtain a highly flexible taxation type and therefore an important tool for the redistribution of the income and the struggle against inflation or deflation. In addition, this system enables the opportunity to regulate the taxation better in terms of promoting effects, social justice and some other purposes (Yuluğ, 1968).

As is seen, although income tax application in agricultural sector is an ideal and modern method, it has also some troubles for especially underdeveloped countries.

3 Taxation of Agricultural Sector in Turkey Within Income Tax

The largest tax reform of the Republic Period was realized with Income Tax System in 1950. ITL, Corportae Tax Law (CTL) and Tax Procedure Law (TPL) are parts of this reform (Yücelik, 1971). However, here, agricultural incomes were excluded from income tax. Technical difficulty in direct taxation of agricultural incomes, the desire for not decreasing the success of income tax reform and the political thoughts during the transition to multi-party system had an effect in this decision (Yaşa, 1965). That being said, excluding the agricultural sector forming more than 40 % of the national income was severely criticized especially in terms of tax competition.

Agricultural sector was firstly included in the system by the Law numbered 202 in 1963. Therefore, agricultural sector was not subject to income taxation in the period until 1925 after abolishing the tithe, ignoring the fact that Soil Products Tax was not regarded as a successful application between 1943 and 1946 (Önder, 1988).

Besides real taxation system, some methods such as small farmer exemption, agricultural income exception, lump sum basis, lump sum basis for the determination of expenses and tax cut were applied within different taxation policies in taxation of agricultural incomes since the past (Taşdelen, 2010). According to the Law numbered 4369 enacted with the amendment made in 1998, the method of stoppage at source (tax cut) is adopted today as the main principle in taxation of agricultural incomes. However, real taxation system was accepted for the farmers who exceeded the business size and the standard of agricultural motor vehicle. In order to tax a farmer according to real taxation system, it is sufficient to exceed one of the standards: the business size or the agricultural motor vehicle (Şenyüz, Yüce and Gerçek, 2013).

Tax stoppage at source is included in 94th Article of ITL. Accordingly, while tax responsible persons and corporations are purchasing their agricultural products, they have to make stoppage at source at the rate indicated in the law during the payment on account or in cash. However, there is no tax cut in the purchases for special needs. Tax cut rates from different agricultural activities are regulated in 11th item of 94th article of ITL as following:

- a) For animals and their products and hunting and fishing products:
 - i) 1 % for purchases by registering in commodity exchange markets
 - ii) 2 % for the others left
- b) For other agricultural products:
 - i) 2 % for agricultural products purchased by registering in commodity exchange markets

- ii) 4 % for the others left
- c) For the services within agricultural activities:
 - i) 2 % for the services such as afforestation, maintenance, woodchopping, collection, transportation of the products, etc., and for the forest administration and the corporations which make a commitment to forest administration
 - ii) 4 % for other services
- d) 0 % for direct income support to farmers and alternative product payments

As can be seen, lower stoppage rates are applied in sales on product exchange markets as compared to the sales on fields out of the exchange markets. This application can be interpreted as to direct farmers who do not want to pay more taxes to the product exchange markets and therefore try to register in the sector as much as possible (Tokatlıoğlu, Selen and Leba, 2018).

Business size standards presented in 54th article of ITL vary according to the activity field of the business. In agricultural activities on land acreage and the number of tree, the number of cattle and small cattle, in agricultural activities out of the land and hunting and fishing nets and fish screens, etc. in aquaculture and total boat length in fishing, the number of hives in beekeeping, the number of boxes in silk worm breeding are taken into consideration. Business size standards were remarkably increased by the Law numbered 4369.

A farmer can have an agricultural business alone or have a partnership with other businesses. In this case, farmers's shares in partnership businesses will also be considered in calculating the business size (Taşdelen, 2010). Hence, it is stated in 53rd article of ITL that the business size will collectively be considered in the businesses which belong to a partner living with the head of the family and the children under custody and the businesses with partnership. It is based on the sum of partnership shares in identifying business size standards of the ones who have partnerships with different people in more than one agricultural business. The fact that businesses are in different places will not change the situation in partnerships. While the farmers whose sum of the shares of their own businesses and partnership businesses exceed the standards, although both the personal business and the partnership business are under the business size standard separately and are exposed to real taxation system, other partnerships whose sum of shares do not exceed the standards will be taxed by means of stoppage. Share cropper is considered as partnership. In share cropper, the owner of the land does not cultivate his/her land and leaves the land (in some cases the seeds with the land) to another person. Here, the owner of the land gets a share from the product without participating in the production. While identifying the business

size standards of a farmer, the agricultural activities such as share cropper is considered as partnership, and the business size standard or standards are calculated accordingly (Akıntürk, 2002; Mutluer, Herekman, Heper and Dönmez, 2005; Taşdelen, 2010).

In the event of the agricultural activities included in some of the agricultural groups stated in 54th article, business sizes belonging to maximum two groups will be taxed in real taxation system when the halves of determined limits are exceeded for these groups (ITL. Article. 53).

As mentioned before, tax cut is commonly used in taxation of agricultural incomes. Tax cut is also applied for the people taxed in real taxation system (Taşdelen, 2010). In other words, in terms of tax cut it is not important whether the incomes of farmers are determined in real taxation system or not. However, while tax cut is the final taxation for the farmers who are not taxed in real taxation system, the farmers who determine their incomes in real taxation system have the opportunity to set off the taxes paid by means of tax cut from the income tax calculated upon declaration (muhasibetr.com, 2013).

That being said, there will be no tax cut in the case that farmers sell their agricultural products to non-tax responsible persons. Therefore, the farmers who are not taxed in real method – since they will not submit a declaration – will not tax due to their agricultural incomes (Şenyüz, Yüce and Gerçek, 2013).

Only the farmers who are not taxed by means of tax cut do not keep books and submit a declaration. However, these farmers must take and keep producer receipt due to their sales and services (Odak, 2001). A producer receipt is the invoice for sellers and purchasers. According to 253rd Article of TPL, it is necessary to keep this document for five years from the beginning of the calendar year following the related year (Taşdelen, 2010). According to the provisions of TPL, the farmers who do not comply with getting, giving, keeping and present liabilities of documents cannot benefit from the in-kind and in-cash supports such as advances, loans, subsidies and premiums given by various public institutions and corporations (ITL. Article. 53).

However, according to the method of agricultural exploitation or balance sheet method, the farmers taxed in real method must keep books and submit declarations, get and give documents for their sells and purchases and keep these documents (Odak, 2001). According to 85th Article of ITL, the people with agricultural incomes taxed in real method are responsible for submitting declaration even if they do not have incomes from agricultural activities.

The farmers to be taxed in real taxation system are left to their preferences in terms of the fact that they keep book in the method of agricultural exploitation or balance sheet method. A farmer has the right to choose between the

two methods mentioned above (Odak, 2001; Şenyüz, Yüce and Gerçek, 2013). However, the farmers choosing the balance sheet method cannot change it before two years (ITL. Article. 59).

According to the method of agricultural exploitation, agricultural income consists of the positive difference between the revenues collected as cash in accounting period or accrued as borrowings and the borrowed or paid expenses. In determining the agricultural income according to the business account method, the value of the products in the beginning of the accounting period is added to expenses, and the value of the product at the end of the period is added to revenue (ITL. Article. 55).

In determining the agricultural income according to the balance sheet method in 59th article of ITL, it is stated that by taking the provisions of 56th, 57th and 58th articles into consideration, the provisions of business income in this issue will be applied. Accordingly, agricultural income in the balance sheet method is determined through the comparison of equities in the beginning and end of the period as in the determination of business income in the balance sheet method. The agricultural income here is the difference on behalf of end-of-period equity values between the equity values found by grounding upon the balance sheets of the agricultural business at the beginning and end of the accounting period. The values, if any, added during the period are subtracted from this difference, and the taken values are added (Şenyüz, Yüce and Gerçek, 2013).

The revenues and expenses in the determination of agricultural incomes according to real taxation system are presented in 56th and 57th articles of ITL. Accordingly, all types of income obtained according to collection or accrual methods as a result of the agricultural activities compose of the revenues. Moreover, the expenses for the maintenance and continuity of agricultural activities are also considered as expenses. However, referring to 41st article of ITL, the payments for the farmer himself, the partner and children of the farmer and all kinds of penalties cannot be deducted from agricultural income. However, according to 58th article of ITL, the consumptions from the products in the agricultural business by the farmer and farmer's partner and children under custody (self-consumption) are kept out of the non-deducted expenses (Şenyüz, Yüce and Gerçek, 2013).

According to 5th item of 9th article of ITL, the farmers who sometimes transport goods and people for a fee without making the transportation a habitual activity by some vehicles or sandals such as animals used in agricultural businesses, carriages, motor vehicles and tractors are exempted from income tax.

According to the repeating 20th article of ITL, in order to promote the fully responsible real persons who are firstly appointed as income taxpayers for their

agricultural activities and are not 29 years old yet as of the beginning date of the responsibility, a part of the income up to 75,000 TL that they obtain throughout three taxation periods since the calendar year of the activity are excepted from income tax.

However, the exception of incentive premiums and rewards is included in 29th article of ITL. Therefore, the incentive premiums and rewards given in order to promote agriculture, animal breeding and other beneficial businesses and activities for the country are excepted from income tax.

Unitary characteristics of income tax necessitates the agricultural incomes subject to real taxation system to be gathered in an annual declaration along with other income items that need the declaration according to the Law numbered 193. Income tax is calculated according to progressive tariff which is included in 103rd article of ITL and is the base for the taxation of income subject to income tax. The lowest rate in the tariff is 15 % and the highest rate is 35 %. As stated before, the people subject to this procedure can reduce the tax cut by the purchaser from the calculated income tax upon the declaration during the payment for their sale in the calendar year (Taşdelen, 2010). As mentioned above, today agricultural sector is mainly taxed through the tax cut out of declaration. This can be appreciated as the fact that tax burden of agricultural sector is low because they get over the progressive income tax tariff. However, tax cut upon gross earning leads to higher actual tax burden (Tokatlıoğlu, Selen and Leba, 2018).

Tax payment is included in 117th Article of ITL. Accordingly, the annual declaration is submitted from the beginning of March following the year of obtained income until the evening of 25th March, and the accrued income tax is paid in March and in July as two equal installments. In addition, through General Communiqué of Income Tax with serial number 250, the Ministry of Finance set November and December as the payment time for farmers with 75 % or more of the agricultural income within the total of their annual declarations (resmigazete.gov.tr, 2013).

However, real persons (including ordinary company partners) operating the agricultural businesses are considered as farmers according to 52nd article of ITL. Unlimited companies, ordinary companies or limited partnerships divided into shares are not considered as farmers even if they are engaged in agricultural activities. The dividends of the partners of unlimited companies engaged in agricultural activities and the unlimited partners of the limited partnerships are regarded as business incomes. The dividends of the limited partners are considered as movable assets income.

Capital companies are the subject of CTL. Although agricultural income items are common income items for CTL and ITL (Taşdelen, 2010), the provisions of

ITL about business incomes are applied in the determination of net corporate profit when obtained by the corporate. However, the provisions of ITL about the determination of agricultural incomes are also taken into consideration when determining the incomes of the corporates engaged in agricultural activities (ymm.net, 2013).

Agricultural activities may sometimes lead to confusions by nesting with commercial activities and real property incomes. Therefore, it is necessary to clarify the difference (Şenyüz, Yüce and Gerçek, 2013).

4 Evaluation and Conclusion

When both the place of agricultural sector in Turkey's economy and its share in employment are considered, it can be thought that agricultural sector is significantly benefited in terms of taxation. However, the data indicate that there is a sudden decrease in the shares of persons declaring profits for agricultural incomes within income taxpayers from the beginning of 1999, and this tendency generally goes on in the following years. It can be said that, on the one hand, the practices within the World Trade Organization Agreement on Agriculture enacted on 25th February 1995 and, on the other hand, the reconstruction policies for agriculture started with the letter of intent given to International Monetary Fund (IMF) on 9th December 1999 have an effect on this situation. In addition, the problems about agricultural structure have an important role in the decrease in income from the agricultural sector. The main problems can be listed as: unfair land distribution and common small businesses in agricultural sector in Turkey, significant economic losses due to the disintegration of agricultural land, low level of agricultural added value per farmer and low labour-land productivity. Widespread small businesses in agricultural sector in Turkey complicate the use of technology in agriculture and reduce the productivity and weaken the competitive capacity of this sector. Due to the unproductive structure in agriculture, costs increase and the obtained income decreases. Today stoppage at source is adopted as a main principle in taxation of agricultural incomes according to the law numbered 4369 enacted in 1998. The fact that business size considered as a standard in real taxation is determined at high levels leads the tax cut to become widespread in taxation of agricultural incomes within income tax. Therefore, agricultural sector in Turkey cannot be efficiently taxed within income tax.

As a conclusion, it will not be sufficient to focus only on taxation in order to tax the agricultural incomes efficiently in Turkey. It is also necessary to consider the problems about the policies applied for agriculture and the agricultural structure and evaluate them together.

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193 Sayılı Gelir Vergisi Kanunu.

Mehmet Sengur

The Relationship Between Income Inequality and Access to Technology: The Case of Turkey¹

1 Introduction

We can classify people as poor or not poor according to a limit value of income determined by any method. However, in social life, individuals need more than just being below or above a certain limit value of income.

When income is not distributed equally to individuals justly, the concept of income inequality emerges. Income inequality leads to the inability of individuals to receive an equal part in the social-cultural and economic spheres.

If the fair distribution of income can be achieved, access to social and economic goods and services of individuals in society will also increase. This access will include indicators representing the average level of prosperity in society, along with basic needs such as, education, nutrition and housing. Many needs are also important for individuals, such as being able to participate in a social event or to organize a social event, access to education, health services and technology. Meeting these needs, however, depends on the equal and fair distribution of income to all individuals in society. Therefore, it is important for people not to be deprived of goods and services relatively compared to other people in society, in addition to their basic needs.

In this study, firstly, income, wealth and income inequality are explained conceptually. Then, the relationship between income inequality and technology for Turkey is examined. The relationship between internet access representing access to technology and monthly average income, education level, gender, age and household size is analyzed with data provided by Turkish Statistical Institute (TÜİK).

2 Income and Wealth

People sometimes think of their pre-tax wages as their real income, although the money they receive every month is rarely the same. Instead, it would be more useful to take the disposable income after taxes and transfers are deducted. In

1 This study is an improved version of the abstract presented at the 6th International Congress on Social Sciences, China to Adriatic.

this way, how much money people need to spend on rent, food, clothing, etc. will be clearly revealed. Disposable income is usually determined by subtracting the taxes (expenses) from the money (income) that is earned by a household. Generally, the income of the household consists of wages or salaries, rent from properties and profits derived from investments. Expenses, on the other hand, are usually a number of payments, such as taxes paid to the government and social security payments (Brian, 2015). Income is a flow variable, which means that it can only be measured for a specific period depending on the time. Income is derived from assets owned, and all types of income provide a certain benefit to individuals. For example, there are types such as interest income, capital income, profit income, wage and salary income (Rycroft, 2014).

The measure of wealth is very complex, and not all countries do it in the same way. Some countries, for example, include retirement benefits in the calculation, but some do not. Because of such differences, it is necessary to know what is included and what is excluded for a good measure of well-being. Because wealth accumulates over time, on average, it is higher than normal income. Another aspect of wealth is that it increases more unevenly than income, which means that wealth inequality increases more rapidly than income inequality. As wealth or fortune can generate income within itself, wealth inequality will increase steadily. Because wealth will be a source for investments, the gap between the poor and the rich will grow steadily. This difference will give an advantage to the rich in accessing opportunities to convert their capabilities into investment (Brain, 2015).

3 Distinguishing Income Inequality from Inequality

There is a lot of debate going on about inequality. The concept of inequality is much more complex, as opposed to what is believed, because it has a different meaning for different people. First, we need to make a distinction between monetary and non-monetary inequality. Monetary inequality depends on the economic activities of the individual or the household, such as income, consumption expenditure and wealth. Non-monetary inequalities, on the other hand, takes into account the wider aspects of economic life, such as welfare or capacity (Atkinson and Bourguignon, 2014).

Basically, to assess different subgroups and different parts of the population, which contribute to inequality, common inequality criteria can be used. For example, average income may vary from region to region, and this alone means that there are some inequalities between groups. Moreover, when inequality varies within each region itself, the intragroup component will be added to total

inequality as well. It would be useful to differentiate sources of income inequality for policy purposes. For example, if the majority of income inequality stems from interregional inequality, a special effort is needed to help the poorer regions. In this case, policies will need to focus on regional development. Household income, in general, is determined by a number of geographical factors, including urban and regional factors, in addition to personal characteristics such as education, gender, profession, etc. (Haughton, et al., 2009).

Income inequality takes different living standards into account for all individuals living in society. Since income inequality affects all individuals in society, it is affected by many factors, such as average income and level of education. As a result of the resulting inequality, poverty indicators will also be affected negatively. Income inequality is an obstacle for individuals in society to have equal opportunities. In addition, income inequality is an important indicator of crime, violence and conflict, which are the source of social unrest (McKay, 2002).

For income inequalities, differences in the nations' level of development are important. Inequality is closely related to access to employment and education opportunities, health and preventive care services, nutrition inequalities, security and conflict-free stability, goods and services not being distributed equally to a certain percentage of the population. Social and economic instability and the lack of fair distribution of resources are the direct causes of poverty (Odekon, 2006).

Inequality can be explored in a variety of ways, each with different meanings about how economic resources are spread across society and around the world. It is quite difficult to present inequality with a single figure. Many approaches have been developed in this regard over the years, but the one developed by the Italian economist and statistician Corrado Gini in the 20th century is the most well-known of these approaches (Brain, 2015).

4 Method: Logistic Regression Analysis

Regression analysis is concerned with examining the dependence of a dependent variable on one or more explanatory variables. Here, the mean or average value estimation or calculation is made with known or constant values of the dependent variable (Gujarati, 2004). Logistic regression can be defined as estimating the relationship between a dependent variable receiving a binary value and a set of independent variables that can be measured or not measured. Logistic regression models are generally referred to as Logit analysis. Logistic regression is also a combination of multiple regression and multiple discriminant analysis. This method is similar to multiple regression analysis because a single dependent

variable is estimated with one or more independent variables. The feature that differentiates logistic regression from multiple linear regression is that the dependent variable cannot be measured as in discriminant analysis. Despite the fact that the dependent variable being on an unmeasurable scale requires differences with the estimation method and the assumptions of the distribution, in many other respects, it is very similar to multiple regression analysis (Hair, et al., 2014). When continuous or categorical variables are included in the established model, the assumption of normality in multivariate analyses is disrupted. In this case, logistic regression, which does not make any assumptions about the distribution of independent variables, should be used (Sharma, 1996).

However, logistic regression analysis is less affected than discriminant analysis when the normality assumption is not met for the variables that are one of the basic assumptions. In addition, regression analysis can be performed with variables that cannot be measured, but can also be applied with dummy variables. Logistic regression is limited because it predicts the dependent variable for only two group values. Therefore, it is more appropriate to use discriminant analysis if the dependent variable receives three or more group values (Hair, et al., 2014).

Since all regression models are not linear, a distinction is made between linear and non-linear regression models. However, there are some important points to be considered here. The parameters of some models may not appear linear, but the parameters in these models may be self-linear. Because, with the appropriate conversion methods, these models can be converted into a linear form. The probit and logit models, which are linear transformable models, are very similar except for differences in function. When the analysis results of the models are examined, it is seen that they are very close values (Gujarati, 2004).

$$P_i = E(Y=1|X_i) = \beta_1 + \beta_2 X_i \quad 1$$

$$P_i = E(Y=1|X_i) = \frac{1}{1 + e^{-(\beta_1 + \beta_2 X_i)}} \quad 2$$

$$P_i = \frac{1}{1 + e^{-Z_i}} = \frac{e^z}{1 + e^z} \quad 3$$

Equity which takes its form in Equation 3 is the cumulative logistics distribution function. In an equation that reflects a non-linear relationship, while z_i takes values between $-\infty$ and $+\infty$, p_i takes values between 0 and 1 (Gujarati, 2004).

Logistic regression is a special type of regression that is used to predict and explain a two-group categorical variable, rather than a measurable dependent measure. The structure of logistic regression is very similar to the structure of multiple regression. Logistic regression analysis determines the relative effect of each explanatory variable on the dependent variable with coefficients similar to those in the regression. Although there are some distinguishing elements between the two methods, there are also many similarities. When the basic assumptions of both methods are met, each gives comparable, predictive and categorical results, and uses similar diagnostic measures. When the basic assumptions for the two methods are met, it gives comparable estimates and classifications. Moreover, the methods work with similar testing criteria. Logistic regression is commonly used when identifying the group to which an element belongs. Logistic regression applications generally estimate anything if the result takes binary values such as yes or no. The success or failure of a newly launched product, whether or not a loan is granted to anyone, or whether a firm will be successful, can be given as an example for these predictions. In each case, the variables are divided into two groups in order to estimate which group they belong to by using a series of independent variables selected by the researcher (Hair, et al., 2014).

In logistic regression analysis, the estimation of the relationship between a dependent variable and one or more independent variables is performed in a similar way to other regression methods. Although it is mainly based on the principles of linear regression, logistic regression includes differences in estimation and calculations (Hosmer et al., 2013). Therefore, when the independent variable is correctly determined by an appropriate estimation method, the basic factors taken into account in multiple regression are also used in logistics regression. As the logistic regression analysis is compatible with any type of independent variables and does not require a multivariate normality assumption, it differs from other methods of analysis (Hair, et al., 2014).

Logistic regression is termed in different ways according to the number of independent variables. If the number of independent variables is one, there is one variable logistic regression. However, if the number of independent variables is two or more, there is a multivariate logistic regression. Furthermore, logistic regression is used in different forms by classifying the dependent variable according to the number of categories. If the dependent variable has only two possible values, this type of analysis is called binary logistic regression analysis. If the students' academic achievement will be analyzed according to whether they are successful or unsuccessful, binary logistic regression is used. If the number of the dependent variable is more than two, multinomial logistic regression is used. In addition to these, if the dependent variable is ordinal, ordinal logistic

Tab. 1: Data Descriptions

Type	Code	Explanation
Dependent Variables	access to technology	have internet access 1, not have 0
Independent Variables	Education	at least high school 1, not 0
	age	15–99
	gender	female 1, male 0
	employment status	own business 1, others 0
	average income	at least 1500 ₺ (Turkish Liras), not 0
	household size	1–20
	Literacy	yes 1, no 0

regression analysis is used (Stephenson, 2008). The regression models for binary outputs are obtained by establishing ordinal and nominal variables. Probit and logit (logistic) regressions are also included in these. Logistic regression is used to create models for discrete variables in two or more categories. Dual output regression models allow the researcher to investigate how each explanatory variable affects the likelihood of occurrence of an event (Freese and Long, 2006).

5 Data Set and Variables

In the application for determining the effect of income inequality on access to technology in Turkey, “Information Technology Usage Survey” data for 2014, obtained from TÜİK, was used.

Tab. 1 shows the dependent and independent variables used in the regression model. Access to technology (internet) was chosen as the dependent variable. For the dependent variable, having internet access took a value of “1”, whereas having no internet access took a value of “0”. Similarly, the independent variables in the model were given the value of “1” for the occurrence of the event and “0” for vice versa.

“1” was given for the case that total income in the household is ₺1500 and above, and “0” was given for the case that total income is under ₺1500. The education variable, which expresses the level of education, took the value of “1” for at least having a high school diploma and the value of “0” for a lower education level than high school. The age variable is between 15 and 99, and the household size is between 1 and 20. For gender, “1” was given for male and “0” for female. For the status of job, “1” was given for being self-employed and “0” for being employed by a company. In addition, being literate was given “1”, whereas being illiterate was given “0” (Tab. 1).

Tab. 2: Summary Information

Variable	Information	Percent
age	average (15 +) (%)	33
education	high school (%)	30
	secondary education (%)	70
gender	female (%)	55
	male (%)	45
average income	1751 ₺(Turkish Lira)	
own business	yes (%)	70
	no (%)	30
average household size	4	

Looking at the summary information about households, the number of observations is 36,227 and the mean age is 33 years. While 70 % have an education level of high school and above, 30 % have an education level of middle school and below. The average monthly household income was ₺1,751 in the relevant period. Fifty-five percent of individuals are women, while 45 % are men. The average number of individuals in the household is four, and the ratio of self-employed or employers is 70 % (Tab. 2).

6 Empirical Results

With the Wald-chi2 (Chi-square) probability value, the overall significance of logistic regression is tested. As can be seen in the table, Wald-chi2 probability values are significant at 5 %. The power of logistic regression analysis is tested by Receiver Operating Characteristic (ROC) Analysis or Classification Table values. As shown in the table, the ROC value is 76 % and the Classification Table value is 72 %. Furthermore, it was observed that there was no multiple linear connection problems due to the calculation of variance-covariance values for the model. As a result of Pearson-chi2 test, it was found that the model was consistent with data because the probability value ($p > 0.05$) was greater than 5 %.

The results of the analysis of the model established to determine the effect of income and other factors on access to technology are given in Tab. 3. According to the results of the analysis, it was determined that all the independent variables included in the model were statistically significant at 5 % ($p < 0.05$). However, the direction and intensity of the effect of all variables in the model are not the same. Gender, age, employment status and household size variables have negative effects on internet access. On the other hand, literacy, income and education levels have positive effects on internet access (Tab. 3).

Tab. 3: Model Estimation

Variables	Odds Ratio	Model
Age	.979	-0.021*** [0.001]
Gender	.884	-0.122*** [0.024]
Literacy	1.921	0.653*** [0.031]
Education	1.132	0.124*** [0.028]
Employment status	.577	-0.549*** [0.033]
Income	4.684	1.544*** [0.027]
household size	.933	-0.069*** [0.006]
constant	1.908	0.646*** [0.064]
observations		36,227
Pseudo -R2		0.36
Wald chi2- (p)	5305.16	0.00
ROC		76 %
correctly Classified		72 %

*** p<0.01, ** p<0.05, * p<0.1

Household income of ₺1,500 and above increases the probability of internet access by 4.7 times compared to a lower income level. In this case, it is observed that the increasing level of income facilitates access to technology.

Being literate increases the likelihood of internet access by 1.92 times compared to being illiterate. Graduating from a high school or higher education institution increases the likelihood of internet access by 1.13 times, compared to graduation from a lower level education institution. As with increasing income, access to technology increases with the increase in the level of education.

The increased age was observed to reduce the likelihood of having internet access by approximately “1” fold. In addition, women are 0.98 times less likely to have internet access than men.

As the number of people living in the house increased, the probability of having internet access decreased by 0.94 times. The increase in the number of individuals in the household limits the access to technology by limiting means. It was also found that being a casual employee or full-time employee reduces the possibility of internet access by 0.57 times.

7 Conclusion and Evaluation

In this study, the effects of income and other factors on access to technology were examined. As a result of the analysis, it was observed that the increased income increased access to technology by about 5 times, and the increased education level increased it by 1.13 times. Accordingly, it was determined that the individuals with lower income had limited access to technology than those with higher income. Individuals in the lower income group were restricted in many areas together with access to technology compared to higher income groups. Given that almost half of the income (47.5 %) in Turkey is earned by 20 % of the population representing the highest income group, the main reason for this unequal income distribution can be clearly explained.

The reduction of the 7.5-fold difference between the lowest income group and the highest income group in receiving a share from the total income in Turkey will solve individuals' problem of having limited access to many goods and services. With increased income, the level of satisfaction will increase as individuals demand goods and services. Many policies and practices for low-income groups in Turkey have been implemented. As a result of these practices, the difference between the lowest income group and the highest income group decreased from 7.7 times in 2017 to 7.5 times. This is a positive development, but it is insufficient. The fight against income inequality and the resulting poverty must be carried out with more effective economic and political means. The tax burden of individuals in the low-income group can be alleviated, and the current amount and content of social support and spending can be increased. Goods and services can be classified for each income group without compromising on quality.

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Asli Guler

Coordination Issue Between Monetary and Fiscal Policies: Case of Turkey in the Context of Interaction Between Fiscal Discipline and Price Stability

1 Introduction

Economic policies are often subject to a distinction in the form of monetary policy and fiscal policy. This raises a choice on economic policy to achieve the desired macroeconomic targets. Because tools used by each policy and thus the abilities of the policies to achieve macroeconomic goals is different. The monetary policy is governed by the central bank and the fiscal policy by governments. In fiscal policy, it is tried to reach goals such as economic growth and income distribution justice by using tools such as taxes and public expenditures. The central bank, affecting savings and consumption decisions through short-term interest rates, tries to maintain and sustain price stability. Regardless of the aim and policy chosen, the effectiveness of the policy depends on the extent to which it is supported by the other policy. Most of the time, aims of monetary and fiscal policies may conflict with each other. For example, the impact of the tight monetary policy on inflation could be significantly reduced by an expansionary fiscal policy. Therefore, the existence of a strong coordination mechanism between the two policies is very important for better functioning of the general economy.

The fiscal authority may affect the consequences of the monetary policy by the size of the fiscal deficit and the financing method of this. The monetary authority may influence the government's capacity to finance the fiscal deficit by extending or limiting access to financial resources or by affecting the cost of the debt service rate (Hanif and Arby, 2003: 1). For this reason, in order to ensure coordination between policies, monetary and financial authorities should make common efforts by implementing the necessary legal and institutional arrangements.

This study aims to examine the institutional and legal regulations that facilitate the implementation of monetary and fiscal policies in a coordinated manner for the success of monetary policies aiming to reduce inflation. In this context, firstly, debates on why coordination between money and fiscal policy are necessary will

be examined, and then the related institutional and legal arrangements related will be discussed. Finally, the development of the said regulations in Turkey and the results will be discussed.

2 Rationale of Coordination Between Money and Fiscal Policies

Policy coordination can be defined as the harmonization of the objectives, instruments, and institutional and operational arrangements of public debt and money management to ensure economic stability (Ekpo and Omoruyi, 2013:7). Even if monetary and fiscal policies are carried out independently, it is impossible for them to be independent in terms of their effects. The change in any policy often places an additional burden on the other policy and damages the independence of policies. Co-ordination will reduce the negative impact of the independent policy authorities on each other and prevent the reduction of economic performance (Çoban and Deyneli, 2014:16).

Blinder gives this example of the rationale of coordination in his paper published in 1982:

“Imagine an automobile design that candidate drivers will learn to use the car. The car will have two steering wheels and two brake sets. One way to achieve “coordination” is to design the car to be a single control system that the teacher can control. This is the right thing to do. But now let’s assume, we don’t know in advance who will seat in control seat. Or what if the teacher, while a superior driver, has terrible eyesight? Under these conditions, it is no longer obvious that we want one set of controls to be able to override the other. On the grounds that a stalemate may be better than a fierce collision, we can decide that it is best to design it with two rival control sets that partially balance each other”.

In Fig. 1, monetary bliss and fiscal bliss points represent the most preferred limited results of two policymakers. Each oval around the points of bliss is one of a family of indifference lines of the appointing authority, and the points in the inside of the line are preferred to the points outside it. Lines M and F show the combination of policies that meet the most preferred total demand for each. The M line shows the combination of R and S that yields the optimal aggregate demand for the monetary authority while the F line does that for the fiscal authority. The heavy line between the bliss points is the cooperative or contract curve. The bliss points are determined by both the fiscal surplus level, that determines the growth rate, and the optimal demand level, affecting both unemployment and inflation. The macroeconomic outcome of the monetary and fiscal

policies will depend on the degree to coordination or independence between two policies. The most desirable macroeconomic outcome will occur when full cooperation is made. In Fig. 1, the thick line between the bliss points of both policies represents the contract curve, which consists of the combination of the most favorable interest rates and fiscal positions resulting from the joint implementation of monetary and fiscal policies. Common policies are shaped by a consensus between the views of both authorities. However, if the government governing fiscal policy outweighs, it is likely that the monetary authority will accept the leadership of the government. In this case, the balance will be close to the point of fiscal bliss, which will include relatively higher inflation and fiscal deficit composition (Nordhaus, 1994:146–148).

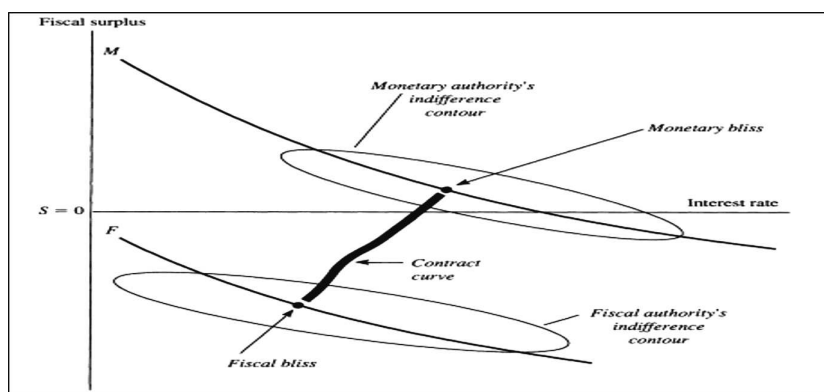


Fig. 1: Reaction Functions and Noncooperative Equilibrium. Source: Nordhaus, 1994:147

More specifically, the objectives of the policy mix could be summarized as following (Alcidi and Thirion, 2016:3):

1. To reduce/remove trade-off between monetary and fiscal stance to better control monetary growth and inflation expectations.
2. To control rapid increase in the ratio of public expenditure to GDP.
3. To eliminate the impact of “crowding out” of private investors caused by a high budget deficit. For this, the amount of public sector borrowing should be reduced to lower interest rates.
4. To solve the growing public sector debt service problem caused by a high real interest rate.

3 Institutional and Operational Coordinating Arrangements

3.1 Central Bank Independence

Economists interested in monetary policy, whether it is the theoretician or the practitioner, believe that making the central bank independent from the political authority will have a positive effect on macroeconomic indicators, such as the expansion of money and credits and thus inflation and the budget deficit (Cukierman, Webb, and Neyapti: 1992: 353–354). In the theories of dynamic inconsistency developed by Kydland and Prescott (1977) and Barro and Gordon (1983), it is proposed that more independent central banks will reduce the inflation rate. Appointing a president of a central bank who is more inflation-averse than society serves as a commitment tool to low inflation rates. High inflation creates negative effects on economic performance by creating deterioration in economic activity, encouraging rent-seeking and raising risk premium. An independent central bank isolated from political pressure acts in a more predictable manner, reducing the risk premium in real interest rates and making the economy more stable. In this respect, the independence of the central bank can be expected to improve economic performance by causing inflation to fall (Alesina and Summers, 1993:152).

However, the fact that a central bank has some kind of independence does not mean that it can conduct its activities in a completely independent manner from the fiscal authority. Absolute independence brings along a number of problems. So, to what extent should the central banks be made independent of political authority? A central bank could be independent in terms of policy instruments that are used in pursuit of monetary policy goals but should be dependent on political authority in setting those goals. Independence requires a more powerful coordination mechanism with the fiscal authority. The two authorities (monetary and fiscal authorities) whose objectives and tools are different will produce ineffective results in the absence of coordination. In this case, the success of the monetary policy implemented by an independent central bank, which provides a strong commitment to price stability, will depend on the degree of coordination with the financial authority (Bernanke, 2010; Laurens and Piedra, 1998: 17–18).

3.2 Restriction of Direct Central Bank Credit to the Government

The fact that the government constantly resorted to central bank resources to finance the fiscal deficit is accepted to be a chronic source of inflation. When a central bank issues new money to finance government debt, money supply increases in the market. Therefore, monetizing government debt is obviously

and unquestionably inflationary (Rashid, Farooq, and Nawaz, 2016). To avoid this problem, in the early stage of the coordination process, countries are more likely to implement arrangements prohibiting direct credit of central bank to the government. For this purpose, in many countries, central bank legislation has been reformed. The limitation of central bank financing plays an important role on shaping inflation expectations due to increasing the credibility of the central bank, a key element in ensuring monetary policy effectiveness (Jácome, Matamoros-Indorf, Sharma, and Townsend, 2012: 3).

It is important to note that the effectiveness of limiting direct central bank financing depends on not circumventing it. Because indirect central bank financing is not forbidden, the central bank can buy government securities in the secondary market via open market operation for the purpose of managing liquidity in the market. Therefore, indirect borrowing should be used in line with the purpose of prohibiting direct borrowing. The Maastricht Treaty is a good example of an institutional arrangement designed along these lines. It imposes some regulations on the government's borrowing from the central bank. The treaty prohibits borrowing and lending facilities from the European central bank and national central banks of the European states. In addition, the European Central Bank and national central banks are prohibited to purchase government debt instruments in the primary market. In addition, secondary market purchases of central banks are monitored to prevent circumvention of the prohibition of direct lending to the government (Laurens and Piedra, 1998: 19)¹.

3.3 Sustainable Budget Balance or Non-inflationary Fiscal Deficit

As a parallel to the restriction of direct credit of the central bank, it should be mentioned that institutional arrangements to promote fiscal discipline is essential. Unlimited fiscal policies can create inflationary pressures that the central banks will have to overcome. The need to consolidate public finances led to the introduction of fiscal rules. Thus, national and international arrangements regarding fiscal rules were made for countries to ensure fiscal disciplines such as the Maastricht Treaty and The Stability and Growth Pact (SGP).

In the Maastricht Treaty, the stages of the Economic and Monetary Union, the economic and monetary policies to be followed in these stages, and the institutional changes necessary for the implementation of these policies are detailed. A prominent feature of the Maastricht agreement is that it imposes significant restrictions on fiscal policy. According to the provisions, excessive budget deficit

1 See *The Maastricht Treaty* (Article 104).

and public debt are not allowed to reach the level that would endanger unity (Von Hagen and Eichengreen, 1996: 134). According to the relevant articles of the Maastricht Treaty, the ratio of Member States' fiscal deficits to GDP should not exceed 3 %, and the ratio of public debt to GDP of Member States should not exceed 60 % (Article 121 (1)). In addition, Article 104 (6) states that the fiscal sustainability will be ensured in the absence of a budget deficit. The provisions of the excessive public deficit procedure aim at ensuring that countries participating in the Economic and Monetary Union avoid borrowing in such a way as to cause inflationary pressures in the Euro area.

4 Evaluation of Coordination in Turkey

The increasing public deficits after the 1970s and the financing of these deficits from the Central Bank of the Republic of Turkey (CBRT) were the main reasons for the chronic inflation until 2002. In the relevant period, the central bank working under political authority has played a supporting role in the economic policies of the government. Increasing economic instabilities and successive crises have revealed that transformation is imperative. Thus, in 2001, significant reforms were made in both the independence of the central bank and fiscal discipline. Monetary policy in Turkey transformed parallel to the independence of the central bank. Inflation targeting strategy has been started to enforce implicitly in 2002. The primary objective of inflation targeting is to maintain price stability without ruining stability of output gap. The fact that fiscal policy dominates the monetary policy makes it difficult for the monetary policy to achieve its goal of price stability. Therefore, to ensure price stability and fiscal sustainability, fiscal rules have been introduced in subsequent years.

4.1 Independence of Central Bank in Turkey

The CBRT has been organized as a private central bank in the form of a non-governmental and joint stock company by Law No. 1715 entered into force on June 11, 1930. Law No. 1715 was describing a very independent central bank. However, this law has been amended 22 times from 1931 until Law No. 1211 was adopted on January 14, 1970. With the amendments made to the Law No. 1211, the independence of the bank has been significantly damaged. With this amendment, the way to provide financing to the government was opened, and the CBRT was able to give loans to the government. With the ease of financing provided, the government has applied heavily to the CBRT resources in financing the increasing budget deficits. In addition, it was determined that the bank should determine its money and credit policy by considering its development and

annual plans. Thus, in the planned economy after the 1960s, the CBRT became an institution supporting the economic policies of the government. This clearly indicates that the central bank and its monetary policies are entirely under the control of the political authority in this period (TCMB, 2012).

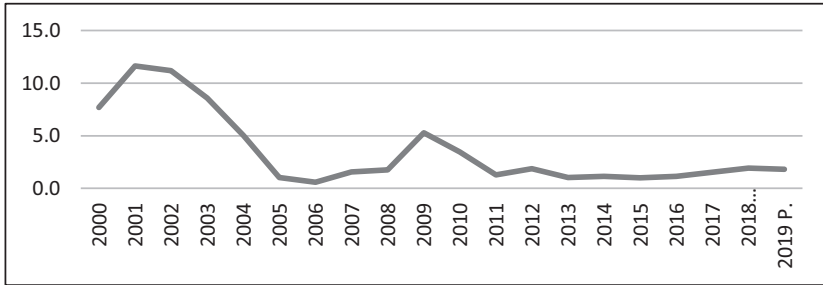
The 1980s were the years of significant changes in terms of CBRT as well as Turkey's economy. After the January 24, 1980 decisions, Turkey's economy has entered a process of structural transformation, a series of regulations have been implemented in order to establish an economic structure which is dominated by the free market economy. In this period, the CBRT continued to play an active role in financing the government's budget deficits and on the other hand, had a significant institutional change process. In 1989, the CBRT and the Treasury agreed to limit short-term credit to the Treasury. Subsequent to, it was agreed that the Treasury would not use short-term advances from the CBRT effective from 1998 onwards. In this way, the government would not resort to the Bank's resources in financing the budget deficits, the Bank's balance sheet would not grow unnecessarily, and the Bank could focus on core duties.

The year 2001 is a year of significant developments for the CBRT in terms of independence and communication design. With the amendment made to the Law No. 1211 on April 25, 2001, it is ensured that the primary objective of the CBRT is to maintain and sustain price stability. Moreover, it is stated that the CBRT's monetary policy and monetary policy instruments will be determined directly by the CBRT. In addition, the independence of the CBRT is ensured by emphasizing that the CBRT is the sole authority responsible for the implementation of monetary policy. Most importantly, with the amendment made in Law No. 1211 in 2001, the CBRT was prohibited to give advance to the Treasury and other public institutions, to issue a credit, and to buy the borrowing instruments issued by the Treasury and other public institutions from the primary market. As a result of this legal arrangement, the CBRT was prevented from supporting the fiscal policies of governments as a source of fiscal funding. In addition, monetization of fiscal deficits by being used CBRT resources has also been prevented. Today, the CBRT continues to implement monetary policy to ensure price stability as a primary objective and financial stability as a supportive purpose.

4.2 Fiscal Discipline in Turkey

The high rate of chronic inflation, covering the period 1980–2001, was the expense of poor fiscal discipline and high budget deficits. The frequent use of central bank's resources in financing fiscal deficit has led to constant monetary

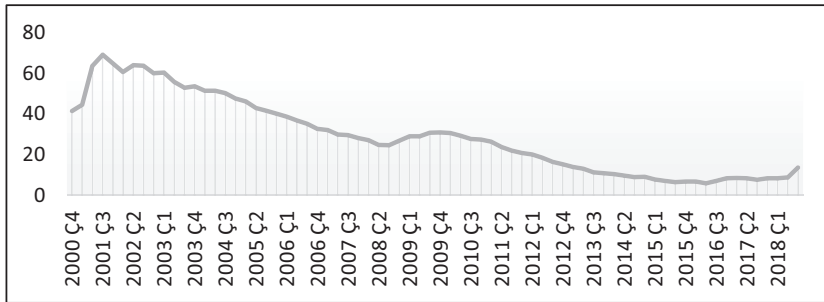
expansion, which has led to high inflation and high interest rates. The depreciation of the national currency because of inflation led to an increase in exchange rates (devaluation in the fixed exchange rate system), currency substitution, and deterioration in both resource allocation and income distribution, thus increasing economic fragility. This situation is clearly observed in 1994, 2000, and 2001 crises (Gülmez, Yardımcıoğlu, And Beşel, 2017).



Graph 1: Budget Deficit/GDP (%). Source: General Directorate of Budget and Fiscal Control – Main Economic Figures

As the problems arising from fiscal discipline have reached irreparable dimensions, financial discipline has been tried to be achieved through practices that increase the quality of public finance to reduce public debt and ensure its sustainability. One of the most fundamental elements of fiscal discipline is budget discipline. Economic problems, such as chronic inflation caused by ever-increasing budget deficits and the financing of these deficits by unsustainable financing methods, necessitated the reform requirement to ensure fiscal discipline (Karakurt and Akdemir, 2010: 228). With fiscal discipline, it is aimed to provide budget financing by tax revenues and debt at the “low cost and reasonable risk level” by reducing the public debt ratio to the lowest level possible. During the 1980s were the years of increasing fiscal deficit. In 1989, while the ratio of the budget deficit to GDP was 2.45 %, the inflation rate increased to 63 %. The 1990s also revealed bad consequences in terms of budget deficit and inflation. In 1991, while the ratio of budget deficit to GDP was 3.93 %, the inflation rate was 66 %. By the end of the 1990s, the ratio of the budget deficit to GDP was close to 10 % and the inflation rate was more than 100 %. In 2001, the ratio of the budget deficit to GDP increased to 12 %. Being considered that the rate of the budget deficit to GDP should not exceed to %3 according to Maastricht criteria, the size of fiscal indiscipline in Turkey could be envisioned (Graph 1).

Because of the worsened economic situation caused by fiscal indiscipline, a number of legal regulations and reforms aimed at improving the income and expenditure composition was performed in public finances in Turkey since the beginning of the 2000s. One of them is Public Financial Management and Control Law No. 5018. In line with the principles of fiscal transparency and accountability, this law introduced a performance management approach for effective, efficient, and economical use of public resources. With the reforms implemented, the ratio of the budget deficit to GDP declined steadily. The fact that the ratio declined from 12% in 2002 to 1.9% in 2018 is proof that fiscal discipline is achieved (Berkay, 2016:46, 49).



Graph 2: Public Net Debt Stock/GDP (%). Source: TR. Treasury and Finance Ministry, Public Finance Statistics

As can be seen in Graph 2, the fact that the rate of Public Net Debt Stock/GDP, which should not exceed 60 % according to the Maastricht criteria, is well below the Maastricht criteria is another indicator of strong fiscal discipline.

5 Conclusion

Monetary and fiscal policies could not be completely independent of each other in terms of their effect. There is an interaction or trade-off between them. Coordination ensured between policies is highly effective in the success of these policies. This study aims to examine the results of the policy interaction that is between a central bank and the treasury on the basis of price stability. For this purpose, firstly, the rationales for the coordination between monetary and fiscal policies were revealed, and then the basic regulations required to prevent the dominance of fiscal policy on the monetary policy were discussed. In the last chapter, institutional and legal developments in question that play an important

role in increasing the effectiveness of policies to achieve price stability via inflation targeting in Turkey are evaluated.

Turkey has made significant progress in terms of ensuring price stability and fiscal discipline since 2001. First of all, public finance has been disciplined to a great extent by providing “Public Debt Management Based on Risk Management”. In accordance with the Maastricht criteria, the ratio of public debt and fiscal deficit to the GDP was significantly reduced. In addition, with the amendment made in 2001, the independence of the CBRT was increased, and the fact that its the primary objective is price stability was ensured. On the other hand, parallel to Maastricht agreement the CBRT is prohibited from lending to the state and from buying the treasury instruments in the primary market.

Both theoretical explanations and provisions in practice reveal that policy coordination is both an operational and institutional need for monetary and financial stability as a whole. In this sense, Turkey recorded significant progress in monetary, fiscal, and debt management practices in adhering to national and international regulations. However, it is clear that there are still steps to be taken in order to ensure policy coordination more effectively. One of these steps should be arrangements made to eliminate the gap between real independence of the central bank and its legal independence. In addition, although the borrowing of the treasury from the central bank is prohibited, the communication between the central bank and the debt management authority should be strengthened, and the timing of the debt should be adjusted in line with the sensitivity required by the policy objectives.

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Salih Ozturk and Harun Demir

Costs to the Developing Country Economies of FED and ECB Monetary Policy Decisions (Turkey Example)

1 Introduction

The decisions taken by the monetary authorities of developed countries in today's world, where global markets are increasing in their interdependence, can cause various effects on developing countries. For example, an upward trend in the interest rates of a financially developed economy may have an impact on the interest rate and inflation of developing countries. Because the increasing interest rates of developed markets (such as FED and ECB) is caused by the appreciation of the currencies such as the Dollar and Euro. Therefore, in terms of developing countries with floating exchange rates¹, this situation may cause an increase in input costs and thus, cost inflation can be seen in the economy. Especially in countries like Turkey, which has a significant share in the production process and the rise in prices of energy resources obtained through the foreign currency can be considered as the main indicator of this situation.

In some studies in the literature (Berk, 1998: 148–149; ECB, 2000: 43; Cengiz, 2009: 228; Arabacı and Baştürk, 2013: 19), it is indicated how the operation of various channels of the monetary transmission mechanism is realized. In the first two stages of monetary transmission mechanism, monetary policy decisions of the central bank affect the financial markets and the real sector in the long term. As a result of these two stages, are affected the production (output) and prices general level. The main focus of this study, show the effects of the FED and the ECB on monetary policy decisions and inflation in Turkey. The monetary policy decisions of the FED and ECB may cause some problems in terms of developing countries.

1 Failure to obtain unresolved banking sector corrective financial measures as a result of the problem, the current accounts and increasing political challenges has been forced to fluctuations the exchange rate in February 2001 in Turkey (Fischer, 2001: 19).

Economic agents, according to rational expectations², in inflation forecasts do not only look at the past inflation rates. They will also take into account the existing information on monetary and fiscal policies (Krugman and Wells, 2011: 481). Increased interest rates may cause losses for banks due to their long-term liabilities. Recently, the rise trend of world interest rates may cause to bankrupt the bank whose reserve structure is not strong. For this cause, banks may decide to narrow down the loans they have received by prioritizing Basel 3 criteria³, as they will act with a rational anticipation and anticipate this situation.

The changes in the monetary policy implemented by the central bank not only affect market interest rates, but also the financial positions of debtors. Thus, the tightening monetary policy may weaken the balance sheets of debtors (Bernanke and Gertler, 1995: 36). A monetary contraction leads to decrease in the net worth of firms for two reasons. The first is that the tight monetary policy causes interest rates to rise, leading to more expensive debts and a decrease in the cash flow of firms. Secondly, decreases in asset prices are observed with rising interest rates. As a result, the value of marketable collateral decreases, and these negative developments cause existing external financing to be under tighter conditions (Guender, 1998: 244).

Another case is that, as a result of Tobin's q theory, the market value of firms will be lower according to the capital cost due to contractionary monetary policy. Therefore, firms will not be able to buy or not to want buy new investment goods. Thus, investment expenditures will decrease (Mishkin, 1996: 6). This case means continuous presence of firms operating in more stringent financial conditions in the economy. In short, declines in the stock market may be observed as a result of monetary tightening on a global scale, and thus central banks of developing countries may have to raise the interest rate by harmonizing them with the world. In addition, central banks cannot ignore price stability, which is their main purpose. Thus, due to the lack of inflation at low levels in Turkey, interest rates may shown increase more than according to developed countries.⁴

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- 2 In 1961, John F. Muth ("Rational Expectations and The Theory of Price Movements") laid the foundations for the theory of rational expectations. Muth's idea was adapted into the macro-based theories led by Robert Lucas, about ten years later.
 - 3 In order to eliminate the shortcomings caused by the global crisis, regulatory changes were created called Basel 3. By regulation changes called Basel 3; increasing the durability of the banking system against financial and economic shocks, corporate governance and aims at achieve objectives such as the development of risk management practices (Cangürel et al., 2010: 1).
 - 4 Taylor (1993) has stated that such a rule has the desired properties and in particular, it will stabilize inflation when the coefficient of inflation difference exceeds unity. This

Another case is that SMEs have a share of more than 95 % in enterprises in many countries. Therefore, the problem of SMEs in a country and the continuation to its existence without risk concerns can provides contributes positively to the economy (Çınar et al., 2014: 96). Economic agents do not make profit maximization as in basic microeconomics, and they make stock value maximization. SMEs generally take decisions according to their confidence index and debt payment capacities. For such reasons, market conditions and monetary policy decisions should be appropriate for SMEs, and sustainable growth should be turned into an aim by providing confidence in the economy.

In this study, it is assumed that countries have effects on each other's financial sense as a result of globalization. Therefore, the monetary policy decisions implements of the Fed and the ECB may vary Turkey's monetary policy. Due to this case, effects on economic parameters can be observed in Turkish economy. For this purpose the CBRT, FED and ECB of short-term interest rates variables how it reflected in the prices in Turkey was analyzed by Eviews 9 package program. The aim here is to eliminate this deficiency due to the inadequacy of studies on this subject in the literature.

2 Literature Review

The studies in the literature are generally in the form of the relationship of the decisions taken by the countries' own monetary authorities with inflation (Granville and Mallick (2004), Işık and Acar (2006), Oktar and Dalyancı (2011), Yücememiş et al. (2015)). However, it is almost impossible in today's conditions for the central banks to choose their monetary policy without considering the decisions of other countries. In the literature, limited number of studies is seen based on the assumption that FED and ECB monetary policy decisions can be effective on countries (Edwards (2010), Goczek and Partyka (2018), Verga et al. (2018)). The studies in the literature in more detail are summarized in Tab. 1.

situation, which is known as the Taylor principle, emphasizes that the nominal interest rate should be increased more than the increase in inflation. In other words, inflation will be kept under control only when real interest rates rise in response to the rise in inflation (Mishkin, 2011: 7–8).

Tab. 1: Literature Summary

Author	Period	Country	Method	Finding
Granville and Mallick (2004)	1900–2000	United Kingdom	Johansen Cointegration Test	It was determined that there is a long-term relationship between inflation and nominal interest rate.
Işık and Acar (2006)	1987: Q1–2002:Q4	Turkey	VAR	In the study, variables such as inflation, exchange rate, money supply and interest rate are used. It is determined that the interest rate causes effects on all other variables. The variables that affect inflation most are exchange rate and interest rate.
Edwards (2010)	2000:04 2008:09 (Weekly)	Brazil, Chile, Colombia, Mexico, Indonesia, Korea, Malaysia, Philippines	GLS and GMM	The results of the analysis show that the changes in the rate of Federal Funds are strongly and rapidly transferred to interest rates in Latin American countries. This effect is the same in Asian countries in the long term, too. However, while the Latin American Fed responds immediately to changes in policy rates, the adjustment is gradual in Asian countries.
Oktar and Dalyancı (2011)	2003: M1–2011:M6	Turkey	Johansen Cointegration Test and Granger Causality Test	The change in inflation causes an effect in the short and long term same direction on the CBRT's policy rate. In addition, CBRT's policy rate affects inflation in the same direction in the long term. However, it was determined that the changes in the policy rate in the short term did not affect inflation.

Tab. 1: (continued)

Author	Period	Country	Method	Finding
Özmen and Koçak (2012)	1994: Q1–2011:Q4	Turkey	ARDL Bounds Test	There is a positive relationship between inflation and money supply in the short and long term. In the long run, it was determined that one unit increase in money supply caused an increase by 42 % in inflation.
Şahin and Karanfil (2015)	1980–2013	Turkey	Johansen Cointegration Test and Granger Causality Test	Long-term relationships were identified between money supply, real effective exchange rate, budget deficit and inflation variables. However, no causality relationship could be determined between inflation and money supply.
Yücememiş et al. (2015)	2003: M1–2014:M12	Turkey	Maki Multiple Structural Break Cointegration Analysis	It has been determined that the CBRT's interest rates decisions are effective on inflation. Moreover, it is proven to be a monetary policy instrument, that is, the determinant of inflation.
Hajek and Horvath (2018)	2001: M1–2016:M1	European Union, United States	GVAR	The ECB shadow policy rate decreases economic activity and, to some extent, also decreases prices in all non-Euro EU countries. Also, changes to the Fed shadow policy rate yield similar quantitative effects as found for Euro area monetary shocks.

(continued on next page)

Tab. 1: (continued)

Author	Period	Country	Method	Finding
Goczek and Partyka (2018)	2000: M1–2015:M9	EEA Countries	Westerlund Panel Cointegration Test	The findings of the study, which investigated the effect of ECB monetary policy on interest rates of EEA countries (the Czech Republic, Hungary, Norway, Poland, Romania, Sweden and the UK), revealed that domestic interest rates follow Euro interest rates.
Verga et al. (2018)	2000: M1–2016:M12	Euro Area and United States	Johansen Cointegration Test, Granger Causality Test, GMM	The result that interest rates are primarily driven by US rates confirm that the causality acts mainly from the USA to the Eurozone.

3 Empirical Analysis

3.1 Data and Methodology

In this study, short-term interest rates of Turkey, the USA and the EU have been tested regarding the impact on Turkey’s inflation. In the analysis, the monthly data covering the period 2005:M1–2018:M4 were used, and detailed information about the variables is presented in Tab. 2. In this study, short-term interest rates of Turkey, the USA and the EU are preferred. In order to show the effect of monetary policy decisions on the general level of prices, consumer price index for Turkish economy has been included in the model.

Because seasonal effects were observed in INF_TR variable, it was seasonally adjusted by Census X-13 method. The trends according to the time for the data sets used in the study are given in Fig. 1.

Short-term interest rates of Turkey, the USA and the EU (INT_TR, INT_US and INT_EU) seem to decline after 2008 (see. Fig. 1). The main reason is to minimize the effects of the global crisis, and the country stems from the purpose of reaching to their old growth levels. However, although the FED has increased its money supply constantly after the global crisis of 2008,⁵ it cannot be said

5 FED’s assets increased from \$ 850 billion in 2007 before the crisis to 4.5 trillion dollars in 2015. Similar increases occurred in BoJ and the ECB. M2 rose from \$ 7.4 trillion to \$ 12.3 trillion (Stiglitz, 2017: 13).

Tab. 2: Variables Used

Variables	Explanation	Source
INF_TR	Consumer Price Index (2003=100) [Annual Rate of Change (%)] (TURKEY)	Tiik
INT_TR	Short-Term Interest Rate (TURKEY)	Eurostat
INT_US	Short-Term Interest Rate (UNITED STATES)	OECD
INT_EU	Short-Term Interest Rate (EURO AREA (19 Countries))	OECD

to be very successful in reaching the growth which lost momentum (Stiglitz, 2017: 12–13). In this period, due to the monetary expansion on a global scale, interest rates showed a drop in Turkish economy, too. In another case, for reasons such as recently show an increase on inflation in Turkey (see. Fig. 1 INF_TR), FED's interest rate increase (see. Fig. 1 INT_US), short-term interest rates in Turkey is trending upwards. Considering these relationships between variables in this study, a model was established as follows:

$$\text{INF_TR}_t = \beta_0 + \beta_1 \text{INT_TR}_t + \beta_2 \text{INT_EU}_t + \beta_3 \text{INT_US}_t + \varepsilon_t \quad (1)$$

In Equation (1) the constant term β_0 , symbolize ε_t error terms. In the equation, the relationship between short-term interest rates and inflation is examined. Also in this model, have the global impact central banks of monetary policy decisions are intended to identify impact on inflation in Turkey.

3.1.1 ARDL Bounds Test Approach

The existence of a long-term relationship between the variables used in this study is evaluated within the framework of the ARDL approach suggested by Pesaran et al. (2001). Among the most preferred methods in the literature, such as Engle and Granger (1987), Johansen (1988) and Johansen-Juselius (1990), there are tests that require all variables in the model to be stationary ($I(0)$) or aware ($I(1)$) stationary (Pesaran et al., 2001: 289–290). The most important advantage of the ARDL test is that it prevents pre-test problems and can be applied regardless of whether the variables are at fixed/non-stationary levels (Baharumshah et al., 2009: 234). Considering the ARDL model, an equation can be established for the variables used in the study as follows:

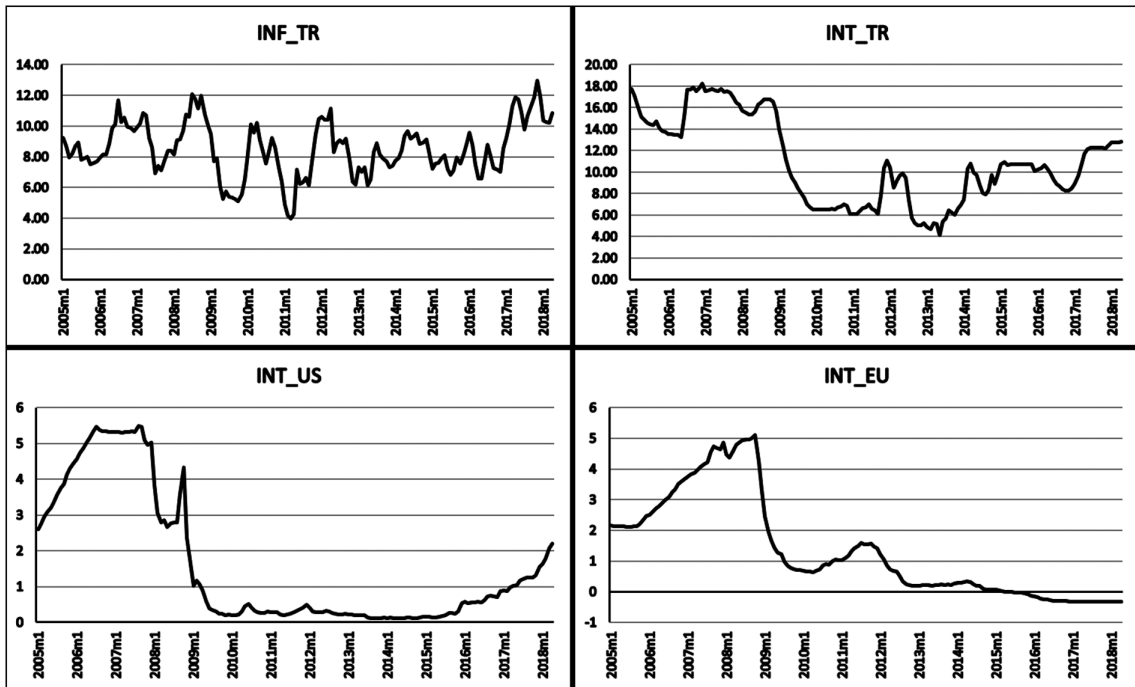


Fig. 1: Trends of Variables According to Time

$$\begin{aligned}
\Delta \text{INF_TR}_t = & \alpha_0 + \sum_{i=1}^p \alpha_{1i} \Delta \text{INF_TR}_{t-i} + \sum_{i=0}^q \alpha_{2i} \Delta \text{INT_TR}_{t-i} \\
& + \sum_{i=0}^q \alpha_{3i} \Delta \text{INT_EU}_{t-i} + \sum_{i=0}^q \alpha_{4i} \Delta \text{INT_US}_{t-i} \\
& + \beta_1 \text{INF_TR}_{t-1} + \beta_2 \text{INT_TR}_{t-1} + \beta_3 \text{INT_EU}_{t-1} + \beta_4 \text{INT_US}_{t-1} + \varepsilon_t
\end{aligned} \tag{2}$$

Equation (2) represents the model established within the framework of the ARDL approach and symbolizes the difference process applied to the series Δ , the constant term α_0 and the error term ε_t . In order to implement the ARDL approach to a model, first it is necessary to determine the model with the optimal lag length by means of information criteria by estimating alternative lag lengths. F value obtained from test results, compared with the asymptotic lower and upper limit values proposed by Pesaran et al. (2001) are determined whether a long term relationship between the variables. If the test statistic is above the upper critical value, the null hypothesis that there is no long-term relationship is rejected. Alternatively, if the test statistic is below the lower critical value the null hypothesis is accepted. Also, if the test statistic value falls between the lower and upper limits, the result is not precise (Atkins and Coe, 2002: 257).

3.1.2 Toda-Yamamoto Causality Analysis

In order to predict the time series, the series must be stable and must provide the assumption of series stability for statistical conclusions. However, it is more appropriate to use the Toda-Yamamoto test if in case the variables contain different levels of stability. Because Toda-Yamamoto (1995) stated that even if the series were not stationary, the VAR model with the level values of the series could be estimated and the Wald test could be applied. For this, k (lag length) and $d_{\max}$ (maximum cointegration level) should be determined first. Then by calculating $k + d_{\max}$, the causality can be estimated by VAR model and Wald test. Toda-Yamamoto (1995) causality analysis related series stationary, stationary around the trend or allows the analysis to be carried out whether they are cointegrated or not. In addition, Rambaldi and Doran (1996) have proven that this method can be used with SUR (Semiingly Unrelated Regression) technique.

$$\begin{pmatrix} \text{INF_TR}_t \\ \text{INT_TR}_t \\ \text{INT_EU}_t \\ \text{INT_US}_t \end{pmatrix} = A_0 + A_1 \begin{pmatrix} \text{INF_TR}_{t-1} \\ \text{INT_TR}_{t-1} \\ \text{INT_EU}_{t-1} \\ \text{INT_US}_{t-1} \end{pmatrix} + A_2 \begin{pmatrix} \text{INF_TR}_{t-2} \\ \text{INT_TR}_{t-2} \\ \text{INT_EU}_{t-2} \\ \text{INT_US}_{t-2} \end{pmatrix} \\
+ A_3 \begin{pmatrix} \text{INF_TR}_{t-3} \\ \text{INT_TR}_{t-3} \\ \text{INT_EU}_{t-3} \\ \text{INT_US}_{t-3} \end{pmatrix} + \begin{pmatrix} \mu_{1t} \\ \mu_{2t} \\ \mu_{3t} \\ \mu_{4t} \end{pmatrix} \quad (3)$$

In this study, the model for INF_TR, INT_TR, INT_EU and INT_US variables was determined as $k = 2$, $d_{\max} = 1$. Therefore, the 3 lagged VAR equation is established as in Equation (3). Finally, the presence of causality is determined by applying Wald test to the k lagged series.

3.2 Findings

In order to determine the stability levels of the variables used in the study, the ADF proposed by Dickey and Fuller (1981) and the PP unit root test proposed by Phillips and Perron (1988) were used. The results are presented in Tab. 3. For other variables other than the INF_TR variable, the test statistic values were determined to be less than the absolute value of the critical table. Therefore, it was found that the variables did not provide the stability condition in the INT_TR, INT_EU and INT_US level values. It was determined that the first degree of stagnation was achieved by applying difference procedure to these series.

Classical unit root tests do not take into account structural breaks due to fluctuations in the economy. As the period covered in the study includes the 2008 global crisis, the results of the structural breaking unit root test proposed by Vogelsang and Perron (1998) are given in Tab. 4. According to the test results, the breaking dates for the INF_TR and INT_TR variables were determined as 2008:M9 and 2008:M11, respectively. In addition, these variables were found to be stable at level values. Breaking dates for INT_EU and INT_US variables were determined as 2007:M9 and 2009:M6, respectively. However, it has been determined that these variables have unit root in the level values. Thus, the difference process was applied to these variables and the first rating stability condition was provided.

After determining the stability of the series, an appropriate lag length was determined from VAR. Akaike of the information criteria given in Tab. 5 shows

Tab. 3: Unit Root Test Results for Variables

Variables	ADF	PP
Panel A: Level		
INF_TR	-3.714282***	-3.530346***
INT_TR	-2.022918	-1.934199
INT_EU	-1.337792	-1.031349
INT_US	-1.142941	-1.074584
Panel B: The First Difference		
ΔINF_TR	-	-
ΔINT_TR	-7.731716***	-7.541946***
ΔINT_EU	-5.087791***	-5.184489***
ΔINT_US	-9.522431***	-9.502345***

Note: The results reflect constant model results. Values denote t-statistic value and *** indicates 1 % significance.

Tab. 4: Vogelsang-Perron AO Model Structural Break Unit Root Test Results

Variables			
Panel A: Level		Break Date	t-statistic
INF_TR	2008:M9	-4.937592 (1)*	I(0)
INT_TR	2008:M11	-4.914458 (1)*	I(0)
INT_EU	2007:M9	-3.684392 (6)	-
INT_US	2009:M6	-4.771026 (12)	-
Panel B: The First Difference			
ΔINF_TR	-	-	-
ΔINT_TR	-	-	-
ΔINT_EU	2008:M10	-7.908074 (3)***	I(1)
ΔINT_US	2009:M9	-10.82217 (0)***	I(1)

Note: The results show constant-trend model results. The lag length was determined according to the SIC. ***, ** and *, respectively, represent the significance level of 1 %, 5 % and 10 %.

that the appropriate lag length is three, while Schwarz and Hannan-Quin indicate that the appropriate lag is two. In the study, appropriate lag was accepted as two according to Schwarz information criterion.

Since the variables used in the study were not stationary at the same level, the cointegration relationship with the ARDL bounds test was analyzed, and

Tab. 5: Determination of Appropriate Lag Length

Lag	LogL	LR	FPE	AIC	SC	HQ
0	-1090.517	NA	31.16385	14.79077	14.87177	14.82368
1	-196.1137	1728.373	0.000218	2.920455	3.325483	3.085017
2	-117.9509	146.8192	9.42e-05	2.080417	2.809469*	2.376629*
3	-95.08712	41.71094	8.59e-05*	1.987664*	3.040738	2.415526
4	-89.40901	10.05178	9.90e-05	2.127149	3.504246	2.686660
5	-76.88268	21.49790	0.000104	2.174090	3.875211	2.865252
6	-57.79820	31.72149	0.000100	2.132408	4.157552	2.955219
7	-45.31245	20.07843	0.000106	2.179898	4.529064	3.134359
8	-26.70491	28.91713	0.000103	2.144661	4.817850	3.230772
9	-5.098135	32.41016*	9.71e-05	2.068894	5.066106	3.286654
10	12.70995	25.74952	9.63e-05	2.044460	5.365695	3.393870
11	27.70323	20.86903	9.96e-05	2.058064	5.703323	3.539124
12	38.45715	14.38699	0.000110	2.128957	6.098239	3.741667

Note: * indicates the most appropriate lag (INF_TR, INT_TR, INT_EU and INT_US).

the results obtained are presented in Tab. 6. Since the obtained F -statistic value exceeded the upper and lower critical values in 5 % significance, the existence of a cointegration relationship was determined between the variables. In addition, ARDL model diagnostic test results; the model shows that it is a suitable model, there is no error in the model installation, it has a normal distribution and It shows that the model does not include any changing variance and autocorrelation problem.

The results of Toda-Yamamoto causality analysis between the INF_TR, INT_TR, INT_EU and INT_US variables are presented in Tab. 7. Results reveal the existence of a causal relationship from short-term interest rates of the USA and the EU toward Turkey's inflation rate. In addition, on Turkey's short-term interest rate, Turkey's inflation rate has been found to be effective. Besides, the EU short-term interest rate has been determined to be effective on Turkey's short-term interest rates. The US short-term interest rate has been found to be effective on the EU short-term interest rate.

Generally, in developing economies, the effectiveness of the monetary policy may be weak due to the lack of financial depth, high inflation level and external dependence on inputs used in the production process. Therefore, the negative effects of an external shock occurring may be felt more in developing economies.

Tab. 6: ARDL Bounds Test Results

Structural Break Date 2008:M9		
Model	ARDL Lag Length	Calculated <i>F</i>-statistic
<i>F</i> (<i>INF_TR</i> , <i>INT_TR</i> , <i>INT_EU</i> , <i>INT_US</i>)	[2,1,0,0]	4.615689**
Pesaran et al. (2001) Critical Values: Unrest. Constant Model		
Significance level	Lower bounds, <i>I</i> (0)	Upper bounds, <i>I</i> (1)
1 %	4.29	5.61
5 %	3.23	4.35
10 %	2.72	3.77
Diagnostic Test Results	F-statistic (Prob.)	
R ²	0.802301	
F-statistic	86.96109 (0.0000)	
Durbin-Watson Statistic	2.066283	
Breusch-Godfrey LM Test	1.048071 (0.3532)	
Heteroskedasticity Test: ARCH	1.746525 (0.1778)	
Jarque-Bera Normality Test	0.043266 (0.978599)	
Ramsey RESET Test	0.189214 (0.6642)	

Note: ARDL bounds test results were determined according to SIC max. (2,2). ***, ** and *, respectively, represent the significance level of 1 %, 5 % and 10 %.

Tab. 7: Toda-Yamamoto Causality Test Results

Causality	Direction	k + dmax	Chi-square	Prob.	Causality
INT_TR	=>. INF_TR	3	1.639675	0.4405	No
INT_EU	=> INF_TR	3	7.145029	0.0281**	Yes
INT_US	=> INF_TR	3	4.888725	0.0868*	Yes
INF_TR	=> INT_TR	3	5.675125	0.0586*	Yes
INT_EU	=> INT_TR	3	5.301784	0.0706*	Yes
INT_US	=> INT_TR	3	0.467870	0.7914	No
INF_TR	=> INT_EU	3	3.552553	0.1693	No
INT_TR	=> INT_EU	3	0.280707	0.8691	No
INT_US	=> INT_EU	3	17.82448	0.0001***	Yes
INF_TR	=> INT_US	3	3.077150	0.2147	No
INT_TR	=> INT_US	3	0.051211	0.9747	No
INT_EU	=> INT_US	3	1.373819	0.5031	No

Note: The causality analysis was determined using the SUR technique. => indicates the direction of causality. ***, ** and *, respectively, represent the significance level of 1 %, 5 % and 10 %.

4 Conclusion and Evaluation

As a result of the globalization of financial markets, the monetary policy decisions of the central banks of countries may cause effects on each other. Considering this situation in the study, for the period 2005:M1–2018:M4, interactions between FED, ECB and CBRT's monetary policy decisions and these parameters' effects were investigated on inflation of Turkish economy. In the analysis part of the study, ARDL bounds test was used to determine the long-term relationship between the series. In addition, the relationship between variables was investigated with Toda-Yamamoto causality test.

The results obtained in the study prove the existence of long-term relationship between the variables used. In addition, results reveals the existence of a causal relationship from the USA and the EU short-term interest rates to Turkey's inflation rate. So, among the causes of inflation in Turkish economy has concluded that the monetary policy decisions of developed country economies is effective. This case can be caused by many different reasons. In particular, the US and the EU monetary policy decisions may have an effect on the Dollar and the Euro. Many of the inputs used in the production process in Turkey are provided in exchange for foreign currencies. Therefore, in terms of cost, a change in the Dollar and Euro may cause effects on prices. In addition, many household goods are supplied in foreign currencies, such as the Dollar/Euro, and this may have an impact on prices.

Another result is that the EU's short-term interest rate is effective on Turkey's short-term interest rate. In addition, the short-term interest rate of the USA has been determined to be effective on the short-term interest rate of the EU. This case, the monetary policy decisions of the developed economies directly or indirectly reveals that effective over monetary policy decisions taken in Turkish economy. In addition, Turkey's inflation has been found to be effective on Turkey's short-term interest rate. Thus, the FED and the ECB's monetary policy decisions are effective on both Turkey's inflation and the short-term interest rate. So, the developed economies monetary policy just as increase CBRT's the impact on monetary policy decisions, also is influential on the economic parameters in the economy of Turkey.

Finally, economic agents take into account elements such as country sophistication and price stability, while making the decision to invest. In addition, they take into consideration the economic variables such as short-term interest rates and inflation rate. In addition, creating expectations, they take into consideration the economic variables such as short-term interest rates and inflation rate. They do not ignore the risk factor when creating their expectations. Therefore, the interest rate policy of a developed market (such as the US economy and the

Eurozone) will be important for investors. Because, reliability is important for economic units, and they consider a risk-free developed economy as a reliable port. On the contrary, borrowing/funding can be provided at a higher cost in a country that is at a risky and high inflation level. In order to minimize these problems or to produce positive results in the national economy, financial markets should be deepened and the risk factor should be minimized. In addition, productive, high quality and production of high added value products should be ensured in the economy, the share of educated and experienced population symbolizing human capital should be increased in the country. Thus, the country's economy will be more resistant to external shocks.

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Hicran Kasa

The Relationship Between R&D Expenditures, High-Technology Export and Economic Growth: The Case of OECD Countries

1 Introduction

After the French Revolution took place in 1789, structural change from the feudalism to the capitalism and from the social system to the capitalist system was called the transition to modernism. The study of the production, distribution and consumption of the wealth that existed with the transition to modernism revealed that the science of economics and capitalism were laid in this period. In Europe, beginning with the second half of the 18th century and extending into the 19th century, scientific methods and rational thinking techniques have been developed, and new discoveries and technological advances have been introduced. The abandonment of traditional production vehicles and the discovery of steam-powered machines have led to a significant increase in capital accumulation in Europe as a turning point in achieving increased production.

The mentioned process in 18th century is called “Industrial Revolution”. The concept of globalization emerged in conjunction with the conclusions of the Industrial Revolution can be described as “the expansion of markets and the accumulation of capital, along with the growth of production, and the fluidity of labor, the triggering of cross-border trade and the realization of the integration process”. Some of the other important consequences of the Industrial Revolution, beginning with the effects in Europe and covering the whole world, can be summarized as follows: the increase in the level of social welfare with the increase in production, the increase in the migration from the villages to the cities, the emergence of the concept of unemployment by the emergence of the industrial cities, the decrease in the mortality rate and the increase of the population with the improvement of the living conditions partly and the problem of not finding the market for the sale of raw materials and processed goods. All these consequences have triggered colonialism and the birth of World War II.

Globalization, defined as the interaction of many economic, cultural, social and political activities with each other in the international arena through individuals and markets, especially with technological connections, without recognizing regional borders, has gained momentum after World War II. After this

period, the destructions created by World War II began to be eliminated in the Western European countries, and prosperity levels of the countries have been examined in depth while the economic balances are changing.

When it comes to the 20th century, it is seen that the industrialization policies supported by technological developments are inadequate for the countries to achieve sustainable economic growth, and for this reason many countries had undergone a transition from the industrial society to the information society.

In the study "The social, economic and technological dimensions of the new economy in the industrial society", Bayraç (2003) stated that as a result of innovations in information and communication technologies, the obvious and lasting effects of the economy necessitated the redefinition of many concepts at micro- and macro-level. While this situation led to a mass consciousness, the types of goods and services demanded have been differentiated, and therefore the types of economic activities have been changed too.

The reasons for failing to achieve sustainable economic growth of undeveloped and developing countries are lagging behind global competition, unable to produce policies for innovation and having difficulties in making investments for innovation. Because of these reasons, the countries lag behind information and communication technologies. In order to achieve long-term sustainable economic growth, the increase of R&D expenditure plays an important role in the increase of innovation and high-tech product production. Thus, economic growth has been accelerating. According to the studies, the resources used for the production of high-tech, high value-added products have played an active role in economic growth and led to a steady increase. Countries such as South Korea and Taiwan, where R&D spending in GDP is much higher than other countries, have started to leap to the status of High-Income Economies (Advanced Economies), which is over USD 12,476 per capita national income. The GDP ratio of Turkey's R&D spending in 2017 was 0.96 %. However, considering the developing countries including Turkey, for many years the resources allocated to R&D are observed to be almost at the same level. This rate is much lower than many countries that have achieved sustained growth.

Innovative policies are the result of technological and managerial developments, the most fundamental dynamics of globalization ideology and global competition. Therefore, the need to be able to compete in the international arena and become a promise making innovative politics are inevitable. The current technology capacity and the ability to demonstrate technological innovation in terms of countries trying to survive in the global world are the most important elements in bringing competitive advantages in global competition and foreign trade. Therefore, in this study, the relationship between R&D expenditures,

high-tech product exports and sustainable growth has been examined. Firstly, the literature related to the subject is reviewed and then comparisons are made among OECD countries.

2 Theoretical and Empirical Framework

It is observed that there is no consensus on econometric models where the relationship between R&D, high-technology product exports and growth is analyzed and the methods used can differentiate. Panel data analysis is generally used in studies. Thus, in cases where it is not possible to study with only horizontal series or time series, panel data analysis is used to obtain results from many countries and observation periods (Baltagi, B., 2008). However, although different analysis methods are used in different country groups, the results obtained are similar. It is observed that R&D expenditures have a positive effect on economic growth and high-technology product export. This section, which examines the relevant literature, is summarized in Tab. 1. This table is prepared according to the publication year of the study, the model used, authors, range of the years studied and results obtained.

Technology, knowledge and technological level are abstract concepts due to their nature. Therefore, direct measuring of technology and comparison by reducing it to units is possible only through the concepts that represent it. For this purpose, many new growth theories use data such as number of engineers and scientists employed in R&D activities, number of patents and R&D expenditures to represent technology and knowledge. As a result of literature review, it has been observed that a panel data analysis is used throughout the articles on the subject. In each study on the subject, it was observed that similar results were obtained despite the use of different economic associations among member countries. The results obtained support that there is a positive relationship between R&D expenditures, high-technology product exports and economic growth.

3 Comparison of Variables on OECD Countries

The budget allocated to R&D, which are defined as increasing the knowledge of culture, the human being and society, and obtaining creative works using this accumulation to create new processes, applications and systems, shows the competitiveness of countries in science and technology. This budget or R&D expenditures, which are considered as the most important indicator supporting innovation, are very important in terms of showing the ratio of human resources

Tab. 1: Literature Review

Year of Publication	Author/Model	Term/Countries	Results
2003	Gavin Cameron, James Proudman, Stephen Redding/Panel Data Analysis	1971–1992/14 countries in the United Kingdom	The increase in R&D expenditures leads to an increase in technology transfer rate in international trade.
2004	Rachel Griffith, Stephen Redding, and John Van Reenen/Panel Data Analysis	1971–1990/ Canada, Denmark, Finland, France, Germany, Italy, Japan, Norway, Sweden, United Kingdom, United States	In this article, the authors tried mapping the effect of R&D to two variables. As a result, R&D is statistically and economically important in both technological catch-up and innovation. In addition to this result, human capital also effect of productivity growth as positively and effect very small to trade.
2009	Mustafa Özer, Necati Çiftçi/ Panel Data Analysis	1993–2005/OECD Countries	The aim of the study is to investigate the relationship between R&D expenditures and general exports, information and communication technologies export and high-technology export. In the analysis conducted using panel data technique, there is a positive and high correlation between R&D and exports for OECD countries.
2013	İsmet Göçer/ Panel Data Analysis	1996–2016/11 Asian countries	The effects of R&D expenditures for 11 Asian countries on exports of high-tech products, export of information communication technologies, total exports and economic growth were examined. As a result, it is determined that R&D expenditures have a statistically positive effect on the mentioned variables.

Tab. 1: (continued)

Year of Publication	Author/Model	Term/Countries	Results
2012	Ertuğrul Yıldırım, Ferdi Kesikoğlu/ GMM and Wald Tests	1996–2008/Turkey	The causality relationships between R&D expenditures and exports were investigated using the panel data set covering the 1996–2008 period and 25 sub-sectors. One-way causality relationship has been found from R&D expenditures to exports by using GMM-system estimation and Wald test-based causality analysis. This result has shown that R&D policies can be an important factor in increasing exports.
2014	Ahmet Şahbaz, Rüstem Yanar, Uğur Adıgüzel/ Cointegration and Panel Causality Tests	1996–2011/17 Europe Countries	The relationship between R&D expenditures and export of advanced technology goods was tested. According to the results, it was found that there is bi-directional Granger causality between R&D expenditures and export of high technology.
2014	Cüneyt Kılıç, Yılmaz Bayar, Halil Özekicioğlu/ Panel Data Analysis	1996–2011/G-8 countries	Findings of the study is that R&D expenditures and real effective exchange rate have a positive effect on high-technology product exports. In addition, there is a two-way causality between R&D expenditures and high-tech product exports and real effective exchange rate while there is a one-way causality from high-tech product exports to real effective exchange rate.
2016	Onur SUNGUR, Halil AYDIN, Mehmet EREN/ Time Series Analysis	1990–2013/Turkey	The three results obtained by Granger Causality Test are as follows: first one-way causality relation from number of patents to growth and second one-way causality relation from export to R&D expenditures were found. Lastly according to Hatemi-J asymmetric causality analysis, in Model 1, positive components towards growth from patent to growth, negative components from growth to patent and one-way relationship between R&D and growth towards negative components were determined.

(continued on next page)

Tab. 1: (continued)

Year of Publication	Author/Model	Term/Countries	Results
2016	Rahmi Çetin/ Panel Data Analysis	1996–2013/ Mexico, Brazil, Thailand, Malaysia, China, South Africa and Turkey	The study examines the impact of R&D expenditures on high-technology exports during the period 1996–2013 by using panel data analysis. According to the results of the analysis, R&D expenditures had positive impact on high-technology exports. Moreover, there was one-way causality relationship running from R&D expenditures to high-technology exports.
2017	Gökçen Özkan, Hüseyin Yılmaz/Panel Data Analysis	1996–2015/12 member countries of the European Union and Turkey	It is concluded that R&D expenditures affect high-technology product exports and GDP positively.
2017	Ümit Yıldız/ Panel Fixed Effects and Panel	2005–2014/BRICS countries and Turkey	Each common result of two forecasting methods is that increase of high-tech exports affect BRICS countries and Turkey's economic growth positively.
2018	Bahadır Ayar, Tayyip Sabri Erdil	Turkey	The effect of innovation and R&D activities on export performance have been examined. The study's data was collected from 313 exporting enterprises operating in Turkey with CATI method. As a result of data analysis, a significant and strong relationship was found between R&D and innovation activities.

employed in the field of information and technology. Innovation, which contributes to economic growth by increasing productivity and competitiveness worldwide, is seen as an important factor in shaping economic activities (Şahinli and Kılınç, 2013, s. 329–356). While the number of patents is an indicator of innovation, it increases the production of high value-added goods and services and thus affects the exports of high-technology products of countries. Countries that use technology effectively in production increase the export of high technology, and thus welfare of the countries with high R&D expenditures is also

Tab. 2: High-Technology Exports (%of Manufactured Exports)

Country	1995	2000	2005	2010	2015	2016	2017
Turkey	1.21	4.83	1.47	1.93	2.16	2.03	2.53
Korea, Rep.	26.03	35.07	32.48	29.47	26.84	26.58	14.18
Japan	26.55	28.69	22.98	17.97	16.78	16.22	13.8
China	10.43	18.98	30.84	27.51	25.65	25.24	23.81
Germany	13.71	18.63	17.42	15.25	16.66	16.91	13.66
USA	30.3	33.72	32.74	19.97	18.99	19.96	13.81
Israel	16.02	19.35	14.03	14.66	19.66	18.38	13.03
Brazil	4.89	18.73	12.84	11.22	12.31	13.45	12.28
Greece	5.81	13.75	10.58	10.1	10.99	11.44	10.35
Chile	3.34	3.41	6.94	5.73	6.1	6.95	6.08
Mexico	15.17	22.45	19.64	16.94	14.69	15.29	15.16
Poland	2.58	3.36	3.79	6.69	8.78	8.46	7.73
High-income countries	21	25.49	22.72	17.67	18.58	17.81	16.56
Middle-income countries	13.86	19.88	19.11	17.16	18.62	18.19	16.65

Note: These data were obtained from the World Bank database.

increasing. According to the World Bank, definition of high-technology exports is the export of products where R&D spending is used at the highest level. These products are generally manufactured according to industry 4.0 requirements; it consists of technological goods such as artificial intelligence, cyber security, modeling, cloud computing, augmented reality and 3D software.

The high-technology export data that is shown in Tab. 2, compiled from the World Bank database, observes that the technological weight of the exports of high-income countries was higher than that of middle-income countries.

Based on Tab. 2, the countries with the highest technology intensity are Korea, China, USA, Israel, Germany and Japan. The proportion of high-tech products exported by these countries as of 2016 exceeds 16 % on average. Countries where the majority of exported goods are composed from low value-added goods are Poland, Chile, Greece, Brazil and Turkey. Compared with other countries in Tab. 2, it has been seen that Turkey's low R&D expenditure does not support the production of high-technology goods and services adequately; thus, its high-tech exports remained at 2.3 %. This percentage is the lowest level among sample

countries. Although struggling with a middle-income trap like Turkey, Mexico's technology exports are about six times higher than from Turkey.

The basis of the innovation ecosystem is based on research and development activities supported by companies, universities and policy-makers (Acemoglu et al., 2016, s. 52–104). The developing countries can achieve innovation in two ways: the first one is to import from technology producing countries and the second through supporting national R&D activities with domestic savings (Korkmaz, S. 2010, s. 3320–3330). However, since technology imports will have a negative impact on the growth rates of developing countries, the ability of these countries to produce and export technology is important in increasing growth rates. Therefore, the ratio and efficient use of national income allocated to R&D expenditures play a key role in increasing the growth rates of countries. In Tab. 3, ratios of R&D expenditures in GDP have been shown in order to compare developed and developing countries.

Tab. 3: Ratio of R&D Expenditures in GDP (%GDP)

Yıllar	OECD	Turkey	Mexico	Estonia	Russia	Korea	Japan	Finland	USA
2000	2.12	0.47	0.33	0.60	0.98	2.18	2.91	3.25	2.621
2001	2.16	0.53	0.35	0.70	1.09	2.34	2.97	3.20	2.638
2002	2.14	0.51	0.39	0.72	1.16	2.27	3.01	3.26	2.550
2003	2.14	0.47	0.39	0.77	1.19	2.35	3.04	3.30	2.553
2004	2.11	0.50	0.39	0.85	1.07	2.53	3.03	3.31	2.490
2005	2.14	0.57	0.40	0.92	0.99	2.63	3.18	3.33	2.506
2006	2.17	0.56	0.37	1.12	1.00	2.83	3.28	3.34	2.550
2007	2.21	0.69	0.43	1.07	1.04	3.00	3.34	3.35	2.627
2008	2.28	0.69	0.47	1.26	0.97	3.12	3.34	3.55	2.767
2009	2.33	0.81	0.52	1.40	1.16	3.29	3.23	3.75	2.819
2010	2.29	0.80	0.54	1.58	1.05	3.47	3.14	3.73	2.740
2011	2.32	0.80	0.52	2.31	1.01	3.74	3.24	3.64	2.770
2012	2.31	0.83	0.49	2.12	1.03	4.03	3.21	3.42	2.689
2013	2.34	0.82	0.50	1.72	1.03	4.15	3.31	3.29	2.725
2014	2.36	0.86	0.54	1.45	1.07	4.29	3.40	3.17	2.734
2015	2.36	0.88	0.53	1.49	1.10	4.22	3.28	2.90	2.740
2016	2.35	0.945	0.50	1.28	1.10	4.24	3.14	2.75	2.744
2017	2.36	0.961	–	1.28	1.109	4.55	3.20	2.75	2.78

Note: OECD (2018), Gross domestic spending on R&D (indicator).

When the R&D expenditures of the selected developed and developing countries are examined by the year 2000, it is observed that there is a 0.23 % change in the average R&D expenditures of OECD countries. South Korea is the country that has highest R&D expenditures and Japan, Finland and the USA follow it respectively. Mexico is the country that has the lowest R&D expenditures and Turkey, Russia and Estonia are to follow it respectively. Turkey's R&D expenditure increased by 0.46 % for the years 2010–2016. Although this rate is well below the developed countries, it is also an indication of increasing importance of R&D in our country.

When the literature on the economic growth is examined, it is understood that R&D expenditures, high value-added product production, increasing total factor productivity and increasing high-technology exports are most important factors affecting sustainable economic growth today.

4 Conclusion

The Industrial Revolution in which technology and scientific methods are used in production is seen as the starting point of industrialization. The changing world trade volume with the Industrial Revolution has effected economic balances and economic systems of the world. In the same period, concept of economic growth has begun to be discussed among less developed and developing countries. Today, it is still one of the most important goal of the countries to increase economic prosperity. In this study, the main purpose is the effect of technological developments on economic growth. However, it is not possible to measure technology, information and technological developments due to abstract concepts. Therefore, the role of high-technology exports and R&D expenditures in increasing economic growth has been investigated. In the future, where industry 4.0 and artificial intelligence will prevail, technology, innovation and R&D activities would be among the most important factors affecting economic growth.

In the survey conducted for Turkey and other OECD countries, it was observed that similar methods were used for measurement of high-technology export and R&D expenditure effects on growth. The obtained results are also similar. Although positive results were found in parallel with expectations, it would be appropriate to select methods which give results for each country separately. Therefore, new generation panel data model, which is called Dynamic CCEGM (cross-correlated effects mean-group estimator) model, takes into account both cross-sectional dependence and slope heterogeneity, and can be used for future studies. In this method, general coefficients can be obtained for each

panel, and coefficient values will be found separately for each country. Thus, the behaviors and differences of the countries in the panel can be analyzed and results can be closer to reality.

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Fusun Celebi Boz

The Relationship Between Political Stability, Fixed Capital, Employment, and Economic Growth: An Analysis on MENA Countries

1 Introduction

As the growth theories before 1980 fell short in explaining the developmental differences in countries and could not provide a solution to the problems of underdeveloped countries, a necessity for new growth models emerged. In connection to the exogenous nature of technology, the perception that all countries can achieve economic growth depending on capital stock and employment volume was in time replaced by the effectiveness of institutional and social factors in this process, and it was seen that other factors rather than economic factors would be influential in terms of achieving economic growth. In the scope of institutional factors, the political regimes of countries were aimed to be analyzed, and accordingly, strong institutional structures were associated with democracy. Thus, it may be stated that, while democratic regimes are dominant in developed societies, antidemocratic regimes are encountered in developing societies. In this context, the type of the political regime is considered to be the main variable for economic growth. As the type of the political regime was inadequate in completely explaining economic growth, the concept of political stability started to be considered in the literature (Telatar, 2003:74–75).

The concept of political stability is important in terms of determination of economic policies in a country for the long-term rather than the short-term and for rational decisions by managers in the private sector. Moreover, political instability may lead to property issues by bringing about weaknesses in the legal system. This situation may affect the marginal effectiveness of capital and income to be gained from investments. In addition to these issues, political instability may cause shortened terms for debts and increased interest costs due to increased uncertainty towards the future. As a result of increased borrowing costs in the private sector, investments decrease, and economic growth slows down. Another issue that is affected by political instability is the exodus of the capital in the domestic field to outside. As capital assets and borrowing costs increase due to the leaving capital, the economic development of the country is affected negatively. With intrinsic growth model, human capital is increasingly

becoming more important. However, as a result of trained labor migrating to different regions in countries without political stability, the production volume of the country will decrease, while the quality-related development of the society will be disrupted (Şanlısoy, 2010:198–199).

Analysis of the relationship between political stability and economic growth is important in terms of being able to explain the developmental differences among countries. In particular, in all political regimes, political stability leads to accelerated economic growth and higher levels of prosperity in the society due to reduced uncertainties. This study investigated the relationship between political stability and economic growth in MENA countries. Additionally, the effectiveness of the model was aimed to be increased by including explanatory variables such as fixed capital and employment in the model. After the introduction section, the second section contains a literature review regarding the concept of political stability, the third section explains the dataset and methodology of the study and discusses the effects of political stability on economic growth, and the last section makes general comments and concludes the article.

2 Literature Review

While some studies on the relationship between political stability and economic growth reported statistically significant positive or negative relationships, some others did not find a significant relationship. Studies that became more abundant after the 1990s are summarized in Tab. 1. This study is original in terms of considering the Political Stability Index by the World Bank as an indicator of political stability and examining the effects of the fixed capital and employment variables in addition to these variables by including them in the model. The study also aimed to assess the period of 1996–2017 by using different econometric methods.

3 Analysis

3.1 Dataset

This study included the logarithmic forms of the data of eight MENA countries¹ for the period of 1996–2017 on economic growth (GDP per capita constant

1 Algeria, Egypt, Jordan, Iran, Morocco, Oman, Tunisia, and Saudi Arabia. As there were no sufficient data on other MENA countries regarding the series, these countries were selected. The real fixed capital in Saudi Arabia, Oman, and Tunisia were calculated and included in the model.

Tab. 1: Literature Summary

Authors	Countries	Data Period	Method	Conclusion
Alesina et al. (1996)	113 countries	1950–1982	GLS	Political instability has a negative effect on economic growth.
Feng (1997)	96 countries	1960–1980	Three-Stage Least Squares	Economic growth has a negative effect on regime change.
Asteriou and Price (2000)	The United Kingdom	1961–1997	LS and GARCH	Political instability has a negative effect on economic growth.
Fosu (2001)	Sub-Saharan African Countries	1960–1986	Principal Components and LS	Political instability has a negative effect on economic growth based on the marginal productivity of capital.
Fielding (2003)	Israel	1988–1998	Johansen Cointegration Test	Political instability has a negative effect on investments.
Jong-A-Pin (2006)	98 countries	1984–2003	Panel Data Analysis	Political instability has a negative effect on economic growth.
Aisen and Veiga (2006)	100 countries	1960–1999	Panel Data Analysis	Political instability causes high inflation.
Tosun et al. (2008)	MENA	1987–2003	Factor Analysis	There is a negative relationship between macroeconomic variables and political risks.
Arslan (2011)	Turkey	1987–2007	Johansen Cointegration Test	There is a one-way causality relationship from Gross Domestic Product towards political instability.
Gür and Akbulut (2012)	19 developing countries	1986–2003	Panel Data Analysis	While there are positive relationships between political instability and economic growth in most Asian countries, there is an opposite relationship in Latin American countries.
Tang and Abosedra (2014)	24 countries	2001–2009	GMM	Political instability affects tourism activities and growth negatively.

(continued on next page)

Tab. 1: (continued)

Authors	Countries	Data Period	Method	Conclusion
Kalay and Çetin (2016)	52 African countries	2000–2011	Panel Data Analysis	There is a one-way relationship from economic growth towards political instability.
Kartal and Öztürk (2017)	Turkey	1946–2016	Principal Components Analysis	Economic growth was affected negatively in the periods with high political instability in Turkey.
Yalçinkaya and Kaya (2017)	20 largest economies	1996–2015	Panel Data Analysis	Political stability is more effective in comparison to the degree of integration of physical and human capital with the outer world in terms of determining the developmental differences among countries.
Akkuş (2017)	33 countries	1994–2013	GMM	Political uncertainty in the USA and three different dimensions of political instability in developing countries are strongly and negatively effective on growth in developing countries.
Alper (2018)	BRICS-T	1996–2016	Panel Data Analysis	Political stability has a positive effect on economic growth.
Al and Belke (2018)	MENA	1991–2016	Panel Data Analysis	There are one-way and two-way causality relationships from the Internal and External Conflict Index towards economic growth except for Iran and Israel.

2010 \$), political stability (percentile rank²), gross fixed capital formation (constant 2010 \$), investments, and employment (thousand) and estimated the relationships among these by panel data analysis. In the estimation of the panel data model, the model below was utilized to investigate the relationship

2 Percentile rank among all countries (ranges from 0 (lowest) to 100 (highest) rank).

among economic growth, political instability, employment, and gross fixed capital formation:

$$lnkgdp_{it} = \alpha_i + \beta_1 lnpsc_{it} + \beta_2 lncl_{it} + \beta_3 lngfcf_{it} + u_{it} \quad (1)$$

In the model above, $lnkgdp_{it}$, the per capita real GDP of country i for the period of t was the dependent variable, while the independent variables of the model were $lnpsc_{it}$, the political stability of the country i for the period of t , $lncl_{it}$, the employed labor force of the country i in the period of t , and $lngfcf_{it}$, gross fixed capital formation of the country i in the period of t . The data on per capita real GDP and gross fixed capital formation were derived from the database of the World Bank, while the data on political stability were obtained from the Global Governance Indicators of the World Bank, and the data on employment were collected from the Conference Board (Total Economy Database).

3.2 Method

In the study, firstly the dependence among the cross-sections that formed the panel (countries) was investigated by the method of LMadj (Adjusted Lagrange Multiplier) which was developed by Breusch-Pagan (1980) and adjusted for deviation by Pesaran et al. (2008). The stationarity of the series was calculated by using the Smith Bootstrap panel unit root test that considers cross-sectional dependence. The homogeneity of the cointegration coefficients was analyzed by the Slope Homogeneity Test proposed by Pesaran and Yamagata (2008). Moreover, the presence of cointegration relationships among the series was analyzed by the Westerlund (2007) ECM bootstrap test which considers cross-sectional dependence. As a result of not finding a cointegration relationship, the causality relationship among the series was analyzed by the panel causality relationship test developed by Kónya (2006). The analyses in this study were carried out by using the Gauss 10.0 software and the codes that were developed for this software.

3.3 Checking for Cross-Sectional Dependence

Before testing the presence of cointegration among the series, cross-sectional dependence needs to be assessed. Analyses that are carried out without assessing cross-sectional dependence affect the results substantially (Breusch and Pagan, 1980; Pesaran 2004). Selection of the unit root and cointegration tests to be carried out requires consideration of cross-sectional dependence (Nazlıoğlu et al., 2011:6618). The presence of cross-sectional dependence among the series is tested by the Breusch Pagan (1980) CDLM1 test when $T > N$, meaning that the

Tab. 2: Cross-Sectional Dependence CDLM Test Results

Tests/Variables	CDLM1 (Breusch, Pagan 1980)	CDLM2 (Pesaran 2004 CDLM)	CDLM3 (Pesaran 2004)	LMadj (Pesaran et al. 2008)
LNKGDP	49.220(0.008)	2.836(0.002)	-2.006(0.022)	19.900(0.000)
LNGFCF	59.339(0.000)	4.188(0.000)	-2.854(0.002)	21.314(0.000)
LNPSE	45.860(0.018)	2.387(0.009)	-2.151(0.016)	11.752(0.000)
LNEL	84.303(0.000)	7.524(0.000)	-0.761(0.223)	18.296(0.000)
MODEL	68.790(0.000)	5.451(0.000)	-0.133(0.447)	14.085(0.000)

*The CDLM1, CDLM2, CDLM3, and LMadj test results on the series are for the model that contains both constant and trend-related data. According to %5 statistical significance levels, there was cross-sectional dependence.

time dimension is greater than the cross-section dimension; with the CDLM test when $T < N$, meaning that the time dimension is smaller than the cross-section dimension; and with the CDLM2 test when $T = N$, meaning that the time dimension and the cross-section dimension are equal (Pesaran, 2004:4). As this study included 8 countries ($N=8$) and 22 periods ($T=22$), the CDLM1 test was utilized. The Breusch Pagan test provides results with deviations in the case that the group mean is zero, but the individual means are non-zero. Pesaran (2008) adjusted this deviation by adding the variance and mean to the test statistic. The adjusted form of the test is LMadj instead of CDLM (Pesaran et al., 2008:109). The initial form of the CDLM test is as shown below.

The test statistics and probability values that are needed to test these hypotheses are calculated by using bootstrap. In this study, the presence of cross-sectional dependence in the variables and cointegration equation was analyzed by the CDLM test, and the results are given in Tab. 2.

3.4 Panel Unit Root Test

At the next stage of the study after examination of cross-sectional dependence, the stationarity levels of the variables were analyzed. The presence of cross-sectional dependence in the variables requires using second-generation panel unit root tests. So, the unit root test developed by Smith, Laybourne, Kim, and Newbold (2004) was used to determine the stationarity levels of the variables. In this test based on bootstrap, five statistics such as the IPS (t), Max, LM, Min.LM, and WS statistics are calculated.

Tab. 3 shows the unit root tests of the variables in the study for the constant and constant-trend models. According to the results, as the null hypothesis is

Tab. 3: Smith et al. (2004) Unit Root Test Results

	Constant Model		Constant-Trend Model	
LNKGDP				
Test Name	Level	First Difference	Level	First Difference
IPS Statistic (Probability)	-1.822(0.141)	-2.958(0.000)*	-1.405(0.946)	-3.425(0.000)*
Max Statistic (Probability)	-0.290(0.885)	-2.882(0.000)*	-1.262(0.794)	-3.000(0.000)*
LM Statistic (Probability)	3.903(0.214)	7.021(0.000)*	2.783(0.975)	-8.628(0.000)*
Min.LM Statistic (Probability)	2.767(0.119)	6.790(0.000)*	2.327(0.890)	7.484(0.000)*
WS Statistic (Probability)	-0.465(0.954)	-3.063(0.000)*	-1.775(0.877)	-3.478(0.000)*
LNGFCF				
	Constant Model		Constant-Trend Model	
IPS Statistic (Probability)	-0.931(0.872)	-3.004(0.000)*	-1.409(0.963)	-3.261(0.001)*
Max Statistic (Probability)	-0.420(0.870)	-2.834(0.000)*	-1.177(0.894)	-3.002(0.000)*
LM Statistic (Probability)	1.408(0.969)	7.612(0.000)*	3.329(0.923)	8.576(0.001)*
Min.LM Statistic (Probability)	0.685(0.968)	7.095(0.000)*	2.627(0.828)	7.764(0.000)*
WS Statistic (Probability)	-0.520(0.957)	-3.192(0.000)*	-1.811(0.863)	-3.410(0.000)*
LNPSE				
	Constant Model		Constant-Trend Model	
IPS Statistic (Probability)	-2.091(0.061)	-4.944(0.000)*	-1.707(0.853)	-5.309(0.000)*
Max Statistic (Probability)	-1.915(0.004*)	-4.822(0.000)*	-1.595(0.636)	-5.123(0.000)*
LM Statistic (Probability)	4.522(0.066)	11.755(0.000)*	3.523(0.869)	12.622(0.000)*
Min.LM Statistic (Probability)	3.980(0.005)	11.483(0.000)*	3.135(0.675)	12.243(0.000)*
WS Statistic (Probability)	-2.057(0.007)	-5.066(0.000)*	-2.081(0.570)	-5.475(0.000)*

(continued on next page)

Tab. 3: (continued)

LNEL				
	Constant Model		Constant-Trend Model	
IPS Statistic (Probability)	-1.939(0.122)	-4.330(0.007)*	-2.393(0.213)	4.524(0.007)*
Max Statistic (Probability)	-0.857(0.473)	4.280(0.000)*	-1.989(0.143)	-4.368(0.000)*
LM Statistic (Probability)	4.627(0.069)	9.957(0.000)*	5.562(0.252)	10.800(0.000)*
Min.LM Statistic (Probability)	3.583(0.015*)	9.792(0.000)*	4.378(0.171)	10.243(0.000)*
WS Statistic (Probability)	-0.741(0.838)	-4.624(0.000)*	-2.555(0.084)	-4.899(0.000)*

Note: *The stationarity of the series may be observed on the significance level of 5 %.

higher than 5 % for the first differences of all three variables for both models (constant and constant-trend models), the variables were found to be stationary for all variables, I(1).³

3.5 Testing the Homogeneity of the Cointegration Coefficients

The Delta Test which is used to determine whether or not the slope coefficient in the cointegration equation is homogenous was proposed in 1970 by Swamy and modified by Pesaran and Yamagata (2008). According to the test, in the cointegration equation in the following form:

Here, N refers to the number of cross-sections, S is the Swamy test statistic, k is the number of explanatory variables, and $\nu(T,K)$ represents the standard error. As seen in the homogeneity test results given in Tab. 4, because the probability values of the statistics that were calculated were smaller than 5 %, the H_0 hypothesis was rejected, and it was determined that the constant term and slope coefficients in the cointegration equation were heterogenous. Therefore,

3 The variable LNPSE was I(0) based on the Max statistic in the constant model and I(1) based on the IPS and LM statistics either the constant or constant-trend model. The variable LNEM was I (1) based on all statistics except for the Min.LM statistic.

Tab. 4: Homogeneity Test Results

	Test Statistic	Probability Value
$\check{\Delta}$ Test	12.432	0.000
$\check{\Delta}_{adj}$ Test	14.053	0.000

* Shows that the H_0 hypothesis was rejected on the level of 5 %.

it is possible to make interpretations about the countries in the panel based on these results.

3.6 Panel Cointegration Test

In this study, the presence of cointegration among the series was analyzed by the ECM bootstrap panel cointegration test developed by Westerlund (2007). As this test utilizes bootstrap probability values, it also considers cross-sectional dependence. In the Westerlund cointegration test, firstly the model below was estimated for being able to calculate the panel statistics:

$$\Delta Y_{it} = \delta_i d_t + \gamma_i x_{it-1} + \sum_{j=1}^{pi} \alpha_{ij} \Delta Y_{it-1} + \sum_{j=0}^{pi} \Delta x_{it-j} + e_t \quad (2)$$

$$Y_{it-1} = \delta_i d_t + \gamma_i x_{it-1} + \sum_{j=1}^{pi} \alpha_{ij} \Delta Y_{it-1} + \sum_{j=0}^{pi} \Delta x_{it-j} + \varepsilon_t \quad (3)$$

The group statistics for being able to determine whether or not there is cointegration in the panel dataset are calculated as the following:

$$G_T = \frac{1}{N} \sum_{i=1}^N \frac{\alpha_i}{se(\alpha_i)} \sim N(0,1) \quad (4)$$

$$G_T = \frac{1}{N} \sum_{i=1}^N \frac{T\alpha_i}{\alpha(1)} \sim N(0,1) \quad (5)$$

The mean group statistics are zero, and alternative hypotheses are expressed as follows:

$H_0 : \alpha_i = 0$; There is no cointegration for all cross-sections.

$H_0 : \alpha_i < 0$; There is cointegration for all cross-sections.

Tab. 5: Panel Cointegration Test

Test	Statistics	Constant		Test	Statistics	Constant-Trend	
		Asymptotic Probability Value	Bootstrap Probability Value			Asymptotic Probability Value	Bootstrap Probability Value
g_tau	-0.693	0.244	0.939	g_alpha	-2.920	0.002	0.904
g_alpha	4.189	1.000	1.000	g_alpha	5.106	1.000	0.999
p_tau	-1.898	0.029	0.623	p_tau	-8.014	0.000	0.214
p_alpha	2.858	0.998	0.994	p_alpha	4.043	1.000	0.994

Note: For the Westerlund 2007 test, leads and lags were taken as 1, and the test used a bootstrap distribution with 10,000 replications.

At the next stage, the error correction coefficient for all the panels and its standard errors are calculated:

$$P_T = \frac{\alpha_i}{se(\alpha_i)} \sim N(0,1) \quad (6)$$

$$P_\alpha = T\alpha \sim N(0,1) \quad (7)$$

The panel statistics are zero, and alternative hypotheses are expressed as follows:

$H_0 : \alpha_i = 0$; There is no cointegration for all cross-sections.

$H_0 : \alpha_i < 0$; There is cointegration for all cross-sections.

While g_tau and g_alpha show group statistics, p_tau and p_alpha indicate panel statistics. Due to the presence of cross-sectional dependence and the heterogeneity of the panel, group statistics and bootstrap probability values are checked. As the bootstrap probability values were higher than 5 %, the main hypothesis could not be rejected, which means there was no cointegration relationship among the series.

3.7 Bootstrap Panel Causality Test

As the least squares method does not provide significant results due to cross-sectional dependence and heterogeneity of the panel, the test that was used was the panel causality test proposed by Kónya (2006). The least squares estimator is not effective as it cannot utilize extra information in cases of simultaneous correlation among countries. Therefore, the causality test proposed by Kónya

(2006) uses the SUR estimator instead of the least squares estimator. This test, which is based on the Wald test statistic, uses bootstrap critical values and does not require testing unit roots and cointegration procedures. Each equation in the system qualifies different predetermined variables in cases where the error terms are simultaneously correlated. Each equation system uses the SUR estimator rather than VAR. The equation system below represents the bootstrap panel causality test:

$$\begin{aligned}
 y_{1,t} &= \alpha_{1,1} + \sum_{i=1}^{ly_1} \beta_{1,1,i} y_{1,t-i} + \sum_{i=1}^{lx_1} \delta_{1,1,i} x_{k,1,t-i} + \varepsilon_{1,1,t} \\
 y_{2,t} &= \alpha_{1,2} + \sum_{i=1}^{ly_1} \beta_{1,2,i} y_{2,t-i} + \sum_{i=1}^{lx_1} \delta_{1,2,i} x_{k,2,t-i} + \varepsilon_{1,2,t} \\
 &\vdots \\
 y_{N,t} &= \alpha_{1,N} + \sum_{i=1}^{ly_1} \beta_{1,N,i} y_{N,t-i} + \sum_{i=1}^{lx_1} \delta_{1,N,i} x_{k,N,t-i} + \varepsilon_{1,N,t} \quad \text{and} \quad (8) \\
 x_{k,1,t} &= \alpha_{2,1} + \sum_{i=1}^{ly_2} \beta_{2,1,i} y_{1,t-i} + \sum_{i=1}^{lx_2} \delta_{2,1,i} x_{k,1,t-i} + \varepsilon_{2,1,t} \\
 x_{k,2,t} &= \alpha_{2,2} + \sum_{i=1}^{ly_2} \beta_{2,2,i} y_{2,t-i} + \sum_{i=1}^{lx_2} \delta_{2,2,i} x_{k,1,t-i} + \varepsilon_{2,2,t} \\
 &\vdots \\
 x_{k,N,t} &= \alpha_{2,N} + \sum_{i=1}^{ly_2} \beta_{2,N,i} y_{N,t-i} + \sum_{i=1}^{lx_2} \delta_{2,N,i} x_{k,N,t-i} + \varepsilon_{2,N,t} \quad (9)
 \end{aligned}$$

In this equation system that is estimated by SUR, different lag values are assessed for the series $\ln large$ and $\ln htex$ (ly_1, ly_2, lx_1, lx_2), and the lag values among countries are not included. The lag values in the equation are determined based on the Schwarz criterion. If $\delta_{1,j,i}$ is not equal to 0 but $\beta_{2,j,i}$ is equal to 0, there is a one-way causality from X towards Y, while there is a causality from Y towards X when $\delta_{1,j,i}$

Tab. 6: Panel Causality Test Results

		$H_0: \text{lnpse} \neq \text{lnkgdp}$				$H_0: \text{lnkgdp} \neq \text{lnpse}$			
		Bootstrap Critical Values				Bootstrap Critical Values			
Countries	Wald	1 %	5 %	10 %	Wald	1 %	5 %	10 %	
Algeria	8.349	24.289	14.480	10.773	0.005	12.110	6.522	4.478	
Egypt	8.737*	5.663	3.054	2.060	0.328	19.104	11.300	8.409	
Iran	1.590	22.427	13.928	10.435	0.287	12.492	6.599	4.353	
Jordan	1.480	10.098	5.592	3.798	2.357	17.352	9.048	6.165	
Morocco	0.195	14.879	8.192	5.982	0.668	5.897	3.322	2.251	
Oman	0.156	6.814	3.529	2.334	6.199*	9.335	4.754	3.267	
Saudi Arabia	0.730	13.741	7.642	5.386	0.204	13.092	7.112	5.003	
Tunisia	0.027	13.453	7.756	5.608	0.374	14.861	9.473	7.199	

Note: * and ** show that the null hypothesis was rejected on the significance levels of 5 % and 10 %, respectively. The critical values were obtained by 10,000 bootstrap cycles. The Akaike information criterion was used. The maximum lag value was taken as 3.

Tab. 7: Panel Causality Test Results

		$H_0: \text{lngfcf} \neq \text{lnpse}$				$H_0: \text{lnpse} \neq \text{lngfcf}$			
		Bootstrap Critical Values				Bootstrap Critical Values			
Countries	Wald	1 %	5 %	10 %	Wald	1 %	5 %	10 %	
Algeria	4.862	16.760	10.352	7.650	0.047	11.122	5.819	3.924	
Egypt	0.367	7.584	3.980	2.823	2.233	25.788	14.353	10.545	
Iran	2.384	18.362	11.131	8.363	0.305	15.495	8.454	6.010	
Jordan	5.590	23.950	14.419	10.854	0.507	14.726	7.673	5.185	
Morocco	0.307	10.928	6.244	4.178	0.002	6.028	3.216	2.202	
Oman	0.001	19.700	11.636	8.649	1.767	19.438	12.813	10.158	
Saudi Arabia	14.655*	11.300	6.528	4.609	0.022	11.851	5.920	3.906	
Tunisia	13.684*	9.883	5.334	3.537	13.019	33.038	20.832	16.047	

Note: * and ** show that the null hypothesis was rejected on the significance levels of 5 % and 10 %, respectively. The critical values were obtained by 10,000 bootstrap cycles. The Akaike information criterion was used. The maximum lag value was taken as 3.

Tab. 8: Panel Causality Test Results

	$H_0 : \text{Impse} \neq \text{Incl}$				$H_0 : \text{Incl} \neq \text{Impse}$			
	Bootstrap Critical Values				Bootstrap Critical Values			
Countries	Wald	%1	5 %	10 %	Wald	%1	%5	%10
Algeria	0.905	7.539	3.743	2.505	0.068	12.483	6.672	4.560
Egypt	2.065	5.929	3.133	2.181	0.063	20.054	11.827	8.861
Iran	2.316	22.883	12.180	7.964	0.409	38.132	18.254	12.113
Jordan	3.970	10.405	6.147	4.349	0.023	12.340	6.691	4.496
Morocco	0.142	11.174	6.164	4.317	0.600	6.346	3.271	2.238
Oman	7.787	22.797	13.840	10.641	0.031	14.121	8.836	6.813
Saudi Arabia	0.890	12.109	6.909	4.890	2.995	22.845	9.936	6.064
Tunisia	0.053	6.718	3.778	2.624	0.005	14.638	8.933	6.483

Note: * shows that the null hypothesis was rejected on the significance levels of 5 %. The critical values were obtained by 10,000 bootstrap cycles. The Akaike information criterion was used. The maximum lag value was taken as 3.

is equal to 0, but $\beta_{2,j,i}$ is not equal to 0. If neither of $\delta_{1,j,i}$ and $\beta_{2,j,i}$ is equal to 0, there is a two-way causality between X and Y, while there is no causality relationship between X and Y if both $\delta_{1,j,i}$ and $\beta_{2,j,i}$ are equal to 0 (Kónya, 2006:981, Kar et al., 2011:689).

Considering Tabs. 6 and 7, it is seen that there were relationships from political stability towards economic growth in Egypt and from economic growth towards political stability in Oman. There were relationships between political stability and fixed capital only in Saudi Arabia and Tunisia, and there was no relationship from fixed capital towards political stability.

According to the results shown in Tab. 8, there was no relationship between political stability and employment in any direction.

As seen in Tab. 9, while there was no relationship from fixed capital towards employment there were causality relationships from employment towards fixed capital in Oman and Tunisia.

As seen in Tab. 10, there was a two-way causality relationship between economic growth and employment in Jordan, while there were one-way causality relationships from economic growth towards employment in Saudi Arabia and from employment towards economic growth in Oman.

Tab. 9: Panel Causality Test Results

	$H_0 : \text{In}gfcf \neq \text{In}el$				$H_0 : \text{In}el \neq \text{In}gfcf$			
	Bootstrap Critical Values				Bootstrap Critical Values			
Countries	Wald	%1	5 %	10 %	Wald	%1	%5	%10
Algeria	0.446	58.484	38.297	30.832	9.078	38.044	23.820	19.278
Egypt	13.555	75.625	50.501	41.804	6.127	70.686	47.048	37.614
Iran	1.120	10.427	4.962	3.548	0.115	15.562	8.264	5.843
Jordan	3.285	24.191	15.527	12.347	2.652	21.910	15.441	12.668
Morocco	30.280	84.863	59.169	49.386	1.333	79.063	49.605	38.936
Oman	2.589	22.041	14.281	11.227	25.799**	41.878	26.895	21.619
Saudi Arabia	1.980	8.826	5.073	3.734	0.094	15.880	8.374	5.676
Tunisia	0.663	6.533	3.506	2.295	5.801*	4.133	2.752	2.152

Note: * and ** show that the null hypothesis was rejected on the significance levels of 5 % and 10 %, respectively. The critical values were obtained by 10,000 bootstrap cycles. The Akaike information criterion was used. The maximum lag value was taken as 3.

Tab. 10: Panel Causality Test Results

	$H_0 : \text{In}kgdp \neq \text{In}el$				$H_0 : \text{In}el \neq \text{In}kgdp$			
	Bootstrap Critical Values				Bootstrap Critical Values			
Countries	Wald	1 %	5 %	10 %	Wald	1 %	5 %	10 %
Algeria	40.100	77.115	53.700	44.728	0.113	23.582	13.141	9.348
Egypt	5.861	45.438	34.130	29.679	0.001	29.509	18.796	14.804
Iran	1.564	9.690	5.039	3.562	2.757	19.871	11.034	7.331
Jordan	4.331**	8.439	5.463	4.103	9.912*	10.195	6.460	4.930
Morocco	5.491	150.213	99.601	81.276	2.387	109.109	69.870	56.575
Oman	1.125	11.997	7.215	5.269	7.803*	3.953	2.207	1.552
Saudi Arabia	6.707**	13.498	7.723	5.687	2.384	15.891	8.145	5.466
Tunisia	6.696	51.343	35.352	28.659	1.097	26.080	16.874	13.312

Note: * and ** show that the null hypothesis was rejected on the significance levels of 5 % and 10 %, respectively. The critical values were obtained by 10,000 bootstrap cycles. The Akaike information criterion was used. The maximum lag value was taken as 3.

Tab. 11: Panel Causality Test Results

Countries	Wald	$H_0: \ln gfcf \neq \ln kgdp$				$H_0: \ln kgdp \neq \ln gfcf$			
		Bootstrap Critical Values				Bootstrap Critical Values			
		%1	5 %	10 %		%1	%5	%10	
Algeria	1.465	17.168	10.625	7.970	19.921**	23.874	16.980	13.873	
Egypt	1.830	55.809	35.343	27.226	4.550	74.130	48.089	39.422	
Iran	0.747	80.722	52.887	42.307	44.104**	75.136	49.250	39.257	
Jordan	14.660*	13.639	7.616	5.450	9.577	31.045	21.974	18.492	
Morocco	5.606	50.057	34.049	27.817	5.590	81.441	54.154	43.524	
Oman	7.774	3.787	2.173	1.503	7.394**	9.539	5.339	3.769	
Saudi Arabia	5.388	38.046	24.917	19.954	3.041	41.772	26.320	20.505	
Tunisia	0.030	11.674	6.230	4.348	9.158*	3.771	2.251	1.612	

Note: * and ** show that the null hypothesis was rejected on the significance levels of 5 % and 10 %, respectively. The critical values were obtained by 10,000 bootstrap cycles. The Akaike information criterion was used. The maximum lag value was taken as 3.

As seen in Tab. 11, while there was a relationship from fixed capital towards economic growth in Jordan, there were relationships from economic growth towards fixed capital in Algeria, Iran, Oman, and Tunisia.

4 Conclusion

Besides physical capital, social and human factors are also important for the economic development of a country. Societies that improve their social and human elements have increased prosperity levels, and they are able to make healthier decisions as they face the future with safety. Therefore, fewer uncertainties in such societies are a factor that accelerates their economic development. One of the most important factors among social factors is achievement of political stability. Economy-related decisions in societies with political stability may be made for a long term and rationally. Moreover, in order for investments to increase and the private sector to be promoted in a country, institutions should operate effectively, and the political system should be trusted. As the borrowing costs for the private sector increase in cases of political instability, investments decrease. Due to decreased investments, production volume and national income also decrease. As the rates of unemployment will increase in the case of decreased national income, this situation will lead the part of the society known as qualified labor to migrate to developed countries.

This study aims to investigate the relationship among the variables of economic growth, political stability, fixed capital, and employment for MENA countries in the period of 1996–2017. Firstly, the presence of cross-sectional dependence in both the series and the model was tested, and the bootstrap panel unit root test by Smith et al. (2004) which considers cross-sectional dependence was applied. As the series were $I(1)$, the presence of a long-term relationship was examined by panel cointegration test. As there was no cointegration relationship among the series, panel causality test proposed by Kónya (2006) was carried out to determine the direction of causality relationships. The causality test revealed a relationship from political stability towards economic growth in Egypt and a relationship from economic growth towards political stability in Oman. This situation suggests that economic growth slowed down in Egypt due to political instability, and they need to make decisions towards achieving political stability. In Oman, as the economy grew, that is, as the level of prosperity increased, political stability was achieved. In this context, it is important to take precautions that increase economic growth. Considering economic growth in terms of fixed capital, it was seen that political stability affected economic growth in Tunisia and Saudi Arabia. Thus, ensuring the effectiveness of institutions for political stability and creating legal policies towards this aim is important for achievement of economic development. There were relationships from fixed capital towards economic growth in Jordan and from economic growth towards fixed capital in Oman, Tunisia, Iran, and Algeria. As national income increases, this situation allows supplying more machinery-equipment and leads to increase in production volume. This way, in relation to increased production volume, the number of people who are employed may also rise.

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Emrullah Mete and Can Usta

The Blue Economy in Turkey

1 Introduction

Blue economy, which increases economic growth and welfare by using marine living resources in almost all countries with coast around the world, has begun to attract attention.

Blue economy is a phrase which includes the aforementioned green economy and circular economy. Although green economy was first brought to agenda at the United Nations sustainable development summit in Rio de Janeiro in 2012, it was mainly perceived as terrestrial-based especially by small island countries, and it was concluded not to be sufficiently marine-based. Circular economy is an idea which proposes that resources should be used within the circle, each waste should be used in another production process, and thus resources will not disappear and pollution will not occur. Blue economy, which involves and develops both approaches, has become widespread in most countries by proposing the prevention of environmental pollution and the further use of marine economy.

The idea of using marine for economic gains is not new. However, blue economy aims to increase the use of marine and to create as much economic value as possible from the marine environment by including new technologies and public policies in line with the market needs. It also aims to protect marine resources and the ecosystem in a sustainable manner as well as economic gains. The relationship between marine and humans will be reshaped along with the implementation of a blue economy, which is successful and has been able to reach the desired level, and both sides will gain advantage from it in terms of the productivity of the seas.

Recent studies suggest that total balance is over 24 trillion dollars in a global economy including the oceans and marines and that the economic output from marines within this number is over 2.5 trillion in a year. According to the European Commission, the EU blue economy offers employment opportunities for 5.4 million people and also creates gross value added of approximately 500 billion Euros in a year. While these numbers indicate the traditional economic value criteria that express the monetary values of the goods and services traded in the market, they cannot express the social benefits of marines and oceans. This increase in the marine-oriented economy emerged in parallel with the increase in marine-based sectors. Marine tourism, productions of fisheries, maritime

transport, and the extractions of offshore fossil fuel are expected to increase at least twice until 2030. It is also predicted that the sectors, such as offshore wind, seabed mining, and biotechnology, which are now expressed with small values, will show a significant growth (WWF 2015, s. 4–5).

The contributions of blue economy to a sustainable economy by its sectors and the share received by Turkey from these contributions are evaluated in this study.

2 Sectors of the Blue Economy

Economic activities focused on marine and ocean constitute a significant basis against the increasing consumption, food, energy, and mineral resource needs with the increasing world population. Offshore wind capacity is predicted to be the leading energy generation technology until 2030 (IRENA 2016). Furthermore, it is also expected that the marine trade will be quadrupled until 2050 (ITF 2015). As migration to cities and coasts increases along with the ocean economy, there will be a movement in the investments in coastal infrastructure, industry, and tourism.

Blue economy is composed of the sectors such as fishing, aquaculture, coastal and marine tourism, maritime transport, shipbuilding, overseas energy and seabed mining. Moreover, it also includes the activities related to conservation and restoration of coastal and marine area management along with ocean monitoring and surveillance in and around the ocean. The activities related to industry and sectors within the scope of the blue economy are summarized in Tab. 1.

2.1 Fisheries and Aquaculture

A sustainable marine fishery can be a principal component of a successful blue economy. Marine fishery, which is a key source of economy and food security, provides a means of livelihood for 300 million people in the sector and helps to meet the nutrition requirements of 3 billion people as an important source of animal protein, an essential micronutrient, and omega 3 (FAO 2016). The role of fisheries is important in most of the world's poorest communities especially because it is a critical source of protein and provides a kind of social security network. Women fulfill the majority of secondary activities related to sea products, such as fishery labor and marketing. While it causes economic activities to continue by enabling young people to stay in their communities since it provides employment opportunities in many places, it also increases the status of women in developing countries. Healthy fisheries for billions of people among the poorest in the world, the growing fisheries sector, and inclusive trade

Tab. 1: Activities and Sectors of Blue Economy

Type of Activity	Activity Subcategories	Related Industries/ Sectors	Drivers of Growth
Harvesting and trade of marine living resources	Seafood harvesting	Fisheries	Demand for food and nutrition
		Secondary fisheries and related activities (e.g., processing, net and gear making, ice production and supply, boat construction and maintenance, manufacturing of fish-processing equipment, packaging, marketing, and distribution)	Demand for food and nutrition
		Trade of seafood products	Demand for food, nutrition, and protein
		Trade of nonedible seafood products	Demand for cosmetic, pet, and pharmaceutical products
	Aquaculture		Demand for food, nutrition, and protein
Extraction and use of marine nonliving resources (nonrenewable)	Usage of marine living resources for pharmaceuticals and chemicals	Marine biotechnology and bioprospecting	R&D and usage for health care, cosmetic, enzyme, nutraceutical, and other industries
	Extraction of minerals	(Seabed) mining	Demand for minerals
	Extraction of energy sources	Oil and gas	Demand for (alternative) energy sources
	Freshwater generation	Desalination	Demand for freshwater
Use of renewable nonexhaustible natural forces (wind, wave, and tidal energy)	Generation of (offshore) renewable energy	Renewables	Demand for (alternative) energy sources

(continued on next page)

Tab. 1: (continued)

Type of Activity	Activity Subcategories	Related Industries/ Sectors	Drivers of Growth
Commerce and trade in and around the oceans	Transport and trade	Shipping and shipbuilding	Growth in seaborne trade; transport demand; international regulations; maritime transport industries (shipbuilding, scrapping, registration, seafaring, port operations, etc.)
		Maritime transport	
		Ports and related services	
Indirect contribution to economic activities and environments	Coastal development	National planning ministries and departments, private sector	Coastal urbanization, national regulations
	Tourism and recreation	National tourism authorities, private sector, other relevant sectors	Global growth of tourism
	Carbon sequestration	Blue carbon	Climate mitigation
	Coastal protection	Habitat protection, restoration	Resilient growth
	Waste disposal for land-based industry	Assimilation of nutrients, solid waste	Wastewater management
	Existence of biodiversity	Protection of species, habitats	Conservation

Source: World Bank 2015

mean more work, more food safety and well-being, and resistance to climate change. Overfishing and harmful fishing applications cause negative effects on the marine environment, especially the degradation of natural environment and biodiversity. This situation not only affects fisheries commercially but also causes negative effects on the marine ecosystem as a whole. The phrase “sustainable” has been used regarding the elimination of environmental threats within the scope

of the blue economy, and procedures for overfishing and harmful fisheries have been fisheries.

Aquaculture, which includes fish, shellfish, and the production of aquatic plants, is the world's fastest growing sector in the food industry. Nowadays, fisheries and aquaculture provide a significant part of the daily animal protein requirements in many developing countries. Food and protein demands are expected to increase if the world population reaches 9.6 billion until 2050. Furthermore, when it is considered that aquaculture supplies 58 % of fish to global markets (FAO 2016), the activation of this sector will provide food security for the poorest people in the world and also will provide added value in social and economic terms.

2.2 Coastal and Marine Tourism

The rapidly growing tourism sector provides employment for 1 out of every 11 people around the world. Therefore, tourism is an important source of foreign exchange and affects the social, economic, and environmental well-being of many countries. Marine and coastal tourism are the vital sectors of the economy in many countries with coast. Marine and coastal tourism involves many activities such as diving tourism, marine archeology, surfing, ship tours, ecotourism, and recreational fishing operations (World Bank 2017).

Sustainable marine tourism constitutes an important part of the blue economy with respect to protecting the marine environment and species, generating income for local communities and ensuring the protection of their culture and traditions.

2.3 Marine-Based (Offshore) Energy

Sustainable marine-based (offshore) energy may also have a vital role in climate adaptation and the reduction of pollution, as well as in social and economic development. Offshore wind power has become more common especially in Europe. Wave and tidal energy, which are the other forms of marine energy extraction, and ocean thermal energy conversion are still experimental and, in most cases, have not yet developed on a commercial scale. Renewable marine-based (offshore) energy, which involves offshore wind power and ocean energy, is increasingly contributing to both meeting energy needs and reducing greenhouse gas emissions. Offshore wind farms are expanding rapidly. The total installed capacity in Europe is estimated to produce approximately 14 % of the EU's total electricity consumption until 2030. However, the problem here is that it is difficult to produce technology that will ensure the use of unlimited energy sources

by using tidal currents, waves, and temperature differences. Although technological developments are still at an early stage and require many technological improvements and innovations, there are tidal power plants in different regions of the world, especially in China, Russia, South Korea, and North America. The seabed mining industry focusing on the extraction of oil and gas in offshore areas is rapidly growing due to the increased consumption of fossil fuel. One fourth of oil reserves, as well as the production of offshore gas which constitutes one third of world production, are estimated to be present in offshore areas. Furthermore, the fact that 30 % and 13 % of undiscovered gas and oil, respectively, are located in arctic regions has been indicated in scientific studies (Ehlers, 2016, s. 191–192).

2.4 Maritime Transport Shipbuilding

Maritime transport constitutes the main source of transport on a global scale for transport, raw materials, consumer goods, basic foodstuffs, and energy supplies. With this respect, shipping facilitates global trade and also contributes to economic growth and employment. According to the studies carried out within the scope of shipping and international maritime trade, the volume of international maritime trade is estimated to be doubled until 2030. According to the International Transport Forum, port volumes are estimated to be quadrupled until 2050 (ITF 2015).

Nowadays, a world economy without maritime transport cannot be associated with global trade. Because more than 95 % of intercontinental trade of goods and 40 % of trade in the European Union are based on maritime transport.

3 Blue Economy and Turkey

Nowadays, the oceans and seas have become the engine of the blue economy, which includes the factors such as the regulation of global climate, maritime transport and tourism, health and fisheries, and therefore blue growth. The status of benefiting and potential of Turkey from marine economy, which has a great importance in meeting the food and energy consumption demands brought along by the growing world population, also have a particular importance. Turkey's seas are connected to the oceans through the Suez Canal with Istanbul and Dardanelles straits. Moreover, Turkey is the only country with coasts in both Mediterranean and Black Sea. Furthermore, it has the Marmara Sea, which connects the three continents and also serves as a hatchery for aquatic living beings and a biological corridor for the surrounding seas. Turkey with such a potential should significantly benefit from blue growth. Within the scope of blue growth sectors, Turkey was evaluated within the context of fisheries, aquaculture,

coastal and marine tourism, and maritime transport and shipbuilding sectors. Since the studies and initiatives related to the sectors such as offshore energy production and biotechnology are very new, they could not be evaluated.

Turkey has rich aquaculture potential due to its location in the world. Furthermore, the seas with different ecological characteristics constituting the most of the fishing area have a coastline of 8333 km, natural lakes of 178 thousand km² which can be used as the aquaculture production area, and dam lakes of 3442 km². There are 247 fish species in the Black Sea, 300 fish species in the Aegean Sea, and 500 fish species in the Mediterranean Sea. The number of species with economic importance in the country is around 100. Turkey has a share of approximately 0.04 % in world aquaculture production. In 2017, fisheries increased by 5.7 % and aquaculture increased by 9.1 %. While the production through fishing was 354318 tonnes, aquaculture production was 276502 tonnes. While aquaculture production increased by 6.9 % compared to the previous year, inland aquaculture production decreased by 5.1 %. About 37.6 % of aquaculture production was in inland waters, and 62.4 % of it was in seas. Eastern Black Sea Region was on the first rank in the production through marine products fishing by 49 %. This region was followed by the Western Black Sea by 24.2 %, Aegean by 14.8 %, Marmara by 7.7 %, and the Mediterranean Region by 4.3 %.

Turkey is very suitable for aquaculture production both in terms of inland water and marine resources and has a great potential. Total sources of aquaculture are more than forest areas and almost equal to agricultural areas with respect to surface area. Nowadays, aquaculture production consists of sea products of approximately 73.9 %, inland sea products of 6.7 %, and aquaculture products of 19.5 % (DTO 2018).

While fresh, chilled fish constituted a large part of Turkey's aquaculture export in previous years, canned fish constitutes a large part of the export structure at the present time.

When the numbers of the current aquaculture export are examined, it is seen that export continues increasingly. European countries also stand out as the countries with the highest exports. In Turkey, the share of processed products in import is gradually increasing by years. The import of fish is performed especially in the European Union countries such as the Netherlands, England, and Norway, in some African countries such as Ghana Ivory Coast, and in Far East countries like Singapore and Thailand. Frozen and semi-processed products have a significant share in import. Increasing amounts of frozen tuna are imported to provide raw materials for canning. Turkey's production constitutes 0.43 % of the world's aquaculture production. While China, which ranks first in global aquaculture

Tab. 2: Productions of Fisheries and Aquaculture in Turkey.

Year	Sea Products (tonnes)	Aquaculture Production (tonnes)	Freshwater Products (tonnes)
2005	380 381	118 277	46 115
2006	488 966	128 943	44 082
2007	589 129	139 873	43 321
2008	453 113	152 186	41 011
2009	425 046	158 729	39 187
2010	445 680	167 141	40 259
2011	477 658	188 790	37 097
2012	396 322	212 410	36 120
2013	339 047	233 394	35 074
2014	266 078	235 133	36 134
2015	397 731	240 334	34 176
2016	301 464	253 395	33 856
2017	322 173	276 502	32 145

Source: Ministry of Food, Agriculture, and Livestock.

Tab. 3: Import and Export Performance of Fisheries and Aquaculture Products

Year	Export (\$)	Import (\$)
2010	312.935.016	133.829.563
2011	395.306.914	173.886.517
2012	413.917.190	176.402.894
2013	568.207.316	188.068.388
2014	675.844.523	198.273.838
2015	692.220.595	250.969.660
2016	790.303.664	180.753.629
2017	854.731.829	230.111.248
2018	952.001.252*	188.951.045*

Source: TURKSTAT *2018 data are temporary.

production, has a production of 45 million tonnes, the total aquaculture production of the European Union is around 6.428 million tonnes for the same year. With respect to the amount of aquaculture production, Turkey ranks 35th in the world and 7th among EU countries (DTO 2018).

Tab. 4: Data on the Shipbuilding Sector

Year	Ships delivered (piece)	Ships delivered (tonnage)	Number of Employment in Shipyards	Export (thousand \$)
2009	76	679908	19719	1831905
2010	52	483164	21449	1118462
2011	35	352079	20560	1321678
2012	18	192804	21769	810936
2013	21	173661	23000	1163591
2014	16	130720	21332	1271866
2015	21	120064	29699	1029863
2016	20	73644	29610	972177
2017	21	98940	27189	1338158

Source: Ship and Yacht Exporters Association, Ministry of Transport, Maritime Affairs and Communications, and DTO.

The shipbuilding industry has been a heavy branch of industry which forms a rapid development and provided foreign exchange inflow in the affiliated sub-industry sectors; increases the qualified workforce in its region; supports the development, growth and strengthening of regional trade; increases the welfare and cultural level of people living in the region; and creates a significant employment potential along with the sub-industry, in all countries where it is supported and developed. Turkey is the largest ship manufacturer in the shipbuilding industry outside South Asia and China. It is also the world's fifth largest ship recycler. While Turkey constituted 98 % of the demolition volume in the Mediterranean region between the years 2006 and 2015, it was followed by Spain (1.22 %), Greece (0.22 %), Portugal and France (0.17 % for each), and Egypt (0.15 %). Most of the ships recycled in Turkey are foreign flagged, mainly from the European Union, and are too small to be economically unsuitable for access to recycling facilities in South Asia.

In 2008, the size of ship and yacht building exports was 2.6 billion USD. In 2009, the effect of the crisis was felt especially in export, and export was 1.83 billion USD. The ship sub-industry export, which was 10.6 million USD in 2008, also decreased to 3.1 million USD with a decrease of 71 % in 2009. The sector with an export volume of 1.16 billion USD in 2013 reached a ship and yacht export of 1.27 billion USD with a little increase in 2014, and according to the data for the year 2015, it decreased by 19 % with an export of 1.03 billion USD. While the export value was 972 million USD in 2016, it was 1.33 billion USD

with 37.6 % increase in 2017. The 2008 global crisis also affected Turkey's ship-building industry in an extremely negative way, as in most industries. While the number of directly employed staff in the shipyards that contribute greatly to employment was 33.480 people in 2007, this number was 23.000 along with the cancellations of the orders due to the economic crisis in 2013. The number of employees in the sector was determined as 27.189 people by the year 2017.

In Turkey, which is surrounded on three sides by the sea, Marine Tourism is one of the branches of tourism with the highest demand and constitutes approximately 20 % of tourism revenues. While there are 379 marinas on 6500 km long coasts of Italy, 356 marinas on 4964 km long coasts of Spain, and 159 marinas on 5835 km long coasts of Croatia, there are only 86 yacht mooring areas on our 8333 km long coasts that are much more beautiful, different, and rich in all aspects. There are 19.000 marinas around the world, and there are 5000 marinas only across Europe. While there were 25 marinas by the end of 2002 on the coasts of our country, this number reached 86 by the end of 2017 and an increase of 150 % was achieved in 15 years. The fastest growing sector of the travel industry is cruise tourism. The annual growth rate has been 8 % since the 1980s. The fast-growing cruiser ships transport 3000–3500 tourists, and thus, they also gain tourists who have not afforded a cruise by reducing costs per person, and they make them very attractive with the opportunities they offer. Today's cruise industry gives the opportunity to visit the cities of many countries with short (3–4 days) and long (15 days and more) cruise opportunities. In the recent year, 14 million people have traveled with cruise ships around the world. While nearly 10 million (9.747) of them are from America, 4 millions of them are from Europe and other countries. Most of the cruisers travel in the Caribbean. Turkey had two cruise ships; however, along with the transfer of these ships to a foreign-owned company, there is no cruise ship belonging to a Turkish company carrying the Turkish flag. Cruise tourism has also been the rising star of marine tourism in terms of the revenue it provides. The average amount of money that a cruise tourist spends during a 7-day voyage on ship-board is 1719 dollars. In other words, the volume of cruise tourism only in terms of ships has reached 36 billion dollars. A cruise tourist spends an average of three times more than a normal tourist. Turkey's rise in cruise tourism, which began to grow again with the healing of the wounds of the 2008–2009 crisis in the world economy, is going on. When the last 11 years are considered, the number of cruise tourists coming to Turkey with a decline only in 2009 exceeded 2.2 million in 2013, and the share it receives from the world increased to 11 %. The growth rate has been 285 % over the past decade. The number of cruise tourists coming to Turkey is also expected to increase by 2–3 % in 2019 and to reach 2.3 million (DTO 2018).

4 Conclusion

Blue economy, which has adopted the concept of “sustainability” with respect to ensuring economic growth and eliminating environmental threats by including green economy and circular economy, has attracted the attention of all countries with coast, especially the island countries. It opens a new door especially for developing countries with appropriate geographical location by promising to get more contributions for economic growth from the marine-oriented sectors such as fisheries, aquaculture, shipbuilding, and marine tourism. Blue economy, which tries to prevent activities that disrupt the structure of the ecosystem, such as environmental wastes, excessive and harmful fishing while performing these sectoral activities, is also known as a “sustainable growth”.

In this study, to what extent Turkey can benefit from the advantages offered by blue economy was evaluated. At this point, while Turkey should be an assertive country in terms of getting share from the blue economy due to its geographical location, it has come to the forefront only in fisheries and aquaculture sectors. Although it has been able to make its mark in the shipbuilding and recycling sectors, it appears that is still quite inadequate. It has remained quite distant from its potential, although it is a geographically and historically suitable country for tourism, especially cruise tourism. Turkey’s initiatives in the energy-sized sectors of blue economy, such as seabed mining, offshore wind energy, and wave energy, are still very new. With respect to benefiting from the blue economy, Turkey should consider the relevant sectors as the leading sectors and should provide the necessary support.

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Edip Durmaz

Edirne's Industry in the Early Years of the Republic

1 Introduction

At the beginning of the 1800s in the Ottoman Empire, there was an advanced industry which manufactured in small ateliers and was organized in the form of tradesmen organizations (guilds) but which was ahead of its time. This local and national industry both met the needs of the country and exported the surplus production of the textile sector. This relatively advanced industrial infrastructure in the Ottoman Empire collapsed after 1839 in the Tanzimat Period. The most important reason for this was the fact that the Ottoman Local Industry, working with manpower, could not compete with the modern industry of Europe at that time, which was based on machine production¹.

The capitulations given by the Ottoman Empire to the European states and the trade agreement with the USA², as well as the socio-economic conditions in which the empire existed, did not allow for the establishment of a modern industry despite all attempts and efforts. This situation caused the Ottoman Empire to begin to meet almost all of the products it needed by importing from outside, starting from the 1850s.

When the Republic was founded, modern industrial enterprises, which were active in the early 1920s, gathered around two large groups. The first group included state-owned factories such as Feshane³, Hereke Carpet and Silk Weaving Factory, Zeytinburnu Weaving Factory⁴, Zeytinburnu Acid and Ether

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- 1 Ömer Celal Sarc, “*Tanzimat ve Sanayimiz*”, *Tanzimat I*, Milli Eğitim Yayını, Maarif Matbaası, İstanbul, 1940, s. 433.
 - 2 Selda Kayapınar, “Osmanlı ABD Ticaret Antlaşması öncesi Amerika'nın Diploması Girişimleri”, *Dumlupınar Üniversitesi Sosyal Bilimler Dergisi*, Sayı 51, Ocak 2017, s. 55.
 - 3 Rifat Önsoy, *Tanzimat Dönemi Osmanlı Sanayii ve Sanayileşme Politikası*, Türkiye İş Bankası Kültür Yayınları, Ankara, 1988, s. 50.
 - 4 Mustafa Kurt, Kemalettin Kuzucu, Baki Çakır; Kemal Demir, “19. Yüzyılda Osmanlı Sanayileşmesi Sürecinde Kurulan Devlet Fabrikaları: Bir Envanter Çalışması”, *Ankara Üniversitesi Osmanlı Tarihi Araştırma ve Uygulama Merkezi Dergisi* 40/Güz 2016, s. 256–258 (245–277).

Factory⁵, which mainly worked for the military needs of the country. In the second group, there were some privately owned factories dealing with textiles, olive oil, cement, etc.⁶

In the last years of the Ottoman Empire, a number of attempts were made by governments to transform the country's economy, which is largely dependent on imports from Europe, into a production-based structure. One of the most important of those similar attempts to develop and strengthen the industrial infrastructure in the country was "Industry Promotion Law" issued in 1913⁷ under the rule of "İttihat ve Terakki Party". This law was revised in the Republican Period and updated in 1927 according to the conditions of the day⁸, and the existing industrial infrastructure of Republic of Turkey was tried to be strengthened⁹.

However, between 1923 and 1929, despite the goal of "development through industrialization", not only no significant changes were made either quantitatively or qualitatively in the industrial infrastructure taken from the Ottoman Empire, but also a significant and satisfying industrial move could not be realized¹⁰. On the other hand, another important problem that negatively affected the industrial infrastructure was the decrease in the urban population in 1912–1927. This decrease reached to the highest rate of %44 in the western part of the Adapazarı - Muğla line, which also includes Edirne.

Especially the wars between 1912 and 1927 caused the general population and the urban population to decrease and the existing ethnic structure to change¹¹.

5 Mustafa Özdemir, *Osmanlı'dan Cumhuriyet'e Endüstriyel Mirasımız*, İTO Yayınları, İstanbul, 2011, s. 317.

6 Memduh Yaşa ve komisyon, *Cumhuriyet Dönemi Türkiye Ekonomisi*, Akbank Kültür Yayını, Apa Ofset Basımevi, İstanbul, 1980, s. 13.

7 Vedat Eldem, *Harp ve Mütareke Yıllarında Osmanlı imparatorluğu'nun Ekonomisi*, Türk Tarih Kurumu Yayınları Ankara, 1994, s. 77–78.

8 Tunç Tayanç, *Sanayileşme Sürecinde 50 Yıl*, Milliyet Yayınları, İstanbul, 1973, s. 58–59.

9 Only the investors, whose domestic capital was at least 25 %, would benefit from the "Industry Promotion Law" adopted in 1927. The government committed to recognize some privileges in transportation, supply free land, to bring exemption to taxes and duties, and to give 10 % bonus over production figures. The state, requiring public institutions and municipalities to use domestic goods, also forced them to work with industrial businesses that benefited from this law.

10 Korkut Boratav, *Türkiye İktisat Tarihi (1908–1985)*, Gerçek Yayınevi, İstanbul, 1998, s. 40.

11 Yahya Sezai Tezel, *Cumhuriyet Döneminin İktisadi Tarihi (1923–1950)*, Tarih Vakfı Yurt Yayınları, İstanbul, 2002, s. 100–101.

Tab. 1: The industry in Edirne province in 1927. Source: Sanayi Sayımı 1927, (1969), s. 31.

Branches of industry in 1927	Edirne province	
	Number of businesses	Number of employees
Extractive industry	2	2
Agriculture, livestock, fishing, game animals industry	409	1.175
Textiles industry	77	215
Timber and herbal products industry	158	460
Paper and cardboard industry	7	29
Metallurgical and machinery industry	158	310
Construction industry	3	11
Chemical industry	4	12
Unspecified industry	2	5
All of the industry branches	820	2.219

All this impacted Turkey's economy negatively. While Turkey was adversely affected by all the goings-on, it can be said that Edirne was affected still more negatively. Because the population of Edirne province was 1,106,005 when it reached its peak before 1912, but it fell to 627,500 in 1914. Following the new administrative divisions in the early years of the Republic and the changes in the ethnic structure with the aforementioned wars, the population of Edirne province in the 1927 General Census fell to 150,889¹². The decline in the population of Edirne was much more severe. The central population of Edirne was 299,147 in 1907, but in 1927 this number decreased to 68,279¹³. The negative effects of the migration or death of about 77 people out of 100 living in the center of Edirne on the industry of Edirne were inevitable.

2 Edirne Industry During the Years 1927–1935

The situation of Edirne province in industry is summarized in Tab. 1, according to the first industrial census organized by the Republic of Turkey, in the autumn of 1927.

12 Kemal H. Karpat, "*Osmanlı Nüfusu (1830–1914), Demografik ve Sosyal Özellikler*", Tarih Vakfı Yurt Yayınları No: 133, İstanbul, 2003, s. 204–208.

13 Türkiye Cumhuriyeti Başvekâlet İstatistik Umum Müdürlüğü, *28 Teşrinievel 1927, Umumi Nüfus Tahriri*, Hüsniyatıbaat Matbaası, İstanbul, 1929, s. VII.

As can be seen from Tab. 1, in the autumn of 1927 in Edirne province, there were 820 businesses active in the field of industry during the census, and 2,219 employees were employed in these businesses. The leading industry branch was agriculture, livestock, fishing, game animals industry, which had half of all businesses on its own. Furthermore, this industry branch accounted for about 53 % of the population working in all branches of industry in Edirne province. Another important industry branch was timber and herbal products industry. In this industry branch, 158 businesses were active and 460 people were employed. Another important industry branch was metallurgical and machinery industry. The number of businesses operating in this industry branch was 158 and the number of people employed was 310. Finally, one of the major industry branches of Edirne province was the textile industry. The number of the businesses operating in this industry branch was 77 and the number of the people employed was 215. When viewed proportionately, employment in the four most important industrial branches was respectively 52.95 % in agriculture, livestock, fishing, and game animals industry; 20.73 % in timber and herbal products industry; 13.97 % in metallurgical and machinery industry; and 9.68 % in the textile industry.

While this was the situation in the industry of Edirne province in 1927, we confront the table below when we get a bit more detail in Edirne city center and Edirne countryside.

As shown in Tab. 2, when the number of businesses is examined, 64.39 %, almost two third, of the industry of Edirne province was clustered in Edirne city center. In Edirne countryside, it is seen that 35.61 %, almost one third, of the industry was clustered. Looking at the number of employees working in industry 67.95 % of the employees worked in Edirne city center and 32.05 % work in Edirne countryside. The number of people working in industry branches in Edirne city center was 1,508, while the number of people working in industry branches in Edirne countryside was 711. The total number of people working in the industry in Edirne province was 2,219. In Edirne city center, there were 2.86 employees per business, and in Edirne countryside there were 2.43 employees per business. While 667 of the industrial businesses in Edirne province have a maximum of four employees, 153 of them have more than four employees¹⁴. It appears in Tab. 2 that, of all the industry branches shown in Tab. 2, in extractive industry and construction industry Edirne countryside comes to the forefront, and in all other branches of industry Edirne city center is in the forefront.

14 İsmail Husrev Tökin, *İktisadi ve İctimai Türkiye Rakamlarla*, Cilt III, T. C. İstatistik Genel Müdürlüğü Yayını, Ankara, 1946, s. 39.

Tab. 2: The industry in Edirne province, Edirne city center and Edirne countryside in 1927. Source: Sanayi Sayımı 1927, (1969), p. 31.

Branches of industry in 1927	Edirne city center		Edirne countryside		Edirne province	
	Number of businesses	Number of employees	Number of businesses	Number of employees	Number of businesses	Number of employees
Extractive industry	-	-	2	2	2	2
Agriculture, livestock, fishing, game animals industry	259	784	150	391	409	1.175
Textiles industry	61	162	16	53	77	215
Timber and herbal products industry	105	353	53	107	158	460
Paper and cardboard industry	3	11	-	-	3	11
Metallurgical and machinery industry	93	182	65	128	158	310
Construction industry	2	4	5	25	7	29
Chemical industry	3	7	1	5	4	12
Unspecified industry	2	5	-	-	2	5
All of the industry branches	528	1.508	292	711	820	2.219

The general appearance of the industry in Edirne province by 1935 changed a lot. The most important change in Edirne industry during these 8 years is that the industry spread from Edirne city center to Edirne countryside. While the number of the businesses in Edirne countryside was about one third of

the businesses in Edirne province in 1927, the situation was reversed in 1935 and Edirne countryside became the home of more than half of the industrial businesses. When we go into more detail, we are faced with Tab. 3.

As can be seen from Tab. 3, in the autumn of 1935, a total of 6,267 people were employed in the businesses of industry which were active at the time of the census. The industry branch which was prominent in Edirne province was the woodworking industry. This industry branch employs 1,318 people and holds approximately a quarter of the working population in the industry of Edirne province. Another important industry branch was the metallurgical industry where 1,091 people were employed. Another important industry branch was the garment industry. The number of people employed in this industry branch was 965. Finally, one of the important industry branches of Edirne province was food, beverage, and tobacco industry. The number of people employed in this branch was 866. When viewed proportionately, employment in the four most important industry branches was 24.27 % in woodworking industry; 20.09 % in metallurgy industry; 17.77 % in garment industry; and 15.94 % in the food, beverages, and tobacco industry. Apart from the aforementioned industry branches, construction and furniture industry can also be mentioned as a considerable industry branch where 606 people were employed. The weight of this industry branch in the industry of Edirne province was about 11 %. It is seen that 89.24 % of the employees working in the industry of Edirne province were employed in these five industry branches.

The situation in the industry of Edirne city center and Edirne countryside in 1935 was as follows. As shown in Tab. 3, when the number of employees in all industry branches is examined, 57.23 % of the industry in Edirne province was clustered in Edirne countryside and 42.77 % of the industry in Edirne province was concentrated in Edirne city center. While the number of people employed in the industry branches was 3,108 in Edirne countryside, it was 2,322 in Edirne city center. Among all the industry branches shown in Tab. 3, Edirne countryside was prominent, and Edirne city center was in the forefront only in machinery and tool manufacturing; paper and printing; food, beverages, and tobacco; and garment industries. In Tab. 3 column 14, seasonal workers whose branch cannot be identified is indicated in bold. The total number of such seasonal workers in Edirne province was 837.

All numbers and ratios given in Tab. 3 are given without taking into account 837 seasonal industry workers. It is highly probable that these workers are employed only in the peak season as seasonal industry workers. Although the time of the season and the total number of days active in the industry branch varies between branches, it is worth remembering that the seasonal workers were

Tab. 3: Number of employees in the branches of industry in Edirne province, Edirne city center and Edirne countryside in 1935. Source: Genel Nüfus Sayımı 20 Ekim 1935 Edirne Vilâyeti Volume: 19, (1937), p. 23–53.

Branches of industry in 1935	Number of employees						
	Edirne city center			Edirne countryside			Edirne province
	Man	Wom.	Tot.	Man	Wom.	Tot.	Total
1. Extraction and exploitation industry	3	7	10	31	-	31	41
2. Stone and soil industry	18	-	18	131	5	136	154
3. Construction and furniture industry	153	-	153	453	-	453	606
4. Woodworking industry	450	28	478	556	284	840	1318
5. Metallurgical industry	278	-	278	737	76	813	1091
6. Machinery and tool manufacturing industry	78	-	78	21	-	21	99
7. Chemical industry	2	-	2	2	-	2	4
8. Textiles industry	28	19	47	8	48	56	103
9. Paper and printing industry	23	-	23	7	-	7	30
10. Leather rubber, bone, horn, amber goods industry	69	-	69	47	-	47	116
11. Food, beverages, and tobacco industry	485	38	523	325	18	343	866
12. Garment industry	510	96	606	278	81	359	965
13. Electricity, gas, and water production and distribution industry	13	-	13	3	-	3	16
14. Unspecified industry workers (seasonal worker)	271	86	357	267	213	480	837
15. Mixed and various industry	2	-	2	2	-	2	4
16. Technical and administrative stuff in industry branches	12	-	12	5	-	5	17
All of the industry branches	2.405	274	2.679	2.863	725	3.588	6.267
All industry branches after removing seasonal workers	2.134	188	2.322	2.596	512	3.108	5.430

employed in the peak season for 2–3 months and they were not employed in the rest of the year. Given this situation, it can be very roughly calculated that this seasonal employment of 837 people corresponds to permanent employment of 176 people¹⁵.

Taking a brief look at the situation of the large industrial businesses supported by the government, benefited from the “Industry Promotion Law”, and had a machine power of more than 10 horsepower within Edirne province will be useful to make some evaluations in terms of volume and size of the businesses owned by the industry of Edirne province. The situation of large industrial businesses between 1932 and 1935 is summarized in Tab. 4.

As can be seen from Tab. 4, between 1932 and 1935, the number of industrial businesses, which can be called as large industrial businesses and was supported by the government in Edirne province, varies between 26 and 15. The number of employees employed in these large industrial businesses ranged from 202 to 155. By 1935, it was not possible to determine the number of employees shown with * in the columns in Tab. 4. The reason for this is that in 1935, a census method based on a workday was applied to industrial businesses. However, it is seen that, in 1934, 155 people were employed in 20 large industrial businesses, and in 1935, the number of employees in 15 large industrial businesses fell below 150. At the same time, the “Industry Promotion Law” is a law that aims to support the certain size of industrial businesses for a certain period of time. Industrial businesses lost the support they received from the government when the determined period was over. Therefore, in Tab. 4 some large industrial businesses do not appear after 1933, but it is not because of the fact that they closed down, but because they lost their government support. There were approximately 7.8 employees per enterprise in these large industrial businesses active in Edirne province in 1932 and 1934. It is understood that a large part of the large industrial businesses in Edirne province was clustered in flour industry and two third of the large industrial businesses were active in flour industry.

3 Conclusion

When the situation of Edirne province in 1927–1935 period is evaluated generally, it is useful to underline some issues. In the years when the Republic of Turkey was founded, although Edirne province was in a disadvantageous

15 It is assumed that temporary workers work an average of 2,5 months in a year, it is estimated that 2,5 months are equal to 21 % in 12 months, and it corresponds to permanent employment of 176 people.

Tab. 4: Large industrial businesses in Edirne province. Sources: Sanayi İstatistikleri Teşviki Sanayi Kanunundan İstifade Eden Müesseselerin 1932 ve 1933 Seneleri Faaliyetleri, (1934), p. 62–197; Sanayi İstatistikleri Teşviki Sanayi Kanunundan İstifade Eden Müesseselerin 1932 ve 1933 Seneleri Faaliyeti, (1939), p. 104–322.

Branches of industry	1932		1933		1934		1935	
	Number of businesses	Number of employees	Number of businesses	Number of employees	Number of businesses	Number of employees	Number of businesses	Number of employees
Flour industry	16	123	14	108	12	108	10	*
Vegetable oil industry	1	5	3	11	4	14	2	*
Chocolate and confectionery industry	3	26	2	31	1	5	-	-
Mixed flour and olive oil industry	1	6	-	-	-	-	-	-
Tricot (knitted) industry	1	4	1	3	-	-	-	-
Car industry (woodworking)	1	6	1	3	-	-	-	-
Mixed industry (mixed industry)	1	13	1	9	1	10	1	*
Ice and cold storage industry	1	7	1	7	1	6	1	*
Electrical industry	1	12	1	12	1	12	1	*
All industry branches	26	202	24	184	20	155	15	*

position for the issues that were necessary for the development of the industry, after 1923, it is seen that the negative conditions changed positively and Edirne province grew very strongly with the government's guidance, and immigration policies followed. With this growth, while the number of people employed in the industry of Edirne province was 2,219 in 1927, it rose to 6,267 with seasonal workers in 1935. It is possible to say that the number of people employed in the industry in Edirne province increased almost two-fold (182 %) with seasonal workers in a short period of eight years. However, in the same years, industrial employment across Turkey experienced a growth rate of almost one-fold (119 %). The increase in industrial employment across Turkey is much lower than the increase in industrial employment in Edirne province in the same years. However, this record level increase in industrial employment in Edirne province could not be achieved in women's industrial labor force participation. In Turkey, in 1935, while the proportion of women employed in the industry was at the level of 19.66 %, this rate remained at 15.94 % in Edirne¹⁶.

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Esra N. Kilci

A Study on Debt Sustainability in Fragile Five: Brazil, Turkey, India, Indonesia, and South Africa; Analysis with Fourier Approach

1 Introduction

External sustainability means that a country is able to fulfill its current and future external liabilities of both private and public sector without any delay, any recourse to debt-rescheduling, and finally any significant adjustment in balance of payments. In external sustainability as there is in the fiscal sustainability, there should be sufficient trade surplus to keep the external debt ratio constant or to reduce it between two periods. It means that there should be a net transfer of resources abroad equal to the difference between net foreign capital inflows and external debt interest payments. The amount of surplus required increases with the external debt ratio and the growth-adjusted real interest rate of external debt. Although the trade surplus required is not directly linked to the policies implemented unlike the primary budget balance, it is influenced by several factors including the exchange rate, the growth rate, and the activities of imports and exports (Akyuz, 2007).

In general, emerging market economies like Turkey, Philippines, Mexico, Indonesia, and Brazil have taken important lessons from the crises of 1990s and 2000s, leading to a more cautious approach towards external debt exposure and fiscal imbalances. Hence, while developed market economies experienced a strong jump in debt to GDP ratios, almost all of the emerging market economies were able to overcome the 2008–2009 Global Financial Crisis without experiencing severe financial and fiscal deterioration. However, there has been an upward trend in the debt to GDP ratios of emerging market economies in the post-2013 period. It signals to the vulnerability of commodity exporters of emerging market economies to commodity price shocks, the damage which negative politics might have on economies (as seen in Brazil and South Africa) and the expansive fiscal policy response which was needed in China to protect the domestic and global economy against a hard landing (Evans, 2017).

In some emerging market economies, the risks stemming from debt are in rise and show significant differences between countries. Increasing vulnerabilities reflect higher public debt levels and also increased debt portfolio risks which the

shift in the debt composition caused. High reliance on debt has contributed to increase in debt service costs, refinancing, and interest rate risks. Average public and external financing needs seem still very important in spite of the low global interest rates. In the case of financial shocks such as sharp increases in global interest rates, financing needs might increase quickly. Although it is observed that some of these emerging market economies have increased their resilience due mainly to prudent policies, the implementation of sound debt management strategies, and the buildup of external and fiscal buffers, some emerging market economies might experience a limited ability to carry out countercyclical policy since high debt ratios restrict fiscal space and some of these might face capital outflows and currency depreciations (IMF and World Bank Development Committee, 2018).

These developments have raised concerns about the outlook for emerging markets and raised questions as to whether these countries would be subject to repetition of previous crisis experiences, such as the East Asian Crisis emerged at the end of the 1990s. The East Asian Crisis following the devaluation of Thai baht in July 1997 led to reassessment of risk and rapid contagion of fragilities to many other countries in Asia. These economies faced sudden stops in capital inflows and severe recessions, which bring balance of payment problems along with weaknesses in the financial sector. While the East Asian Crisis had several causes, a key determinant was reliance on short-term external debt denominated in foreign currency coupled with fixed exchange rate regimes and financial sector fragilities. Historically, emerging market economies crises have tended to coincide with crises and periods in which the US dollar appreciation was experienced that exposed countries to large-scale unhedged currency mismatches between their US dollar-denominated assets and liabilities (Chifu and Quint, 2018).

It is obvious that many emerging market economies have managed to reduce their external vulnerabilities and to improve their policy frameworks when compared to 20 years ago. Some of these economies recorded improvement in their current account positions even experienced current account surpluses. Nevertheless, current account deficits of these economies have widened slightly again since the 2008–2009 Global Financial Crisis. It is seen that policy frameworks have also evolved in these economies when compared to the past. For instance, a large proportion of these economies have more flexible exchange rate regimes and most of them have adopted inflation-targeting monetary policy frameworks that can help to meet inflation expectations and stabilize business cycles. Furthermore, many emerging market economies have accumulated reserves in the period following the East Asian Crisis and have continued to maintain it since the 2008–2009 Global Financial Crisis. The International Monetary

Fund (IMF) risk-weighted metric of reserve adequacy, which compares reserves not only with traditional measures such as imports and short-term external debt but also with broad money to take into consideration the risk of capital flight, suggesting that many emerging market economies are now in a better situation to cope with sudden stops or sudden flight shocks. However, the current conditions with the ongoing US monetary policy normalization and the strengthening US dollar unfortunately have raised concerns that similar fragilities could expose some emerging market economies to the balance of payments problems in the near future (Chiřu, L. and Quint, 2018).

The objective of this study is to analyze the debt sustainability of emerging market economies that are called fragile five. In a significant part of the studies focusing on debt sustainability, it is concluded that borrowing is sustainable if the debt stock to GDP ratio is stable and constant in the long term. For instance, Kremers (1989) stated that if the ratio of debt stock to GDP is stationary, fiscal policies could be sustainable. This condition tries to analyze whether the GDP growth rate covers the debt level or not even the borrowing increases. Parallel to this approach, in this study, the debt sustainability of the countries including fragile five is examined by testing the stationarity of the series related to debt service ratios of these countries. Debt service ratio (private nonfinancial sector) which is used in this study denotes the ratio of principal and interest payments to GDP in private nonfinancial sector. We will conclude that the debt is sustainable if the series are stationary. That is, the debt service ratios remain constant when the principal and interest payments and incomes increase at the same rate. Therefore, it means that incomes have the ability to compensate the increase in debt. Otherwise, if the series do have unit root, it implies that debt service ratios are not constant and stable so debt sustainability is not possible in these emerging market economies.

In this study, Fourier KPSS Test is being employed to test the stationarity of the series by using monthly data belonging to the period of 2002:Q1–2018:Q3. The Fourier KPSS Stationary Test developed by Becker et al. (2006) can detect not only sudden changes but also slow changes, and it seems that the position, number, and form of structural changes do not affect the power of the test. The study has the following structure: After mentioning about fragile five countries and the debt service ratios of these in Section 2 and a brief literature review in Section 3, Section 4 represents the data set and the methodology for the empirical model, which we applied to test the stationarity of the series, which denote the debt service ratios in fragile five including Brazil, Turkey, India, Indonesia, and South Africa. Section 5 introduces the results. Finally, Section 6 outlines the main conclusions.

2 The Fragile Five: Brazil, Turkey, India, Indonesia, and South Africa

In the economy report presented by Morgan Stanley in 2013, it introduced a group called fragile five in which Brazil, Turkey, India, Indonesia, and South Africa were included. The fragile five, a term to define these countries' vulnerabilities to capital outflows, have experienced several challenges in this period. The classification of these emerging market economies was made after the US Federal Reserve (FED) announced that it would reduce bond purchases in the upcoming period. Following FED's announcement, the emerging market economies called as "fragile five" experienced depreciation in their currencies. The reasons for evaluating these economies within fragile five are stated as high current account deficits, high inflation rates, and decreases in growth performances. It has been estimated that these economies would have a significant increase in their external financing needs and might have a stressful period in providing external borrowing in the following period in case of FED's starting monetary tightening (Egilmez, 2013).

Although monetary accommodation and growth performances have hidden the main challenges of these economies by attracting more capital in the form of portfolio investments, events and developments in Turkey and Argentina have revealed that these weaknesses arose in the last period. The tightening in global liquidity, which was led by the US Federal Reserve (FED) increasing interest rates and slowing down its bond purchases, resulted in a decrease in capital inflows and an increase in the cost of borrowing in terms of these economies. Trade tensions, sanctions, the deterioration of the global institutional structure, and increasing geopolitical risks are the major factors that exacerbate the stress in the recent period. The weaknesses in the real economy and the financial system are linked to each other on a vicious cycle. Capital withdrawals from these economies lead to depreciation of the currencies and decrease in the value of both real and financial assets such as stocks, bonds, and properties. Reduced funding availability and rising funding costs trigger banking problems by increasing pressure on overextended borrowers. Moreover, investment downgrades by credit rating institutions extend this adverse cycle (Financial Post, 2018).

Analyzing the external debt positions in this group as of 2018, it is observed that Turkey has the highest ratio of external debt stock to GDP as %52 and Brazil following Turkey with the ratio of %48. The external debt to GDP ratios of the other three economies remain low compared to these economies. Turkey has gradually increasing external debt burden and growing external debt service although the Turkish economy has more reasonable outlook in terms of net foreign debt

burden (Egilmez, 2018). High levels of external borrowing might be a source of financial and economic instability particularly in emerging market economies. In the case of default, the results may affect several countries because of the contagion effect. In addition to interest rate risk, the foreign debt of emerging market economies is very sensitive to exchange rate fluctuations considering the fact that about 75% of the public or publicly guaranteed external debt is denominated in US dollars. While fixed exchange rate regimes of the 1990s reduce the turbulence risk associated with exchange rates, flexible exchange rates do not force countries to increase their interest rates to defend their currencies against speculative movements. In this regard, monitoring the changes in external debt and the other indicators related to external debt as reserves and current account deficits of these emerging market economies has great importance. When analyzing the external debt positions and reserves of fragile five, it is seen that while Brazil, Turkey, and Indonesia have all high levels of external debt, Brazil and India have substantial reserves covering at least 60% of their external debt. On the other hand, Turkey does not seem to have adequate reserve levels to meet their financing needs. A permanent current account deficit is a good indicator of a country's need for foreign capital. While most of emerging market economies have accumulated such current account deficits, it is seen that some countries such as Brazil and South Africa have improved their conditions. In addition, other risks such as increasing protectionism, which could further reduce export revenues and potentially hinder the free flow of capital, may increase the risk of insolvency over the next few years. Further exchange rate fluctuations might make the adverse conditions more complex and difficult (Desjardins, 2017).

3 Literature Review

Karam and Hostland (2005) employed stochastic simulation methods to evaluate debt sustainability in emerging market economies and tried to provide measures for projections of the external and public debt burden over the medium term. They showed that fiscal policy could act in a preemptive manner to prevent the debt burden from rising significantly over the medium term, and this required flexibility in fiscal planning, which many emerging market economies lacked. Therefore, emerging market economies faced a difficult trade-off between managing the risk of a debt crisis and pursuing other important fiscal policy objectives. Goktan (2008) investigated that whether the financial policies carried out in Turkey after the year of 1999 satisfied the sustainability of borrowing by employing the unit root and co-integration tests. After using the data belonging to the period of 1999–2006, the findings implied that the debt sustainability

condition, which was introduced by Hamilton and Flevin (1986), was satisfied for Turkey in the relevant period. On the other hand, it was found that, the sustainability condition of Kremers (1989), in which the ratio of debt stock to GDP was taken, could not be proved as the ratios had unit root in the relevant period. Finally, the sustainability condition set forth by Hakkio and Rush was tested, and the findings did not support the existence of debt sustainability in the period of 1999–2006.

Mahmood and Rauf (2012) examined the debt sustainability in Pakistan by using the present value of budget constraint approach. Their results indicated that the series of government expenditure, revenue, and discounted debt had unit root, thereby signaling that the necessary conditions for debt sustainability were not met, and debt has remained unsustainable in the period of 1971–2011. The results also showed that the problem of debt sustainability stemmed from persistent fiscal indiscipline. Therefore, if a major correction in fiscal policy was not made, the debt profile of the country would remain under pressure. Unalmis (2015) tried to derive debt sustainability conditions for emerging and developing countries by taking into account both domestic and foreign debt. Using the setup to evaluate the short-term fiscal stance of a government, he analyzed the post crisis fiscal adjustment in transition countries, namely Armenia, Azerbaijan, Georgia, Kazakhstan, Kyrgyz Republic, Moldova, Tajikistan, Turkmenistan, Ukraine, and Uzbekistan. The results suggested that country performances were quite mixed, and the pace of debt accumulation in some of these countries should be monitored closely.

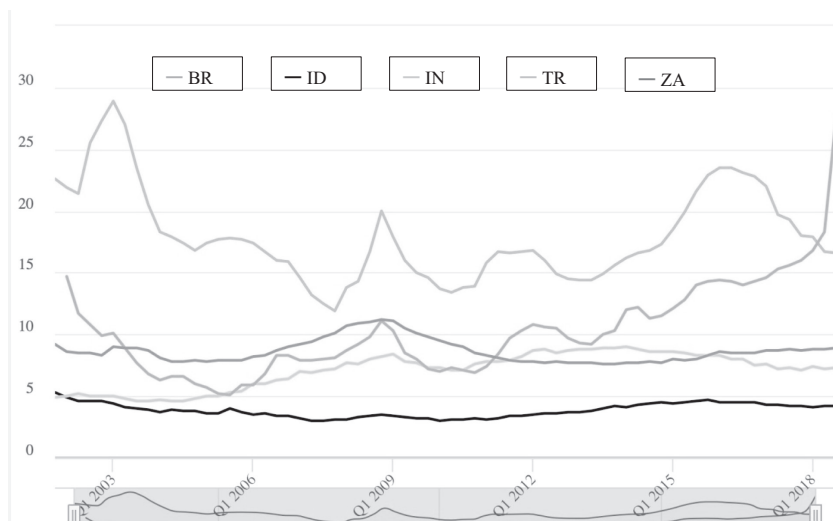
Brière et al. (2016) employed contingent claims analysis (CCA) to assess public debt sustainability of five Asian economies including Indonesia, Malaysia, the Philippines, the Republic of Korea, and Thailand and forms in their study which formed part of a research project exploring risk-adjusted public debt sustainability analysis for Asian economies. They estimated the sovereign consolidated balance sheet and computed a risk-based fiscal sustainability indicator, which they then subjected to scenario analysis to assess the implications of specific shocks to countries' macroeconomic and policy environment and used bootstrapped projections of exchange rates and the CCA framework to assess public debt sustainability in the countries studied. They found that none of them would appear to be particularly vulnerable to sovereign debt distress, not in normal times and not when facing the less favorable macroeconomic scenarios considered. However, CDS spreads and distance to distress were highly volatile throughout the period of observation, and none of the countries considered was immune to sudden drops in market perception of risk as reflected in higher CDS spreads and eroding assets value.

Paret (2017) carried out a probabilistic approach to simulate the medium-term public debt courses of various major emerging market countries by extending the standard debt sustainability analysis framework so as to more faithfully reproduce these countries' economic reality in two aspects. At first, he allowed them to differ in the cyclical stance of their fiscal policy and in their degree of fiscal responsiveness to debt. Second, he explicitly integrated the specific risk premium paid by each country when borrowing in foreign currency. The results led to three policy recommendations which were stated as follows. Firstly, a country should consider decreasing its exposure to currency risk only in extreme cases (in the example of Argentina). However, greater fiscal responsiveness (for example, stronger fiscal tightening whenever there is a debt increase) could enhance sustainability to a much greater extent. Finally, countries with low responsiveness to debt or a poor fiscal consolidation track record should be cautious with countercyclical fiscal policies, as they might trigger an unsustainable debt course in the economic cycle. Dag and Kizilkaya (2018) analyzed the external debt sustainability in Turkey by using the quarterly data belonging to net external debt stock to GDP in the period of 2004:Q1–2017:Q4. After using Fourier KPSS unit root test developed by Becker et al. (2006), the results showed that net external debt stock to GDP was not stationary so external debt in Turkey was not sustainable in the relevant period.

4 Empirical Analysis

4.1 Data and Methodology

In case of debt unsustainability, the emerging market economies are more subject to financial crises especially when there is experienced high volatility in foreign exchange rates. Particularity in the post-2013 period, after, the central banks of developed economies as US Federal Reserves' announcement concerning monetary tightening, the emerging market economies which are called fragile five experienced several difficulties but not so serious including increases in external borrowing costs and depreciation in their currencies. Therefore, in this study, the debt sustainability in fragile five including Brazil, Turkey, India, Indonesia, and South Africa is analyzed in the period of 2002:Q1–2018:Q3. The period covers the periods in which the volatility increased in the financial markets as a result of 2008–2009 Global Financial Crisis and 2010–2014 European Debt Crisis. In investigation of the stationary of series related to debt service ratios (for private nonfinancial sector) in these economies, monthly data which are obtained from CBRT have been used. Below is the graph and the table which show the debt service ratios of fragile five and the variables and measures, respectively.



Graph 1: Debt Service Ratios (private nonfinancial sector) of the Fragile Five – 2003:Q1–2018:Q3. Source: CBRT Electronic Data Delivery System.

Tab. 1: Variables and Measures

Variables	Measures
Debt sustainability of Brazil	Debt service ratio of Brazil
Debt sustainability of Turkey	Debt service ratio of Turkey
Debt sustainability of India	Debt service ratio of India
Debt sustainability of Indonesia	Debt service ratio of Indonesia
Debt sustainability of South Africa	Debt service ratio of South Africa

In the analysis, Fourier KPSS Stationary Test has been employed to test the stationary of the variables. The Fourier test developed by Becker et al. (2006) can detect not only sudden changes but also slow changes, and it seems that the position, number, and form of structural changes do not affect the power of the test. The methodology developed by Becker et al. (2006) seems strong to notice sharp and u-shaped breaks as well as smooth breaks near the end of a series, and it performs well when breaks are gradual. Becker et al. (2006) employ

trigonometric terms to capture unknown nonlinearities. The test they developed is a KPSS-type stationary test (Becker et al., 2006).

4.2 Analysis and Results

Since there is generally no specific information relating to the definite nature of the breaks and no practical knowledge about the location and the number of breaks to use in testing for stationary, using an incorrect specification for the number or form of breaks might create some dilemma such as ignoring the breaks together. Becker et al. (2006) developed a stationary test in which a chosen frequency component of a Fourier function was used to estimate the deterministic components of the model. A Fourier series could effectively model the behavior of an unknown function even though this function is not regular. In this stationary test, the Fourier function capturing the movements of the unknown function is the primary reason for applying it. The Fourier test developed by Becker et al. (2006) can detect not only sudden changes but also slow changes, and the position, number, and form of structural changes do not affect the power of the test. For this reason, the performance of the tests was substantially robust to several kinds of structural breaks often seen in economic analysis, including the breaks of opposite signs (Tsong, 2016).

Becker et al. (2006) take into account the following DGP:

$$y_t = X_t' \beta + Z_t' \gamma + r_t + \varepsilon_t \quad (1)$$

$$r_t = r_{t-1} + u_t,$$

where ε_t is a stationary error and u_t is independent and identically distributed with variance σ_u^2 . It is chosen $Z_t = [\sin(2\pi kt/T), \cos(2\pi kt/T)]'$ to capture a break in the deterministic term, where k represents the frequency and T is the sample size. Here, to test whether y_t is stationary or not, it is defined as $X_t = [1]$ for a level-stationary process for y_t and $X_t = [1, t]'$ for a trend-stationary process.

At the start, one of the models described below is estimated and the residuals are obtained, in order to acquire the t-statistic required for testing the null hypothesis ($H_{0=} \sigma_u^2 = 0$);

$$y_t = \alpha_0 + \gamma_1 \sin\left(\frac{2\pi kt}{T}\right) + \gamma_2 \cos\left(\frac{2\pi kt}{T}\right) + \varepsilon_t \quad (2)$$

$$y_t = \alpha_0 + \beta t + \gamma_1 \sin\left(\frac{2\pi kt}{T}\right) + \gamma_2 \cos\left(\frac{2\pi kt}{T}\right) + e_t \quad (3)$$

The following test statistics is obtained:

$$\tau_\mu(k) \text{ or } \tau_\tau(k) = \frac{1}{T^2} \frac{\sum_{t=1}^T \tilde{S}_t(k)^2}{\tilde{\sigma}^2} \quad (4)$$

where $\tilde{S}_t(k) = \sum_{j=1}^t \tilde{e}_j$ and $\tilde{e}_j$ denote the OLS residuals from the regression (2) for

$\tau_\mu(k)$ or (3) for $\tau_\tau(k)$. As in KPSS, a nonparametric estimate $\tilde{\sigma}^2$ of the long-run variance could be acquired by selecting a truncation lag parameter l and a set of weights $w_j, j = 1, \dots, l$

$$\tilde{\sigma}^2 = \tilde{\gamma}_0 + 2 \sum w_j \tilde{\gamma}_j, \quad (5)$$

where $\tilde{\gamma}_j$ refers to the j th sample auto-covariance of the residuals $\tilde{e}_t$ from Equation (2) or (3).

To determine the optimal number of k , the value which gives the minimum sum of residuals (OLS) is being selected. In case the nonlinear trend is not present in DGP, the standard KPSS test can be applied to it in order to obtain increased power. As it seems useful to test for the absence of a nonlinear trend, Becker et al. (2006) proposed that F-test statistic be used. Therefore, the following F-test statistic for this hypothesis (absence of a nonlinear trend (i.e., $\gamma_1 = \gamma_2 = 0$) can be calculated against the alternative of a nonlinear trend with a given frequency k .

$$F_i(k) = \frac{(SSR_0 - SSR_1(k)) / 2}{SSR_1(k) / (T - q)} \quad i = \mu, \tau, \quad (6)$$

The F-test could be used only if the null of stationary is rejected. The standard KPSS test statistic would be used in case trigonometric terms are not significant. In comparison with the critical values related to Fourier Test which are shown in the study of Becker et al. (2006), the results of the stationary test are stated below:

As seen in the Tab. 2, the variables are stationary at their level. According to the F-Test results which are used to test the significance of the trigonometric terms, it seems that trigonometric terms for all variables are significant when

Tab. 2: Fourier KPSS Unit Root Test (T=67)

Variable	Frequency	MinSSR	FKPSS	F-Statistic
DebtServiceRatioofBrazil	1	435.2867	0,1659	35.49
DebtServiceRatioofTurkey	1	487.6728	0,3076	31.18
DebtServiceRatioofIndia	1	25.5606	0,2745	135.06
DebtServiceRatioofIndonesia	1	2.4831	0,1739	208.35
DebtServiceRatioofSouthAfrica	2	27.0196	0,5464	38.257
FKPSSCriticalValues(1and2)		F-StatisticCriticValues		
1 %	0,2699	0,6671		6,730
5 %	0,1720	0,4152		4,929
10 %	0,1318	0,3150		4,133

the values are compared with the F-Statistic Critical Values which are shown in the study of Becker et al. (2006). Therefore, since all the series seem constant and stable, the analysis results indicate that the debt service ratios of Brazil, Turkey, India, Indonesia, and South Africa in the period of 2002:Q1–2018:Q3.

5 Conclusion

The emerging market economies including Brazil, Turkey, India, Indonesia, and South Africa have been identified as fragile five by Morgan Stanley in 2013 by focusing some macrofinancial fragilities of these economies. On the other hand, there has been observed no significant financial deterioration in these countries until the year of 2018 except for a slight contradiction in economic activity in some of them. However, especially in the recent period, it has been emphasized by researchers that the interest rate increases in parallel with the process of monetary policy tightening implemented by FED, which might affect these emerging market economies negatively in two ways. Firstly, the cost of external borrowing in emerging market economies increases as a result of appreciation seen in US dollar. It has been observed that the ratios of external debt to GDP ratio of these economies are already high and have an upward trend in time. Moreover, negative movements might be seen in capital flows in these economies with the increase in interest rates in advanced economies such as the USA or some EU countries.

In this study, the debt sustainability in fragile five including Brazil, Turkey, India, Indonesia, and South Africa is analyzed in the period of 2002:Q1–2018:Q3. In the analysis, the stationary of the debt service ratios of these economies are

tested by employing Fourier KPSS unit root test. The results of the analysis indicated that debt is sustainable in these economies in the relevant period. However, as mentioned above, the increase in external borrowing costs and negative movements in capital flows might lead to the deterioration of this outlook. Considering the balance of payments problems, appreciation of US dollar against the currencies of these economies, and difficulties in external debt, it is of great importance to take precautionary measures to reduce these vulnerabilities in these economies in the future. In this regard, Turkey can minimize its vulnerability by reducing external borrowing and finance its high current account deficit with net foreign investments instead of external borrowing.

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After EU Enlargement: Development or Deterioration

1 Introduction

In terms of the old communist small countries, globalization was a more substantial problem due to their starting position. While their markets were not even ready for basic rules of capitalism, the concept of being competitive globally was a faraway issue for them. Their unique conditions separated them from rest of the world. Klyacova mentioned these conditions as “ethical values, basic principles in the role of the state in the society and the economy, in mutual relations between citizens and state, in mutual relation between state and the entrepreneurial sphere, in character and structure of laws, in form of their exaction, in the size and structure of sanctions for misdemeanour of single legal norms” (Kirch: 2002, 184).

It is obvious that a deep and fundamental change was necessary for the transition economies. However, it was a critical watershed that the question of how this performance transformation would embody. It could appear spontaneously thanks to the external dynamics, but it was unrealistic to expect that the application process will finish in a short time and there will not be experienced any resist about not to change (Kirch: 2002, 186). Alternatively, countries could adjust their internal institutions as a process of accession by the help of an “external political anchor” (Kirch: 2002, 188).

Apart from these economic reasons, joining to EU was also substantial for Central and East Europe Countries (CEECs) due to the political problems and for their security. Although the Soviet Union collapsed, the new country, Russia, was regarded as big enough to be a threat. They wanted to be a part of the modern and capitalist world as independent and individual countries. Locating between two dominant players, Russia and EU, was regarded as a threat; and only EU was providing them capitalist world’s doors, a modernization process and an opportunity to be represented on the decision-making level.

1 It is derived from our presentations in “European Conference of Social Science Research”, IASSR, Marmara University, June 19–21, 2013, İstanbul, Turkey Congress.

On the EU side, even though the evidence that the accession would be beneficial for them was not clear, European citizens and governments supported the enlargement with the exception of Germany and Australia's reluctance. Due to their near location to the CEECs, they thought they could suffer from a huge amount of immigration from their neighbours. But, nevertheless the integration regarded as the way of assuring the stability for entire continent. It was required for building the shared values of integrated Europe. Besides, for last, it was necessary for sustaining the economic superiority against the new emerging countries which has cheap costs advantage. Even though the researches say the old 15 members had a modest benefit from the integration, it is hard to measure the real effect of not to integrate, especially for long term. Actually, the new members were already integrated to the EU markets, thanks to the European firms who invested in these countries before 2004 and their citizens who already migrated to the EU-15 countries.

So, it can be claimed that it was a natural process for liberal capitalist Europe to integrate not only in economic terms but also politically.

Thus, it can be asked now whether or not the enlargement has proceeded as a win-win game. On the EU 15 side, as it was stated in the House of Lords paper, the impact of integration has been "limited" but "positive" (House of Lords Report: 2006, 76). Germany and Australia which had significant trade value with CEECs and the Netherlands and France which had the biggest investment share on these countries have been the "net winners from enlargement" (House of Lords Report: 2006, 77). On the new member side, it can be said that there have been some apparent gains from the enlargement, but also due to some arguable indicators it is needed to be more careful about the final effect. Therefore, this article will discuss these indicators in order to study the total impact of the Eastern integration of European Union by exercising eight CEECs' growth rate, net trade, inflation rate, Foreign Direct Investment (FDI) and unemployment rates.

Before going on, it could be helpful to mention the economic environment of the CEECs and the differences of these enlargement. As it was stated in the Enlargement Papers of European Commission, firstly, unlike the previous enlargement (the Mediterranean Enlargement: Greece, Spain and Portugal), the eastern enlargements' countries were planned social economies. Therefore, the integration process was not only about EU but also capitalism. Secondly, "the average GDP per capita (in PPS) of Greece, Portugal and Spain was 66 per cent of level in EC-9. By contrast, the average GDP per capita of 12 candidates stood, in 1998, at only 38 per cent of level in EU-15" (Enlargement Paper: 2001, 7). Also, if it is compared to the population of these two enlargements' new members: the

population of candidates in 1998 was 28 % of population of EU-15, while in 1980 three candidates population was 22 %. Plus, in terms of GDP, it was obviously a different integration; 12 candidates' total GDP was 5 % of EU-15's in 1998, whereas three candidates' was 10 % in 1980 (Enlargement Paper: 2001, 6). This means, by the Eastern enlargement, the union admitted a huge population with a relatively low personal income level whose total GDP amount was quite small. So, all analyses about these countries should consider their unique conditions and relatively weak economic parameters.

Therefore, when it is regarded to their GDP performance after the accession to the union, a rapid increase can explain both their previous worse conditions' recovery and of course the facilities coming from the integration. It is also difficult to monitor the real impact of the enlargement on CEECs because, as mentioned before, the impact had already started to emerge in the 1990s.

2 Selected Macroeconomics Indicators in EU and CEEs Countries

In Tab. 1, old EU members' and new ones' joined in 2004 to the EU, GDP annual change per cent is given. New members' and old ones' growth performance can be seen in the table clearly. Also old and new members of EU economies' average growth rates and enlargement effect on that growth rate can be studied from this table.

The tables demonstrates that even though countries already reached high growth level in 1990s and especially after 2000, their growth experience after 2004 is significant. The average values of two periods, before and after enlargement, illustrates that the integration increased EU's GDP 0.1 % while the CEECs growth rate rose more than five times. Therefore, it can be easily claimed that the enlargement had much more positive impact on the new members rather than the old ones.

Performance of CEECs did not indicate to a significant fluctuation, because they belonged to Eastern Block. Decline in the period between 1990 and 1995 indicates the ERM crises. These countries' growth indicators after their membership illustrate an increment compared to their previous term. The major decline in the graph shows the crisis of 2007/2008.

With Graph 2, CEECs' and EU countries' growth performance have been compared. As it is clear from the graph new members' and EU countries' growth lines are parallel.

It is a fact that CEECs have reached high growth rate. So now, the sources of these rates can be studied in detailed. Artis et al. combined this "broad based

Tab. 1: GDP Annual Change (%). Source: World databank for 1990–2019 and Eurostat for 2019 figures

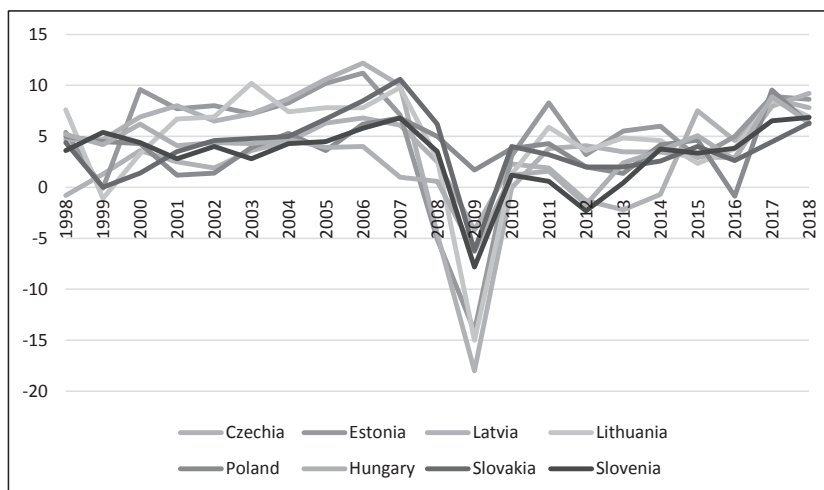
Year	EU	Czechia	Estonia	Latvia	Lithuania	Poland	Hungary	Slovakia	Slovenia
1998	3	-0,8	5,4	4,7	7,6	5	5,2	4,4	3,6
1999	3,1	1,3	-0,1	4,7	-1,1	4,5	4,2	0	5,4
2000	3,9	3,6	9,6	6,9	3,3	4,3	6,2	1,4	4,4
2001	2	2,5	7,7	8	6,7	1,2	4,1	3,5	2,8
2002	1,3	1,9	8	6,5	6,9	1,4	4,4	4,6	4
2003	1,3	3,6	7,2	7,2	10,2	3,9	4,3	4,8	2,8
2004	2,5	4,5	8,3	8,7	7,4	5,3	4,7	5	4,3
2005	2	6,3	10,2	10,6	7,8	3,6	3,9	6,7	4,5
2006	3,2	6,8	11,2	12,2	7,8	6,2	4	8,5	5,8
2007	2,9	6,1	7,1	10	9,8	6,8	1	10,6	6,8
2008	0,6	2,5	-5,1	-4,6	2,8	5	0,6	6,2	3,5
2009	-4,2	-4,2	-14,1	-18	-15	1,7	-6,3	-6,2	-7,8
2010	1,8	2,3	3,1	-0,03	1,3	3,8	1,2	4	1,2
2011	1,4	1,9	8,3	3,8	5,9	4,3	1,6	3,2	0,6
2012	-0,6	-1,3	3,2	4,1	3,6	2	-1,7	2	-2,3
2013	1,12	-2,2	5,51	3,49	4,83	1,37	2,4	2,01	0,45
2014	2,26	-0,68	6,01	3,45	4,6	4,16	3,58	2,58	3,76
2015	3,67	7,5	2,94	2,97	2,36	4,64	5,07	4	3,35
2016	2,76	4,6	4,99	2,94	3,78	-0,86	2,7	2,63	3,84
2017	3,9	8,7	8,91	7,96	8,6	9,55	8,9	4,46	6,54
2018	3,36	7,8	8,64	9,21	6,92	6,23	6,35	6,3	6,85

Tab. 2: Average GDP (%). Source: Prepared by writers

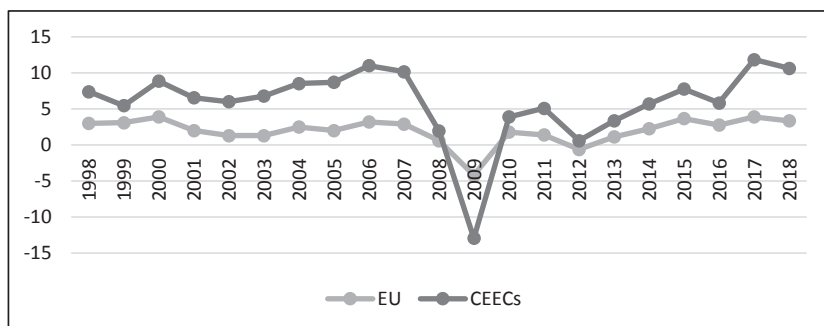
	Average 1998–2003	Average 2004–2012	Average 2013–2018
European Union	2,43	1,07	2,84
CEEC (8 Countries)	4,41	3,04	4,68

growth” with “strong demand in the traditional Western European export markets with steady growth in private consumption” and with “gradual recovery in domestic capital spending” (Artis et al.: 2006, 9). Indeed, CEECs’ growth can be explained with some sources below:

1. Export: Especially after 1990 these countries managed to reach huge export values, in particular Poland: thanks to the weak currency; Slovakia; Czechia



Graph 1: Graph 1: CEECs' GDP (%). Source: Calculated by the authors



Graph 2: Graph 2: EU and CEECs' GDP (%). Source: Created by authors using Google Public Data.

(Artis et al: 2006, 9–12). Besides, these export growth had some spill-over effect, like "...commodity diversification, improved trade flexibility and improvements in competitiveness, first of all in manufacturing sector" (Kirch: 2002, 60).

2. Domestic demand: Thanks to the free trade, domestic consumers have achieved more diversity and much more amount of product. This demand

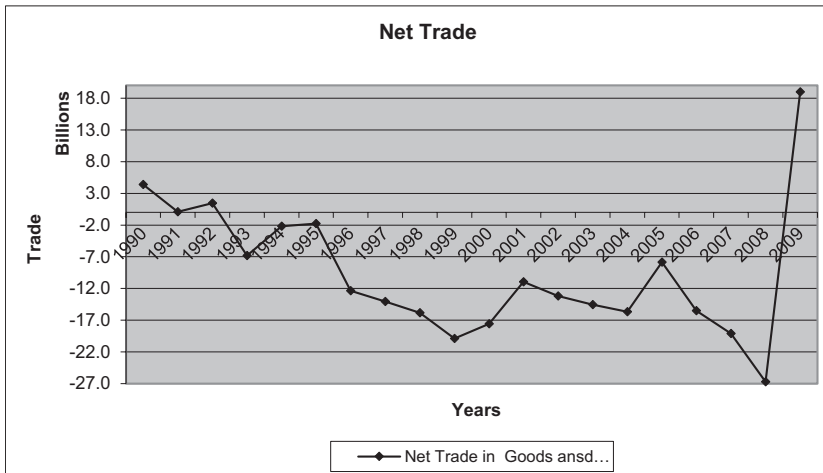
Tab. 3: Average GDP Growth Rates of CEECs. Source: World databank

	1990–2004	2005–2009	2010–2018
European Union	2,1	3,50	3,18
Czechia	1,1	1,86	5,73
Estonia	1,1	2,04	4,21
Latvia	-0,4	2,64	4,65
Lithuania	-0,3	4,66	3,91
Hungary	1,4	0,64	3,34
Poland	3,6	5,16	3,46
Slovenia	2,2	2,56	2,70
Slovakia	1,2	5,15	1,65

has been generally created by the private sector, and public had a diminishing role on markets (with the exception of Estonia, and the inflation risk has rose in Latvia due to the domestic demand) (Artis at al: 2006, 17–40).

3. Foreign investment: Nearly for all CEECs but especially for Poland, Czechia, Hungary, Slovakia and Lithuania, foreign direct investment (FDI) has been the key factor for growth. Where domestic capital accumulation is not enough (it is true for the transition economies like CEECs), FDI could become the main driver of the economy. Or, in some cases (Latvia and Lithuania), it can be used in order to finance the deficit (Artis at al: 2006, 17–40).
4. Investment spending: Particularly in Hungary, Poland and Czechia this expenditure has accelerated the economy. However, this spending also created the risk of deficit and inflation.

Tab. 3 demonstrates that eight new members had better growth rates between 2004 and 2008, than they did between 1998 and 2004. However, it rose to 0.62 in the period 2010–2018. According to the figures, Slovakia has experienced the modest growth rate while and Czechia reached its highest level. Hungary seems have some difficulties during the transition period. As it was stated in Foti's paper, due to the regional disparities, high inflation and residual of planned economy's price control habit, country could not reach the intended growth rate (Kirch: 2002, 92). It has the highest inflation rate (4.6) among the 10 new members in 2009 and the second highest central government debt proportion of GDP (81.7 %). Despite these structural problems, it had a greater rate than the EU average. On the other hand, our analyses do not include the figures of 2009 due to the crises. The figures changes dramatically, so they are not suitable to monitor the long-term indicators. At the same time, these abnormal figures



Graph 3: Net Trade. Source: World Development Indicators (WDI) & Global Development Finance (GDF) < <http://databank.worldbank.org/ddp/home.do> > [accessed 05.05.2011]

illustrate that because all these small economies became liberal, and because the foreign investment dominated the growth and the domestic markets, they also became vulnerable.

In that context, the structure of foreign trade and investment is needed to be studied with more detail. “The EU and the CEECs started to dismantle bilateral trade barriers in the early 1990s. By 1997, the EU had abolished all tariffs and quotas for imports from the candidate countries – with the exception of food products, some ‘sensitive’ items and services” (House of Lords: 2006, 71). On the other side, CEECs’ markets totally opened to the EU in 2002. Thanks to this liberalization, after 10 years Hungary increased its export 380 % and Czechia 280 % (House of Lords: 2006, 72). “In other words, long before membership, they were trading more with the EU than many of the EU countries were trading with each other” (House of Lords: 2006, 72). Therefore, can it be said that the EU had created a significant opportunity for the CEECs; but did CEECs manage to develop and catch-up with the EU by using this opportunity properly?

The answer is clear from the chart above. The figures are total amount of net trade value in goods and services of new members of the EU. It demonstrates that, particularly after the liberalization of the markets, during 1990s, their external trade balance turns to negative. Interestingly, after the accession, 2004,

this negative feature deteriorated. However, the crises years, 2007/2008, were the period when this imbalance was tended to recovery. So, it can be said that although the trade volume was redounded rapidly, thanks to the free trade, in general CEECs have had two main external trade problems:

1. Their import has been greater than their export.
2. Their export has been based on their import.

Due to the lack of capitalist institutions and the fact that they are small countries, they could not manage to develop their markets enough. Even though their investment reached huge amounts and they received enough foreign investment, capitalism is more than capital accumulation. Some institutions which enable to create added value by the help of innovation cannot occur in such a short time. They entail not only time, but also capacity for entrepreneurship and incitements. Especially, CEECs were deficient in knowledge, technology, experience and adequate production size in order to produce something advantageous relatively to the EU, with the exception of cheap labour. Therefore, "in fact by the time of accession the EU sold much more to the newcomers than it bought in return" (O'Brennan:2006, 134).

In order to analyze why a gap between export and import has occurred, reducing barriers can be the initial point. Indeed, abolishment of trade barriers has two effects: firstly, the relative price of intermediate input and final goods change, which means both the manufacturer and the consumer could reach cheaper products. This relocates production in domestic markets. Namely, the domestic firms who did not have price advantage had to face the risk of closing. "In most countries, transnational companies (and their subcontracting networks) can be identified as the main winners of accession, while small- and medium-sized domestic firms could only partially exploit the opening up of a huge market (mainly remaining concentrated on cross-border trade with clear geographic, financial and logistic constraints)" (Avery et al: 2009, 94). Secondly, the terms of trade values could be effected. Especially, the abolition of export subsidies will reduce the supply of these products and hence increase producer prices (Lejour et al: 2001, 20). However, in the case of CEECs, this price augmentation caused disadvantages in terms of domestic firms. Because, when the subsidies abolished, EU firms became cheaper in global markets. Besides, because of the first effect, EU firms became cheaper in domestic markets too.

As it is clear from Tab. 4 above, in general, CEECs have had a negative external balance. After enlargement, a significant change in the countries' net foreign trade did not occur. However, only Czechia's net foreign trade remained positive after enlargement.

	Czechia	Estonia	Latvia	Lithuania	Poland	Hungary	Slovakia	Slovenia
1997	4938429	-1124113	-847907	-1147487	-9822	-1327552	-2084484	-724800
1998	-2646960	-1115236	-1130200	-1518325	-12836	-1884962	-2350727	-792100
1999	-1902400	-877503,6	-1027000	-1404565	-15072	-2170078	-1109020	-1234700
2000	-3095199	-781461,5	-1043900	-1103768	-12307	-2912815	-894535	-1139900
2001	-3078056	-774676,7	-1335500	-1108025	-7661	-2237199	-1530072	-619600
2002	-2239991	-1117987	-1478900	-1314865	-7249	-2118856	-2165609	-247800
2003	-2519262	-1566973	-2002800	-1704181	-5725	-3278510	-648715,6	-622200
2004	-528976	-1947572	-2780700	-2382558	-5961	-3861272	1556974	-1257800
2005	2071640	-1919820	-3018300	-2916000	-3095	-3138308	-2363344	-1258100
2006	2584233	-2906291	-5130800	-4209267	-7372	-3084050	-2548660	-1458395
2007	2337152	-3618865	-6897900	-5873831	-19066	-962802,9	-908741	-2314390
2008	1706387	-3123817	-6013700	-6231365	-30659	-1752617	-898885,9	-3880506
2009	4576632	-775249,9	-1182600	-1181176	-7617	-3335034	1374277	-996315
2010	2804133	-330687,9	-1169300	-1684103	-1810	-4291895	181662,9	-1601835
2011	4276540,1	1328980,7	-1436505,7	-1104246,3	-11334,0	8676177,8	-462952,2	611035,3
2012	6338764,8	361295,8	-1304303,4	359862,6	-2781,0	8595934,5	3767389,3	1825844,7
2013	8526688,8	706953,7	-1122976,8	575717,1	9689,0	9409490,6	4474266,7	3229762,9
2014	10720948,2	943649,9	-456376,0	918198,6	7755,0	8964277,9	3942724,2	3819662,3
2015	7644146,8	917654,7	-138863,3	-273612,4	14575,0	9975359,5	1376387,5	3776392,7
2016	10720948,2	980707,6	248052,2	518661,3	18721,0	12636699,5	2349414,2	4197482,0
2017	10125049,8	1224311,4	-408380,8	1350883,5	21843,0	10451095,0	1750761,3	4840276,0

Tab. 4: Net Trade. Source: EUROSTAT, External and Intra-EU trade – A Statistical Yearbook

There has been a problem of unemployment which could not be absorbed by the manufacturing and the services sectors. So, apart from the structural change, the effect of FDI is needed to be studied in detail in order to understand the weakness of the CEECs' economies.

Foreign interest started before the accession process and continued to rise during the transition period. It was regarded as "a fast way of improving competitiveness" and "an easier entry into foreign markets" (Kirch: 2002, 64). Especially, in the case of lack of domestic sources (not only capital but also technology) FDI's way looked as the best way of remedy. However, "FDI have in many cases created enclaves surrounded by backwardness, capacity destructions and lay-offs of workforce" (Kirch: 2002, 65). It is because due to the fact that they generally came for takeovers instead of greenfield investments (Avery et al: 2009, 10). Besides, the investment entered the CEEC markets in order to take advantage of privatization while the economies transforming their markets central-planned to liberal markets. However, the integration, of course, facilitated foreign firms' decision-making process. Another fact about the FDI flows to the CEECs is

Tab. 5: FDI % of GDP. Source: Prepared by writers

Year	Czechia	Estonia	Latvia	Lithuania	Poland	Hungary	Slovakia	Slovenia
1998	5,79	10,38	5,39	8,22	3,68	6,97	1,92	0,99
1999	10,15	5,35	4,77	4,43	4,33	6,85	1,18	0,48
2000	8,48	6,82	5,27	3,31	5,45	5,97	7,15	0,68
2001	8,76	8,70	1,59	3,67	3,00	7,48	9,22	2,46
2002	10,83	3,89	2,72	5,03	2,08	4,54	11,85	7,17
2003	2,12	9,34	2,71	0,96	2,12	2,61	1,22	1,03
2004	4,37	8,03	4,63	3,43	5,03	4,20	5,42	2,46
2005	8,92	22,49	5,06	4,58	3,64	7,71	4,89	2,72
2006	3,72	13,17	8,54	6,82	6,30	16,60	5,90	1,77
2007	5,88	15,59	9,43	5,95	6,01	51,90	4,63	3,98
2008	2,92	7,88	4,63	4,04	2,84	48,62	4,16	3,34
2009	1,45	9,76	-0,17	0,24	3,34	-2,34	1,84	-0,70
2010	3,08	10,89	1,80	1,93	3,63	-16,07	0,76	1,35
2011	2,48	1,97	5,32	3,38	2,97	6,88	3,81	1,65
2012	4,50	7,80	3,80	1,30	1,50	8,30	1,90	0,10
2013	3,50	4,40	3,30	1,50	0,20	-2,80	1,00	0,20
2014	3,90	6,70	3,00	1,00	3,60	9,20	-0,40	2,00
2015	0,90	-3,10	3,10	2,30	3,20	-4,50	1,70	4,00
2016	5,60	3,90	0,90	2,20	3,90	54,80	5,30	3,20
2017	4,30	5,90	3,70	2,50	2,00	-9,60	6,20	2,20

that their structure did not change significantly after the enlargement (Avery et al: 2009, 95).

Tab. 5 illustrates the ratio of countries' FDI to GDP. In terms of before and after enlargement the remarkable point is in four out of eight countries (Czechia, Lithuania, Slovakia and Slovenia) FDI decreased after enlargement and increased in other four (Estonia, Poland, Latvia and Hungary). When it is looked at the average of this eight countries' the ratio of FDI to GDP increase 4,94% to 6,3% after joining to EU..

The diagram below shows ratio of FDI to GDP. As it is clear from the diagram after joining to EU, FDI to Estonia and Hungary rose considerably. After the crisis in 2007/2008, FDI to Estonia did not diminish dramatically while FDI to Hungary decreased sharply. Due to the low-tech structure of FDI and chronic problems which came from transition period, unemployment has been a critical issue for CEECs.

Tab. 6 illustrates countries' unemployment rates. It is obvious from the graph that unemployment rates in these eight countries stayed within a band until EU membership. After joining the EU, a substantial decline can be seen in the figures. However, during the crisis at 2007/2008 the rates soared.

Before the enlargement, EU's average unemployment rate was 9,14 while eight countries' average was calculated as 11,16. After the enlargement in 2004, EU's average fell to 8,81 while eight countries' figure fell to 9,17. As it can be seen from Tab. 6 inflation targeting was implemented in four out of eight countries. Among the new EU member countries, only Slovakia started implementing inflation targeting in 2005.. These countries' average inflation rate became 2,80 after enlargement while it was 5,34 before enlargement. The effect of inflation targeting will be investigated in the empiric analysis part in detail.

According to the graph 6, countries reached a low and stable inflation rate after the enlargement while they had a high inflation rate in the 1990s. The inflation rates rose only in Estonia, Lithuania and Latvia during the crisis.

3 Empirical Analysis

With this study EU's new eight members' performance before and after enlargement were analyzed by using their macroeconomic data for the period 2003 and 2018. Macroeconomic variables such as GDP, Net Trade, Unemployment rate, Inflation rate and FDI were used. The aim of this empirical analysis is to illustrate how these eight countries' economic performance changed after they joined the EU for the period 2003–2018. Furthermore, apart from EU membership, whether or not these countries were affected by the crisis in 2007/2008 was investigated by adding a new dummy variable into the model. EViews program was employed for analysis.

In the regression models, variables are admitted to be continuous for the period studied. However, during that period variables cannot follow a steady path due to crisis or circumstances like crisis. Therefore, Dummy Variables Method was employed in order to distinguish diffractions like crisis from normal times. Dummy Variables Method is a way that enables us to discuss interesting and various issues.

There are four different models in the literature that investigate two valued dependent variables in dummy variables method. These models are classified as Linear Probability Model, Logit Model, Probit Model and Tobit Model. Linear Probability Model is the most basic model that is employed when dependent variable is 1 and 0. If Linear Probability Model is written as $Y = b_0 + b_1X + u$, while dependent variable (Y) takes the value 1 and 0, the error terms take values $1 - b_1X$ and $-b_1X$, respectively.

	Czechia		Estonia		Latvia		Lithuania		Poland		Hungary		Slovakia		Slovenia	
	UR	IR	UR	IR	UR	IR	UR	IR	UR	IR	UR	IR	UR	IR	UR	IR
1998	6,5	10,63	9,5	8,21	14,5	4,66	13,7	5,07	10,7	11,73	7,8	14,18	6,5	6,7	7,6	7,91
1999	8,7	2,14	11,6	3,3	13,8	2,36	13,4	0,75	12,5	7,28	7	10,03	8,7	10,57	7,4	6,15
2000	8,8	3,9	13,1	4,03	14,2	2,65	15,9	1,01	16,1	10,06	6,4	9,78	8,8	12,04	7,2	8,88
2001	8,1	4,5	12,4	5,6	13,1	2,5	16,8	1,6	18,3	5,3	5,6	9,1	19,5	7,2	6,2	8,6
2002	7,3	1,4	9,4	3,6	13,2	2	13	0,3	20	1,9	5,6	5,2	18,8	3,5	6,3	7,5
2003	7,8	-0,1	10,7	1,4	10,6	2,9	12,9	-1,1	19,7	0,7	5,8	4,7	17,7	8,4	6,7	5,7
2004	8,3	2,6	10	3	9,9	6,2	11,3	1,2	19	3,6	6,1	6,8	18,4	7,5	6,3	3,7
2005	7,9	1,6	7,9	4,1	8,9	6,9	8,3	2,7	17,8	2,2	7,2	3,5	16,4	2,8	6,5	2,5
2006	7,1	2,1	5,9	4,4	6,8	6,6	5,6	3,8	13,9	1,3	7,5	4	13,5	4,3	6	2,5
2007	5,3	3	4,7	6,7	6	10,1	4,3	5,8	9,6	2,6	7,4	7,9	11,2	1,9	4,9	3,8
2008	4,4	6,3	5,5	10,6	7,4	15,3	5,8	11,1	7,1	4,2	7,8	6	9,6	3,9	4,4	5,5
2009	6,7	0,6	13,8	0,2	17,1	3,3	13,7	4,2	8,2	4	10	4	12,1	0,9	5,9	0,9
2010	7,3	1,2	16,9	2,7	18,7	-1,2	17,8	1,2	9,6	2,7	11,2	4,7	14,5	0,7	7,3	2,1
2011	6,7	2,1	11,7	5,1	14,6	4,2	14,3	4,1	10	3,9	11	3,9	14,1	4,1	8,7	2,1
2012	7	3,3	10	3,9	15	2,3	13,4	3,1	4,9	3,6	11	5,7	14	3,6	8,9	2,6
2013	7	1,4	8,6	2,8	11,9	0	11,8	1	5,4	1	10,2	1,7	14,2	1,4	10,1	1,8
2014	6,1	0,3	7,4	-0,1	10,8	0,6	10,7	0,1	5,6	0,1	7,7	-0,2	13,2	-0,1	9,7	0,2
2015	5,1	0,3	6,2	-0,5	9,9	0,2	9,1	-0,9	5,7	-0,9	6,8	-0,1	11,5	-0,3	9	-0,5
2016	4	0,7	6,8	0,1	9,6	0,1	7,9	0,9	6	-0,7	5,1	0,4	9,7	-0,5		-0,1
2017	2,9	2,5	5,8	3,4	8,7	2,9	7,1	3,7	5,5	2,1	4,2	2,3	8,1	1,3	6,6	1,4

Tab. 6: Unemployment Rate/Inflation Rate. Source: Prepared by writers

AAAA: Date of Transition to Inflation Targeting, UR: Unemployment Rate; IR: Inflation Rate

In our study, three out of eight countries' GDP increased after membership while five out of eight decreased. The biggest augmentation was experienced in Slovakia; Poland and Czechia followed it. After membership, the largest loss in GDP was calculated in Hungary and then in Latvia, Estonia, Lithuania, and Slovenia, respectively. Our model that includes not only EU membership effect but also crisis effect on GDP concluded as follows.

It has been calculated that holding EU membership enhances GDP in six countries (Czechia, Estonia, Latvia, Lithuania, Poland and Slovakia), while it decreases in two countries (Hungary and Slovenia). On the other hand, crisis in 2007/2008 reduced GDP in seven countries and rose in Poland. Briefly, among these eight countries only Poland and Slovakia's GDP increased after EU membership. Besides, it has been seen that 2007/2008 crisis had a significant impact on these poor performances.

The results for the analysis that estimate countries' net foreign trade performance are as follows. In all countries net foreign trade augmented against the country after enlargement. After 2007/2008 crisis added to the analysis it was seen that net foreign trade increased in Poland after enlargement and in Hungary during the crisis. Eight countries' average foreign trade went up from -1.218.459,3 billion \$ before enlargement to -1.767.720,8 billion \$ after enlargement. The crisis in 2007/2008 raised net foreign trade in these countries on the average -1.667.112,1 billion \$.

Findings related to unemployment rate, another variable in the study, are as follows. When looking at the effect of EU membership on selected eight countries, it can be found that after enlargement, unemployment rate increased in Hungary, while a significant difference could not be calculated in Slovakia and Slovenia. In all other countries, unemployment rate decreased. Poland became the country whose unemployment rate diminished the most. When effect of crisis in 2007/2008 added into the model that was run in order to calculate EU membership's effect, results changed like this: during crisis, unemployment rates tended to decrease in all selected countries. Finally, when looking at the effect of EU membership and crisis together, it is observed that joining to EU increased and crisis decreased unemployment rate in Hungary, Slovakia and Slovenia. In other all countries, both joining to EU and crisis had an effect that decreased unemployment rate.

Another variable employed in order to calculate EU membership's effect on countries' macroeconomic performance is countries' inflation rates. In the model that EU membership used as the only variable, it is estimated that inflation rates augmented in Estonia, Latvia and Lithuania, and decreased in other countries, after the enlargement in 2004. When the effect of crisis added to the

model, it is found that inflation rate in only Slovakia decreased, and it increased in the rest of the selected countries. In the model, when both EU membership and the crisis were used in 2007/2008, it was observed that EU membership increased the inflation rate in Latvia and Lithuania and the inflation rate decreased in the rest of the selected countries. Czechia, Poland, Hungary and Slovakia started to implement inflation targeting strategy after enlargement in different times. Results for the models that were run in order to find the effect of inflation targeting strategy after enlargement are here: Model for Czechia was calculated as $CZE=10,63+0,062EU-8,26IT$. The inflation targeting strategy in Czechia was much more effective on decreasing the inflation than EU membership. Model for Poland was calculated as $POL=9,50-1,50EU-5$. It is found that inflation targeting had also much effect on decreasing inflation in Poland. Model for Hungary was calculated as $HUN=10,77-0,0011EU-5,82IT$. Likewise Poland and Czechia, inflation targeting was more effective on decreasing inflation in Hungary than EU membership. Finally, in Slovakia the model was calculated as $SR=8,068-2,918EU-2,544IT$. In this model, it is concluded that EU membership's effect on decreasing inflation is bigger than inflation targeting. The reason why Slovakia separates from other countries that implemented inflation targeting can be explained by the later date (2005) when Slovakia started to implement inflation targeting. Yet Czechia started to implement inflation targeting in 1998, Poland in 1999 and Hungary in 2001. Namely, only Slovakia started to implement inflation targeting strategy after enlargement.

In the model that we study regarding effect of EU membership, we found that FDI mounted in Estonia, Latvia, Poland and Hungary, while it went down in the rest of selected countries. When the effect of crisis in 2007/2008 added to the model, it is reached that FDI to Hungary and Slovenia increased during the crisis while FDI to others decreased. Different from the first model, effect of EU membership enhanced FDI in Estonia, Latvia and Poland and decreased in Hungary. In the other countries (Czechia, Lithuania, Slovakia and Slovenia), effect of EU membership decreased FDI like the first model.

Conclusion

In this study eight Central and East European Countries' GDP, Net Trade, Unemployment rate and FDI variables were employed. The study aimed to investigate the effect of EU enlargement in 2004 on new memberships. It can be said that in general EU membership had a positive effect on new members' macroeconomic performance.

When the effect of EU membership is studied in detail, it was reached that in three out of eight countries GDP increased and in five out of eight countries it decreased. The effect of crisis in 2007/2008 was calculated negative in seven out of eight countries while it was positive only in Poland. On the other hand, their net foreign trade performance progressed against these eight countries after enlargement. When the effect of crisis added to the model it is seen that net foreign trade progressed in favour of Poland after enlargement and in Hungary during crisis. According to another macroeconomic variable, unemployment, membership had no effect on two countries, a negative effect on five of them and only a positive effect on Hungary. During crisis in all eight countries, unemployment tended to diminish. This can be because these countries had cheap labour.

Another macroeconomic variable that was employed in this study was inflation rate. After enlargement, inflation rate in Estonia, Latvia and Lithuania enhanced, and it decreased in the rest of them. During the crisis, only Slovakia's inflation rate diminished while all others went up. When looked at the effect of countries' inflation targeting strategy, we reached these results: in countries which implemented inflation target strategy like Czechia, Poland and Hungary, the strategy had much more effect on decreasing inflation than EU membership. In Slovakia, which started to implement inflation targeting strategy after EU membership, the effect of EU membership on reducing inflation was calculated to be more effective than the effect of inflation targeting strategy. Lastly, FDI was studied as a macroeconomic indicator. After enlargement in 2004, the amount of FDI to Estonia, Latvia, Poland and Hungary rose, and FDI to other selected countries diminished. During crisis, FDI to Hungary and Slovenia increased while FDI to others declined.

When eight CEECs' five macroeconomic indicators' effect on the country's economic performance was studied, it was seen that GDP and net trade performance were better before the enlargement, FDI was better after enlargement and Unemployment and Inflation rates were better after the crisis in 2007/2008. As a result, it can be claimed that the enlargement in 2004 had no positive effect on selected EU members.

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Serkan Celik

Credits and Credit Analysis in Banking Sector

1 Introduction

The main activity of banks can be defined as providing funds to who needs funds by collecting deposits. Banks, which are a commercial enterprise, have to perform transactions called credit analysis in order to minimize the risk of non-payment within the natural structure of the credit while selling credit and other financial products to their customers as a company that produces and sells any product. Banks provide the most accurate offer to the customers by analyzing the possible risks and positive aspects of their customers with the credit analysis method. In terms of banks, repayment of credits in terms of maturities means that their credit portfolios are healthy and their profitability is high.

In order to ensure the continuity of commercial life of banks and contribute to the sustainable growth of the economies of the country, the credit transactions of the banking sector are important. It means that the contribution to the country's economy by providing the financing needs of the right projects will be reflected positively to many macro-indicators such as unemployment and tax revenues, and also to ensure the return of the wheels in terms of the country's economy. The economic importance of credits has been experienced by the whole world in the last global crisis that emerged on the basis of unpaid credits. Since the shock experienced by a financial actor causes crises in multiple markets at a time, the way in which banks manage the lending process is of great importance both for their profitability and for their contribution to the economy of the country.

This study tries to determine the importance of the credit analysis activities and the importance of the banks in terms of banking sector and global economies as mentioned above.

2 Credit Concept and Elements

2.1 Credit Definition and Elements

The framework for credits, the most important product of the banking sector, is regulated by the relevant articles of the Banking Law No. 5411. According to these articles:

Non-cash credits such as cash credits and letters of guarantee, counter-guarantees, guarantees, aval, turnover, acceptance and other such commitments,

purchased bonds and similar capital market instruments, deposits, or any other form of borrowing, receivables arising from the sale, overdue cash credits, uncollected interest rates, uncollected interests of non-cash credits, receivables from reverse repurchase agreements, futures and option contracts, risks assumed by other contracts, shareholding interests and transactions accepted as credits by the Board. Regardless of the account they are monitored, they are counted as credits in this Law (Banking Law [BK], 2005:48).

In addition to those specified in the first paragraph, the financing provided by the financial leasing of the development and investment banks and the payment of the movable and immovable goods and services of participation banks or the investments of profit and loss partnership, real estate, equipment or commodity supply or financing, financing of goods and documents, joint financing provided by investments, or similar methods are also considered as a credit in this Law.

If it is necessary to make a definition based on the meaning of the word credit, the word credit originated from the word “credere” which means to believe, in ancient Latin. Even after all these years, the word “credit” has remained meaningfully close to the root where it was first derived. In such a case, the lenders give “credit” on a certain interest rate and under agreed terms on the confidence that the borrower will pay the debt back.

As can be seen from the definition of credit process 2 focuses on the main basis:

1. The debtor who uses credit is willing to repay the loan.
2. The debtor using the credit has the power to repay the credit.

The meaning of the willingness of the borrower who uses credits to pay the credit is, of course, the reliability of the borrower as a result of the evaluation made by the lender about the borrower. The second and more important requirement, namely the ability to repay the credit, is the guarantee of the ones who want to use the credit resulting from the evaluation with the assets that have the financial power to repay the credit (Golin et al., 2013:1).

If we look at another definition, it is roughly possible to define the credit as a credit to another party in exchange for a certain provision, the current known interest rate, or under the conditions agreed on without interest (Rosenberg, 1982:312).

If we try to make a definition in terms of banking, as a result of banks' intelligence activities, the real or legal person is given to customers, the existing laws, the bank's own financial facilities and liabilities, and after considering and evaluating them, in return for a certain amount of collateral or unsecured debt field service, money, collateral, etc. the limits and possibilities that are recognized in ways (Akgüç, 2006:7).

3 Credit Analysis (Evaluation) Process

3.1 Credit Analysis Concept

It is possible to define the concept of credit analysis as a concept of risk management in order to avoid the risk of nonpayment of each credit, instead of defining the necessary intelligence about the customer/person applying for the credit, in order to take a positive or negative decision after the credit application. Credit analysis is not only the analysis of the current situation of the person requesting a credit by means of financial ratios. The past situation of the credit requester means that the bank can take decisions that can “protect itself from credit risk rather than taking positive/negative decisions” by examining all the concepts that may be important in relation to its future vision and commercial life (op. cit.:1).

The credit analysis of the banks performed by the relevant credit intelligence personnel is the most difficult and most important stage of the credit process in terms of the bank. The banks should employ trained and experienced personnel to perform these intelligence procedures objectively and accurately in the relevant units. In addition, they should support the necessary development processes of the related personnel of the banks through continuous training (Öztürk, 2015:33).

According to the general belief in the banking sector, credit analysis means assessing the risk in the credit transaction. In the simplest form, “credit analysis” that we can define as managing the credit process is to analyze whether the borrower and the borrower have the necessary conditions to obtain the credit (op. cit.:37).

It is aimed that the bank evaluates itself on credit risk by credit analysis and takes the decision if it is necessary to extend the risk. Each credit carries a risk within itself. There is a possibility that the bank may suffer losses due to the difference in inflation related with the long duration of the credit repayment between the date on which the loan is given and the date on which the loan is paid. In addition, each credit carries the risk of being nonrepaid within itself. No matter how reliable the customer is and how rational credit decision is, credit repayment situation can be caused due to different factors (op.cit.:1). These factors are possible as follows:

- a) To make false assumptions about the future,
- b) Economic crisis caused by disaster that can be experienced due to nature,
- c) Liquidity problems that may be experienced due to the decreasing and increasing in the sales of the firm requesting credit,

- d) Unable to keep up with the competition due to the fact that the company requesting credit cannot keep up with the technology,
- e) Crises in general economic life,
- f) Economic problems that the borrower may experience due to the economic policies of the government such as devaluation,
- g) Sales problems due to seasonal effects,
- h) Bad management of the firm requesting credit,
- i) Bad intention,
- j) The fact that the firm demanding a credit is unable to sell the product it produces at a time due to international conflicts as in the crisis experienced with Russia and the tomato producers,
- k) It may be possible that the credit cannot be paid back due to the reasons such as conflicts with the top management of the company that is requesting the loan.

The credit analysis process starts with the application of the loan application to the bank. A preliminary meeting is held for the realization of the credit relationship. Even the purpose of the business to ask for credit is very important in terms of credit analysis. In order to start the credit analysis process, the documents that the bank will request from the applicant must be prepared and delivered. In order to make a credit assessment, the bank staff assesses the applicants about the credit application in accordance with the methods determined in the credit policy for the determination of the moral and financial situation in the commercial life of the applicant (Orhaner and Şahin, 2011:11).

In the process of credit assessment in developing countries, the provision of data that will be based on the assessment of the bank poses a major problem for the banks. The problem of the subterranean economy in developing countries is a major problem in the assessment process. Understanding the financial structure of the credit applicant because of the unregistered economy may pose a problem for the bank, and due to the fact that the organizational culture is not settled, the financial statements prepared away from professionalism pose a danger to the banks in the assessment process (Çabukel, 2007:8).

As per Article 52 of the Banking Regulatory and Supervisory Agency (BRSA), banks are required to assess/analyze the credit risks of banks, to regularly analyze and monitor the financial strength of the counterparty, to provide the necessary information and documents, and to determine the principles related to these. In this context, credit customers are obliged to submit the required information and documents to the banks on a consolidated and unconsolidated basis.

In addition, in the “Guide For The Credit Management Of Banks” published by the Banking Regulatory and Supervisory Agency (BRSA) in the third section titled “Management of Credit Allocation Processes”, the framework of the evaluation procedures that banks should carry out regarding credit processes has been clearly drawn. According to this:

“Principle 13 - Banks form predefined assessment and approval functions for effective management of credits. Credit approvals must conform to the bank’s written procedures.”

- 75- Credit allocation process is a process that includes assessment and analysis of customer demands, preparation of credit proposal, limit allocation/revision, maturity renewal, modifications in the conditions of use, and credit approval reviews.
- 76- Credit allocation processes are managed by experts and trained people.
- 77- Banks can apply single-signed approval, two-signed approval, or a committee approval process according to the size and structure of the credit. The procedures and principles for the approval process are determined in writing, including the functioning of the committees. Adequate transparency of the bank’s decisions taken during the credit approval process should be established, and the final authority authorized to approve the credit should be appointed.
- 78- The Board of Managements establishes structures and practices that will prevent interference in the operation of the credit evaluation process by shareholders, management, or other related parties.

3.2 Crediting Process and Credit Assessment Elements

3.2.1 Crediting Process

In order to better understand the place and importance of the credit assessment process in the crediting process, it will be appropriate to examine the crediting processes of the banks in steps from the first meeting with the customer to the completion of the credit.

Crediting Process:

- Credit application,
- Pre-interview with the person requesting credit,
- Visit of the bank staff and the first interview,
- Investigation of the credit request,

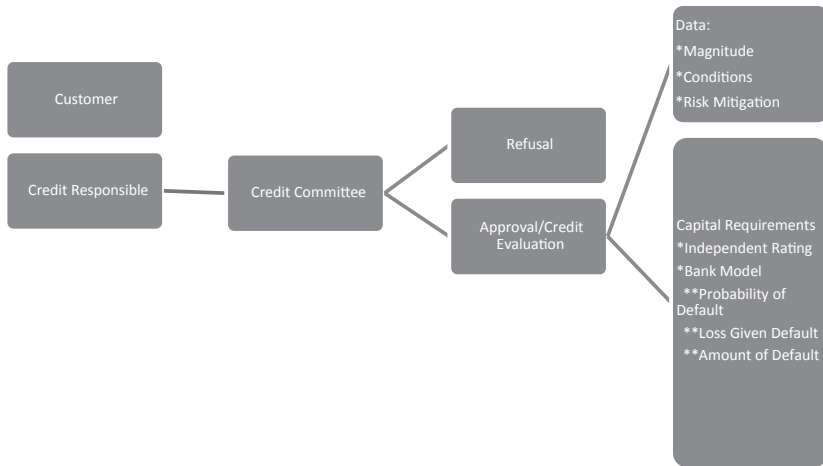


Fig. 1: BASEL II Credits Process. Reference: (op.cit.:26)

- Investigation of the suitability of the credit,
- Credit Risk Assessment,
- Completion of Documentation Procedures for Credit,
- Credit Assessment (Intelligence and Limit Analysis),
- Monitoring of Credit and Prevention of Losses,
- Denial of Credit.

3.2.2 Factors to be Considered in the Credit Evaluation Process

During the credit evaluation process, the related personnel should consider many factors in the light of the information provided by the person requesting the credit and their work when evaluating the credit demand. If analyst personnel ignores any of these factors, the decision phase will have a negative impact on the decision phase (op.cit.:41). If we examine these factors respectively:

- Personal Qualifications:** The personal qualifications of the person requesting the credit are the most important factor affecting the credit evaluation process. Banks will prefer customers to be creditworthy and trustworthy and not to have bad reputation. In addition to this, they will judge their customers' ability in managing their businesses, their asset and management capabilities, and the potential to repay the credit as a preference for the evaluation process.

- b) **Financial Factors:** When assessing a credit request, analysts will of course see the financial potential of the borrower as a guarantee for the repayment of the credit as one of the most important elements. At this stage, the bank will pay attention to two main headings:

- 1) Sufficient equity capital of the company to repay the credit.
- 2) Resource creation capacity.

The purpose of the source is to have the potential to repay the loan with the revenues that will be obtained from the activity as a result of the activity, to have the potential to repay the credit with the debt funds to be found if necessary or to have the return potential of the borrower's assets to repay the credit. The bank has to take these principles into consideration at the assessment stage. A customer with the necessary financial potential will have the opportunity to repay the loan with their assets even if their income is disrupted. Otherwise, the bank will be fully responsible for the risk in the credit decisions it will take, regardless of its financial potential.

- c) **Economic Factors:** Economic factors should not be confused with financial factors. The economic factors to be taken into consideration by the bank are external factors that may disrupt the payment regardless of the financial and characteristic factors of the borrowers who are beyond the control of the borrower. These are:

- 1) Economic conditions,
- 2) Fluctuations in the business line,
- 3) Government policies.

Crises that may be experienced by the economic structure of the customer's market, the economic fluctuations in the economy or only the crises that may occur in the business line where the customer is headed, the problems related to the demand, intercountry crises, or changes in the government's economic policies will be taken into consideration by the bank and assessed at the assessment stage.

- d) **Other Factors:** Even though the personal, financial structure and external factors of the borrower are favorable, the bank must take into account some important issues. The most important ones can be listed as follows:

1. The legal structure of the firm that demands credit, institutional culture,
2. Is the company a one-man business? Does it have a shared management structure?
3. Will credit demand result efficiently in terms of banks?
4. Does the demand for credit parallel the bank's credit policy?

5. Are the bank's financial ratios appropriate to the credit demand?

3.2.3 Objectives of the Credit Assessment Process

Banks, of course, do not aim to protect themselves against risk only when they are carrying out the credit assessment process. The credit assessment process provides banks with a number of significant benefits. Let us briefly summarize what the banks aim for in the credit assessment process.

1. The main purpose of the credit assessment process is to determine whether the customer/potential customer who requests credit is eligible for credit. This occurs at this stage whether the borrower has the power to repay the credit and the moral request to pay it back.
2. Lending is the main profit item of banks. Banks want to increase their profitability and to avoid risky credit approvals by making credit ratings to keep their credit portfolios efficient and profitable.
3. The credit demand of the borrower is examined well by offering the appropriate maturity and interest rates to the customer.
4. The aim of this is to provide the customer with the opportunity to benefit from the different services needs through the correct analysis of the customer's needs and to turn the credit relationship into a continuous customer relationship by providing some kind of financial consultancy services to the creditor.
5. If the credit demand is considered to be reasonable, legal, and profitable within the framework of the pricing of the risks related to the loan demand and if the credit is considered risky if necessary, the decision to refuse the bank in order to keep it out of risk is the decision.

4 Conclusion

The most important stage of the lending process in terms of the banking sector is the credit analysis/evaluation stage in terms of obtaining the credit decision correctly and funding the right investment projects. The banking sector, especially in countries like our country where a large portion of the sector is covered by the banks, is important both for the correct analysis of the credit and for the decision-making process of the banks in the sector and for the positive continuation of the macroeconomic outlook as well as for the absence of any liquidity crisis. It is vital for the sector that banks are prepared for innovations and new risks by continuously educating their personnel about credit analysis and the importance of credit analysis activity and keeping them informed about the authority boundary/framework.

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Selim Tuzunturk

Fundamentals of Sample Survey Research: A Statistical Perspective

1 Introduction

Science is the most trustworthy way of acquiring reliable and valid knowledge about the natural world (Nayak and Singh, 2015: 6). The *science of statistics* is essentially a branch of Applied Mathematics, and may be regarded as mathematics applied to *observational data* (Fisher, 1934: 1). *Statistics* is a mathematical discipline that describes procedures for deriving results about a population from sample data (Romejin, 2011: 751). Data is collected by three techniques: *Experiment*, *observation*, and *survey* (Bülbül, 2006: 58). *Experiment* technique is a scientific test that is done to find out how something reacts under certain conditions, or to find out if a particular idea is true (Longman Dictionary, 2003: 548). Laboratory conditions needed for experiment is used mostly in physical and health sciences. *Observation* is the action of observing something or someone and involves the recording of data. The best way to know how people act and think is to observe them repeatedly and directly over time (Polland, 1998:1). However, it is not possible to follow all people, and to record their behaviors. Moreover, some kinds of behaviors such as attitudes, beliefs, and opinions cannot be observed directly (Polland, 1998:1). In the present case, researchers apply a practical research tool, which is known as a survey. A *survey* is an activity that collects information in an organized and methodical manner about the characteristics of interest from some or all units of a population using well-defined concepts, methods, and procedures, and compiles such information into a useful summary form (Franklin and Walker, 2003: 1). It is a set of questions that you ask a large number of people in order to find out about their opinion or behavior (Longman Dictionary, 2003: 1673). It is often the best way to get information and feedback to use in planning (Hart and et al., 2010: 4).

As a natural consequence of living in an information society, there is a growing demand for *statistical information* about the economic, social, and cultural topics which will enable policy makers to make informed decisions (Bethlehem, 2009: 1). Surveys typically collect three types of *information* (Rea and Parker, 2014: 6): Descriptive, behavioral, and attitudinal. *Descriptive information* is collecting data on variables such as gender, age, education, household

size, and income for the purpose of eliciting the descriptive information or facts about the respondents. These socioeconomic characteristics provide important information that enables the researcher to better understand the larger population represented by the sample. *Behavioral information*, such as the patterns of transportation use, recreation, entertainment, and personal behavior, are the other information that researchers are interested in. For instance, understanding human behaviors may change product designs of the related industry. *Attitudinal information* is about the respondent's attitudes and opinions about the variety of conditions and circumstances collected to predict possible future actions of the respondents.

The collection of such valuable statistical information by using survey research techniques has many uses for practitioners (Franklin and Walker, 2003: 2): (i) It is highly practical for public opinion polls and market research studies; (ii) Planners and administrators use surveys to get baseline information for policy decisions; (iii) Social scientists use surveys to measure voter behavior, psychological influences on the spending and saving behavior of consumers, attitudes, values and beliefs related to economic growth, and correlations of mental health and illness; and (iv) Economists rely on regular consumer surveys for information on family financial conditions and survey of business establishments to measure recent investment outlays.

At the root of *sample survey research* is the discipline of *statistics* (Rea and Parker, 2014: xi). *Statistics* is regarded as (i) the study of populations, (ii) the study of variation, and (iii) the study of methods of the reduction of data (Fisher, 1934: 1). Statistical researches are based on *data*. A survey usually begins with the need for information where no data – or insufficient data – exists (Franklin and Walker, 2003: 1). Two types of data sources are available (Bülbul, 2006: 57): (i) primary data and (ii) secondary data. *Primary data* is a type of data source that is collected first hand and directly from the subjects under the study (Rea and Parker, 2014: 5). Primary data is the information that you collect by yourself for the purpose of your own research study. So, it just satisfies your research needs and objectives. *Secondary data* is the information that is collected by someone else. It was collected for the purpose of someone else or institution research. It consists of compiling and analyzing data that have already been collected and exist in usable form (Rea and Parker, 2014: 5). This secondary information can be obtained from some institutions such as Statistical Institute, Central Bank, State Planning Organization, Security General Directorate, and Stock Exchanges. Also, libraries and private institutions may have this kind of information in their records. Secondary data can serve to satisfy the research requirements of a particular study. The negative side of the usage of secondary

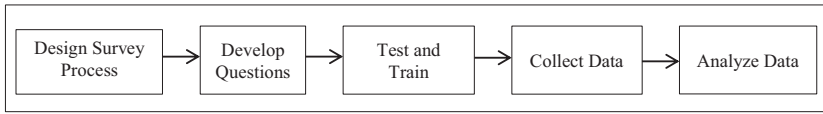


Fig. 1: The survey process (Hart and et al., 2010: 4)

data is the data collection purpose may not match the purpose of your own research study. So, secondary data may not completely satisfy the research's needs and objectives.

Although there are other techniques, there is no better method of research than the *sample survey process* for determining, with a known level of accuracy, detailed and personal information about large populations (Rea and Parker, 2014: 5). *Collecting data*, which is one of the steps in the survey process, is so important that there is no chance to reach the exact information and to give correct decisions without having reliable data (Bülbül, 2006: 57). Up to the collecting data stage of the survey process, attention should be paid to some additional statistical key issues. Therefore, methodologically understanding the significance of the sample survey research process, especially in *statistical point of view*, is the subject matter of this chapter.

2 Sample Survey Process

Sample survey is a term that indicates a *survey* conducted on a sample taken from a target population. The *ultimate goal of sample survey research* is to allow researchers to generalize about a large population by studying only a small portion of that population (Rea and Parker, 2014: 4). The survey process can be summarized with Fig. 1 (Hart and et al., 2010:4):

British statistician Ronald Aylmer Fisher's following quote would help to touch on the significance of the steps of the survey process:

"To consult a statistician after an experiment is finished is often merely to ask him to conduct a post mortem examination. He can perhaps say what the experiment died of."

Not taking into account the sample survey design methods may cause thoughtlessly prepared surveys, absence of research questions, inaccurate population and also improper sample, and ignorance of sampling methods. After all of this, statisticians cannot help anyone. As Ronald Aylmer Fisher said, he can perhaps say, "What the experiment died of." So, researchers should give due importance to the steps of the survey process.

2.1 Designing Survey Process

In this step, *general plans* are drawn by the researcher. At the beginning, finding the right *research topic* is a vital decision for researchers. The decision for choosing the right research topic is directly related to the *researcher's field of interest*. One may be interested in business topics; another may be interested in economics and so on. Researchers may start finding the right research topic by thinking of several subjects and writing them on a paper. Then, he or she may choose one of them that interest him or her most. The decision for choosing the right research topic is related to two more features. It should be an *interesting* and *contemporary* topic. For instance, choosing a business research topic such as *digital marketing* can be an interesting and contemporary topic in today's world. Digital marketing is the marketing of products or services using digital channels such as internet and mobile phones to reach customers. Advertisements are displayed to potential customers on these digital platforms. This research topic can be interesting for all segments of society. Academicians, business world, and consumers can be attracted and may have valuable information depending on the research result.

Sometimes it is hard to find a research topic for a researcher. In such a situation, the researcher may apply background information, namely literature. Library sources, such as books, articles, and dissertations are overviewed in the literature review. These documents may give new ideas of interesting topics. Another source may be online sources such as the internet. Besides the many facilities of the internet, browsing through online media sources can be creative. All these initiatives will help the researcher to find out an idea, or a general thought that can be turned into a research topic.

After choosing a research topic¹, two steps are followed: (i) *Finding background information* about the research topic and (ii) *developing research questions* of the study. Background information can be found in text books, articles and dissertations, encyclopedias, and dictionaries. Such information needs detailed literary review. Relevant information such as terminology and developments are studied. The background information step provides a good overview of the research topic and helps in forming *research objectives*. Research questions are clear and focused questions centered on the *research topic and objectives*. However, sometimes a research topic can be broad in scope. For example, researching a broad topic such as *marketing* can be difficult. It involves many marketing types. In this situation, this broader topic can be narrowed

1 The purpose of the study, population and the statistical technique that is going to be used also helps the researcher as indicatives of identifying the borders of the study.

by reviewing related literature. For instance, more narrow topics such as *digital marketing* can be chosen. Then focusing on such questions below is easy to search: “What factors influence people’s choice of digital marketing?” “How do people feel about digital marketing?” “Who is currently buying products or services from digital marketing?” “What is the gender proportion of digital marketing customers?” “What are the most purchased products or services from digital marketing?” “What is the average income of the average customer who buys products from digital marketing?” “What are the advantages and disadvantages of buying products or services from digital marketing?”

It has to become clear which population will be investigated. Consequently, this is the population to which the conclusions apply. For instance, a survey research can be designed to understand the consumer behavior in digital marketing. Millennials (Generation Y) born between 1980 and 2000 and influenced by digital media can be the target population. On the other hand, Post-Millennials (Generation Z) born 2000 and later years, and grow up with a highly sophisticated media and computer environment, can also be the target population. A researcher can choose Post-Millennials as a target population because they are more internet savvy and expert than Millennials. For instance, a survey research can be designed to understand the consumer behavior in buying luxury cars such as BMW and Mercedes. The target population may be meaningful to choose consumers whose socioeconomic class is high.

2.2 Developing Questions

After the research objectives have been identified, a process called *developing questions*², which is one of the crucial points in the survey, the research process begins. A *survey question* is a measuring device for things that are not directly observable (Hart and et al., 2010: 4). For useful and meaningful survey results, questions that you ask must have two characteristics (Hart and et al., 2010: 4): *reliability* and *validity*. Reliability is the extent to which repeatedly measuring the same property produces the same result. Each survey question will mean the same thing to everyone, including those administering the survey.

2 Because the focus of this chapter is in statistical point of view, the detailed information about developing questions is excluded from this text. On the other hand, it should be stressed that one way of developing questions is using focus group technique. For detailed information, following resources are recommended for interested researchers: Fowler (1995), Brace (2008), Bethlehem, (2009), Hart and et al., (2010), Rea and Parker, (2014), Saris and Gallhofer (2014).

Validity is the extent to which a survey question measures the property it is supposed to measure. Without satisfaction of reliability and validity, research results would be affected negatively. In the end, meaningless statistical analysis results leads to impractical information.

2.3 Testing and Training

Writing a survey is an iterative process (Hart and et al., 2010: 12). Reviewing, testing, and revising the questions are the parts of this process. The scientific way of doing this is applying a *pilot study*, which is sometimes called a *pilot test*. It is a critical component of questionnaire design (Iarossi, 2006: 11). It is one of the important stages in a research project and is conducted to identify potential problem areas and deficiencies in the research instruments and protocol prior to implementation during the full study (Hassan and et al., 2006: 70). A pilot study is conducted prior to the actual survey, particularly in order to (Hassan et al., 2006: 73): (i) evaluate the feasibility of the study and identify weaknesses in it; (ii) test the prepared questions in terms of adequacy and comprehensibility³; (iii) test the appropriateness of the selected questions for the target population; (iv) test the appropriateness of the selected data collecting tool (mobile survey, online survey, face-to-face or mail survey); (v) test the data collection process (time taken to complete questionnaire and willingness to participate in the study); (vi) test the appropriateness of the data entry, coding items, and statistical tests; and (vii) obtain preliminary data for the primary outcome measure, *in order to calculate a required sample size*.

The word “training” simply means *interview training*, which is conducted for interviewers⁴. In training, the *survey manager* explains the purpose of the survey, the research aim, explains the questions, how the questions meet the goal, and the importance of the interviewer’s role in achieving survey quality (Iarossi, 2006: 162). Besides the well-designed questionnaires, experienced interviewers have positive impacts of the quality on the collected data; training has positive impact on both the quality of responses recorded and on the quantity of interviews completed (Iarossi, 2006: 160).

3 Similarly, the training sessions often helps identify problems with wording and translation (Iarossi, 2006: 11).

4 Age, experience, and education are the interviewers’ three most important attributes in business surveys (Iarossi, 2006: 159).

2.4 Collecting Data

The next step in the process is data collection. Traditionally, in many surveys, *paper questionnaires* were used (Bethlehem, 2009: 3). Survey information can be collected by implementing *mail-out surveys*, *online surveys*, *mobile surveys*, and *face-to-face surveys*.

In *mail-out surveys*, printed questionnaires are disseminated through mail. They are less expensive than mobile surveys and face-to-face surveys. However, comparatively, they require long time periods and have lower response rates than mobile surveys. In *online surveys*, questionnaires are emailed to respondents. They are practical, cheap, and trained interviewers are not required. However, response rates and the quality of the data are low because of the lack of the interviewers' positive impacts. *Mobile surveys*, also called *telephone surveys*⁵, collect information through the use of telephone interview. The cost is less than face-to-face surveys and rapid data collection is available. However, the respondent can easily end the interview. In *face-to-face surveys*, respondents are visited by interviewers. The quality of the data tends to be good.

2.5 Analyzing Data

After the collection of data, the *data entry* is made on the computer. Particularly, the data entry is made to a data file of a statistical program such as MINITAB, SPSS, STATISTICA, or STATA. The next step is the *data cleaning step*⁶ that is for the preparation of your data analysis. Data cleaning refers to the process of *detecting*, *correcting*, or *deleting* (removing) inaccurate records from the data set for a better quality data set.

Missing data, *incorrect data entry*, and *outliers*⁷ can be the essential causes of poor data sets. One way to determine the missing data and incorrect data entry is looking at the data set with the naked eye. But it is not easy in huge data. The other way, which is easier than the naked eye, is *drawing frequency tables* or *bar graphs* for each one of the *categorical variables* separately. This time, researcher

5 Recently, computer-based telephone interviewing is in demand. It allows speeding up the survey process, improving the quality of collected data, and simplifying the work of interviewers (Bethlehem, 2009: 3). One of the programs that is used is CATI (Computer Assisted Telephone Interviewing). The other two are CAPI (Computer Assisted Personal Interviewing) and CASI (Computer Assisted Self Interviewing).

6 See Tüzüntürk (2018 (a)).

7 Outliers are observations that are distant from other observations, their distance are abnormal from other values of the variable.

looks to the frequency tables with the naked eye, which takes a short time. When you look at the table, you notice that there exists irrelevant category entered. So, it should be an *incorrect data entry*. The identification of the related sample unit should be determined from the data file, and then the respondent's questionnaire should be checked. After all, the researcher decides to correct or delete that respondent's entered data from the data file.

Outliers can be detected by *drawing box-plot diagrams* or *histograms* for each one of the *continuous variables* separately. Outliers can cause serious problems in statistical analysis, which lead to wrong decisions. When an outlier is detected and if its value is not acceptable and *does not follow the nature of the variable*⁸, it should be deleted from the data set. After data cleaning, data analysis can be applied with a better quality data set.

In statistical perspective, the goal of the study can describe a population, estimating population parameters, making comparisons between groups, and determining the relationships between variables. When the goal is describing a population, tables such as frequency, percentage, and contingency tables; graphs such as bar chart, pie chart, histogram, and box-plot; and summary statistics such as central tendency measures and dispersion measures can be used. Asymmetry and kurtosis measures and distribution shapes can also be used. When the goal is estimating population parameters, point estimation and interval estimations can be used. Parametric and nonparametric hypotheses tests⁹ can be used to compare groups in the univariate analysis. Hotelling T² and MANOVA testing can be used to compare groups in the multivariate analysis. Correlation and regression analysis can be used in determining the relationships between variables. Moreover, multivariate statistical methods¹⁰ such as principal component analysis, factor analysis, discriminant analysis, etc., can also be applied.

3 Sampling Design

By combining surveys with *scientific sampling*, the researcher is using the only method of gaining the detailed and personal information about large populations

8 For instance, suppose that an individual whose age is 80 years old is accidentally entered into the Millennials (Generation Y) data set in the digital marketing example. While Millennials' ages range between 19 and 38, the value of age variable which is 80 would be an outlier.

9 See Tüzüntürk (2018 (b)).

10 See Tatlıdil (2002).

to a well-known accuracy (Rea and Parker, 2014: 5). Sample unit, sample size, and sampling method must be properly determined for the required accuracy of the results.

Sample unit is about whom to survey. Sampling units can be individuals, children, adults, households, animals, plants, things, products, institutions, organizations, securities quoted on the stock market, towns, cities, countries, etc. In the context of business research topic *digital marketing*, which was exemplified in section 2.1, the sampling unit is an individual person. Sampling units are chosen from the target population and are used to research, analyze, and draw conclusions.

Sample size, which is symbolized with “n”, is about how many people should be surveyed. Large sample sizes hypothetically give more accurate and representative results. However, when surveying large populations, it is not always the best choice. The following formula can be used in the determination of the sample size for *large (infinite) populations*¹¹ (Rea and Parker, 2014: 167):

$$n = \left(\frac{Z_{\alpha} \sqrt{p(1-p)}}{ME_p} \right)^2 \quad (1)$$

where ME_p is the margin of error¹² in terms of proportions, Z_{α} is the Z score for various levels of confidence (α), and p is the expected proportion of an attribute that is presented in the population. If the population is *not large (finite)* or *small*, the finite population correction should be included into the formula, and this will yield the following formula¹³ (Rea and Parker, 2014: 169):

$$n = \frac{Z_{\alpha}^2 [p(1-p)]N}{Z_{\alpha}^2 [p(1-p)] + (N-1)ME_p^2} \quad (2)$$

A researcher wants to determine the *purchasing preference from digital marketing channels* (prefer to purchase or not) of the 5000 students who live at the university dormitory. In this situation, because of the census time constraint and

11 A population size of 100,000 or greater can be considered large, and populations of fewer than 100,00 could be considered small (Rea and Parker, 2014: 171).

12 Margin of error is the desired level of precision.

13 Here, “N” is the population size.

high costs, the researcher decided to work with an appropriate sample. A proper sample size needs to be calculated. To do so, the researcher must establish the values of Z_α , ME_p , and P , respectively. Z score is most commonly set at 1.96 for the 95 percent level of confidence or 2.58 for 99 percent. The researcher decides to establish a 95 percent level of confidence.. Margin of error is typically set not to exceed 10 percent and is much more frequently set in the 3–5 percent range (Rea and Parker, 2014: 167). The researcher decides to establish a margin of error that does not exceed 3 percent. The true *purchasing preference from digital marketing channels* of population proportion (p) is unknown. The most conservative way of handling this uncertainty is using 0.5 for p value (Rea and Parker, 2014: 167). However, a sample proportion that has been sampled by the researcher prior to actually conducting the survey can also be used. Moreover, if there are published past research results found related to this proportion, and then they can also be taken into consideration. The researcher decides to establish 0.5 for p value.

$$n = \frac{(1.96)^2 [0.5(1-0.5)] 5000}{(1.96)^2 [0.5(1-0.5)] + (5000-1)(0.03)^2} = 880$$

If the researcher wishes to be 99 percent confident of the ± 4 percent margin of error, the following sample size would be required:

$$n = \frac{(2.58)^2 [0.5(1-0.5)] 5000}{(2.58)^2 [0.5(1-0.5)] + (5000-1)(0.03)^2} = 1,350$$

The following table represents the minimum sample sizes calculated with the above formula for selected *not large (finite)* or *small populations* using 0.5 for population proportion:

Sampling method is how respondents should be chosen. If a survey can be conducted on the full set of observation objects which belong to a target population and a complete enumeration of this population is done, then this is called a **census**. However, surveying whole populations may not be executed for some reasons such as when the application takes a long time period, high costs, the employment problems of interviewers, and in some cases the existence of infinite populations. In coping with such problems, *sampling*, which is the process of selecting a subset of the population (sample), is applied. In this case, the reliability of the findings depends on how well the sample is selected from the target population. A *sample* should be a true representative model of the population.

Tab. 1: Minimum Sample Sizes for Selected Finite Populations^a

Margin of Error Population Size	95 % Confidence Level			99 % Confidence Level		
	±3 %	±5 %	±10 %	±3 %	±5 %	±10 %
500	341	217	81	393	286	125
1,000	516	278	88	649	400	143
1,500	624	306	90	828	461	150
2,000	696	322	92	961	500	154
2,500	748	333	93	1,063	526	156
3,000	787	341	93	1,144	545	158
5,000	880	357	94	1,350	588	161
10,000	964	370	95	1,560	624	164
20,000	1,013	377	96	1,693	644	165
50,000	1,045	381	96	1,783	657	166
100,000	1,056	383	96	1,815	661	166

^a Table values were calculated by using formula (2) in Microsoft EXCEL.

So, it should have the same attributes such as gender distribution the population has. And also, it should include various sections of the population.

In general, there are two types of samples (Lind et al., 2002: 265): a *probability sample* and a *nonprobability sample*. Probability sample is a sample that is selected in such a way that each item or person in the population has a known (nonzero) likelihood of being included in the sample. Sampling methods can be categorized as *probability* and *nonprobability* (Weiers, 2002: 139). The distinction is that with probability sampling, each person or element in the population has a known (or calculable) chance of being included in the sample. The selection of samples with some kind of *probability mechanism*, such that each element has the same probability of being selected, produces samples that are *on average representative* with respect to all variables (Bethlehem, 2009: 24).

To select a sample, two elements are required: a *sampling frame* and a *selection procedure* (Bethlehem, 2009: 23). A sampling frame is a list of elements (individuals, households, or institutions) in the target population that exist on a paper or in a computer. It involves information such as name, address, telephone number, or e-mail address. It is used to select the sample from the population. Probability sampling is only performed when a sampling frame exists. If the researcher cannot reach the sampling frame, then she or he can apply nonprobability sampling. But,

the researcher should bear in mind that collecting data by using nonprobability sampling methods has some weaknesses that may cause serious misleading results for their research projects. The types of sampling methods¹⁴ are: (i) probability sampling¹⁵ and (ii) nonprobability sampling¹⁶.

All probability sampling methods have a similar goal, namely, to allow the chance to determine the items or persons to be included in the sample (Lind, et al., 2002: 265). So the randomization for achieving a random sample is provided. On the other hand, in nonprobability sampling, not every unit in the population has a chance of being included in the sample. This process involves at least some degree of personal subjectivity (Weiers, 2002: 146). In such cases, respondents cannot be representative individuals of the population. When every element of the target population does not have an equal chance of being chosen, *data selection bias* occurs.

In *convenience sample* case, the sample is selected on the basis that respondents are readily available and willing to participate (Weiers, 2002: 146). These respondents can be surveyed with street surveys, mail surveys, shopping center surveys, and market place surveys. For instance, a student from the Department of Business is planning to collect data from the Faculty of Economics and Administrative Sciences students for his assignment. He is waiting for possible participants for his survey around 10 a.m. near the café at the university. However, the sample includes only respondents who exist there at that time of the day (e.g., just the Department of Economics students who have a course at that time), not the other department's students who are not there. Besides, these sample units are collected without regard to their demographic properties. The result is an occurrence of data selection bias. In *judgment sample* case, the sample is selected on the basis that the researcher believes the members to be representative of the population (Weiers, 2002: 147). For instance, the manager of a textile company believes that the products produced in machine planting by the 15:00 p.m. – 23:00 p.m. shift are typical of those produced by all three shifts. As

14 Because of the descriptions and theoretical bases of the sampling methods given in text books, they are left out of this chapter. See Runyon and Haber (1982), Lind, et al. (2002), Weiers (2002), Serper, et al. (2013), and Tüzüntürk (2018 (a)).

15 Simple random sampling, systematic sampling, stratified sampling, and cluster sampling are the types of probability sampling.

16 Convenience or accidental sampling and purposive or judgmental sampling are the two types of nonprobability sampling. Typical/model case sampling, expert sampling, quota sampling, heterogeneity sampling, and Snowball sampling are the types of purposive or judgmental sampling.

a result, the representativeness of such a sample is only as good as the judgment of the person who selected the sample. To summarize, statistically generalizing the population on the basis of a *nonprobability sample* is not valid for each of the cases above (Weiers, 2002: 147).

Depending on the nature of the study, two forms of analysis can be performed (Bethlehem, 2009: 4): *an inductive analysis* or *an exploratory analysis*. Inductive analysis is carried out to make inference on the population (e.g., testing hypotheses and estimating population parameters). Exploratory analysis is a kind of approach in which data sets are analyzed in order to summarize the main characteristics of the data. This summarization is often performed with visual methods such as tables, graphics, and also with summary statistics and distribution shape. The detection of possible existing patterns, structures, and relationships in the collected data is aimed for in the exploratory analysis (Bethlehem, 2009: 4). *Nonprobability sampling* is primarily used in *exploratory research studies* where there is no intention of making statistical inferences from the sample to the population (Weiers, 2002: 139).

4 Conclusions

At the beginning of the study, every step should be clear and data should be collected aimfully.

Before the data collection starts, besides the *research questions*, important decisions about the *target population* and *sampling design* (Bethlehem, 2009: 2) and the *statistical methods* that are going to be used in the context of the study have to be made by the researchers. While doing this, a *timeline* can be used in the planning of the survey. Having purposefully collected data in hand, the researcher satisfies all those needs and obtains reliable and valuable results.

As we move through the 21st century, myriad technological and analytical innovations have firmly entrenched the *probability sample* as an indispensable part of life (Rea and Parker, 2014: xi). In *probability sampling*, all units have a chance of being selected, and results can be generalized to the target population. On the other hand, the sample selection procedure is not random in nonprobability sampling. So, there exists a potential sample selection bias that causes an unrepresentative sample of the population. Such sampling can be useful in small-scale, exploratory studies where one wishes simply to gain greater familiarity with the population rather than to reach statistical conclusions about its characteristics (Weiers, 2002: 146). The statistical generalization of the population on the basis of a nonprobability sample is not valid.

The aim of the *statistical analysis method* and what it is used for should be understood prior to the data collection, and parallel to research topic selection and also parallel to the preparation of research questions. Research questions are developed related to the purpose of the study. They address the importance of the research topic and create interest. Whatever the goal is, the researcher should completely command the statistical method that is going to be used. Because, from start to finish, every piece of the sample survey research process is interconnected.

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Imam Bakir Kanli

A Solution Proposal for Car Parking Problems in Cities: “Neighbourhood Car Parking Cooperatives”

1 Introduction

Solidarity has always been a core agenda of human beings. When we lack power, we seek the assistance of a partner with common objectives. It has been the solution in solving problems. Here, cooperatives are the reflection of such consciousness focusing on unity and solidarity. There is no tissue incompatibility between a human being and a cooperation because cooperatives are totally compatible with human nature and values. That is why they are as old concept as the history of humankind, and they have been a social behaviour model in history.

Cooperatives are strategic organisations formed on the basis of a synergetic partnership approach; partaker, participator and solidarity, responding to all kinds of needs and requests including today and tomorrow of a community, offering quality products and services using the resources in a transparent and rational way on the basis of effectiveness, and finally predominantly focusing on human happiness and hence no doubly targeting sustainable development (Mori, 2014:330; Picciotti, 2014:214; Ifenkwe, 2012:21; Zaimova et al., 2012:52; Segu'i-Mas, 2015:365; Casaburi, 2015:286; Verhees et al., 2015:42). Kanlı (2016:4) also claims that “Co-operatives are not enterprises that have only economic objectives but they also have social and cultural objectives using democratic principles in order to establish a new generation social order.” According to International Co-operative Alliance (ICA) (www.ica.coop, 2018): “A co-operative is an autonomous association of persons united voluntarily to meet their common economic, social, and cultural needs and aspirations through a jointly owned and democratically-controlled enterprise.” Hence, they are sustainable business models strengthening local economies and improving quality of life.

Achieving sustainable development in a country has always been a vital issue. This is a phenomenon which involves not only scholars but also policy makers and NGOs. Particularly developed countries have used the tool wisely despite the thought that the cooperative system is tailored for developing and undeveloped countries. They also play a role in solving social problems in a community. Today cooperatives operate in different sectors and in different services varying from

Tab. 1: Types of Parking. Source: Adapted from Ison S. (2014)

Ownership	User		On or Off Street	Charging Regimes
Local Authority	General Public	Kerb-Side or Bays	On Street	Charged and Free
		Surface or Multi-storey	Off Street	Mostly Charged but Can Be free
Privately Owned	Public		Off Street	Charged and Free
	Private Non-residential Parking		Off Street	Free
	Residents		Off Street	Free

health to utilities, from food and agriculture to art, from housing to energy, etc. Human beings have tried to accelerate their velocity throughout history due to its power. Having velocity has been perceived as having power. That is why people were so addicted to it. Transportation is not only a kind of tool in facilitating the acceleration of velocity but also it is a powerful mechanism in transforming the structure of cities. When considering their negative consequences on the community both economically and socially, they are both addictive yet problematic. One negative consequence has been the issue of parking since the increase in the number of cars in cities, probably after the T model manufactured by Henry Ford in 1913. He probably did not foresee the traffic issues.

High percentage of car ownership in both developed and developing countries is one of the major reasons leading to conflict within local communities. Hence parking has always been an issue for people. Today, it is a complex issue and a priority for local authorities due to its contribution to traffic congestion (Cirit, 2014:3; ICPMP, 2016:14). Furthermore, extra travelling to find suitable parking for drivers brings an additional load on traffic flow. This causes both wasting of time and increases the amount of exhaust gas emission leading to air pollution. In addition it leads to global warming in the context of sustainability which came into prominence in the 1990s based on transportation issues (Özalp and Öcalır, 2008:73). In case of not finding a parking lot when driving, drivers may prefer to use the curbs. This reduces the use capacity of roads and slows down the flow of traffic resulting in traffic congestion and urban cost.

Developing parking policies at local levels requires great effort as it is an extremely complex issue (Mingardo, 2016:2; Weinberger et al., 2010:1–4; Marshall, 2014:362–363; Ison and Mulley, 2014:4). There are certain parameters to consider such as the structure of the demographics, mobility patterns, level of car ownership and public transport facilities. As a result, one can easily say

that the policy is directly related to transport and land use policies (Pitsiava-Latinopoulou et al., 2012:897; Rye and Koglin, 2014:158–159). Authorities strive to establish appropriate solutions (UMT, 2014:7–8) such as high parking fees for vehicles wishing to enter the inner city or impose dissuasive penalties for vehicles violating time limits for parking. Although parking problems appear to solely be a land use matter, the problem consists of two major factors. One is the lack of adequate parking space which is about land use and the other is management of parking services including the cost parameter. The new paradigms experienced in recent years bring a new approach to public administration allowing private sectors to fulfil some public services. One of the examples is Chicago (USA). It has pioneered the participation of private sector in providing car parking facilities. Hence the Chicago municipality declared that it would have more than one billion dollars' worth of revenue (Kozalı, 2014:192–194).

The importance of this study is to introduce an innovative point of view with a solution, which is effective, efficient and participatory, to parking problems experienced in inner cities. This also brings a new approach to public administration with the third sector which is the intersection cluster of private, public and non-governmental organisations, “cooperatives”. The aim is to debate the model suggested in an academic manner, and it has been limited to the local level, “neighbourhoods”. SWOT analysis has been used in this study in order to evaluate the model suggested.

2 Neighbourhood Cooperatives

The concept of “Neighbourhood Co-operatives” developed by İ. B. Kanlı is an innovative approach aiming at combining the two terms which are neighbourhood and cooperative. The main objective of the concept is to build the shortest route to arrive in liveable spaces. The concept is based on certain subsystems including Hierarchical Levels, Holistic Legal Framework, Arbitration, Control-Audit, Incitement, Understanding of Cooperatives Focusing on Service Delivery Groups, Lifelong Training and Learning and Governance Based on Full Participation. In this model all, services in the neighbourhood level including economic, social and cultural will be provided by a system named “neighbourhood cooperative”. It is thought that the quality and the cost of services to be delivered by the cooperative would be cheaper due to the features of cooperatives in general. The management level of the cooperative would also become one of the members of neighbourhood administration. Hence it would have the power of participation in the process of making decisions. While the control mechanism provides holistic audits on quality of goods and services provided, the arbitration

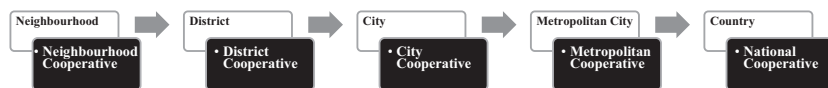


Fig. 1: The Relationship Level of Settlements and Cooperatives

Tab. 2: The Main Services to Be Delivered or Provided by Neighbourhood Cooperatives (Kanli, 2016:26)

Neighbourhood Cooperative				
Services to Be Delivered				
Economic	Social	Cultural	Educational	Technical
Shopping	Healthcare Welfare	Library	Kindergarten	IT
Insurance	Childcare	Music	Primary School	Technical Infrastructure Real Estate
Banking	Aged-Care Disability	Art	Middle School	Security
Finance	Funeral Sport	Exhibition	Vocational	Car Parking

system will facilitate the resolution of legal disputes. These will make a great contribution to neighbourhood administrations which will be at the intersection cluster of these subsystems. In the model suggested, there will be a cooperative organisation at each level starting with neighbourhood, country and international. At each level the cooperatives will serve people who live there considering the demands of the communities (Kanli, 2016:13–27).

The table below shows the types of services to be delivered or provided by neighbourhood cooperatives.

As seen in Tab. 2, car parking services can be delivered by neighbourhood cooperatives or by a car parking cooperative solely established by the local community.

3 The Parking Problem and Its Components

Parking is no longer simply the action of parking cars solely. It deals with sustainability and even technology. According to a survey in 2015 conducted by the International Parking Institute (IPI), technology and the environmental concerns transform parking issues. Based on the survey 47 % of the participants

believed that the desire for more liveable, walkable communities has influenced the parking concept. Technology, as always, continues to drive parking. About 53 % of people consider that innovative technologies are emerging trends in parking. Another finding based on the survey is that 46 % believe that in order to improve sensitivity on sustainability, guidance systems should be developed so that motorists can find parking spaces faster to reduce carbon emissions (IPI, 2018). A parking space is an area where vehicles are brought to a halt for a short-term period (Haldenbilen et.al., 1999:1099). There also exists another explanation to the planning disciplinereferring to the term of land-use (Marsden, 2014:12).

Cities, most importantly, inner cities need more parking spaces due to intensive economic activities and/or the rise of the number of vehicles based on economic wealth. The need for space is increasing (IUTMP, 2011:112–113) when considering people involved in certain activities in certain spaces such as coliseums, shopping centres, financial institutions, medical centres, hotels and cultural and art centres. Parking problems may affect the daily life of people (Ison and Mulley, 2014:2) such as congestion, accidents, pollution and obstruction to emergency vehicle operations, hence, parking problems occur in our daily lives.. We may experience the negative consequences of the problem at least once. Lack of parking space may not only affect locals but it may also affect business activities. According to a report (Banerjee and Associates, 2003:1–2), there are sixteen types of problems identified regarding parking. These are: inadequate information for motorists (parking availability and price); inefficient use of existing parking capacity (legislation may cause oversupply of parking spaces or inefficient use of existing ones); excessive automobile use (dependency on vehicles costs the community); economic, environmental and aesthetic impacts of parking facilities (bearing the costs of unpriced parking directly or through taxes); parking spaces that are an inconvenience (nearby for residents and businesses); demand for handicapped parking spaces (should be close to access ramps if possible); impact of additional parking spaces (on traffic and locals); existing, severe, spill-over problems (not being accommodated on the site of those uses or within the adjacent on-street spaces); out-of-town parking (the vehicles from outside of the neighbourhood); loading and unloading zones (insufficient parking zone for commercial use); inconvenient parking options (reasonable walking distance (3 blocks)); inadequate pricing methods (motorists may face different costs and fines); confusing parking policies (regulations may cause uncertainty); difficulties with parking regulation and pricing (congestion while motorists drive for parking or stop in the lane to wait for a space); lack of sufficient parking at event site (need crowd management based on the type of

event) and low parking turnover rate (parking in the same space for at least 4 hours (on average)).

When considering car parking types, in literature (Marshall, 2014:365–366; Ison and Mulley, 2014:4; McCahill and Garrick, 2014:34) it is possible to come across a few types of parking systems. Basically, it consists of two major clusters which are on-street parking system and off-street parking system. The first system is also known as kerb or curb-parking. It is where cars park alongside the curb on one or both sides of the street. The on-street system is composed of five types such as parallel, 30-, 45-, 60- and 90-degree parking. Regarding off-street parking system, it can be grouped based on ownership structure such as private and public. The system includes surface, multi-storeyed, roof, mechanical and underground car parks. On the other side, parking systems can also be grouped based on their technological structures such as automated systems consisting of vertical, rain, hybrid and rotary and traditional systems including off line parking. Today some parking systems use smart systems which help drivers to find a vacant spot through sensors. It is based on a wireless sensor network technology.

Other detailed parking types can be grouped under five titles (www.parking-net.com, 2018). These are: parking garages, carports, parking spaces on the side of the street, automated parking systems (APS) and semi-automated parking systems. The use of technology in parking systems has some advantages. The advantages for customers are that there is no need to search for availability of parking spaces or to walk through the parking garage, saving time and consistency of parking experience, safety and security (theft/damage) while the advantages for municipalities are as follows: providing space efficiency, being environment friendly, increasing visual impact and public safety and money saving. These can bring both cost and management efficiency. Parking problems basically can cause not only costs but also environmental pollution in cities. According to the Union of Concerned Scientists, a significant amount of air pollutants such as carbon monoxide and nitrogen oxides were emitted into the air in 2013 (www.ucsusa.org, 2018). This type of air pollution can induce health problems such as respiratory system-related illnesses. In addition, motor vehicles may contribute to pollution by emitting carbon dioxide.

4 The Model Proposal “The Neighbourhood Car Parking Cooperative”

One major fact is that the model proposed shelters an innovative approach that has not yet been suggested in academic society. Hence the fundamental aim of

the suggestion is to open the model to debate. It is considered that car parking co-operatives can be located under the technical services of the new concept “Neighbourhood Co-operatives”. The model proposed aims at providing high-quality services using technologies at low cost through co-operatives not only to its members but also others by enhancing security issues for vehicles and developing smart systems for effective management solutions. The basic principles of the model are determined as follows:

- The system will be based on the full participation of the local community with “compulsory membership” from each household residing in the neighbourhood.
- The co-operative will be the legal “strategic partner” of local administration in the process of making local decisions.
- “Neighbourhood Arbitration” system is a sub-system of the model enabling to resolve disputes quickly and efficiently.
- Indispensable part of the system is to establish an effective and transparent “control mechanism”.

5 The SWOT Analysis for the Model

SWOT Analysis is an effective technique used for the identification of a system. It was developed by Albert S. Humphrey in the 1960s. The term “SWOT” is the acronym of the words “Strengths, Weakness, Opportunity and Threats”. It is a framework for analysing and evaluating internal and external conditions of a system, plan, project, product, place or even a person. SWOT Analysis is also considered as the part of a strategic plan.

It may be possible to develop a synergetic system by increasing *strengths*, decreasing *weaknesses*, benefiting *opportunities* and avoiding *threats* with this technique (Toksoy et al., 2009:15). Based on the information given above, the model suggested can be analysed as follows.

When considering the synergy through the model cumulatively it can be said that there are nearly 32,000 neighbourhoods in Turkey based on the data provided by the Ministry of Interior of Turkey (www.e-icisleri.gov.tr, 2018). The number of these neighbourhoods obviously may bring hope in solving parking problems in urban areas when considering the basic principle that major problems may be solved quicker when broken up into small pieces such as neighbourhoods. Strengths of the model can prove that there are particular advantages to be gained when implemented. These benefits can be achievable in short- and mid-term. Perhaps the most significant outcome of these advantages is to bring the

Tab. 3: The SWOT Analysis

Strengths (S)

- Becoming part of the administrative system (S1)
- Local governments may encourage the co-operatives to solve parking and its related problems (S2)
- Accessibility to find appropriate parking areas to develop with the partnership of municipalities due to its local administrative power (S3)
- A powerful control and audit mechanism from local citizens due to providing semi-public service (S4)
- Ready market and almost a monopoly (S5)
- Low cost, low price but high-quality service and high competitiveness (S6)
- High profile customer care (S7)
- Tax advantage (S8)
- High-tech use in services (S9)
- Providing security and safety for the vehicles (S10)

Weakness (W)

- Hardship in finding a parking facility area in built-up urban spaces and putting them into service (W1)
- Less business diversity (W2)
- Low profit rate (W3)
- The cost of the use of technological equipment (W4)
- Limited income-generating activities (W5)
- Inadequacy of capital and financial capabilities (W6)
- Untrained workforce (W7)

Opportunity (O)

- Ensuring solidarity and prosperity in social structure based on the cultural background (O1)
- Achieving sustainable development with a sustainable business model (O2)
- Employing its members (O3)
- Caring for and protecting the environment (O4)

Threats (T)

- The lack of appropriate parking areas in local urban development plans (T1)
 - Unaffordable expropriation cost (T2)
 - The regulations complicating management system of co-operatives (T3)
 - Technological dependence externally in the context of both hardware and software or maintenance (T4)
-

Tab. 4: Strengths

S	Advantages	Term (Short, Mid, Long)	Opportunities for the Future
S1	Rapid response in making decisions	Short and Mid	Sustainable Business Eco-system at Local, Regional and National Level
S2	and implementations, support from		
S3	local governments		
S4	Increasing reliability amongst public administration, community and co-operative		
S5	Less commercial risks		
S6	Competitiveness in the market		
S7	Customers “Community” oriented high-quality service		
S8	Reducing the costs		
S9	Increasing customer satisfaction		
S10	and providing market advantage		

environment enabling sustainable business eco-system at all levels in a country. Tab. 4 illustrates the benefits of the strengths of the model proposed.

Weaknesses of the model, as undesired facts, should be removed or reduced if possible due to the reason that they may cause business problems. In order to prevent these facts there are some corrective and preventative actions which need to be taken. Tab. 5 shows these actions, its terms and authorities to be responsible.

Regarding the opportunities, as the desired facts have formative effects and consequences particularly on social structure starting from local to national level. These facts can play a strategic role in achieving sustainable development in co-operatives. Tab. 6 explains the details of the effects based on levels.

The threats which can be the most external undesired facts of a system that could have negative consequences on the environment should be removed or reduced. Tab. 7 shows the corrective and preventative actions which need to be taken and its terms and the responsible authorities to implement these actions.

6 Conclusion

It is known that the two different terms “co-operatives” and “car parking” issues in inner cities are both popular. It is also known that co-operatives are the major actors in today’s developing and developed countries. It is considered that

Tab. 5: Weaknesses

W	CAPA (Corrective and Preventative Action)	Term (Short, Mid, Long)	Authority
W1	Expropriation through the revision in the local plans with resource transfer from the central government	Short and mid	Metropolitan and local municipalities and the related ministries
W2	Vehicle-related businesses	Short	Co-operative management and related institutions
W3	Collective sale or service	Short	Co-operative management
W4	Using appropriate financial credit	Short and mid	Financial agencies and co-operative management
W5	Product and or service diversification	Mid and long	Co-operative management
W6	Finding low-cost and long-term credits or aid from the government or developing partnership network with similar co-operatives	Mid and long	Co-operatives network, governments, financial agencies or institutions
W7	Lifelong Learning System	Short and mid	Co-operative management, governments, universities

Tab. 6: Opportunities

O	Multiplier Effects and Consequences	Term (Short, Mid, Long)	Level
O1	Social Unity, Liveability, Sustainability	Mid and Long	Local, Regional and National
O2	Liveability, Sustainable Development		
O3	Liveability, Sustainable Development		
O4	Liveability, Sustainable Environment		

they, as strong economic and social tools, may play a strategic role in administrative systems in order to provide synergetic and sustainable formations. Hence car parking co-operatives, as a new model, may undertake the responsibility of solving problems related to parking issues by providing socio-economic advantages to locals at the same time. These co-operatives may also produce effective services at low cost just as Robert Owen did in the past.

The parking problem is a phenomenon in most cities in the world. The consequences of the problem may be drastic and even fatal. The problem mainly consists of two basic elements including lack of capacity of management and business. Co-operatives to be established at local level “neighbourhood” may

Tab. 7: Threats

T	CAPA (Corrective and Preventative Action)	Term (Short, Mid, Long)	Authority
T1	Revision in the plans from the level of local to national	Short and mid	Metropolitan and local municipalities and the related ministries
T2	Resource transfer from the central government	Short	Related ministries such as finance
T3	Comprehensive and holistic approach	Mid	Political power, government
T4	Research and Development activities	Mid and long	Government, universities

play a role in solving the problems as the intersection cluster of two elements. The co-operatives may have some advantages and benefits for local people. Perhaps the most significant one of these is to have sustainable development. Sustainable development is a vital concept referring to liveability. Consequently, it can be said that car parking co-operatives using appropriate level of technology can provide: customer satisfaction, low-cost yet quality services, participation environment to their members in making decisions at local level, security for the customers' vehicles, effective business opportunity at local and then national level, employment opportunity, positive social contribution to local solidarity and unity by using co-operation, a network between locals and administrative level and overall contribution to the development of the local civilisation.

In addition, car parking co-operatives can provide co-operation based on "family" and full participation based on "compulsory membership". They can resolve disputes *based on* "arbitration" system at local level while they can establish effective "control mechanism" based on technological interface. They can also provide training based on a "lifelong learning" principle in order to achieve the objected consequences. On the other hand, car parking co-operatives using intelligent technologic systems can play a strategic role on the road to smart cities. For instance, with these systems they can decrease parking time, help the protection of the environment, be cost-effective in mid- and long-term, provide maximum accommodation for vehicles in minimum space (vertically), provide security and efficiency, prevent vandalism in inner cities and save fuels.

In conclusion, "neighbourhood car parking co-operatives" as an innovative approach to parking problems in inner cities of today's world can play a great

role in facilitating sustainable structures. They can undertake a comprehensive mission from economic to social. The cooperatives can solve parking problems not only in residential areas but also in central business districts in inner cities. Particularly in heritage zones in urban areas where parking problems are at an extreme, “neighbourhood car parking co-operatives” can be the strategic partnership of the solution for local authorities.

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Selcuk Yalcin

Strategic Cost Management Process

1 Introduction

The developments that globalization has brought about in the world economy is changing the business environment of the businesses. In this environment, the customer's sensitivity to the quality, price, function and timing of products and services is increasing. On the other hand, there are serious decreases in the price and life cycle of products and services due to increasing competition. These changes force businesses to survive in a competitive environment that is rapidly changing and unforgiving of mistakes and delays. Organizations that can adapt to changing conditions instead of seeing the change as a threat or even manage it will be able to survive in competitive markets. Increasing competition and the pressure of the customer preferences on the businesses have removed decisiveness of the enterprises on product prices, quality and functions. Nowadays, products and services that do not have the desired functions at the price levels demanded by customers are not successful in the market. Success of sustainable competition depends on offering products with similar features at lower prices or products with superior quality, features and functions from competing products at similar prices. Stated reasons have led companies to manage their production processes and product costs. Strategic management and strategic cost management will help businesses in this regard.

2 Strategic Management Process

The competitive firm incorporates the emerging and anticipated changes in the contemporary environment of business into its business planning and practices. The competitive firm is customer driven; uses advanced manufacturing and information technologies when appropriate; anticipates the effect of changes in regulatory requirements and customer tastes and recognizes its complex social, political, and cultural environment (Blocher et al., 2009: 10). Strategic management process in competitive markets is useful in providing a sustainable competitive advantage. Strategy is defined as "a plan of action designed to achieve a long-term or overall aim" and "the art of planning and directing overall military operations and movements in a war or battle" in the Oxford Dictionary. One way that a firm can choose its strategies is through the strategic management process.

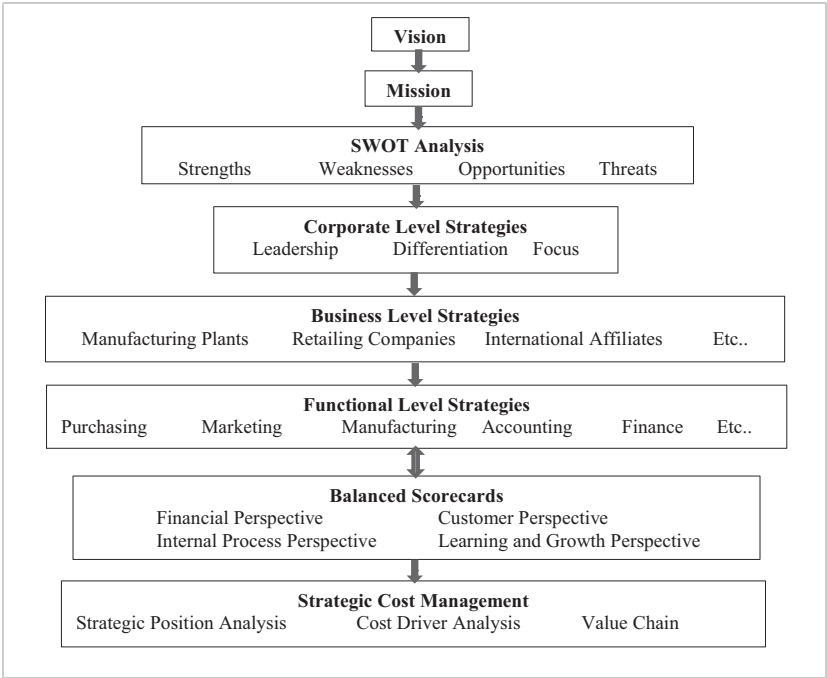


Fig. 1: The Strategic Cost Management Process

This process is a set of analyses and decisions that increase the likelihood that a firm will be able to choose a “good” strategy, that is, a strategy that will lead to a competitive advantage (Barney and Hesterly, 2014: 44). Strategic cost management model which is formed by associating the existing methods in the literature in this study is presented in Fig. 1.

2.1 Mission

The strategic management process begins when a firm identifies its mission, or its long-term purpose (Barney and Hesterly, 2014: 45). The mission is the distinctive expressions from the other businesses. It determines the scope of the product and the target market of the enterprise. This mission is often written down in the form of a mission statement (Barney and Hesterly, 2014: 45). A mission is an open-ended statement of the firm’s purposes and strategies. Strategic objectives translate the mission into strategic milestones for the business strategy

to reach (Botten and Sims, 2006: 9). In essence, a mission statement describes the scope and purpose of the organization, as it is, in terms of what it does in the present, for whom and where (Brennan and Sisk, 2015:14). Mission statements, by themselves, can have no impact on performance, enhance a firm's performance or hurt a firm's performance (Barney and Hesterly, 2014: 45).

2.2 Vision

Vision is a comprehensive picture of the future. It is the reason why the enterprise exists. Meaningful visions provide coherence across an organization. A vision statement is a unifying concept for employees, managers and executives to understand the strategic direction of the organization (Brennan and Sisk, 2015:12).

2.3 SWOT Analysis

SWOT is an acronym for explaining strengths, weaknesses, opportunities and threats for any business. Strengths and weaknesses are determined relative to the competitors of the enterprise. Opportunities and threats are often factors outside of control of the business. SWOT analysis guides the strategic analysis by focusing attention on the strengths, weaknesses, opportunities and threats critical to the company's success. By carefully identifying the critical success factors in this way, executives and managers can discover differences in viewpoints (Blocher et al., 2009: 35). The SWOT analysis enables management to interpret the information they have gathered about their organization and its environment in relation to how it affects the ability of the business to reach its strategic goals (Botten and Sims, 2006: 68).

2.4 Strategy

Strategies are developed in accordance with the vision and mission definitions and SWOT analysis of the enterprise. A firm's strategy is its theory of how to gain competitive advantages (Barney and Hesterly, 2014: 44). In defining strategies, it is important to support the weaknesses of the enterprise, utilize the opportunities, emphasize the strengths of the enterprise and avoid threats. Strategies are generally determined at corporate, business, and functional levels.

2.4.1 Corporate-Level Strategies

Corporate-level strategies are determined by the senior management of the business. Corporate strategy addresses the questions of how to capture

cross-business synergies, what businesses to hold or divest, which new markets to enter, and how to best enter new markets—by acquisition, by creation of a strategic alliance, or through internal development (Gamble et al., 2014: 26).

2.4.2 Business-Level Strategies

Business-level strategies are actions firms take to gain competitive advantages in a single market or industry (Barney and Hesterly, 2014: 124). In Michael Porter's (1980) model of generic strategies, he identifies three alternatives: competition on cost, differentiation, or competition in a focused market niche (Brennan and Sisk, 2015: 65).

2.4.2.1 Cost Leadership

A firm that chooses a cost leadership business strategy focuses on gaining advantages by reducing its costs to below those of all its competitors (Barney and Hesterly, 2014: 124). In the cost leadership strategy, cost management techniques are applied in order to access cheap raw materials, increase efficiency, and reduce waste. As the cost leader has the lowest costs in the market, it can potentially demand the lowest prices and still make profit. By developing a sustainable cost advantage, the cost leader can offer products that are low in price and functionality. In essence, the cost leader avoids competition by saying, "Don't compete with me. If you do, I'll drop prices even lower and render you unprofitable." (Cooper and Slagmulder, 1997: 4).

2.4.2.2 Differentiation

The enterprise that implements the differentiation strategy competes by providing completely unique and different products or services in the eyes of customers. Differentiation strategy depends on customer perception. If similar products and services are perceived as different by the customer, it creates a competitive advantage. This differentiation increases customer loyalty and prices that customers are willing to pay. Thus, the company can achieve higher market share and profit more than its competitors.

2.4.2.3 Focus

A business implementing the focus strategy uses the advantage of cost leadership or differentiation for a limited customer group or market niche. Special market niches can be separated as geographical regions, customer types, and product groups. Thus, while the customer can reach the products customized to him, the business gains competitive advantage in the special market niche.

2.4.3 Functional-Level Strategies

Functional-area strategies concern the actions related to particular functions or processes within a business. For the overall business strategy to have maximum impact, a company's marketing strategy, production strategy, finance strategy, customer service strategy, product development strategy, and human resources strategy should be compatible and mutually reinforcing rather than each serving its own narrower purpose (Gamble et al., 2014: 27). For example, in a company that implements the cost leadership strategy, functional strategies should be identified for many departments such as procurement, production, marketing, and finance. These strategies include high volume and inexpensive production, unqualified personnel employment, and marketing efforts to increase customer demand.

3 Balanced Scorecard

The balanced scorecard can use a tool as a basis for developing a strategic management system (Clinton et al., 2002: 2). Balanced scorecard is developed by Robert S. Kaplan and David P. Norton. Balanced scorecard is a performance measurement tool that translates a company's strategic goals into consistent performance metrics. Balanced scorecard transforms strategy into behavior. The Balanced Scorecard translates mission and strategy into objectives and measures into four different perspectives [and] provides a framework, a language, to communicate mission and strategy; it uses measurement to inform employees about the drivers of current and future success (Botten and Sims, 2006: 427). The objectives and measures view organizational performance from four perspectives: financial, customer, internal business process, and learning and growth (Kaplan and Norton, 1996: 8).

3.1 Financial Perspective

Financial perspective is an indication of how well a business satisfies its owners and shareholders. Financial performance measures indicate whether the company's strategy, implementation, and execution are contributing to bottom-line improvement (Kaplan and Atkinson, 1998: 368). Financial perspective is related to measures such as long-term shareholder value, increasing asset utilization, enhancing customer value, expanding revenue opportunities, and improving cost structure. Alternative financial objectives can be rapid sales growth or generation of cash flow (Kaplan and Atkinson, 1998: 368).

3.2 Customer Perspective

In the customer perspective of the Balanced Scorecard, managers identify the customer and market segments in which the business unit will compete and the measures of the business unit's performance in these targeted segments (Kaplan and Atkinson, 1998: 368). The customer perspective is an indicator of customer satisfaction. The core outcome measures include customer satisfaction, customer retention, new customer acquisition, customer profitability, and market and account share in targeted segments (Kaplan and Atkinson, 1998: 368).

3.3 Internal Business Perspective

Each business has its unique set of processes for creating value for customers and producing financial results (Kaplan and Atkinson, 1998: 371). Internal processes are the indicators of efficiency and effectiveness of the products and services produced by the company. Internal processes are composed of operations management process, customer management process, innovation process, and regulatory and social processes.

3.4 Learning and Growth Perspective

Learning and growth is an indicator of the business' ability to develop and use human resources to achieve its strategic goals for the present and the future. Learning and growth perspective is composed of human capital, information capital, and organizational capital.

4 Strategic Cost Management

Strategic cost management is the development of cost management information to facilitate the principal management function and strategic management (Blocher et al., 2002: 8). Cost data is used to develop superior strategies en route to gaining sustainable competitive advantage in strategic cost management. The emergence of strategic cost management results from a blending of three underlying themes, each taken from strategic management literature. These are strategic position analysis, value chain analysis, and cost driver analysis (Shank and Govindarajan, 1993: 13).

4.1 Strategic Position Analysis

Strategic positioning analysis determines the business' comparative position in the industry in terms of performance. After determining the business' strategic

positioning, enterprises should also determine the allocation of enterprise resources and the corresponding management operation mechanism. Therefore, business by the strategic positioning analysis specifies the cost management with enterprise strategies and reflects the effect of strategic cost management. In reality, many firms will choose not just one general strategy, but a combination of the three general strategies. Strategic positioning is the process of selecting the optimal mix of these three general strategic approaches. The mix is selected with the objective of creating a sustainable competitive advantage (Hansen et al., 2007: 378).

Cost leadership requires the business to achieve the lowest unit production cost in its industry. A business that follows a differentiation strategy must create the impression that its products or services are better, different, or unique. A business that focuses on a market or customer segment has to offer its customers the best products and services. Understanding the competitive position of a business' products and services is a critical first step in Target Costing. For example, if a business has established a position as a quality leader in furniture sector, it knows that the wide color choice is a very important and value-adding expectation based on customer feedback and will not consider changing these options to achieve target costs.

4.2 Value Chain Analysis

Porter (1985) developed the concept of the value chain and value system to demonstrate how managements can analyze their business in terms of how it generates 'value' for the customer and the shareholder (Botten and Sims, 2006: 68). The value chain for any firm in any business is the linked set of value-creating activities all the way from basic raw material sources of component suppliers through to the ultimate end-use product delivered into the final consumers' hands (Shank and Govindarajan, 1993: 13).

A firm which performs the value chain activities more efficiently, and at a lower cost, than its competitors will gain a competitive advantage (Drury, 2001: 468–469). In particular, an analysis of the firm's value chain helps management discover which steps or activities are not competitive, where costs can be reduced, or which activity should be outsourced. Management can use the analysis to find ways to increase value for the customer at one or more steps of the value chain (Blocher et al., 2009: 12). Also, managing organizational and operational cost drivers to create long-term cost reduction outcomes is an important input in value-chain analysis when cost leadership is emphasized (Hansen et al., 2007: 382).

Business can use the Activity Based Costing distribution of resources and costs in value chain activities, defining cost drivers which regulate each value chain activity and provide sustainable cost advantages. On the other hand, businesses can use Target Costing to achieve equal value with their competitors in value chain management in a more efficient manner than competitors. Especially in standardized products and in products and markets where competition is intense, enterprises can offer similar products or services at lower costs with Target Costing. Thus, obtained cost advantage can be used to propose lower sales prices than competitors, or to obtain a higher dividend per unit.

Continuous improvements should be made as long as the products and services remain in the market. Kaizen Costing is the way to make continuous improvements in production costs while the product is in production. Life Progress Costing approach should be used in order to manage the cost of the product as long as the product remains in the market in comprehensive strategic cost management practices. In addition to the price of the product in the competitive markets, the functionality and quality level demanded by the customer is also important. Therefore, customer demands and the ability of the business to meet these demands must be matched. This can be accomplished by applying Quality Function Deployment, which combines customer requirements with business capabilities in new product design and gives product features at the point where both meet. Thus, with Target Costing, businesses can offer products at the price they want, while Quality Function Deployment enables customers to put the features they want into these products. Quality Function Deployment can be used to combine customer expectations and business capabilities in each of the value chain activities of the enterprise, including raw materials, research and development, production, marketing, distribution and after-sales services.

4.3 Cost Driver Analysis

Based on the analysis of value chain and strategic positioning analysis, businesses can specify the cost management strategy, but in order to further define the emphasis of cost management, enterprises still need to find the driver factors of cost to ensure the validity of the cost management strategy. The cost driver of a product or process often provides great opportunities for cost reduction or process improvement. Cost is driven by different interrelated factors. Understanding cost behavior means understanding the complex interplay of the set of cost drivers at work in any given situation (Shank and Govindarajan, 1993: 19). Cost driver analysis examines, measures and explains the financial effect of the cost driver concerned with the activity. For a firm that competes

on the basis of cost leadership, management of the key cost drivers is essential (Blocher, et al., 2009: 64). The cost for the business that chooses differentiation or focus strategies is in the second priority. However, managing significant cost elements is useful in providing a sustainable competitive advantage.

Activity Based Costing is used to distribute overheads to products and services using cost factors through multiple cost pools. On the other hand, cost driver analysis can be used to determine the target level cost in the target costing process.

5 Conclusion

Traditional cost and management accounting practices cannot meet the information needs of businesses in competitive markets. Therefore, many companies implement strategic management and strategic cost management processes. Strategic management process is composed of determination of vision and mission definitions, SWOT analysis, Balanced Scorecards, and strategy and tactics. Strategic cost management consists of strategic position analysis, value chain analysis, and cost driver analysis. In this context, methods such as Activity Based Costing, Value Engineering, Life Cycle Costing, Target Costing, Kaizen Costing, and Quality Function Deployment can be stated as an auxiliary method in the strategic cost management analysis, strategic positions analysis, value chain analysis, and cost driver analysis.

Use of information as technology in production environments has also changed the cost structures of enterprises. As a result, while the labor expenses of the enterprises have decreased significantly, the overall production expenses have increased significantly. Under these conditions, overhead to be distributed to the operating activities and products produced by the traditional methods of cost accounting leads to the inability to calculate the exact costs. Therefore, Activity Based Costing has been developed.

In free market economies, businesses have to be customer-oriented, and offer products in line with the customers' demands on price and function. The way to do this is to design the products at the prices requested by Target Costing while transferring the demands of the customers about product functions to the designed products and business processes with Quality Function Deployment. Use of Activity Based Costing provides a benefit in this process.

On the other hand, in order to create and maintain a competitive advantage, it is necessary to consider the Life Cycle Costs of the products in the design and subsequent stages of the product. In addition, Kaizen Costing can reduce the production cost in production process. Evaluation of the mentioned methods

within the framework of the strategic cost management practices and the usage of these methods by businesses will be beneficial for the enterprises to provide and maintain a competitive advantage.

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Liquidity Management Performance of Major International Airline Companies¹

1 Introduction

Civil aviation sector is developing in line with global economic developments. Total number of passengers carried on scheduled services rose to 4.1 billion passengers increasing by 7 % in year 2017. While total number of departures increased to 37 million, highest number of all times, global passenger load factor (LF) reached 81 % in 2017. European air carriers recorded a growth of 8 % and accounted for the largest share of 37 % of total international scheduled passenger traffic. Asia Pacific had the second largest share of 29 % growing by 10 % in 2017 (Directorate General of Civil Aviation, 2018). Meanwhile, number of passengers in Turkey rose to 193 million increasing by 11 % for the same year (Turkish Airlines 2017 Annual Report, 2018). Due to this growth, Turkey had a market share of 5 % on a worldwide basis.

The aim of this study is to analyze liquidity management performance of 20 international airline companies, of which two of them are operating in Turkey, between 2011 and 2017 by using liquidity ratios. This analysis thus underlines best airline companies in the world from liquidity perspective.

2 Literature Review

Feng and Wang (2000) analyzed operational and financial performance of five Taiwanese domestic airlines named as Far Eastern Air Transport (FAT), TransAsia Airways (TNA), UNI Air (UIA), Great China (GCA) and Formosa Airlines (FMA). They divided the performance into three categories as production, marketing and execution and formed some indicators to be able to diagnose problems. As the sample size was small, they used grey analysis to determine representative indicators and TOPSIS method to rank airlines at the end. The empirical analysis showed that FAT outranked other airlines in all

1 Presented as a part of symposium paper “Comparative Analysis of Financial Performance of Airline Companies in the World” at 5th Asos Congress, International Symposium on Social Human and Administrative Sciences, 25–27 October 2018, Istanbul, Turkey.

three categories (production, marketing and execution efficiency). This study underlined that financial ratios also should be added to performance evaluation of an airline company in addition to transportation indicators to reach a more comprehensive analysis.

Akkaya (2004) used financial and non-financial data of an airline company to analyze company performance with TOPSIS method. Over 60 ratios were calculated and reduced to 13 under three main titles as operation, marketing and financial parameters.

Li, Oum and Zhang (2004) applied Tobin's q to 27 airline companies from Asia Pacific, Europe and North America over the period of 1989–1999. Their findings proved evidence that the industry on average had a low q ratio, and European airline companies generally had lower q values compared to other airline companies in other two regions. In addition, while there was a general pattern of low correlation between q and other financial measures except the variable of price-to-book-value ratio, there was a significant relationship between q and some operational variables and economic factors of the industry.

Wang (2008) used grey relation analysis to find representative indicators from financial ratios and apply fuzzy TOPSIS to evaluate financial performance of three domestic airlines in Taiwan between 2001 and 2005. The analysis was capable to underline competitive strengths and weaknesses of the companies at the end.

Koçyiğit (2009) examined performance of 14 airline companies between 2005 and 2007 by using Tobin q , and he calculated its correlation with selected financial ratios. The findings underlined that they could not effectively use their assets as their Tobin q was lower than 1. In addition, the airline companies in Asia-Pacific region performed better compared to other companies.

Che-Tsung, Yu-Je and Kuo-Hsien (2009) also studied financial ratio indicators, (financial structure, profitability, solvency and turnover ratios) for performance evaluation of four Taiwanese airline companies between 2004 and 2007. The results indicated that the mixed method used in the study was a reliable and practical method.

Zarb (2010) investigated the relationship between earning power (ROA) and financial health in the airline industry by taking financial data of US and non-US airline companies (469 observations) between 1998 and 2008. He used regression analysis with dependent variable of ROA and independent variables of net profit margin (NPM), current ratio (CUR), total asset turnover (TAT), fixed assets to net worth (PPE), return on investment (ROI), long-term debt to equity (LTD) and earnings before tax (EBT). Among them, NPM, TAT, ROI and LTD were found significant. In addition, he compared earning power of US, European

and other non-US, non-European airline companies by conducting one-way ANOVA analysis. He found out that earning power of US airline companies was different from that of non-US competitors.

Torlak, Sevкли, Sanal and Zaim (2011) used marketing and customer relations data of four Turkish airline companies to analyze company performance with TOPSIS method.

Ömürbek and Kınay (2013) compared financial performance of two airline companies ABC and XYZ which are listed in Borsa Istanbul and Frankfurt Stock Exchange, respectively. They analyzed main financial ratios via TOPSIS approach in order to reach a final result. The analysis proved evidence that ABC Company performed better compared to XYZ Company from a financial point of view.

Öncü, Çömlekçi and Coşkun (2013) analyzed performance of seven airline companies with data envelopment analysis method by selecting some financial ratios of year 2010. They used liquidity ratio, short-term debt/total assets, equity/total assets, cost of goods sold as inputs and return on assets and net income margin as outputs in their analysis. The result of the research showed that only three of the seven companies performed effectively.

Stepanyan (2013) focused on financial performance of US airline companies between 2007 and 2011 using cash flow information as a supplement to traditional ratio analysis. The analysis underlined that airline companies had liquidity problems especially in the economic crisis year of 2008.

Barros and Couto (2013) evaluated productivity of European airline companies by estimating Malmquist index using operational and financial variables from 2000 to 2011. The results showed that most European airlines' productivity did not grow between 2001 and 2011.

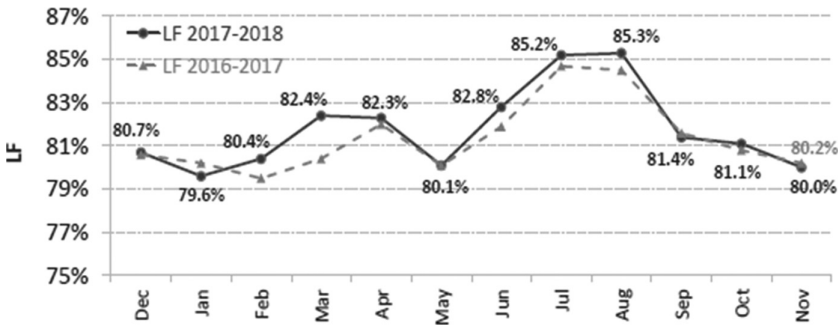
Teker, Teker and Güner (2016) investigated financial performance of 20 biggest airline companies in the world by calculating a harmonic index built by financial ratios between 2011 and 2014. The top three performance companies for 2011 were Delta, International Consolidated Airlines and Cathay Pacific Airlines. Turkish Airlines was ranked as the fourth best performer for 2011. The worst three performance companies for 2011 were listed as American Airlines, Singapore Airlines and Southwest. The top three performance companies for 2012 were Hainan, International Consolidated Airlines and Ryanair. Turkish Airlines was ranked as the fifth best performer for 2012. The worst three performance companies for 2012 were listed as Southwest, EasyJet and JetBlue Airlines. The top three performance companies for 2013 were EasyJet, Alaska Air and Delta. Turkish Airways was ranked as the eleventh for 2013. The worst three performance companies for 2013 were listed as China Eastern Airlines, China Southern Air and Air China. The top three performance companies for

2014 were EasyJet, Alaska Air and Southwest. Turkish Airways was ranked as the thirteenth for 2014. The worst three performance companies for 2014 were listed as Qantas Air, China Eastern Airlines and China Southern Air.

Zarb (2018) continued to focus on airline companies and studied if liquidity, solvency and financial health had an impact on profit volatility. He used cash flow from operations, current ratio, operating profit margin and debt-to-equity over the period 2010–2016. This analysis revealed that debt-to-equity and operating margin are statistically significant with respect to profit volatility in airline companies.

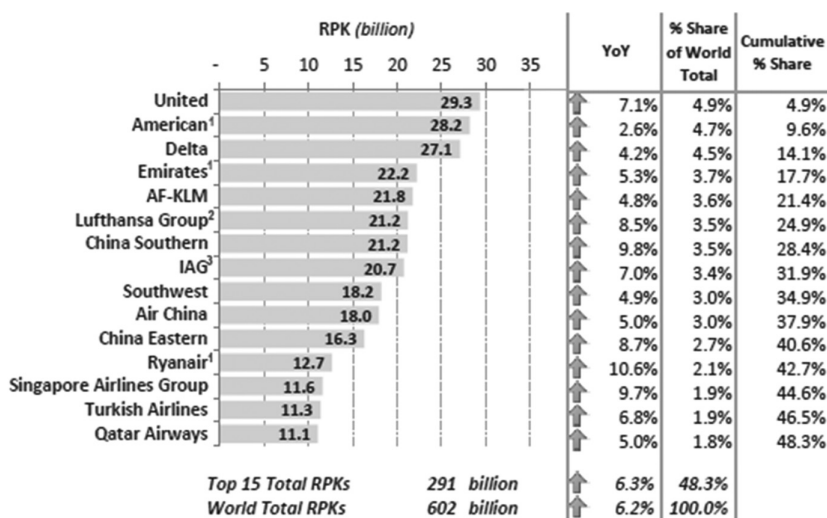
3 Airline Industry

According to monthly monitor published by International Civil Aviation Organization (ICAO), world passenger traffic grew by 6 % year-on-year (YoY) in November 2018. This growth remained solid and was around the average pace observed in the last five months except for the slowdown in September impacted by weather. Africa, Europe and North America accelerated in traffic growth, with Europe demonstrating the most significant improvement and becoming the fastest growing region. Domestic traffic in Russia rose robustly and outstripped India and China for the first time of the year. Capacity of airline companies worldwide increased by 7 % YoY in November 2018 where the passenger load factor reached 80 %. As capacity expansion outpaced the traffic growth, the November load factor was -0.2 percentage points lower than the same period in 2017 as seen in Graph 1.



Graph 1: Load Factor (Source: IATA, 2018)

In terms of revenue passenger kilometers (RPK), top 15 airline groups accounted for 48.3 % of world total RPK in November 2018 and grew by 6.3 % YoY. This growth was 0.1 percentage point higher than the world average growth on scheduled services. All the Top 15 airline groups posted YoY increases as shown in Graph 2. United continued to grow faster than the other North American airlines in the Top 15 and regained the first position with a 7.1 % rise. Following were American and Delta, up 2.6 % and 4.2 %, respectively. Southwest improved one position to 9th with a growth of 4.9 %. Emirates ranked two positions up to 4th and rose 5.3 %. Qatar remained at 15th rank with a growth of 5.0 %. Air France (AF)-KLM grew by 4.8 % and maintained the 5th place. Lufthansa and IAG continued to post stronger growth at 8.5 % and 7.0 %, albeit ranked two positions and one position down to 6th and 8th, respectively. Ryanair recorded the fastest growth within the Top 15 and remained at 12th rank. Turkish Airlines grew at 6.8 % and ranked first position down to 14th. Growth of major airlines in Asia/Pacific continued to be solid. China Southern recorded the second fastest growth at 9.8 %, and ranked first position up to 7th. This was followed by Singapore Airlines with the third fastest growth at 9.7 % and ranked 13th. Air China and China Eastern ranked 10th and 11th with an increase of 5 % and 8.7 %, respectively.



Graph 2: Top Airline Companies by RPK (Source: ICAO and airlines websites, 2018)
 (1. ICAO estimates 2. Lufthansa Airlines, Eurowings, SWISS, Austrian Airlines, Brussels Airlines, Sun Express, and Lufthansa Cargo 3. British Airways, Aer Lingus, Iberia and Vueling)

4 Data

The study covers financial data of 20 airline companies which are Turkish Airlines, Pegasus Airlines, Aeroflot, Air France-KLM, Deutsche Lufthansa, Delta Airlines, United Continental Holdings, Alaska Air, Hawaiian Holdings, American Airlines, Copa Airlines, LATAM Airlines, Air China, Cathay Pacific Airways, China Airlines, China Eastern Airlines, China Southern Airlines, EVA Airways, Asiana Airlines and Korean Air. This data is obtained via Datastream.

Turkish Airlines and Pegasus Airlines are from Turkey; Aeroflot from Russia; Air France-KLM from France; Deutsche Lufthansa from Germany; Delta Airlines, United Continental Holdings, Alaska Air, Hawaiian Holdings and American Airlines from USA; Copa Airlines from Panama; LATAM Airlines from Chile; Cathay Pacific Airways from Hong Kong; EVA Airways from Taiwan; Air China, China Airlines, China Eastern Airlines, and China Southern Airlines from China and Asiana Airlines and Korean Air from South Korea.

To be able to make liquidity management performance analysis of these companies on a comparative basis, main liquidity ratios are used. These are current ratio, acid-test ratio and cash ratio which serve to analyze short-term debt payment capacity.

5 Liquidity Ratios

Liquidity refers to the ease and quickness with which assets can be converted to cash (Westerfield, Ross, Jaffe and Jordan, 2011). Liquidity is an important indicator of the company because it represents the company's ability to meet its short-term liability. The more liquid a company's assets, the less likely the company is to experience problems meeting short-term obligations (Tekler, Tekler and Güner, 2016).

To examine short-term debt payment ability of airline companies, current ratio is calculated by dividing current assets by current liabilities. Current assets normally include cash, marketable securities, accounts receivable and inventories. Current liabilities consist of accounts payable, short-term notes payable, current maturities of long-term debt, accrued income taxes and other accrued expenses (principally wages and taxes) (Brigham and Gapenski, 1996). According to Morrell, a ratio of 1.00 is normally considered for industry in general to be broadly sound. Any ratio falling substantially below this level indicates that the business may not be generating adequate cash to meet short-term obligations as they become due. It is calculated between 2011 and 2017 and the seven-year average current ratio for 20 companies is 0.75 as

shown in Tab. 1. Unfortunately, the proposed sound ratio cannot be achieved during the study period on average but is achieved by only three airline companies that are Pegasus Airlines, Copa Airlines and EVA Airways. The average is changing between 0.70 and 0.76 between 2011 and 2017, except this average rises to 0.84 for year 2013. The company that has highest current ratio is EVA Airways for 2011 and 2012 with 1.26 and 1.31 and Pegasus Airlines between 2013 and 2017 with 1.76 for 2013, 1.75 for 2014, 1.85 for 2015, 1.41 for 2016 and 1.66 for 2017, respectively.

The airline companies whose average current ratio above 0.75 are Turkish Airlines, Pegasus Airlines, Aeroflot, Deutsche Lufthansa, Alaska Air, Hawaiian Holdings, American Airlines, Copa Airlines, Cathay Pacific Airways and EVA Airways. The remainder performs below the average. The average current ratio of Air France-KLM is 0.72; this average ratio is 0.58 for Delta Airlines, 0.70 for United Continental Holdings, 0.61 for LATAM Airlines, 0.35 for Air China, 0.69 for China Airlines, 0.27 for China Eastern Airlines, 0.34 for China Southern Airlines, 0.47 for Asiana Airlines and 0.49 for Korean Air. These companies seem to have some problems especially in the years of 2015, 2016 and partly 2017.

On a regional/country basis, the average current ratio is 1.09 for Turkey, 0.99 for EMEA (Europe, Middle East and Africa), 0.79 for Europe, 0.80 for USA, 0.87 for Latin America and 0.57 for Asia Pacific. Turkish companies stand for best performers contrary to Asia Pacific airline companies that are apparently worst performers.

To focus on the performance of two airline companies from Turkey, both of them perform above the average 0.75 taking into account their seven-year averages. Turkish Airlines records highest current ratio of 1.03 in year 2011 and lowest current ratio of 0.68 in year 2013. Meanwhile, Pegasus Airlines records highest current ratio of 1.85 in year 2015 and lowest current ratio of 0.36 in year 2011. In addition, Pegasus Airlines has the highest current ratio between 2013 and 2017 among the 20 airline companies on a yearly basis.

From the point of liquidity, a second ratio named acid-test is needed to analyze more accurately deducting inventories from current assets then dividing the remainder by current liabilities. Inventories are typically the least liquid of a firm's current assets, so they are the assets on which losses are most likely to occur in the event of liquidation. Therefore, a measure of the firm's ability to pay off short-term obligations without relying on the sale of inventories is important (Brigham and Gapenski, 1996). The seven-year average acid-test ratio for 20 companies is 0.69 as shown in Tab. 2. The average is changing between 0.65 and 0.70 except for year 2013 when this average rises to 0.78. The company that has highest acid-test ratio is Copa Airlines for 2011 with 1.08; Alaska Air for 2012

COUNTRY/REGION	CURRENT RATIO	31.12.2011	31.12.2012	31.12.2013	31.12.2014	31.12.2015	31.12.2016	31.12.2017	AVERAGE
TURKEY	Turkish Airlines	1,03	0,86	0,68	0,77	0,81	0,80	0,85	0,83
	Pegasus Airlines	0,36	0,63	1,76	1,75	1,85	1,41	1,66	1,34
EMEA	Aeroflot	0,99	1,04	1,20	0,68	0,75	1,08	1,19	0,99
EUROPE	Air France - KLM	0,70	0,77	0,73	0,61	0,63	0,75	0,82	0,72
	Deutsche Lufthansa	0,97	1,00	0,88	0,75	0,72	0,93	0,87	0,87
USA	Delta Airlines	0,61	0,62	0,68	0,74	0,52	0,49	0,42	0,58
	United Continental Holdings	0,97	0,78	0,72	0,65	0,63	0,59	0,56	0,70
	Alaska Air	1,06	1,16	1,12	1,05	0,92	0,81	0,79	0,99
	Hawaiian Holdings	1,02	0,87	0,92	0,86	0,96	0,96	0,79	0,91
	American Airlines	0,78	0,76	1,04	0,90	0,73	0,74	0,61	0,80
LATIN AMERICA	Copa Airlines	1,15	1,16	1,34	1,03	0,92	1,19	1,13	1,13
	LATAM Airlines	0,58	0,58	0,77	0,62	0,50	0,58	0,64	0,61
ASIA PACIFIC	Air China	0,38	0,38	0,37	0,35	0,40	0,31	0,29	0,35
	Cathay Pacific Airways	0,85	1,01	0,96	0,73	0,67	0,71	0,80	0,82
	China Airlines	0,67	0,72	0,60	0,67	0,67	0,69	0,79	0,69
	China Eastern Airlines	0,32	0,26	0,24	0,30	0,31	0,23	0,23	0,27
	China Southern Airlines	0,44	0,34	0,42	0,51	0,22	0,20	0,26	0,34
	EVA Airways	1,26	1,31	1,19	0,98	1,00	1,11	1,14	1,14
	Asiana Airlines	0,38	0,52	0,72	0,56	0,38	0,37	0,34	0,47
	Korean Air	0,51	0,54	0,40	0,66	0,39	0,36	0,54	0,49
	AVERAGE	0,75	0,77	0,84	0,76	0,70	0,72	0,74	0,75
	MAXIMUM	1,26	1,31	1,76	1,75	1,85	1,41	1,66	1,57

Tab. 1: Current Ratio of International Airline Companies by Country and/or Region. Source: Datastream, 2018

COUNTRY/REGION	ACID TEST RATIO	31.12.2011	31.12.2012	31.12.2013	31.12.2014	31.12.2015	31.12.2016	31.12.2017	AVERAGE
TURKEY	Turkish Airlines	0,97	0,80	0,63	0,72	0,76	0,75	0,80	0,78
	Pegasus Airlines	0,33	0,60	1,75	1,74	1,84	1,38	1,65	1,33
EMEA	Aeroflot	0,64	0,81	1,13	0,63	0,71	0,99	1,10	0,86
EUROPE	Air France - KLM	0,62	0,70	0,68	0,56	0,58	0,69	0,77	0,66
	Deutsche Lufthansa	0,90	0,93	0,82	0,69	0,66	0,85	0,80	0,81
USA	Delta Airlines	0,58	0,55	0,61	0,69	0,48	0,43	0,35	0,53
	United Continental Holdings	0,91	0,73	0,66	0,60	0,57	0,52	0,49	0,64
	Alaska Air	1,03	1,12	1,08	1,02	0,89	0,79	0,77	0,96
	Hawaiian Holdings	0,97	0,83	0,89	0,84	0,94	0,94	0,75	0,88
	American Airlines	0,71	0,70	0,96	0,83	0,67	0,67	0,52	0,72
LATIN AMERICA	Copa Airlines	1,08	1,10	1,29	0,97	0,86	1,11	1,06	1,07
	LATAM Airlines	0,55	0,55	0,73	0,58	0,46	0,54	0,60	0,57
ASIA PACIFIC	Air China	0,35	0,35	0,35	0,33	0,37	0,29	0,27	0,33
	Cathay Pacific Airways	0,82	0,98	0,92	0,70	0,64	0,68	0,76	0,78
	China Airlines	0,52	0,53	0,49	0,55	0,55	0,57	0,64	0,55
	China Eastern Airlines	0,28	0,22	0,19	0,26	0,28	0,20	0,20	0,23
	China Southern Airlines	0,40	0,31	0,38	0,48	0,19	0,18	0,23	0,31
	EVA Airways	0,89	0,97	0,97	0,83	0,86	0,97	0,99	0,92
	Asiana Airlines	0,32	0,45	0,63	0,48	0,31	0,30	0,28	0,40
	Korean Air	0,44	0,46	0,34	0,61	0,33	0,30	0,44	0,42
	AVERAGE	0,67	0,68	0,78	0,70	0,65	0,66	0,67	0,69
	MAXIMUM	1,08	1,12	1,75	1,74	1,84	1,38	1,65	1,51

Tab. 2: Acid-test Ratio of International Airline Companies by Country and/or Region. Source: Datastream, 2018

with 1.12 and Pegasus Airlines between 2013 and 2017 with 1.75 for 2013, 1.74 for 2014, 1.84 for 2015, 1.38 for 2016 and 1.65 for 2017, respectively.

The airline companies whose average acid-test ratio above 0.69 are Turkish Airlines, Pegasus Airlines, Aeroflot, Deutsche Lufthansa, Alaska Air, Hawaiian Holdings, American Airlines, Copa Airlines, Cathay Pacific Airways and EVA Airways. The remainder performs below the average. The average acid-test ratio of Air France-KLM is 0.66; this average ratio is 0.53 for Delta Airlines, 0.64 for United Continental Holdings, 0.57 for LATAM Airlines, 0.33 for Air China, 0.55 for China Airlines, 0.23 for China Eastern Airlines, 0.31 for China Southern Airlines, 0.40 for Asiana Airlines and 0.42 for Korean Air.

On a regional/country basis, the average acid-test ratio is 1.05 for Turkey, 0.86 for EMEA (Europe, Middle East and Africa), 0.73 for Europe, 0.74 for USA, 0.82 for Latin America and 0.49 for Asia Pacific. Turkish companies again stand for best performers contrary to Asia Pacific airline companies that are apparently worst performers.

To focus on the performance of two airline companies from Turkey, both of them perform above the average 0.69 taking into account their seven-year averages. Turkish Airlines records highest acid-test ratio of 0.97 in year 2011 and lowest acid-test ratio of 0.63 in year 2013. Meanwhile, Pegasus Airlines records highest acid-test ratio of 1.84 in year 2015 and lowest acid-test ratio of 0.33 in year 2011. In addition, Pegasus Airlines has the highest acid-test ratio between 2013 and 2017 among the 20 airline companies on a yearly basis.

Finally, a third ratio that is cash ratio is calculated by adding cash and marketable securities and then dividing the sum by current liabilities. The seven-year average cash ratio for 20 companies is 0.42 as shown in Tab. 3. The average is changing between 0.39 and 0.42 except for year 2013 when this average rises to 0.50. The company that has highest cash ratio is Copa Airlines with 0.80 for 2011 and with 0.91 for 2016; Alaska Air with 0.85 for 2012 and Pegasus Airlines with 1.33 for 2013, 1.00 for 2014, 0.97 for 2015 and 1.08 for 2017, respectively.

The airline companies whose average cash ratio above 0.42 are Pegasus Airlines, Alaska Air, Hawaiian Holdings, American Airlines, Copa Airlines, Cathay Pacific Airways and EVA Airways. The remainder performs below the average. The average cash ratio of Turkish Airlines is 0.37; this average ratio is 0.26 for Aeroflot, 0.42 for Air France-KLM, 0.40 for Deutsche Lufthansa, 0.28 for LATAM Airlines, 0.17 for Air China, 0.35 for China Airlines, 0.07 for China Eastern Airlines, 0.17 for China Southern Airlines, 0.16 for Asiana Airlines and 0.18 for Korean Air.

On a regional/country basis, the average cash ratio is 0.58 for Turkey, 0.26 for EMEA, 0.41 for Europe, 0.54 for USA, 0.57 for Latin America and 0.29 for Asia

COUNTRY/REGION	CASH RATIO	31.12.2011	31.12.2012	31.12.2013	31.12.2014	31.12.2015	31.12.2016	31.12.2017	AVERAGE
TURKEY	Turkish Airlines	0,46	0,42	0,22	0,24	0,27	0,45	0,53	0,37
	Pegasus Airlines	0,06	0,42	1,33	1,00	0,97	0,67	1,08	0,79
EMEA	Aeroflot	0,21	0,23	0,29	0,21	0,21	0,31	0,39	0,26
EUROPE	Air France - KLM	0,33	0,44	0,46	0,35	0,38	0,44	0,51	0,42
	Deutsche Lufthansa	0,45	0,53	0,47	0,29	0,28	0,41	0,36	0,40
USA	Delta Airlines	0,35	0,33	0,32	0,26	0,31	0,24	0,14	0,28
	United Continental Holdings	0,68	0,52	0,43	0,40	0,42	0,36	0,30	0,44
	Alaska Air	0,79	0,85	0,85	0,73	0,74	0,62	0,61	0,74
	Hawaiian Holdings	0,69	0,65	0,70	0,67	0,73	0,77	0,54	0,68
	American Airlines	0,56	0,52	0,75	0,60	0,51	0,50	0,36	0,54
LATIN AMERICA	Copa Airlines	0,80	0,84	1,10	0,78	0,69	0,91	0,89	0,86
	LATAM Airlines	0,26	0,22	0,41	0,28	0,25	0,27	0,29	0,28
ASIA PACIFIC	Air China	0,25	0,22	0,22	0,16	0,15	0,11	0,09	0,17
	Cathay Pacific Airways	0,53	0,67	0,68	0,46	0,41	0,46	0,46	0,53
	China Airlines	0,30	0,31	0,33	0,36	0,38	0,38	0,41	0,35
	China Eastern Airlines	0,16	0,09	0,04	0,02	0,12	0,03	0,06	0,07
	China Southern Airlines	0,22	0,21	0,25	0,29	0,07	0,06	0,10	0,17
	EVA Airways	0,55	0,64	0,70	0,55	0,64	0,75	0,72	0,65
	Asiana Airlines	0,08	0,12	0,34	0,25	0,15	0,09	0,09	0,16
	Korean Air	0,25	0,26	0,15	0,11	0,13	0,14	0,19	0,18
	AVERAGE	0,40	0,42	0,50	0,40	0,39	0,40	0,41	0,42
	MAXIMUM	0,80	0,85	1,33	1,00	0,97	0,91	1,08	0,99

Tab. 3: Cash Ratio of International Airline Companies by Country and/or Region. Source: Datastream, 2018

Pacific. Turkish companies again stand for best performers contrary to Aeroflot and Asia Pacific airline companies that are apparently worst performers.

To focus on the performance of two airline companies from Turkey, Pegasus Airlines performs above the average of 0.42 taking into account its seven-year average of 0.79. Turkish Airlines' seven-year cash ratio average stands at 0.37, which is lower than the overall average of 0.42.

Turkish Airlines records highest cash ratio of 0.53 in year 2017 and lowest cash ratio of 0.22 in year 2013. Meanwhile, Pegasus Airlines records highest cash ratio of 1.33 in year 2013 and lowest cash ratio of 0.06 in year 2011. In addition, Pegasus Airlines has the highest cash ratio for the years 2013, 2014, 2015 and 2017 among the 20 airline companies.

6 Conclusion

The developments in the airline industry are in line with economic growth of countries and jet fuel prices. The 1990s and 2000s were difficult for airline companies as there were economic downturns due to regional political instabilities in the world. After 2010, the industry performance began to improve (IATA, 2018). IATA (2019) forecasts that passenger demand will rise by 6.0 % in 2019, marking the tenth consecutive year of above trend growth in RPKs. Jet fuel price is not expected to increase too much. However, the increasing uncertainty about global economic conditions in 2019 pose downside risks to the passenger demand growth forecast in the upcoming year. For the coming risks mentioned, the liquidity management gains importance day by day.

This study analyzes the 20 international airline companies from the liquidity perspective using main financial ratios such as current ratio, acid-test ratio and cash ratio between 2011 and 2017. The seven-year average current ratio is 0.75, which is well below the 1.00 that is the sound level for this ratio. This average proves evidence that these airline companies may not be generating adequate cash to meet short-term obligations as they become due. The seven-year average acid-test ratio is 0.69 and cash ratio 0.42. Acid-test ratio is again far below the sound level but cash ratio seems to be contrarily high compared to general rule of 0.20. Stepanyan (2013) also found that the airline companies had some liquidity problems between 2007 and 2011. This problem seems to continue according to our findings.

From the current ratio perspective, EVA Airways is the best performer for years 2011 and 2012 and Pegasus Airlines for years from 2013 to 2017. From the acid-test ratio perspective, Copa Airlines is the best performer for year 2011, Alaska Air for 2012 and Pegasus Airlines for years from 2013 to 2017.

From the cash ratio perspective, Copa Airlines is the best performer for year 2011 and 2016, Alaska Air for 2012 and Pegasus Airlines for years from 2013 to 2015 and 2017. On the average basis, Turkish companies stand for best performers contrary to Asia Pacific airline companies that are apparently worst performers.

Turkish Airlines recorded lowest ratios in year 2013 compared to seven-year average ratios. The reason was that the company's net income margin diminished by 50 % in that year. Meanwhile, Pegasus Airlines had very low liquidity ratios in year 2011. The reason was that the company recorded net loss of 14 million TL in year 2011. To overcome, it made significant profits in year 2012. Pegasus Airlines was among the best performers between 2013 and 2017 from the liquidity point of view.

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Murat Karahan

Reporting Key Audit Matters Within the Independent Audit's Report

1 Introduction

The independent audit report on financial statements that was prepared in the field of audit and assurance services is an important communication item that provides an opinion about the financial statements of a company's accuracy, authenticity, and compliance with ethical rules. Independent audit reports used by the information users to make various economic decisions provide reasonable assurance of financial information. Independent audit reports are essential tools for information users in meeting the need to access reliable information. However, the economic crises that emerged since the 2000s have adversely affected the trust of information users in obtaining satisfying and correct information from the independent audit reports. Causes of this problem include: the expectation difference between the auditor and the parties using the audit report, the company scandals (financial crises), and the complexity of financial reporting standards can be counted (IAASB Consultation Paper, 2011, 5–8).

The aim of independent audit is to increase the confidence in the financial statements and the credit of these reports. This process is sometimes misunderstood by the users of the audit report. Many of the users of the audit report think that the responsibility of the auditor is to reveal all errors, fraud, and irregularities (Güredin, 2014, 22; Nwaobia et al., 2016, 120). Information users are expected to identify and be informed of both intentional and unintentional inaccuracies in the financial statements from the independent audit and audit reports. The independent auditors declared that they give reasonable assurance as a result of the audit, that is, they do not guarantee 100 % accuracy of the financial information. The difference in expectations among the auditors and the information users and the information required by the decision makers and the information provided in the audit reports causes a decrease in the confidence in the independent audit and audited financial information. Information users demand more information about the company due to the fact that information provided in the independent audit reports cannot match the request, and it raises doubts about the quality of the audit and decreases the confidence in the independent audit. The fact that makes audit reports transparent, accountable, and informative is possible by

reviewing the independent audit standards that will establish the trust. One of the organizations that make important research on this subject is IAASB.

IAASB has revised the standards in order to restore confidence in the independent audit system and to advance the quality of audits and commenced work by establishing various commissions. Revisions have been made particularly in the auditor's report as a result of the IAASB studies. Furthermore, the standardization of the IAS701 Key Audit Matters that was released by the International Federation of Accountants (IFAC) in the Independent Auditor's Report increases the quality of the auditor's report and is intended to provide detailed and transparent information to users of the financial statements. The standard regulates KAM and the auditor's responsibilities for reporting the matters in the auditor's report. Widely revised standards and newly advanced audit standards are summarized as follows (Karacan and Uygun, 2018: 633–634):

- 1) BDS-260 Communication with the Senior Management
- 2) BDS-570 Business Continuity
- 3) BDS-700 Forming Opinion on Financial Statements and Reporting
- 4) BDS-701 Reporting of Key Audit Matters in the Independent Auditor's Report
- 5) BDS-705 Giving an Opinion Other than the Positive Opinion in the Independent Auditor's Report
- 6) BDS-706 Remarkable Matters and the Paragraph of the Other Matters in the Independent Audit Reports
- 7) BDS-710 Information Related to Previous Periods and Comparative Financial Statements
- 8) BDS-720 Auditors' Responsibilities for Other Information in Documents Containing Audited Financial Statements

Other standards revised in the context are BDS210, 220, 230, 510, 540, 580, and 600. BDS701 is a new standard from the standards listed above while others have been revised and updated with changes made to the format and content of the auditor's reports. Some additional obligations have been imposed by the KGK on the implementation of our country in the Turkish version of the BDS701 standard. BDS701 standards were released in the Turkish Official Gazette dated 09.03.2017 and numbered 30002 in order to be applied under the supervision of the accounting periods that would start from 01.01.2017 on. BDS 701 standards have been applied to the companies in the stock exchange on and after 01.01.2017 and in the companies listed according to Turkish Commercial Code Numbered 6102 on and after 01.01.2018. Standard text applications have been available in our country and (Turkey app) is shown with (T) coded paragraphs.

It is the only document auditor report that occurs as a result of a complex audit process that is examined by financial statement users. IAASB has made significant adjustments in the format and content of the report by revising the ISA 700 in accordance with the demands and expectations of its business partners and report users. Auditor reports are freed from complex expressions, and standard language with the adjustment has been made. The adjustment has gained high document value among the auditors, business management, and partners due to KAM section in the standards.

IFAC's adjustment in existing standards the new standards aim to achieve and the benefits expected by the adjustment are as follows (Karacan and Uygun, 2018: 634; IAASB, 2016; Şirin, 2015):

- a) Developing transparency and communication quality in the auditor's report.
- b) Increasing and improving communication among auditors, investors, and persons responsible for senior management.
- c) Ensuring that more attention is paid to the explanations and references in the audit report to the business management.
- d) Ensuring that the auditor focuses on new areas that will increase professional skepticism.
- e) Attaching importance and giving attention to the footnotes in the financial statements.
- f) Obtaining more information that is specific to the business.
- g) Increasing audit quality and user perception.
- h) Attracting investors' attention to financial statement items and key matters and ensuring that they can better understand the financial statements.
- i) Reducing the difference between expectations and information gaps.

This research aims to provide explanations about the KAM standard regarding reporting of KAM in the auditor's report to provide detailed and transparent information to users of financial statements in order to improve the communication value, quality, and reliability of auditor reports. Since the main subject of the research is limited to BDS 701, the audit has been excluded from other parts of the reports.

2 Reporting Key Audit Matters in the Auditor's Report

BDS701 regulates the responsibility of the auditor to be notified in the audit report of the KAM (BDS701, Md. 1). The aim of reporting in the audit report of KAM is to increase the communication value of the report by providing more transparency regarding the audit conducted (BDS701, Md. 2).

The management may also provide a basis for more targeted engagement with the management and senior management of the target users about the audited financial statements or particular aspects of the audit conducted when KAM is reported in the audit report (BDS701, Md. 3). Reporting of KAM in the auditor's report is within the framework of the auditor's opinion in the financial statements as a whole. Reporting of KAM in the auditor's report (BDS701, Md. 4):

- a) Does not replace the explanations to be made by the management or other explanations that should be made in order to ensure the fair presentation according to the financial reporting framework in the financial statements.
- b) Is not a substitute for the auditor to give an opinion except the positive opinion due to the specific requirements of the audit according to BDS 705.
- c) Does not replace the reporting according to BDS 570 (BDS 570, Md. 22–23) if there is significant uncertainty about the conditions or conditions that may cause serious doubt about the ability of the management to continue its sustainability.
- d) Is not a separate opinion on the exceptional matters.

It is emphasized in the standard provisions that the terms particular to audit are not used too much and that target users will help them understand the focus of the audit in the statement on exceptional KAM (BDS701, Md. A34).

The auditor's opinion is the most important matter in the audit of the financial statements in the current period according to BDS 701 standard expression of KAM (BDS701, Md. 8). In addition, when looking at the objectives of the auditor in the standard, it is necessary to determine the key audit matters and to explain them in the auditor's report after forming an opinion on the financial statements (BDS701, Md. 7). The auditor is required to make an explanation for each of the items specified in the auditor's report.

Report of key audit matters in the independent auditor's report is a product of a risk-oriented audit approach, and the audit is conducted in accordance with the standards. Also, the auditor is responsible for obtaining sufficient and appropriate audit evidence that the identification of the risky areas encountered by the auditor during the audit process and the detailed procedures and the procedures that applied. Emphasizing the professional skeptical approach in the audit of the financial statements and the process of forming views on the key audit matters selected from the subjects to be reported to the management are an indicator that the auditor is using the professional judgment of the auditor (Akdoğan and Bülbül, 2019: 4).

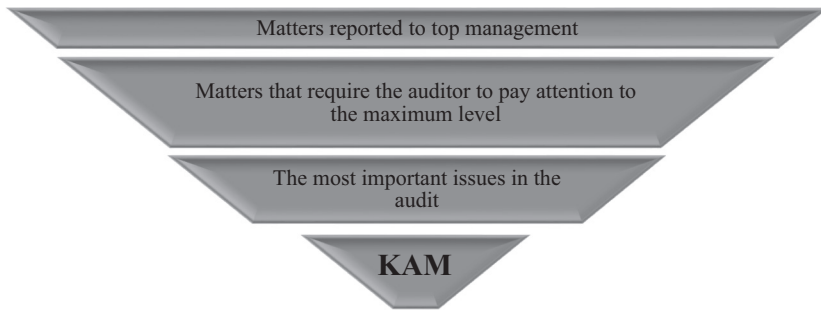


Fig. 1: IAASB 2016b, International Auditing and Assurance Standards Board), Determining and Communicating Key Audit Matters (“KAM”) ve Doğan, 2018, 67.

3 Identification and Reporting of the Key Audit Matters

Independent accounting reports are the most basic source of information about the scope, content, and results of the audit process. From this perspective, key audit matters are an expression of key matters or risky areas of the company by users of financial information. Therefore, the auditor’s professional judgment will determine and report key audit matters, which are the most important issues in the audit, to all parties, as part of the correct information transfer.

3.1 Identification of Key Audit Matters

The auditor determines the matters that require the highest level of attention during the conduct of the audit, among the matters reported to those responsible for the oversight of the audit. The auditor considers the following when making the determination:

- a) “Significant inaccuracy” is the area where the risk is higher or where the risk is determined to be serious according to BDS315.
- b) Essential auditor’s judgment in the financial statement areas containing the important judgments of the management, including the accounting estimates determined to have high estimation uncertainty.
- c) The effects of significant events or transactions in the period.

Examples of such areas contain accounting estimates, applied accounting policies, significant unusual transactions, related party transactions, fraud-related errors, the transition to new information technology in the management, internal control weaknesses, or the possibility of a violation of controls by the

management (BDS701, Md. A9-A-26). Accordingly, the auditor determines which issues are most important in the auditing of the financial statements of the current period and these constitute KAM. (BDS701, Md. 9).

When determining key audit matters, the auditor should pay maximum attention to the following matters:

- 1) Whether the matters are specific to the business.
- 2) Avoiding general and standard statements in the explanation of the matters.
- 3) Not specifying too much KAM for the fear of getting rid of responsibility.
- 4) Not proposing a solution to the matters identified.
- 5) Not extending the audit procedures in detail according to KAM.
- 6) Not disclosing the confidential and critical information of the management.

In addition, the perception that KAM has a conditional or negative opinion as stated in the report should not be created.

3.2 Reporting of Key Audit Matters

The standard brings responsibility to the auditors regarding the subjects and events that they focus most on in their work, and what audit methods and techniques they use to investigate these matters. The auditor explains per KAM in the KAM title under a separate section of the report by using an appropriate subheading within the scope in the context of this responsibility, does not permit the disclosure of the matter to the public and, in the most exceptional cases where it is expected that the negative consequences of public disclosure will exceed the public interest of public disclosure (BDS701, Md. 9).

The statements that are used in the audit report should be carefully selected in order to be understood by the target users of KAM. Therefore, in the expressions:

- * The auditor should not imply that the matter concerned has not been properly resolved while creating an opinion on financial statements.
- * Avoiding general or standard expression and the subject must be directly related to the conditions of the management.
- * If any – the relevant report in the financial statements should take into account how the matter is addressed.
- * It should not contain or imply a separate view of the elements of a financial statement.

Not only the standards that mentioned but also BDS is principle-based standards. As a result, certain statements are avoided in the auditing standards in the quantity, and the determination of the numerical values is usually left to

the auditor's professional judgment. In this direction, no determination is made about the number of KAM, and the matters are also left to the auditor's professional judgment.

The auditor's communication with those responsible for senior management allows the auditor to be aware of the key audit matters that the auditor has decided to report in the report and to obtain additional information on these matters when necessary. A communication in this manner allows the top management to assess whether a new or comprehensive disclosure is useful in the financial statements, taking into account key audit matters to be reported in the report. In addition, in the certification section, the auditor places the following points in the working papers (BDS230, Md. 8–11, A6) (BDS701, Md. 18):

- a) The reasons for the decision of the auditor regarding the issue that requires the auditor to pay attention to the maximum level and whether or not each subject is KAM that related to the matter,
- b) If any – the purpose of the auditor's decision is to determine whether there is no KAM to be reported in the auditor's report, or whether the matters to be reported as KAM are only the matters discussed in paragraph 15,
- c) If any – the reason for the decision of the auditor not to report an issue identified as KAM in the auditor's report.

Working papers require an experienced auditor with no prior contact with the audit to be prepared to allow him/her to understand the above matters and important professional judgments.

4 How to Handle the Key Audit Matters

Key statements related to the handling of the audit matters in the auditor's report is a result of the auditor's professional judgment. For that reason, the provision that "The level of detail of information contained in the auditor's report is the subject of professional judgment in order to explain how the audit is handled" describes the process in question (BDS701, Md. A49). In addition, the auditor may explain the following in accordance with the explanations (BDS701, Md. A46):

- * The approach or provision of the auditor is considered to be the risk of "significant error" or the most relevant aspects of KAM,
- * A brief assessment of the procedures applied or,
- * Indicating the results of the procedures applied by the auditor,
- * Key observations on the subject or some combinations of these elements.

Tab. 1: The key audit matters that are highlighted in auditor reports

Equity	Tax/Deferred Tax	Provisions For Employee Benefits
Live Assets	Construction Projects/Contracts	Evaluation of Pension Fund Obligations
Financial Liabilities	Assets for Sale	Credit and Receivables, Classification and Impairment
Information Technologies	Tangible and Intangible Asset	Determination of Fair Value of Investment Properties
Derivative Tools	Commercial and Financial Receivables	Revaluation of Tangible Assets
Betterment	Revenue Recognition	Supervision of Auditor Transition and Opening Balances
Depreciation	Related Party Transactions	Fair Value of Land and Buildings
Warranty Provisions	Stocks/Inventory Depreciation	Cash Flow Hedge Transactions
Business Combination	Provisions and Contingent Liabilities	Capitalization of Financing Cost
Trade Payables	Activated Development Costs	Fair Value of Financial Assets Available for Sale

All in all, how KAM is addressed contains the characteristics of the auditor's approach, and the audit procedures applied are opened in subheadings. Therefore, detailed observations made by audit procedures, analytical studies, control tests, and investigations on the subject are important to show that key audit matters are identified and reported with a risk-oriented audit approach.

In the literature studies, it can be sorted the key audit matters that are highlighted in auditor reports (Kavut and Güngör, 2018; Akdoğan and Bülbül, 2019; Karacan and Uygun, 2018; Doğan, 2018; Gökgez, 2018; Uzay and Köylü, 2018):

5 Conclusion

Two functions are important for the parties involved in the management results of a business. These are good functioning, healthy accounting and financial reporting system and a qualified independent audit function that investigates the accuracy and reliability of financial information. Given the input and output of each process, the input of the risk-oriented independent audit process is risk assessment and risk response while output is the independent audit report (Sayar

and Ergüden, 2016: 88). In particular, the global financial crisis experienced since 2000, accounting scandals, and subsequent company bankruptcies have led to the questioning of audit quality in financial reporting and independent audit, and further discussion of the appropriateness of the value produced by the audit and the need for audit (Yanık and Karataş, 2017, 2). The independent audit, which is expected to play a role in reducing the asymmetry of information among those responsible for business management and nonbusiness parties, suffered the loss of reputation and trust due to the reasons listed above; required that international and local regulatory bodies carry out reform-oriented studies in professional regulations. The property of the KAM is that it consists of matters that should be considered by the auditor during the audit, which are reported to those who are in charge of the senior management and based on the auditor's judgment. KAM should be identified and reported in accordance with the principles in BDS 701. In this context, continuation of studies and monitoring of KAM over the years will contribute significantly to the sharing of knowledge and experience among independent auditors by identifying the critical and serious risky accounting practices and the ways of monitoring them.

It is necessary to write the subjects in an understandable language, avoiding long narratives and unnecessary explanations and the impression that a separate opinion is given about the subject, and it is necessary to pay attention to the relations with the financial statement explanations in order to obtain the expected benefits from KAM notification application. Thus, transparency and quality in audit, the communication value of the auditor reports, and information users will be able to reach satisfactory information about the management. In other words, users of the financial statements have requested that the auditors not only express their opinions but also include more specific and explanatory information about the management in their reports. In accordance with this demand, IAASB has made revisions to the standards regarding the auditor's report and released new standards in order to match the needs of the users of the financial statements. Such revisions and new standards will continue in the future processes and will be shaped within the framework of current developments.

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Cemal Cevik

An Investigation on the Determination of Financial Literacy Level

1 Introduction

One of the most important goals in the life cycle of the individual is to maintain and improve the current level of prosperity. Every individual independent from the level of existence must make an expenditure in order to meet his/her needs and demands in daily life. People with sufficient monetary power will make their expenditures without any problems and those who do not have the right to sell or borrow their assets. People with more than their income will want to protect the future value of existing assets and to make more profitable and safe investments against unexpected economic and natural risks. An individual needs to have financial literacy knowledge to be able to take these decisions into the finance field effectively and correctly.

It is necessary to increase the need for financial literacy in order to compare individuals' credit card usage alternatives in daily life, to develop preferences among payment methods, to take a lot of financial decisions effectively, from where to invest to what amount to save and where to get the credit in the best conditions (Lusardi, 2008) expressed in the study.

In this study, the aim is to measure the financial literacy levels and the variables affecting the level of undergraduate students in different faculties and colleges within Kırklareli University. The data obtained in this study has been compared with the results of the study conducted in different universities, and the financial literacy levels of the undergraduate students of Kırklareli University were tried to be revealed. It is determined that the level of financial information and literacy of undergraduate students is generally low. Therefore, in the light of these data, it can be said that financial literacy education, which should start in the family, should be provided in a regular manner, especially from the first grade in universities, including secondary education.

The first study examined the concept of financial literacy in the framework of theoretical background and financial education in Turkey are given for information outlined and financial literacy. Then, a field study was conducted on the students studying at Kırklareli University and the findings were evaluated. The study is complemented by a conclusion section that includes various recommendations for improving the level of financial literacy.

2 Financial Literacy: Conceptual Framework and Theoretical Background

Since the 1990s, the concept of financial literacy, which is used more intensively in the media and academic community, has not been developed in the literature in different countries and in different studies, even though a common definition cannot be developed. In some countries, financial literacy is accepted as the equivalent of the term “financial literacy” used in the English language. In the Turkish Language Society dictionary, the word “literacy” is defined as “being literate”. Literacy can be defined as having information about the related subject.

In their studies Hilgert, Hogart, and Bewerly (2003) have defined financial literacy as financial information in the simplest terms. Financial literacy, which is frequently on the agenda in the academic community, media, and financial markets, not only includes the theoretical knowledge ownership in finance, but also plays an active role in financial decision making by reflecting on the individual attitudes and behaviors. The aim of this course is to provide the students with the skills and attitudes that will enable them to make the right decision in the stages of using and managing financial assets besides the information transfer, which is expected from financial education.

In the Global Finance Education Program study (2009) “Financial education will provide people with knowledge, skills, attitudes and behaviors that enable them to make informed and effective decisions about the use and management of money” expressed.

In a further description, Huston (2010) has expressed knowledge is a measure of how much understanding and how well it is applicable. Remund (2010) tells “knowledge about financial concepts, ability to communicate with concepts, ability to manage personal finance, ability to make appropriate financial decisions and planning behavior for future financial needs” 5 key components about financial literacy. In the other explanation, Lusardi (2008) says “To be able to master the basic financial concepts such as simple-compound interest, differences between nominal and real returns and risk diversification” about financial literacy.

Based on the definitions made in different studies on financial literacy, it is possible to define the personal financial issues that affect the financial situation of individuals as the ability to read, analyze, and manage. Vitt and his colleagues (2005), on the other hand, state that by distinguishing financial choices, they can analyze their money and financial problems by taking into account the general economic events, including daily events that will affect financial decisions, and the ability to plan and discuss the future. In this respect, it is possible to say that

financial literacy does not only include assessments of today, but also has an important function in drawing a future perspective.

3 Financial Education and Financial Literacy in Turkey

When the level of financial literacy and financial education comparison to the developed and developing countries to Turkey it is possible to say that there is inadequate in Turkey. On the other hand, the World Bank and the Capital Markets Board, created in cooperation with “Financial Capability Survey in Turkey” which was found satisfied levels of financial literacy. Following the analysis of the current situation, the Financial Stability Committee was formed between 2014–2017 for the solution of the problem as Financial Education and Financial Consumer Protection Action Plan. Within the scope of this plan, many public institutions and organizations have been assigned (Turan İçke, B. 2017: 130).

In Turkey, it is detected that a separate financial literacy course exists in some universities, which are too few in number, although in many universities there are courses such as finance and banking. As far as it can be determined, it is concluded that in the public or private universities such as Selcuk University, Trakya University, Kırklareli University, Adnan Menderes University, and Medipol University, financial literacy courses are included in the curriculum at the associate degree and bachelor’s level.

4 A Research on the Undergraduate Students of Kırklareli University

4.1 Purpose of the Research

The study aims to determine the financial literacy levels of the undergraduate students in different faculties and colleges of Kırklareli University. In this respect, it tries to reveal the similarities and differences in money management and financial decision making by students who are not taking finance and banking courses. In this context, students’ knowledge levels about financial literacy, which sources they use to make financial decisions and on what motives they are acting, how they consider themselves sufficient for financial literacy, the level of knowledge they have about various financial concepts, and the level of financial concept and financial information are investigated.

In this context, the following hypotheses were tested in the context of the investigation of financial literacy levels of undergraduate students studying at Kırklareli University:

H1: According to the section, there is a difference between the perceptions of success in managing the financial situation.

H2: According to the classes, there is a difference between the perceptions of success in managing the financial situation.

H3: There is a difference in how the financial developments are followed according to departments.

H4: There are differences between departments and classes according to the students' level of knowledge about basic financial concepts and the success of using financial concepts.

4.2 Method of Research

The data set of the study consists of 635 students randomly selected from 11045 undergraduate students studying at Kırklareli University. After removing faulty questionnaires from 635 people, it continued to work with 607 datas. The study is based on a literature review of the subject and a quantitative analysis of the data collected from the sample. In the development of the questionnaires, different studies have been used such as Saraç (2014). All of the analysis was carried out with the help of SPSS Statistics 22 (IBM, 2013) program at 95 % trust level. The data obtained for the research were analyzed by using the SPSS package program, two independent sample T tests, and one-way ANOVA test among the parametric tests. The use of parametric tests is due to the fact that the number of samples obtained from the mass is greater than 50 and provides homogeneity of variance.

4.3 Evidences

In this study, frequency analysis was applied to measure the level of difference between financial status management according to the students' department. Students are divided into two categories in terms of the departments they read and the courses they take. Students taking finance, economics, and banking courses (Faculty of Economics and Administrative Sciences, School of Health, and School of Applied Sciences) and those who did not take these courses are listed below (other sections).

According to the results of frequency analysis, it was determined that more than half of the students participating in the survey were from two units – Health School 33.8 % and Faculty of Economics and Administrative Sciences 25.2 %. The participation rates of the other departments were between 3.5 % and 11.5 %. Class distributions ranges are from 23.4 % to 26.9 %.

It can be said that according to the results of the ANOVA test, students' perception of achievement in financial status management is greater than 0.05 for all of the p values determined according to age, division, and class of readings.

Tab. 1: The Unit and Class Distribution of Students.

<i>Faculty</i>	<i>Frequency</i>	<i>Percent</i>	<i>Classes</i>	<i>Frequency</i>	<i>Percent</i>
Science Literature	41	6.8	1st Class	142	23.4
Economics and Administrative Sciences	153	25.2	2nd Class	143	23.6
Engineering	21	3.5	3rd Class	159	26.2
Architechture	55	9.1	4th Class	163	26.9
Tourism	62	10.2			
School of Health	205	33.8			
School of Applied Sciences	70	11.5			
<i>Total</i>	<i>607</i>	<i>100</i>	<i>Total</i>	<i>607</i>	<i>100</i>

Tab. 2: Differences in the Success Perception According to Age, Division and Class in Financial Status Management

	<i>df</i>	<i>F</i>	<i>Sig.</i>
<i>There is a difference between the age groups in the perception of success in managing the financial situation.</i>	11	0.361	0.970
<i>According to the education departments, there is a difference between the mean scores of achievement perception in managing the financial situation.</i>	6	1.068	0.381
<i>There is a difference between the average scores of achievement perception scores in managing financial situation.</i>	3	0.334	0.801

For Financial Management Situation, the ratio of females who said that they managed successfully at medium or higher level was 83.70 %, and this ratio was 84.6 % for males. The rate of those who failed was 16.3 % in females and 15.4 % in males. The rate of those who said “I am not successful” was found to be lower in Temizel-Bayram’s (2011) study with 8.5 % and Çam-Barut’s (2015) studies with a rate of 7.8 %.

The source of money spending and managing information was the family for 39.4 % of females, while this rate was 28.8 % in males. These results are different with Temizel-Bayram (2011) – 84 % – and Alkaya, Yağlı (2015), and Akben (2015) – 40 %. On the other hand, the ratio of males who stated that they had acquired information with their own experience in spending and managing money was 64.1 % and the rate of females was 58.1 %, lower than males. It is quite remarkable that the rate of spending and managing information obtained from higher education is between 0 % and 0.3 %.

Tab. 3: Students' Perception of Success in Financial Status Management, Source of Money Spending and Management Information and Relationship with Gender

<i>Managing Financial Status</i>	<i>Female</i>	<i>Percent</i>	<i>Male</i>	<i>Percent</i>
I'm quite successful	93	24.9	74	31.8
I'm intermediate successful	220	58.8	123	52.8
I'm not successful	61	16.3	36	15.4
<i>Total</i>	<i>374</i>	<i>100</i>	<i>233</i>	<i>100</i>
<i>The Source of Money Spending and Management Information</i>				
From my Family	124	39.4	57	28.8
From my Friends	2	0.6	7	3.5
From Publications such as Books and Magazines	5	1.6	7	3.5
My Own Experiences	183	58.1	127	64.1
From School	1	0.3	0	0
<i>Total</i>	<i>315</i>	<i>100</i>	<i>198</i>	<i>100</i>

These ratios are (Saraç, 2014) between 0.8 % and 1.8 % and (Alkaya-Yağlı, 2015) average 9.2 %, and only for the students of Economics and Administrative Sciences Faculty, Temizel and Bayram's (2011) study should be 5.9 %. It is possible to say that even the students who took the finance courses could not get the practical information they would benefit from in their daily lives in the subjects of spending and managing money.

In the study, the p value of the students found to be different from the class groups according to how the financial developments are followed, as a result of the ANOVA test analysis, is smaller than 0.05, at the 95 % trust level, and it can be said that there is a difference in the average of the students' views on how the financial developments are followed according to the class groups they are in. As a result of the Games Howel test conducted to determine the source of the difference, it is concluded that 2nd class students differ according to 3rd and 4th class students. In this study, the p value of the students was found to be greater than 0.05, for the class groups as a result of the ANOVA test analysis of the frequency of following the economic and financial developments. At the 95 % trust level, it can be said that there is no difference in the frequency of following the economic and financial developments according to class groups.

When we look at the ratio of individual pension account ownership, which is very important for the country's long-term funding, the ratio of ownership rate is very low at 5.3 % and the rate of nonowners is very high at 94.7 %.

Tab. 4: Follow-Up Sources of Economic and Financial Development and Relations Between the Frequency of Follow-Up and Class

	<i>Df</i>	<i>F</i>	<i>sig.</i>
<i>There are differences in how the financial developments are followed according to the class.</i>	3	4.714	0.003
<i>Here are differences in the frequency of following economic and financial developments according to the class.</i>	3	0.561	0.641

Tab. 5: Ownership of Providing Basic Banking Operations Tools

<i>Private Pension Account</i>	<i>Frequency</i>	<i>Percent</i>
Yes	32	%5.3
No	575	%94.7
Online Banking Applications		
Yes	478	%78.6
No	129	%21.4

The ownership rate was 3.5 % in Barış (2015) and 29 % in Saraç (2014). Among the reasons for the low level of ownership of the private pension account, it can be said that the students did not have enough information about the income levels, the family perspective, and the state contribution.

In the face of this, in the use of online banking applications, which is one of the basic banking services, a high percentage of students (78.6 %) have been identified as “yes” and 21.4 % “no”. Developments that lead to high rates are mobile phone ownership and internet access. According to the results of the past years, the increase is considered as normal.

In order to measure the level of knowledge of students about basic financial concepts, only four of the questions included in the Core Questionnaire, which the OECD used to test the basic concepts of finance in each country, were taken as examples. These concepts are simple interest, compound interest, inflation, and time value of money. The ratio of those who say that they know simple interest is 47.9 %. The rate of those who do not know is 25.9 %, and the rate of those who do not know their details is 26.2 %.

The rate of those who know the concept of compound interest is 41.7 %. The rate of those who do not know is 30.1 %, and the rate of those who say they do not know the details is 28.2 %.

The rate of those who know the concept of inflation is 49.1 %. The rate of those who do not know is 19.9 %, and the rate of those who do not know their details is 31 %.

Tab. 6: The Relationship Between the Knowledge of Basic Finance Concepts with Basic Finance Questions

Basic Financial Concept Awareness	<i>I know</i>	<i>I heard but I do not know the details</i>	<i>I do not know</i>	<i>Total</i>
<i>Simple Interest</i>	291 %47.9	159 %26.2	157 %25.9	607 %100
<i>Compound Interest</i>	253 %41.7	171 %28.2	183 %30.1	607 %100
<i>Inflation</i>	298 %49.1	188 %31	121 %19.9	607 %100
<i>Time Value of Money</i>	289 %47.6	176 %29	142 %23.4	607 %100

The rate of people who know the time value of money is 47.6 %. The rate of those who say I do not know is 23.4 %, and the rate of those who say I do not know the details is 29 %.

In total, 46.57 % of the students know basic finance concepts, 28.7 % of them heard but did not know the details, and 24.8 % stated that they did not know.

The students stated that they know the compound interest rate 41.7 % lowest and the inflation 49.1 % the highest. These results show that more than half of the students do not have enough knowledge about basic finance concepts, and their basic financial knowledge levels are low. On the other hand, these findings are differently described by Wagland and Taylor's (2009), Ergün et al.'s (2014), and Özdemir et al.'s (2015) research results. Moreover, the rates of those who knew simple and compound interests and inflation concepts were quite low compared to the results of the Peace (2015) survey, whereas the time - to - rate ratio of money was higher than the same study result.

In this study, it was determined that there is a difference in the relationship between class groups and basic finance concepts. It can be said that there are differences in the averages of the answers related to the recognition of these four basic finance concepts according to the classes they read.

As a result of the Tukey test conducted to determine the source of the difference, it was observed that the first class students differed according to the other students. The fact that they do not take finance, economics, and banking courses in the first class is the most important data source that supports this result.

In this study, the results of the ANOVA test, which shows that the relationship between the students' education units and the basic finance concepts differ, as a result of simple interest, compound interest, inflation and time value of money for each of the four basic financial concepts in the table, p values are less than 0.05. It

Tab. 7: The Relationship Between the Students' Reading Class and Basic Finance Concept Awareness

	df	F	sig.
With the awareness of the concept of simple interest and the average of the answers given by students, there is a difference according to the class groups they read.	2	10,261	0,000
Recognition of the compound interest concept and the average of the answers given by the students according to the class groups they differ.	2	3,986	0,000
With the awareness of the concept of inflation and the average of the answers given by the students, there is a difference according to the class groups they read.	2	11,345	0,000
With the awareness of the concept of time value of money and the average of the answers given by the students, there is a difference according to the class groups they read.	2	15,633	0,000

Tab. 8: The Relationship Between Fundamental Finance Concept Awareness and the Other Departments of the Students' Department of Reading (FEAS, Faculty of Health Sciences, School of Applied Sciences)

	df	F	sig.
Recognition of the concept of simple interest and the average of the answers given by the students according to the education section there is a difference.	6	21,756	0,000
Recognition of the compound interest concept and the average of the answers given by the students according to the education section there is a difference.	6	20,083	0,000
Recognition of the concept of inflation and the average of the answers given by the students according to the education section there is a difference.	6	9,886	0,000
With the awareness of the concept of time value of money and the average of the answers given by the students, there is a difference according to the education section of the students.	6	11,730	0,000

can be said that, at the 95 % trust level, there is a difference in the average of the answers related to simple interest, compound interest, inflation, and awareness of time value concepts of money according to the education unit they read.

As a result of the Tukey test conducted to determine the source of the difference, it was concluded that the students of Faculty of Science and Literature differed according to the students of Faculty of Science, Faculty of Architecture, School of Health, and School of Applied Sciences.

Tab. 9: The Distribution of the Score (over 100) of the Students According to the Answers Given to the Questions That Measure the Basic Finance Information

Education Departments	Number	Average	Standard Deviation
Science Literature	41	21,3415	20,58593
Economics and Administrative Sciences	153	59,8039	27,98544
Engineering	21	22,6190	17,5080
Architechture	55	45,4545	27,25519
Tourism	62	28,2787	27,18586
School of Health	205	51,2195	29,36794
School of Applied Sciences	70	59,2857	27,64010
<i>Total</i>	<i>607</i>	<i>48,4736</i>	<i>30,26640</i>
Classes	Number	Average	Standard Deviation
1st Class	142	33,2746	27,24362
2nd Class	143	59,0909	29,15256
3rd Class	159	51,7405	29,06684
4th Class	163	49,2331	29,97212
<i>Total</i>	<i>607</i>	<i>48,4736</i>	<i>30,26640</i>

When the correct answers to the questions asked in the measurement of basic financial information were analyzed, it was determined that the lowest rate was 21 % from the Faculty of Arts and Sciences. The highest percentage of correct response rates (between 51 and 59 %) belonged to the students of the Faculty of Economics and Administrative Sciences, School of Health, and the School of Applied Sciences. These results are similar to those of Barış (2015) and Başarır (2017), which were restricted with the results of the FAES.

These ratios are one of the important results that will help to support the hypothesis that the success rate of the students in the departments that take finance, economics, and similar courses is higher than those who do not take courses.

On the one hand, when correct response rates are analyzed on the basis of classes, the correct response rates of the 1st classes are at the lowest level of 33.3 %, and on the other hand it was found that the correct answer rates of the 2nd and 3rd classes were the highest rates of about 51–59 %. Finance-related courses are often included in the 2nd and 3rd class programs, which can be seen as another remarkable conclusion confirming the results in the table.

In the study, the p value for the basic finance information as a result of the ANOVA test analysis of the difference in the average of the points they received as

Tab. 10: The Relationship Between the Points They Received According to the Answers to the Questions that Measure the Basic Financial Information (over 100) and the Their Department

In the average, according to the correct answers given to the questions that measure basic finance information, there is a difference according to the departments they study.	Df	F.	sig.
	6	21,704	0,000

Tab. 11: The Relationship Between the Students' Average Score (over 100) and Their Current Class According to the Correct Answer to the Questions That Measure Basic Financial Information

There is a difference according to the class they score in the average of the points they score according to the correct answers given to the questions that measure basic finance knowledge.	df	F.	sig.
	3	20,207	0,000

a result of their answers to the questions measuring the basic financial information between the departments in which the students receive education (Faculty of Economics and Administrative Sciences, School of Health and School of Applied Sciences) and others. Since it is smaller than 0.05, it can be said that there is a difference in the basic financial information of the students according to the departments they read at 95 % trust level. As a result of the Games Howell test conducted to determine the source of the difference, it was concluded that the students of the Faculty of Arts and Sciences differ according to the students of the School of Economics and Administrative Sciences, School of Health, and the School of Applied Sciences. These values are similar to the results of Dvorak and Hanley's (2010) and Kılıç and Ata's (2015) surveys.

In the study, as a result of the ANOVA test analysis, whether or not the students' scores were different in terms of their answers to the questions they measured and the financial information they measured, p value for measuring financial information is less than 0.05, at 95 % trust level; it can be said that there is a difference in basic financial information according to the class they study. As a result of the Tukey test conducted to determine the source of the difference, it was concluded that the answers of the 1st class students differed according to the other classes.

5 Conclusion

The lower the tradition of saving in societies, the smaller the volume of financial markets in those countries and the slow progress of their development. The tradition of making financial savings may occur as a result of the education and

financial awareness and behavior of the society. For this purpose, a financial education strategy at national level should be determined. According to Visa Europe (2011), Turkey is the only country that does not have a national strategy about financial education within the 15 members of International Cooperation Platform in Financial Education by OECD. According to an action plan prepared by the CMB (2014), it was recommended to review the course contents from the curriculum at the primary and the lower secondary level to the university, and to add financial issues to the curriculum as content.

The importance of the subject for the country and the fact that it has a high rate of young population have been tried to determine the financial literacy levels of undergraduate students at different faculties and colleges studying at Kırklareli University. Students living outside the family have to decide independently on many financial issues. Students who have financial knowledge and experience will be able to make a right decision in managing the financial situation and increase their welfare levels and savings, and those with lack of knowledge will experience loss of assets as a result of wrong decisions.

In this study, it has been tried to determine in what respect the lack of knowledge of the students and also the suggestions of what kind of changes will be beneficial for young people in the education curricula will be presented.

According to the results of the study, it was determined that there was no significant difference in the perception of success of students in the management of financial status according to their age, class, and gender. The rate of those who fail is 15.5 %.

About %61 of them stated that they have acquired knowledge through their own experiences as a source of information in money management and expenditure. This result shows that students are at high risk of exhibiting incorrect financial behavior by trial and error method. About 34.4 % of the family as a source of information shows that parents also have training needs in financial information. Very low (0 % – 0.3 %) information obtained from the school is a remarkable data. It will be useful to review the content and application methods of the financial and similar courses given at the undergraduate level.

According to the classifications of the students in terms of the resources they follow, the financial development of the 2nd class and the 3rd and 4th class was found to be a significant difference. In terms of the frequency of following economic and financial developments, no significant difference was observed. In the case of basic banking transactions, private pension ownership was very low by 5.3 %, whereas online banking ownership rate was determined as 78.6 %.

The ratio of those who say “I know” about the recognition of basic finance concepts is totally below the 50 %. And than it comes respectively who say

I know inflation, simple interest and time value of money and compound interest concepts. These results show that students' level of knowledge of basic financial concepts is not high. When the students were analyzed in terms of class and department, it was found that there was a significant difference in the average of the answers related to the basic level of financial concept. It was observed that these differences were formed between 1st classes and other classes, especially between Faculty of Economics and Administrative Sciences, Faculty of Architecture, School of Health and School of Applied Sciences, and other departmental students. These results support the hypothesis that the success rate of the students in the departments of finance, economics, and banking courses is higher than the students who do not take the courses.

Similarly, in the answers given to the questions measuring the basic financial information, it was found that the success rate of the students in the departments taking finance, economics, and banking courses was higher than the other department students who did not take the courses.

As a result, the educational needs of students in financial literacy cannot be denied. It is important to provide training on spending, borrowing, risk, and savings in the use of financial assets using appropriate content and information techniques that will help students to make more informed and accurate financial decisions in their daily lives. Young people's early education in these issues will play an important role in raising their welfare levels in their future lives. Considering the size of young people in the population, it can be said that in the future the society will gain more knowledge and accumulation and will contribute positively to the financial and economic development of the country.

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Strategic Management Accounting Practices from Marketing Perspective

1 Introduction

Decision-making is based on the rationality approach, which is calculated as the traditional management accounting wisdom, as one of the main reasons for management accounting (Nielsen et al. 2015). In this context, strategic planning is referred to as a terminology that addresses the impact of the multifunctional unit and often used to classify planning decisions in organizations to be compatible with the environment of an organization but does not earn the recognition it deserves (McNamee, 1988).

Management accounting which requires forward-looking information due to the nature of business decisions related to the future (Horngren et al., 2012) is defined by IMA (Institute of Management Accountants) as “the process of defining, measuring, accumulating, analyzing, preparing, interpreting and communicating the financial information used by Management to plan, evaluate and control an organization and to ensure appropriate use and accountability for its resources”. However, the field of managerial accounting has improved significantly since the publication of this definition (IMA, 2019).

Managerial accounting includes developing the systems to meet needs such as determining the information needs of management and planning, forecasting, budgeting, cost and revenue management and performance measurement (CPA, 2019). In the definition of management accounting by CIMA; “Formulating the strategy, making decisions to plan and control activities, efficient resource utilization, corporate governance for performance improvement and value development, internal control and identification of relevant information that will ensure the protection of tangible and intangible fixed assets; an integral part (CIMA, 2019).

The term “strategic management accounting” was first used by Simmonds (1981), who investigated the accounting perspective in competitor evaluation, which represents a significant divergence from the traditional intrinsic-oriented orientation of accounting (Cadez & Guilding, 2008). According to Simmonds, strategic management accounting as a classification is defined as providing and analyzing management accounting data related to an enterprise and its competitors for use in the development and monitoring of business strategy.

Focusing on providing a firm with information on the assessment of its competitiveness in a sector, it is referred to as the object of management accounting analysis outside of customers and competitors. Some authors emphasize the interface between accounting and marketing (Roslender & Hart, 2003), while others focus on links to strategy (Ma & Tayles, 2009). Therefore, there has not been a common consensus on the definition of strategic management accounting (Cuganesan et al., 2012).

Strategic management accounting is defined as a general approach to accounting for strategic positioning, which is defined as an attempt to integrate opinions from management accounting and marketing management within a strategic management framework (Roslender & Hart, 2003). Strategic management accounting, which focuses on the market by pulling away from introversion, brings together information on management accounting and marketing within the strategic accounting framework (Dunković et al., 2010). And accounting is also for strategic management; it also recommends using accounting information about cost factors, business strategy, markets and competitors for strategic management at the same time. Although it is said that the use of accounting information in accounting and marketing literature is useful, the interaction, communication and information exchange between departments has not reached the desired level (McManus & Guilding 2008).

2 Change in Management Accounting

The quality of information that decision-makers can access does not allow to optimize it, because decision-makers have inborn talent to use information to optimize their decisions. In the studies on the specific decision-making role of managerial accounting, it is observed that accounting in institutions generally compromises domestic policy following administrative power groups (Nielsen et al., 2015).

The change occurs when people respond to the internal dynamics of repetitive transactions while they are provided by people, and by their actions on organizational and environmental factors, including organizational and contradictions. In the case of a change in managerial accounting, the possible contradictions will arise from the experience of the participants' beliefs, desires and motivations in the interface of the technical character of change. These experiences can serve by assuming that new techniques are later accepted as part of the organization's implementation of rule-based practices (McLarena et al., 2016).

The main differences between traditional managerial accounting and strategic management accounting are given by Collier (2006); reporting unit; focus

(internal, external); profitability analysis (products, customers and markets); approach to cost analysis (cost control with department/product costing systems, periodic production costs and monthly segmental budgeting); performance evaluation (monthly basis analysis, three-/six-month multidimensional analysis); investment evaluation (financial analysis with strict criteria, strategic analysis based on multiple models to promote judicial decisions) and ownership (being under the control of the accountant, being a part of the team during the strategic review process) are summarized in seven scopes (Dunković et al., 2010).

Strategic management accounting does not only assume the role of information provider, but also takes its place as an active player who has the power to achieve its goals in the strategic management process. Strategic management accounting differs from traditional management accounting and strong external financial reporting fundamentals with the information it tries to provide on costs, prices, sales volumes and market shares, cash flows and resource consumption (Roslender & Hart, 2002).

3 Strategic Management Accounting

The current position of strategic management accounting is the advancement of many management accounting pioneers (Horngren, Porter, Simmonds, Johnson & Kaplan, Shank, Seigel, Atkinson, Cooper et al.) and thought leadership and innovative management institutes (IMA, CIMA, CMAC) has been through research studies (Guay, 2014).

Strategic management accounting has been defined by many scientists in different ways. For example, while Simmonds sees it as a way of analyzing its business and its competitors to develop its business strategy, Bromwich sees it as spending and operating policy strategies of the financial analysis method of product markets, businesses and competitors. While Innz defines as the presentation of information that supports the organization's strategic solutions, Ryan defines as the most reliable way to follow up the financial and accounting issues. While Bogataya et al. describes business activities as a system for providing efficiency to business managers and investors in their decision-making by combining financial and managerial accounting, Yurasova et al. refers to strategic management as a necessary system to provide information needs. Miles and Snow (Dunković et al., 2010) described strategic goals and evaluation of external environmental opportunities with internal forces and resources in deciding on a set of action plans to achieve these goals (Vetrov et al., 2017).

Strategic management accounting consists of three factors such as strategic planning, strategic analysis and strategic control (Vetrov et al., 2017). Strategic

planning aims to develop long-term plans to create a long-term business development program. Strategic analysis aims to determine the economic activity of the organization for a certain period of time and to evaluate the strategic objectives and objectives of its practices. The objective of strategic control is to monitor the process of achieving strategic goals and to design and implement the return.

4 Strategic Management Accounting Applications in Marketing Decisions

Strategic management accounting practices include environmental or marketing orientation, competitor orientation and prospective long-term orientation elements (Guilding et al., 2000). Managerial accounting practices or managerial accounting standards are tactical and strategic tools and techniques necessary to develop, evaluate and make decisions in-house operations. Management accounting institutes (CMAC) categorize management accounting practices in three ways: costing, cost use and strategic management control (Guay, 2014). In another classification, six titles are identified: strategic management, risk management and governance, performance management, performance measurement, financial management and financial reporting.

Strategic management accounting practices are part of the definition of contemporary strategic management accounting. In Guay's study, strategic management accounting practices are described as follows (Guay, 2014): feature costing, brand value budgeting, brand value tracking, competitor cost assessment, competitive position tracking, competitor evaluation based on published financial statements, life cycle costing (LCC), quality costing, strategic costing, strategic pricing, target costing (TC), value chain cost and evidence-based decision-making, financial simulation, cash flow management, target costing, activity-based management, activity-based costing, budget and pro forma management financial statements, strategic cost management, Kaizen costing, product mix decisions, uncertainty and capital budgeting, activity-based costing, balanced scorecard, total quality management, total quality assurance, continuous quality improvement, just-in-time (JIT) and costing integrated into enterprise resource planning system improvements.

Guilding et al. (2000) has identified 12 strategic cost management applications from the literature has identified the technique (technique), Cravens and Guilding (2001) added three more techniques to these applications. These applications are classified into five broad categories such as: (1) costing; (2) planning, control and performance measurement and (3) decision-making, and other two categories such as "competitor accounting" and "customer accounting" (Cadez & Guilding, 2008).

4.1 Costing

Monden and Hamada (1991), the practice of contemporary Japanese management accounting practices, identified the target cost and Kaizen costs in Japanese automobile companies. In addition to the target cost, quality costing and other Japanese management accounting practices were also discussed by Ito (1995) and by European predecessors. Shank (1989) proposed a strategic cost management framework including value chain analysis, cost driver analysis and competitive advantage analysis (Malmi, 2016).

Until the value chain analysis is delivered to the customer from the main raw material sources in a final product process, it offers a comprehensive external focus on the measurement of the added value of the activities available, in contrast to value added in a sector. In the other way, strategic positioning analysis focuses on the methods preferred by firms in competition. While activity-based costing management sees business process and resource consumption from the activity perspective, target costing management emphasizes the foundations of market-oriented philosophy (Roslender & Hart, 2002).

The terms and characteristics of a business related to its products or offerings are concrete values. The descriptive qualities of the products evoke their attractiveness, while the benefits evoke the ability of customers to meet their emotional needs. This approach considers products or services as a package of features that differentiates the costs. Instead of focusing on what needs to be reflected, exploring the more subjective aspects of products or proposals will be accepted as objective attributes or features (Roslender & Hart, 2002).

A customer life cycle is a series of transactions between a client and an organization during his or her stay in the firm. It is measured as the present value of the future net cash flows expected to be received during a customer's lifetime, and the revenue from the customer reduces the cost of attracting, selling and servicing the customer (Brownlie, 1999).

Horngren speaks of four types of quality-related costs. These are prevention costs, appraisal costs, internal breakdown costs and external fault costs. While most of the prevention costs occur in the R&D phase of the value chain, most evaluation and internal failure costs occur when the product is being manufactured. External failure causes an increase in customer service costs or may cause sales to fall due to an unhappy customer (Horngren et al., 2012).

The target cost was introduced in the late 1980s (Sakurai, 1989; Monden and Sakurai, 1989) to the attention of Western administrative accounting as *genka kikaku*. Kaizen costing together with target costing is the basis of an effective total cost approach. This approach rejects cost-based or cost-plus pricing

(Roslender & Hart, 2002). Target costing is defined as a systematic approach to determine the cost to produce a desired product with a certain quality and functionality to achieve the expected profit from the foreseen sales price. This approach, which is based on achieving high profit starting from the design phase of the product in order to achieve the targeted revenue after product sales, is a useful and effective tool in achieving the goal under intense competition conditions. Market-oriented pricing, customer focus, intensification in the design process, collaboration between specialized sectors, orientation within the product life cycle and participation in the value chain are the basic principles of target costing (Sharafoddin, 2016).

4.2 Planning, Control and Performance Measurement

The idea of benchmarking came to the fore in the planning literature due to the pioneering work of Xerox Corporation and the impact of Total Quality Management. It is a continuous comparative process covering all areas of organizational activities from strategic development to transactions, customer service and satisfaction (Brownlie, 1999).

Performance measurement consists of the idea of incorporating financial and nonfinancial measures into more complex frameworks based on a balanced set of measures, linking these measures to the strategy. They have taken the form of causal maps showing the operational impacts of different strategies that address both market and cost assessments of strategic importance (Brownlie, 1999).

Integrative performance measurement systems can vary according to the degree to which information is integrated into activities and strategy. It provides feedback on how to link the various aspects of the value chain to understand the level of interdependence caused by product differentiation and between cost-price strategies between business activities and strategies (Chenhall, 2005). Today's integrative performance measurement systems involve responsibility concerns, including environmental and social impacts of corporate activities when making decisions (Brownlie, 1999).

4.3 Strategic Decision Making

Strategic cost management is considered to be a strategic and competitive advantage as a result of its focus on a wider range of topics than traditional managerial accounting and its external emphasis.

For decision-making purposes, different costs and revenues, opportunity cost and submerged cost concepts are very important. Different costs and revenues are costs and revenues that differ between alternatives. Opportunity cost is the

resulting benefit when one alternative is selected. Wreck cost is a cost that has occurred in the past and cannot be changed. Wreck costs are always irrelevant in decisions and should not be taken into account (Noreen et al., 2011).

Simmonds claims that strategic pricing using competition-driven analysis will result in a better informed pricing decision by including factors such as competitive price response, price elasticity, projected market growth, economies of scale, and experience economies (Guiling, 1999).

It is not surprising that the managerial implications of brand valuation attract attention in both the accounting and marketing literature. In the marketing literature, it is believed that companies with strong brands require more measurable information to help with brand management (Cravens & Guiling, 2001). Brand management accounting, which is an interdisciplinary concept such as strategic management accounting, emphasizes customers and branded products by adding a dimension to the target cost philosophy including information from marketing management and management accounting. For example, price/value swap is what determines what attracts customers to a specific brand and how much they will pay against a competing brand (Roslender & Hart, 2002).

4.4 Competitor Accounting

There are five applications of competitor-oriented accounting in the literature. These are (Guiling, 1999): 1. competitive cost assessment, 2. competitive position tracking, 3. competitor assessment based on the financial statements issued, 4. strategic costing and 5. strategic pricing.

The competitive cost assessment approach suggests that businesses should consider not only their own companies but also the cost structure of all businesses and potential participants in the relevant market (Bromwich, 1990).

Simmonds (1986) proposes to analyze the competitive position of the major competitors by analyzing and following the trends of their market share, volume, unit costs and sales. Moon and Bates (1993) Competitive performance assessment analysis, sales and profit levels, as well as monitoring the movement of assets and debts also underlines the fact that the traditionally trained accountants include techniques (Guiling, 1999).

4.5 Customer Accounting Applications

Customer accounting provides a framework to enable customers to choose their customer resources technique based on customer resources interfaces, focusing on financial measurement of customers (Lind & Strömsten, 2006). In the literature, customer accounting practices are presented in four dimensions. These

are: i-customer profitability analysis, ii-customer segment profitability analysis, iii-lifetime customer profitability analysis and iv-valuation of customers or customer groups as assets (Guilding and McManus, 2002).

Customer profitability analysis measures the individual contribution of a client to an operating profit by defining revenues and costs over a period of time (annual or quarterly). By selecting full costing or activity-based costing, it is assumed that by distributing the general production costs to certain customers, the activities consume resources and consume activities in their customers (Lind et al., 2006). In the late 19th century, the largest share in increasing interest in market-oriented management accounting has taken the customer profitability analysis. Establishing reliable figures in customer accounts, identifying resources to maximize customer profitability, customer profitability and customer profitability priorities, in particular relations between customer loyalty, customer loyalty and customer satisfaction, analysis of customer segments and preparation of segment-driven key figures accounting system can be applied and can be used only a few customer and market-oriented approaches (Helgesen, 2007).

The purpose of customer segment profitability analysis is a group of customers or customer segment based on variables that make it different from mass buying behavior, geographic location, demographic variables or other customers. In this approach, it is accepted that customer groups are more important than individual customers, and therefore general production costs should be monitored according to customer segments (especially in banking sector applications) (Guilding & McManus, 2002).

Lifetime customer profitability analysis was analyzed by Foster and Gupta (1994) in the life insurance sector (Foster & Gupta, 1994). Cooper and Kaplan have defined a credit pricing model that foresees incomes and wages and interest income, as well as the life expectancy of customer loans with an expected profitability in retail credit transactions. In this application, customer profitability analysis which is extended to include future years is focusing on all future revenue streams and costs including serving a specific customer (Cooper & Kaplan, 1999).

The first study evaluating customers or customer groups as assets in accounting literature is the study of Foster et al. (1996). The view of customers as assets is noteworthy in the context of marketing literature. Considering customers as assets is a concept closely related to the theory of bir relationship marketing an, which is of great interest among marketing commentators. In this application, the present value of all future profit flows that can be attributed to the asset value of the customers, a particular customer or a group of customers, is calculated (Guilding and McManus, 2002).

5 Cost Drivers

Cost-driver analysis, designed more extensively from the context of activity-based costing, is an element of the strategic cost management framework (Roslender & Hart, 2002).

Cost drivers are two-tiered, namely the first tier (cost pools in which the resources consumed in the activity-based costing system are allocated) and the second tier cost drivers (activity rates used to apply cost to products, customers, and other cost objects). Examples of first stage cost drivers: number of customer orders (bulk level pool), number of product designs (bulk level pool), machine hours with order size (unit level pool), number of active customers and customer relations (customer level pool), and number of products (party level pool) supported by the number of products. The activity rate is a function of the cost driver, the cost driver ratio represents the shareholder, the cost driver (such as the number of orders) (Guay, 2014).

Many considerations should be taken into account when using estimation methods such as full cost, variable cost, and activity-based cost to generate reliable profitability figures for client accounts. Using different approach methods can lead to different estimates in customer profitability (Helgesen, 2007).

6 Conclusion

This study is based on previous studies that consider strategic management accounting as a set of strategically oriented management accounting techniques. The integration of traditional managerial accounting into new managerial accounting techniques has resulted in more efficient cost management accounting systems.

Marketing decisions cannot be distinguished from operating cost information and the impact of marketing strategy on transactions and business profitability. A number of developments can also be envisaged on the basis of interdisciplinary cooperation between marketing managers and management accountants.

Collecting data from competitors requires full cooperation of management accounting and marketing colleagues in price decisions, bargaining power, and cost position evaluation. This information can be provided from strategic management accounting in one place integrated into financial and nonfinancial databases.

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Effect of Corporate Governance on Market and Accounting Based Performance Measures: An Investigation in Borsa Istanbul

1 Introduction

The financial crises due to company scandals all around the world and the lack of trust in the market because of those scandals cause a chaotic situation. In this state of chaos, companies and countries try to figure out ways in order to overcome these hard times. One of the solutions is corporate governance. In general, corporate governance is the approach and principle that aims to provide assurance to preserve the rights and interests of the company shareholders and beneficiaries, and it gives prominence to sustainability for businesses.

Good corporate governance practices are an important step in establishing trust for long-term international investments in the market and creating stability. In addition to providing long-term sustainability to companies, corporate governance also supports them to make important decisions about vision, mission, and processes. The profitability of companies also means that they add value to the society in environmental, social, and economic terms. Many countries and companies consider corporate governance as an important way to improve economic dynamism and to improve overall economic performance. Compliance of corporate governance activities with international standards is one of the most important steps to success for companies.

The first concrete step in corporate governance in Turkey was taken with the publication of OECD Corporate Governance Principles in 1999. After the publication of corporate governance principles, the studies on this subject have accelerated. The need to prepare corporate governance principles, the establishment of corporate governance index, and the ongoing corporate governance rating activities are the best examples of studies concerning corporate governance in Turkey.

Our study consists of four parts. After the introduction, which contains a general evaluation of the subject, in the second section the literature on the subject is

examined. In the third section, the data set and sample are enlightened and analyzed by using the panel data method, and the results are given in the last section.

2 Literature Review

In the literature, there are many studies examining the relationship between corporate governance and basic characteristics of companies. In these studies, one of the most important variables of the basic characteristics of the companies is the financial performance indicators. Some among the studies that investigate the relation between financial performance and corporate governance are: Bianco and Casavola (1999); Judge, Naoumova, and Koutzevol (2003); Sanda, Mikailu, and Garba (2005); Hebble and Ramaswamy (2005); Coleman (2007); Bhagat and Bolton (2008); Kajola (2008); Chidambaran, Palia, and Zheng (2008); Rashid (2008); Brown and Caylor (2009); Ragothaman and Gollakota (2009); Khatab, Masood, Zaman, Saleem, and Saeed (2011); Nur'ainy, Nurcahyo, Kurniasih, and Sugiharti (2013); Amba (2014); as well as Arora and Sharma (2015).

In the studies that were conducted by Bianco and Casavola (1999), Brown and Caylor (2009), and Sanda et al. (2005) although they used valid indicators and data, they could not identify the relationship between corporate governance and financial performance. As for the studies of Judge et al. (2003), Coleman (2007), Bhagat and Bolton (2008), Kajola (2008), Rashid (2008), and Khatab et al. (2011), they generally point towards a positive and significant relation between corporate governance and company performance. One of the most important point that sticks out in these studies is that an independent board of directors having broad powers and increasing the value of the company and separating the roles of the CEO and the chairman of the board have a significant and positive effect on corporate performance [Coleman (2007), Bhagat and Bolton (2008)]. Other than this one of the other important points made by Rashid (2008) is that the processes that affect the company value are different in developed and developing markets. Outside of these studies, Chidambaran et al. (2008) investigated the effects of changes in corporate governance to the companies' future performances. As the result of the analysis, it is found that underneath the changes in corporate governance lies the pressure of merger in the industry. On the other hand, Hebble and Ramaswamy (2005) and Ragothaman and Gollakota (2009) emphasized the impact of company size and financial ratios on corporate governance practices.

Kouwenberg (2006), Javed and Iqbal (2007), and Black, Carvalho and Gorga (2011) studied the relation between the company value and corporate

governance practices. Kouwenberg (2006) researched the impact of voluntary corporate governance practices in developing markets to the value of the company, and as a result of his analysis he found out that the 1 % increase in the standard deviation between the years of 2003 and 2005 caused a 10 % increase in index in accordance with the company value. Another result is some significant effect of the structure of board of directors and ownership and partnership variables to the company performance. Black et al. (2011) determined that the view of corporate governance estimating the market value is strongly supported.

When we look at it from the perspective of Turkey, we see that there are many studies that investigate the relation between corporate governance and company performance. Some of these studies are related to corporate governance ratings. Esendemirli and Erdener Acar (2016) included the years 2013 and 2014 to their analysis and used the TOPSIS multicriteria decision-making method to investigate relations between the corporate governance ratings of the nonfinancial sectoral enterprises and their financial performances. As the result of the study, it is found that the financial performances of the companies and the corporate governance ratings did not move in the same direction for both years. However, in their study, Kayalı and Doğan (2018) evaluated the relationship between the corporate governance rating and the financial success of the enterprises in the manufacturing sector in the Borsa İstanbul (BIST) Corporate Governance Index between the years 2012 and 2016. As the result of the study, it is determined that with the increase of the corporate governance ratings of the enterprises comes the decrease of chances to go bankrupt and therefore causing an increase in their financial success.

Acaravcı, Kandır, and Zelka (2015) examined the relationship between corporate governance and financial performance of the enterprises operating in the BIST Manufacturing Industry during the period of 2005–2011. As a result of the implementation of the panel data analysis, a positive correlation was found between the return on assets and the size of the enterprise and the size of the board of directors, while a negative relationship was found between institutional investor ownership and free float ratio. No relation was found between Tobin's Q ratio and the corporate governance indicator. In the study conducted by Kara, Erdur Acar and Karabıyık (2015), they analyzed the relationship between corporate governance levels and financial performance of enterprises in the Corporate Governance Index in the period of 2006–2012 by applying panel data analysis. As a result of the analyses, a significant and positive relationship was determined only between corporate governance and Tobin Q and leverage ratio.

3 Research Design and Methodology

3.1 Data Set

In our study, the annual data of the 52 companies included in the Borsa İstanbul (BIST) 100 index, whose data can be obtained and included in the evaluation scope, between the years 2013–2017 were used. While determining the companies to be included in the analysis, banks, insurance companies, leasing companies, real estate investment companies, and holdings are excluded from the study.

3.2 The Scope and Objective of the Research

The aim of our study is to examine the effects of some features of corporate governance on financial performance by using the data of 52 companies traded in BIST 100 index. The data gathered about the companies examined were obtained by using the financial statements publicly announced through internet by the Kamuoyu Aydınlatma Platformu-KAP (Public Disclosure Platform). In addition, company annual reports, corporate yearbook, company general information form, and share market data were used to obtain information about the variables. In the study, accounting-based and market-based criteria are used to represent financial performance. The accounting-based measures of the dependent variables represent the return on assets (ROA) and return on equity (ROE), while the market value/book value (MBV) and the price-earnings ratio (PE) represent market-based criteria. The independent variables are the free float ratio (FFR), the size of the board of directors (SBD), the ratio of independent members in the board of directors (IBD), whether the general manager is also a member of the board of directors (GM), and the corporate governance index (CGI). Within these variables, GM and CGI were used as dummy variables.

3.3 Research Methodology and Models

In our study, panel data analysis method was used to determine the effect of corporate governance on financial performance.

Panel data analysis is a method of estimating economic relations by using the cross-sectional data of time dimension. Panel data analysis allows for the establishment and testing of more complex behavior models than the cross-sectional or time series (Greene, 1993: 464).

The panel model with “K” variable that has two main approaches, one being fixed effects model and other random effects model, is generally shown as:

$$y_{it} = \beta_{1it} + \beta_{2it} X_{2it} + \dots + \beta_{Kit} X_{Kit} + \epsilon_{it} \quad (1)$$

Tab. 1: Descriptive Statistics

Variables	Min.	Max.	Mean	Std. Dev.
MBV	0.0084	5.6108	1.8440	1.1009
PE	0.8790	42.4086	2.6412	4.7228
ROA	-0.1842	0.3068	0.0882	0.7476
ROE	-1.2843	0.5321	0.1064	0.2218
FFR	4	82	32.4608	22.5429
SBD	3	16	1.9248	7.3201
IBD	0.0000	0.5248	0.1268	0.1448
GM	0.0000	1.0000	0.2388	0.4017
CGI	0.0000	1.0000	0.3896	0.4628

For the analysis, we used four different models. Those models are listed below:

$$\begin{aligned} \textbf{Model 1: } MBV_{it} = & \beta_1 (FFR)_{it} + \beta_2 (SBD)_{it} + \beta_3 (IBD)_{it} \\ & + \beta_4 (GM)_{it} + \beta_5 (CGI)_{it} + 0\alpha_i + \varepsilon_{it} \end{aligned} \quad (2)$$

$$\begin{aligned} \textbf{Model 2: } PE_{it} = & \beta_1 (FFR)_{it} + \beta_2 (SBD)_{it} + \beta_3 (IBD)_{it} \\ & + \beta_4 (GM)_{it} + \beta_5 (CGI)_{it} + 0\alpha_i + \varepsilon_{it} \end{aligned} \quad (3)$$

$$\begin{aligned} \textbf{Model 3: } ROA_{it} = & \beta_1 (FFR)_{it} + \beta_2 (SBD)_{it} \\ & + \beta_3 (IBD)_{it} + \beta_4 (GM)_{it} + \beta_5 (CGI)_{it} + 0\alpha_i + \varepsilon_{it} \end{aligned} \quad (4)$$

$$\begin{aligned} \textbf{Model 4: } ROE_{it} = & \beta_1 (FFR)_{it} + \beta_2 (SBD)_{it} + \beta_3 (IBD)_{it} \\ & + \beta_4 (GM)_{it} + \beta_5 (CGI)_{it} + 0\alpha_i + \varepsilon_{it} \end{aligned} \quad (5)$$

3.4 Empirical Evidence

EViews package program is used for data analysis. Descriptive statistics of the 5-year (2013–2017) data of the 52 companies that were used in panel data analysis are given in Tab. 1 below.

As it can be seen from the table, the market value/book value ratio in the companies examined within the scope of analysis is 1.8 on average and price/earnings ratio is 2.6. On the other hand, on the basis of accounting-based criteria, the ROA is calculated as 0.09 and the ROE is calculated as 0.10 on average.

Tab. 2: Hausman and F Test Results

	Model 1	Model 2	Model 3	Model 4
Hausman	24,76	52,84	18,23	12,96
Prob Value	0.0012	0.0000	0.0026	0.0018
F- Test	3,67	2,26	6,18	4,48
Prob Value	0.0000	0.0000	0.0000	0.0000

Tab. 3: Wald Test Results

Test Statistics	Value	Probability
Chi-Square	2.945620	0.3628

The corporate governance index, which is one of the independent variables, is approximately 40 %, and it is determined that almost half of the firms participating in the analysis take part in the index.

Test results that are obtained from Hausman test and F test is shown in Tab. 2. Hausman test is a test used when a choice must be made between the Fixed and Random Effects Model to decide which one should be preferred. By this means Hausman test is used in comparing the effects of random effects model and fixed effects model, while F-test is used to test the significance of fixed effects model.

According to F-test statistics, the effect of fixed effects model for every company is significant. The individual effects on all models within the scope of the Hausman test statistic are not random. Therefore, fixed effects is chosen for each of the four models.

At this stage of the study, the reliability of the model is examined before the interpretation of the estimation made under the fixed effects model. The reliability of the results of the model is discussed within the scope of changing variance (heteroscedasticity) and cross-sectional dependence.

The modified Wald test results, which are used to determine the heteroscedasticity problem that has changed due to the constant effect of the model, are given in Tab. 3. The chi-square statistics obtained by the Wald test was 2.945620 and the probability value was 0.3628, therefore the H_0 null hypothesis homoscedasticity was accepted. In this respect, it is concluded that the model does not contain a changing variance problem.

As the cross-sectional dependence between the units in the panel data analysis may cause the bias of the test results, the autocorrelation assumption

Tab. 4: Pesaran Test Statistics Results

Test	Test Statistics	Probability
Pesaran CD	0.842567	0.2972

Tab. 5: Results of the Models (Market-Based Criteria)

Model 1			Model 2		
(MBV)	Coefficient	t Statistics	(PE)	Coefficient	t Statistics
Fixed	5.6874	3.62***	Fixed	0.173	0.14
FFR	-8.2403	-4.54	FFR	4.8423	6.20
SBD	0.7423	2.31***	SBD	2.5217	5.68**
IBD	-6.6486	-1.19	IBD	-2.4650	-2.84
GM	0.3149	4.86	GM	1.2486	3.42***
CGI	1.8765	3.32**	CGI	0.4825	1.76
R ²	0.28		R ²	0.18	
Adjusted R ²	0.20		Adjusted R ²	0.12	
DW Statistics	2.5189		DW Statistics	2.1147	
Observations	260		Observations	260	

Note: * significant at 1 % level, ** significant at 5 % level, *** significant at 10 % level

was investigated with the Pesaran CD test. In the light of the results obtained, the H_0 null hypothesis that there is no correlation between the series was accepted.

Results of the models are given in Tab. 5 and Tab. 6.

Within the context of the model that does not involve changing variance and autocorrelation problem, it was concluded that there is a statistically significant and positive relationship between the size of the board of directors and the corporate governance index and the market value/book value ratio for Model 1. Model 1's explanatory power is 20 %. So it means that independent variables can explain the 20 % of the changes in dependent variables. For Model 2, it is concluded that there is a statistically significant and positive relationship between the size of the board of directors and whether the general manager is also a board member and the price/earnings ratio.

In both of these models, a significant relationship was found between market-based criteria and the size of the board of directors. However, the explanatory power of Model 2 is at a low level at 12 %. In the light of the analysis made, it can be said that with the increase in the size of the board of directors market-based performance criteria also increases.

Tab. 6: Results of the Models (Accounting-Based Criteria)

Model 3			Model 4		
(ROA)	Coefficient	t Statistics	(ROE)	Coefficient	t Statistics
Fixed	4.3142	0.72	Fixed	-9.1463	-0.88
FFR	-12.1204*	-6.34	FFR	-18.2018*	-2.16
SBD	1.4056	2.18	SBD	7.3240*	6.36
IBD	-4.7866	-1.24	IBD	1.7046	1.66
GM	3.2197	5.98	GM	1.2746	4.04
CGI	2.5862**	3.66	CGI	3.3884**	5.66
R ²	0.32		R ²	0.41	
Adjusted R ²	0.26		Adjusted R ²	0.34	
DW Statistics	3.7869		DW Statistics	4.5842	
Observations	260		Observations	260	

Note: * significant at 1 % level, ** significant at 5 % level, *** significant at 10 % level

When Model 3 is examined, a statistically significant relationship is found between free float ratio and corporate governance index and ROA. The free float ratio from the variables negatively affects the ROA. Explanatory power of Model 3 is 26 %. In the evaluation made for Model 4, it is concluded that there is a statistically significant relationship between the free float rate, the size of the board of directors and the corporate governance index and ROE. In Model 4, just as in Model 3, there is a negative correlation between free float ratio and ROA. Model 4's explanatory power is highest among other models with 34 %. When Models 3 and 4 are analyzed, we see that there is a negative relationship with the accounting-based criteria and the free float index, while there is a positive relationship with accounting-based criteria and corporate governance index. This shows that with the increase in the free float ratio, the ROA and the ROE decreases and that corporate governance index increases ROE and ROA.

4 Conclusion

The implementation of corporate governance principles that have been created for a more healthy structure of business management has been made mandatory in some countries, and it is voluntary in others. As one of the pioneer countries in this field, the USA has made significant changes in the field of corporate governance with the Sarbanes-Oxley Law. In Turkey, certain principles concerning corporate governance are formed, corporate governance index has

begun to be calculated, and new laws are enacted by the Turkish Commercial Code to support the development of the institutional structures of the corporate managements.

By this point, the aim of our study is to reveal to what extent the corporate governance structure of the companies affects the companies' financial performance. In our study, four different dependent variables (market value/book value, price/earnings, ROA, and ROE) represent the financial performance of the company as market- and accounting-based criteria. In addition, independent variables considered to affect these variables are the free float ratio, the size of the board of directors, the proportion of independent members in the board of directors, whether the general manager is also a member of the board of directors, and whether it is in the corporate governance index.

As a result of the analyses, it is concluded that there is a relationship between the financial performance of the companies and corporate governance. A significant relationship was found between the market-based criteria and the size of the board of directors in Model 1 and Model 2. While in Model 3 and Model 4, a negative relation between the accounting-based criteria and free float ratio and a positive relation between accounting-based criteria and corporate governance index was found.

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Dilek Kayakiran, Saime Dogan, and Bulent Kilic

The Perspectives of the School of Applied Sciences Students to the Accounting Ethics: The Case of Kırklareli University School of Applied Sciences

1 Introduction

Mobility in capital markets, one of the economic indicators of globalization, is directly related to financial reports consistent with international reporting standards. Economic crises, business insolvencies, and economic scandals in recent years have increased the importance of financial reporting. For this reason, the responsibility of those preparing the financial reports, the presenters, and the auditors has increased. Because of the nature of the profession, a number of issues have become important, especially in terms of the preparation of financial data for the enterprises, keeping the records of books, giving place to equity and impartiality, protection of confidentiality, focusing on the auditing standards and principles, and giving place to independence in the audit (Akbas et al., 2009).

The structures and management approaches of the enterprises are changing with globalization. Therefore, the accounting profession also changes and increases the quality of the information produced. But the irregular increase in the number of professionals and unfair competition (Ömürbek and Türkoğlu, 2013) caused an increase in unethical behaviors in the accounting profession. In order to eliminate unethical behaviors, it is necessary to develop professional ethics in the accounting profession. For this purpose, professional ethics principles and rules of accounting profession have been revised, and new laws have been enacted by professional organizations in the world and in our country. At the international level, the principles laid down by the IFAC (International Federation of Accountants Ethics Committee) and the AICPA (The American Institute of Certified Public Accountants) and at the national level the regulations on professional ethics made by The Ministry of Finance, Capital Markets Board (CMB), and in particular Union of Chambers of Certified Public Accountants and Sworn-in Certified Public Accountants of Turkey (TURMOB) are guiding the ethical behaviors of professionals. The basic ethical principles that must be complied with by all professionals at the code published by IFAC and translated into Turkish by TURMOB in 2007 are explained under the headings of

“honesty”, “impartiality”, “professional competence and care”, “confidentiality”, and “professional behavior” (Turmob 2009). From these principles, honesty is to be honest by all professionals in their professional and business relations; impartiality means that the unjust and improper pressures of third parties do not affect or prevent the professional decisions of professionals; professional competence and care means accounting profession to behave in accordance with professional standards; confidentiality is not to disclose information to third parties or interests unless they have a right or duty to require them to disclose information obtained as a result of their professional relations and that such information is not used for the benefit of the professional or third parties; professional behavior means that the professional member complies with the existing laws and regulations and avoids any behavior that would impair the reputation of the profession (Turmob 2013). When these principles are examined, it is seen that the responsibility of the professional member towards the society is emphasized. Professional and legal regulations represent minimum standards of behavior to professional accountants and to the participants of financial reporting. Ethics build on them (Todorovic 2018). Ethics, as a theory, is the branch of philosophy that explores the nature of morality and virtue, and that evaluates human actions (Stephensen 2016). Ethics can be defined as the whole of the principles and rules that apply in all parts of the world, including the evaluation of the attitudes and behaviors of people by the direction of good or bad, right or wrong in the past, present, and future (Yatkin, 2008). Ethics, in modern society, is related not only to the system of moral standards, but also to personal behavior and responsibility for the consequences of decisions taken as well (Bonaci et al., 2013). Professional ethics is the whole of the principles and rules governing the relations between the individuals that make up a profession group and their relations with the society (Bilen and Yilmaz, 2014). Ethics in accounting can be expressed as the whole of the professional principles created by the professional organization and adopted by the parties related with accounting in the form of rules (Daştan, 2009), which must be complied with in order to provide reliable information by taking into consideration the current value judgments of the society as well as the transactions in accordance with the laws (Yıldız, 2010). Ethics in the accounting profession is an important issue that is directly related to the honesty of the profession and the ability to ensure the community trust (Ağyar et al., 2012). Ethics has a fundamental impact on the formation of the value system of the future generations of economists in the field (Toma 2016). Unethical conduct in accounting causes failures in the economic system (Anzeh, Abed, 2015). Ethical conducts increase the credibility of the profession (Sharma, 2010). In addition to professional knowledge and experience, professional accountants should also

have the ethical structure required by the profession and adhere to the rules of professional ethics.

This study was conducted in order to examine the professional ethics perceptions and the professional ethics perspectives of university students who received accounting education at the undergraduate level. It tries to find out whether students' perception of professional ethics differs according to demographic characteristics. The universe of the study is composed of the undergraduate students of the Department of Banking and Finance and the Department of International Trade and Logistics who study in Kırklareli University School of Applied Sciences and take the general accounting course. Within the scope of the study, questionnaire was applied as data collection method. The first part of the questionnaire consists of questions related to demographic variables, the second part consists of accounting professional ethics judgments, and the third part consists of questions about professional ethics education. In the third chapter, the questions directed to students are examined under subtitles of ethical principles of accounting, which are honesty, impartiality, professional competence and care, confidentiality, and professional behavior. In order to compose the questions of questionnaire, the studies of Ağyar et al. (2012) "The Investigation of The Attitudes of Vocational School Students Studying In Accounting Department Towards Professional Ethics in the Light of Socio-Economic Variables (Akdeniz University Case)" and Gülmez et al. (2016) "Investigation of Viewpoints to Accounting Professional Ethics of University Students: OKU Students Case" were used.

2 Literature Study

In the literature, along with the ethical issues in accounting gain importance, the studies examine the impact of ethics on students studying in the academic field, the opinions of the accounting instructor on ethics and the effects of the ethics on the professionals in the business world have increased. It is seen that researches are dedicated to, in particular, determining the effects of demographic characteristics on ethical perception and the factors that cause different ethical perceptions. Because, this study examines how the ethical perception of students is evaluated, and the studies in this field will be mentioned.

Dastan (2009), in his study related to the place and importance of the ethical training in accounting education, examined the ethics courses of Economics and Administrative Sciences Faculties of universities in Turkey. In conclusion, he stated that the importance given to accounting education in Turkey remained at low levels. He proposed that ethical issues should be presented in main lines

in general accounting and auditing courses, including accounting at the undergraduate level as an elective course and at the graduate level by adding a course to the program called as Accounting Professional Ethics.

In a study conducted by Kurnaz and Gümüş (2010), in order to analyze the perceptions of unethical behaviors of students studying accounting, they found that students in upper class were more sensitive to unethical behaviors compared to students in lower grades, and female students' perceptions about some unethical practices are more sensitive than male students when ethics perceptions are compared according to the gender. In the study, it was found that students perceived economic behaviors to be less unethical than other behaviors due to the economic problems they were in. Therefore, there were economic reasons that caused them to behave in this way, and they tried to put this on a logical basis. Rao et al. (2009) compared the accounting students with business students in terms of business ethics attitudes. Accounting students have taken ethics courses whereas business students have not taken ethics courses. The study showed that accounting students have higher ethical attitudes towards business ethics compared to other students.

Ağyar et al. (2012), in their study, found that ethical judgments differ according to gender, and women are more sensitive to ethics than men. Moreover, they found that those who graduated from vocational high schools and accounting-related departments showed lower ethical sensitivity than those who graduated from other high schools. It is seen that more than half of the students (56.1 %) stated their opinions about a professional ethics course to be added. In the study, it is suggested that more courses about accounting ethics should be given in developing curriculum programs.

Akın and Özdaşlı (2014) conducted a comparative analysis for the perception of the ethical concept of professional accountants and vocational high school students. In this study, they are based on the ethical dimensions such as honesty, impartiality, confidentiality, professional competence and care, and professional behavior contained in the Code of Professional Ethics. As a result of the research, it was found that students who took accounting education thought that professional accountants had violated the principles of honesty and transparency more than the students who received accounting education.

Akyatan and Angay Kutluk (2015) examined the factors affecting the ethical decision-making process of the students who took the accounting course in their research. As a result of the research, it has been concluded that the ethical judgments of women are higher than men, and the students of the department of business have more ethical values and the students who have taken the accounting ethics course have stronger ethical judgments than the students who

have not taken the accounting ethics course, stating that it would be useful for students to take accounting ethics course.

Alkan (2015) examined the students' perceptions of ethics on the associate degree students of Izmir Dokuz Eylül University in his study. The research showed that female students' level of ethics perception is higher than that of male students. The income level of students' families leads to significant differences in the perception of ethics. The decrease in the perception of ethics as the income level increases is one among the interesting results of the study. It was stated that the perception of ethics about duties and responsibilities and the perception of ethics towards professional individual behaviors decreased as their mother's education level increased.

Feil et al.'s (2016) research showed female students who are below 25, working in the financial/accountant field, academically mature, and who have studied ethics have individual factors towards more ethic significant attitudes. The professional law and ethics course in the accounting programme contributed to future professionals' ethic conduct.

Ceylan and Terzi (2016) determined that female students were more sensitive about professional ethics than male students. In addition, it was observed that students taking ethics courses were more sensitive about the principles of accuracy and honesty, impartiality, and professional behavior than the students who did not. It was found that there was no statistically significant difference between the perception of ethics and age, income level and place of residence.

Subramaniam et al. (2017) examined the relationship between ethical behavior of accounting students and intention to make an academic offense. The study took place in Malaysia. According to the study, confidentiality, objectivity, professional behavior, professional competence, and due care have a positive and significant influence on academic offense, but integrity has a negative and non-significant affect on academic offense.

Ahmad et al. (2017) studied Islamic ethics principles effect on accounting students. The study was held in Malaysia. The finding revealed that Islamic ethics principles were suitable in directing the user towards Islamic behavior and positively linked to quality accounting practice and the development of ethical accounting students.

3 Method

Survey model is used in this study. Survey models are research approaches that aim to describe a situation that exists in the past or the present (Karasar, 2012). The research population consists of 684 students attending Kırklareli University

School of Applied Sciences in the spring term of 2018–2019 academic year. In the study, 327 students are sampled by convenience sampling method. The questionnaire developed as a data collection tool is developed by searching the relevant literature by the researchers. This survey consists of the following three parts:

In the first part, the students' social, demographic, economic, and other information is included.

In the second part, a total of five questions are collected in order to determine the opinions of students on professional ethics education.

In the third part, a total of 21 questions are collected in order to determine the opinions of the students about the professional ethics (five categories including Honesty, Impartiality, Confidentiality, Professional Qualification, and Professional Behavior).

Likert-type answers are as in the following:

1 = Strongly Disagree, 2 = Disagree, 3 = Neither I agree nor disagree, 4 = Agree, and 5 = Strongly Agree

The Professional Ethics Education Score and Professional Ethics Score used in the analyses are obtained by taking the answers given to the items in the sections. How the average of each section is evaluated is given below:

In the range of 1.00–1.79 points: very negative, in the range of 1.80–2.59 points: negative, in the range of 2.60–3.39 points: average, in the range of 3.40–4.19 points: positive, in the range of 4.20–5.00 points: very positive.

In order to analyze the data, IBM SPSS Statistics 20 Program was used.

4 Findings

The descriptive statistics related to the PEE and PE are given in Tab. 1.

Mann-Whitney Test was performed to understand whether PEE scores or PE scores differed based on "Gender", "Taking a course related to Ethics", and "Family Type". Mann-Whitney Test was preferred instead of independent samples t Test due to violation of normality (Tab. 2). The results of the Mann-Whitney Tests are as follows:

- There was no statistically significant difference found between scores of PEE ($U=11988$, $p=0.165$) and PE ($U=12138.5$, $p=0.228$) based on "Gender".
- There was no statistically significant difference found between scores of PEE ($U=2214$, $p=0.866$) and PE ($U=2173.5$, $p=0.776$) based on "Taking a course related to Ethics".
- There was no statistically significant difference found between scores of PEE ($U=7228.5$, $p=0.227$) and PE ($U=7995$, $p=0.968$) based on "Family Type".

Tab. 1: Descriptive Statistics of the Views on Professional Ethics Education (PEE) and Professional Ethics (PE)

	Mean($\bar{X}$)*	Std. Dev.(s)
Opinions on Professional Ethics Education (General)	3.13	1.133
Honesty	3.48	1.157
Objectivity	3.75	1.143
Confidentiality	4.16	1.085
Professional Competence and Care	4.15	0.997
Professional Behavior	3.95	1.098
Opinions Related to Professional Ethics (General)	3.93	1.083

* The minimum score is 1 and the maximum score is 5.

High scores show positive attitudes and low scores show negative attitudes.

Tab. 2: Mann-Whitney Test Results of the PEE scores and PE Scores

Variable	Categories	N	Professional Ethics Education				Professional Ethics			
			Mean Rank	Sum of Ranks	Mann-Whitney U	Sig.	Mean Rank	Sum of Ranks	Mann-Whitney U	Sig.
Gender	Men	153	170.65	26109	11988	0.165	169.66	25958.5	12138.5	0.228
	Women	172	156.20	26866			157.07	27016.5		
Ethics Course	Yes	15	163.40	2451	2214	0.866	152.90	2293.5	2173.5	0.776
	No	303	159.31	48270			159.83	48427.5		
Family Type	Nuclear	263	165.52	43530.50	7228.5	0.227	162.60	42764	7995	0.968
	Family									
	Large Family	61	149.50	9119.50			162.07	9886		

Kruskal-Wallis Test was performed to understand whether PEE scores or PE scores differed based on “High School Type”, “Residential Place”, “Mother’s Educational Background”, “Father’s Educational Background”, “Grade Levels”, “Socio-Economic Status”, “Number of Siblings”, “Region of Hometown”, and “Monthly Income”. Kruskal-Wallis Test was preferred instead of ANOVA due to violation of normality (Tab. 3). The results of the Kruskal-Wallis Tests are as follows:

- There was no statistically significant difference found between scores of PEE ($\chi^2_{(7)} = 11.896$, $p = 0.104$) and PE ($\chi^2_{(7)} = 6.657$, $p = 0.46$) based on “High School Type”.

Tab. 3: Kruskal-Wallis Test Results of the Professional PEE scores and PE Scores

Variable	Categories	N	df	Professional Ethics Education			Professional Ethics		
				Mean Rank	Chi-Square	Sig.	Mean Rank	Chi-Square	Sig.
High School Type	Anatolian High School	121	7	175.10	11.896	0.104	170.30	6.657	0.465
	Anatolian Vocational High School	75		162.53			159.53		
	Industrial Vocational High School	12		111.21			169.08		
	General High School	23		180.28			157.87		
	Religious Vocational High School	21		179.31			168.02		
	Vocational High School	28		160.32			175.66		
	Technical High School	13		117.58			103.88		
	Others	33		141.23			159.77		
Residential Place	City	165	2	158.89	0.268	0.875	160.98	0,169	0,919
	Town	127		164.31			163.48		
	Village or Borough	30		163.95			155.97		
Mother's Educational Background	Illiterate	11	5	177.05	3.413	0.637	178.55	4.369	0.498
	Literate-Dropout	11		180.95			151.91		
	Primary School Graduate	161		159.91			172.82		
	Secondary School Graduate	74		174.31			153.82		
	High School Graduate	60		150.81			154.59		
	Junior Technical College Graduate or higher degree	9		185.50			131.61		
Father's Educational Background	Primary School Graduate or Lower Degree	133	3	162.09	4.591	0.204	161.79	4.533	0.208
	Secondary School Graduate	75		154.03			179.10		
	High School Graduate	89		177.38			157.06		
	Junior Technical College Graduate or higher degree	27		139.00			137.80		

Grade Levels	1st Grade	65	3	174.85	6.636	0.084	181.65	11.129	0.011
	2nd Grade	105		145.00			179.58		
	3rd Grade	111		169.50			149.68		
	4th Grade	46		178.75			138.07		
Socio-economic STATUS	Low	54	2	140.97	7.248	0.027	162.67	0.706	0.703
	Intermediate	253		167.13			161.11		
	High	13		112.54			139.58		
Number of Siblings	1 Sibling	16	4	157.41	2.732	0.604	136.81	5.207	0,267
	2 Siblings	155		156.91			158.86		
	3 Siblings	112		168.50			176.41		
	4 Siblings	25		173.56			175.26		
	5 or more siblings	19		188.29			140.87		
Region of Hometown	Marmara Region	138	6	153.84	5.247	0.513	152.05	9.493	0.148
	Aegean Region	22		155.41			165.18		
	Black Sea Region	69		159.69			160.65		
	Central Anatolia Region	32		173.36			179.72		
	Mediterranean Region	14		164.57			207.32		
	East Anatolia Region	40		189.40			180.60		
	Southeast Anatolia Region	9		172.78			118.94		
Monthly Income	Under 1 000 TL	9	7	156.22	6.006	0.539	136.94	5.813	0.562
	1 001–2 000 TL	57		178.05			184.47		
	2 0001–3 000 TL	98		159.73			154.36		
	3 001–4 000 TL	48		136.84			155.18		
	4 001–5 000 TL	45		165.84			150.83		
	5 001–6 000 TL	31		155.92			159.77		
	6 001–7000 TL	13		166.77			146.96		
	7 001 TL or higher	17		145.85			161.56		

- There was no statistically significant difference found between scores of PEE ($\chi^2_{(2)}=0.268$, $p=0.875$) and PE ($\chi^2_{(2)}=0.169$, $p=0.919$) based on “Residential Place”.
- There was no statistically significant difference found between scores of PEE ($\chi^2_{(5)}=3.413$, $p=0.637$) and PE ($\chi^2_{(5)}=4.369$, $p=0.498$) based on “Mother’s Educational Background”.
- There was no statistically significant difference found between scores of PEE ($\chi^2_{(3)}=4.591$, $p=0.204$) and PE ($\chi^2_{(3)}=4.533$, $p=0.208$) based on “Father’s Educational Background”.
- There was no statistically significant difference found between scores of PEE ($\chi^2_{(3)}=6.636$, $p=0.084$) and PE ($\chi^2_{(3)}=11.129$, $p=0.011$) based on “Grade Levels”.
- There was found a statistically significant difference between scores of PEE ($\chi^2_{(2)}=7.248$, $p=0.027$) and PE ($\chi^2_{(2)}=0.706$, $p=0.703$) based on “Socio-Economic Status”.
- There was no statistically significant difference found between scores of PEE ($\chi^2_{(4)}=2.732$, $p=0.604$) and PE ($\chi^2_{(4)}=5.207$, $p=0.267$) based on “Number of Siblings”.
- There was no statistically significant difference found between scores of PEE ($\chi^2_{(6)}=5.247$, $p=0.513$) and PE ($\chi^2_{(6)}=9.493$, $p=0.148$) based on “Region of Hometown”.
- There was no statistically significant difference found between scores of PEE ($\chi^2_{(7)}=6.006$, $p=0.539$) and PE ($\chi^2_{(7)}=5.813$, $p=0.562$) based on “Monthly Income”.

5 Conclusion

Students have “average” attitude scores (3.13 ± 1.133) on Professional Ethics Education and they have “positive” attitude scores (3.93 ± 1.083) on Professional Ethics. It was found that “Gender”, “Taking a course related to Ethics”, and “Family Type” have no statistically significant effects on opinions related to Professional Ethics Education or Professional Ethics. It was found that “High School Type”, “Residential Place”, “Mother’s Educational Background”, “Father’s Educational Background”, “Grade Levels”, “Number of Siblings”, “Region of Hometown”, and “Monthly Income” have no statistically significant effects on opinions related to Professional Ethics Education. However, it was also found that “Socio-Economic Status” has statistically significant effect on opinions related to Professional Ethics Education ($\chi^2_{(2)}=7.248$, $p < 0.05$). It was found that “High School Type”, “Residential Place”, “Mother’s Educational Background”,

“Father’s Educational Background”, “Socio-Economic Status”, “Number of Siblings”, “Region of Hometown”, and “Monthly Income” have no statistically significant effects on opinions related to Professional Ethics. However, it was also found that “Grade Levels” has statistically significant effect on opinions related to Professional Ethics ($\chi^2_{(3)} = 11.129, p < 0.05$).

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Huseyin Senkayas and Melih Can

Industry 4.0 Awareness of Manufacturing Sector in Konya Province

1 Introduction

Firms today look for new ways of innovation to ensure their sustainability and to provide competitive advantage. The most important pillar of these innovations is Industry 4.0, which can be defined as digitalization at all stages of production. Manufacturing businesses are now facing serious challenges with disruptive concepts such as Internet of Things (IoT), Cyber-Physical Systems (CPS) or Cloud Based Manufacturing. Industry 4.0 paves the way for high utilization of technological factors in the production systems, thus making it possible to produce high-tech products in production lines that are fully adapted to automation and utilize robotic equipment in the production stage.

Dujin et al. (2014) put forth the central role and challenges of industry in the European Union countries. It accounts for 15 % of value added and serves as a key driver of research, innovation, productivity, job creation and exports. Industry generates 80 % of the EU's innovations and 75 % of its exports. But for the last decade, European industry faces a big challenge as it has lost many manufacturing jobs and emerging markets.

The challenge of competition posed by certain countries like China and India forced developed countries to focus more on innovation. Therefore, there is a trend of growth in budget allocated for next wave of manufacturing (Kahn and Turowski, 2016).

Manufacturing industry is one of the most effected sectors from radical changes of digitalization in production systems. Especially in Turkey, the manufacturing industry is a sector with high added value and competitiveness. Therefore, governments have tried to increase the share of the manufacturing sector in the gross domestic product.

In Turkey, the manufacturing industry has been transforming especially from 1996 to 2008. It is seen that the highest share of manufacturing industry in total export shares is composed of automobile, white goods, machinery, electronics, petroleum products and rubber-plastic sectors. The comparative advantage of the Turkish manufacturing industry in foreign trade complies with the structures of countries with developed, strong manufacturing structures such as Germany and the USA. (Özen, 2015).

Most of the firms in manufacturing industry in Turkey have not met industry 4.0 concept yet. Therefore, a survey was conducted in manufacturing enterprises operating in the province of Konya in order to measure the awareness of businesses to industry 4.0.

2 Literature Review

Industry 4.0 is digitization of every player in the system by set of new technologies and internet. In the first stage, firms should convert and digitalize their manufacturing processes by collecting data and analyzing systems.

In order to serve increasingly fast-moving markets, Industry 4.0 covers a series of innovative processes and developments that are combining new technologies with industry standards in the manufacturing sector. The internet has fundamentally changed communication and consumer behaviour (Bartevyan, 2015). According to Drath and Alexander (2014), Industry 4.0 is the application of the generic concept of CPS to industrial production systems. Kahn and Turowski (2016) define Industry 4.0 as a revolution that applies advanced technologies at production level to bring new values and services for customers and organization itself. Zhong et al. (2013) think that industry 4.0 links and digitizes all productive units in an economy and depends on new and innovative technological developments: information and communication technology (ICT), cyber-physical systems, collection of big data, network communications, simulation, augmented reality and intelligent tools.

The IoT and Services makes it possible to create networks incorporating the entire manufacturing process that convert factories into a smart environment. Cyber-Physical Production Systems comprise smart machines, warehousing systems and production facilities that have been developed digitally from inbound logistics to production, marketing, outbound logistics and service (Kagerman et al., 2013).

The IoT connects items like buildings, machines, industrial plants, assets, vehicles, transport units, containers, devices, people and animals in order to share, process and utilize data, at the same time interacting with human beings and the virtual world. Big Data refers to large data sets which are hard to manage with traditional systems because of their volume and speed of creation (Flores et al., 2018).

Industry 4.0 can be built over some systems and technologies like Lean Manufacturing, ERP (Enterprise Resource Planning), PDM (Product Data Management), MES (Manufacturing Execution System), RFID (Radio Frequency Identification) and data collection systems from the manufacturing lines.

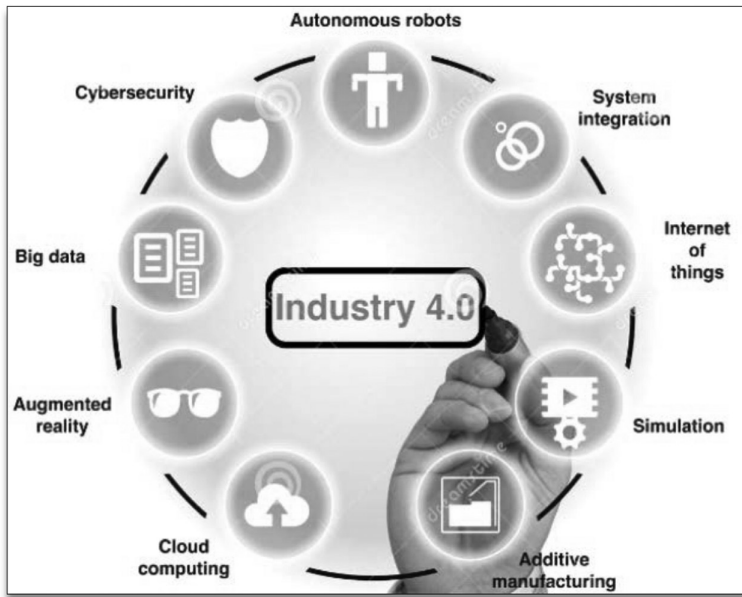


Fig. 1: Main technologies of Industry 4.0 (Guban and Kovacs, 2017)

ERP is a commonly used integration and flow of information tool between business functions inside and outside an organization. It concentrates on the managerial level of decision-making and has not paid attention to the shop floor and real-time data to monitor and control real-time and variable execution processes (Zhong et al., 2013; Ugarte et al., 2009). PDM system provides simple data storage, revision/change/access control, document management, project and workflow management and simple interfaces with other systems (Gao et al., 2003). The MES concept was born due to the demand on the manufacturing enterprise to fulfil the requirements of markets from a point of view of reactivity, quality, respect of standards, reduction in cost and deadlines. MES executes and controls production orders which emanate from ERP (Ugarte et al., 2009).

Due to its powerful ability of real-time data collection, RFID technology has been deployed to various manufacturing objects through different schemes such as individual items, traybased or container-based approach. Objects become smart as they are tagged, and can be traced and tracked. The data carried by them can be collected and are updated when their locations change from time to time (Zhong et al., 2013).

In order to get some opinion about awareness of EU countries to Industry 4.0, Dujin et al. (2014) created an index called “RB Industrial 4.0 Readiness Index”. The matrix roughly divides the European economies into four major groups. The Frontrunners are characterized by a large industrial base and very modern, forward-looking business conditions and technologies (Sweden, Austria and Germany). The Traditionalists are found mainly in Eastern Europe. They still thrive on their sound industrial base, but few of them have thus far launched initiatives to take industry into the next era. The third group, the Hesitators, lacks a reliable industrial base. Many of them suffer from severe fiscal problems and are therefore not able to make their economies future-proof. The Potentialists’ industrial base has been weakening over the past few years. Here we find countries such as France and the UK.

Kagerman et al. (2013) conducted a survey on the “prospects for Industrie 4.0” by the professional associations BITKOM, VDMA and ZVEI and documented the need for fuller and more targeted information. About 47 % of the firms said that they were already engaged with Industrie 4.0, 18 % of them pointed out that they were involved in research into Industrie 4.0 and 12 % claimed that they were already putting it into practice.

Dujin et al. (2014) also propose New Industry 4.0 Roadmap by which Industry 4.0 will be implemented in three steps involving all players: 1) setting conditions for the industry 4.0 ecosystem, 2) boosting industry 4.0 offering (accelerating innovation and setting digital ecosystem), 3) promoting fast adoption as competitive lever.

Gabriel and Pessl (2016) discuss how Industrial 4.0 approach can influence the social and environmental perspective of sustainability in SMEs in Austria. In some companies, first steps of Industry are realized, while the path towards Industry 4.0 is an evolutionary process. Information, automation and production technologies will be more intertwined than ever before. Networking is not only a goal; it is an absolute requirement for Industry 4.0. Approaches of the Industry 4.0 already exist, but it will take a long time until full implementation. Experts disagree and talk about a period of up to 20 years.

In order to provide practice education of the students and training for professionals who are employed by the regional SMEs, an Industry 4.0 Learning factory was invented at the campus Velbert/Heiligenhaus (CVH) of the Bochum University of Applied Sciences. For the aspect of integration between shop floor and top floor is mainly for teaching, with opportunities to develop solutions for third parties. Supported areas are improved performance monitoring in a distributed manufacturing scenario and real-time reaction on problems in production processes. The Learning factory comprises a holistic model of a producing

company, from the top floor ERP level to the shop floor Field Level. Real processes and physical products can be trained, analyzed and optimized in the factory (Faller & Feldmüller, 2015).

Koch et al. (2014) surveyed 235 companies from the German processing industry as well as the information and communications industry which were organized into five industry sectors balancing between large corporations and SMEs. They found that the industrial internet transforms the entire company. In addition, by 2020 European industrial companies will invest €140 billion annually in industrial internet applications. In five years, more than 80 % of companies will have digitized their value chains. The integrated analysis and use of data are the key capabilities for the industrial internet. Digitization of the product and service portfolio is the key to sustainable corporate success.

Flores et al. (2018) aim to understand the level of awareness of Industry 4.0 in firms about its benefits and business impact and identify the key enablers for the successful implementation of IoT and Big Data. According to survey results, 65 % of the firms agreed with the standard Industry 4.0 definition. But only 17 % of the companies already have fully implemented strategy. Increase in efficiency is the main driver for 57 % of the firms and for 47 % it is predicted to save costs. The role of Chief IoT officer is not available for 47 % of them. They mostly apply Industry 4.0 applications in production processes to support the real-time process control, to gather data and make informed decisions and to monitor and control machiners. Big Data is seen as a supporting technology enabling optimized decision-making.

3 Materials and Method

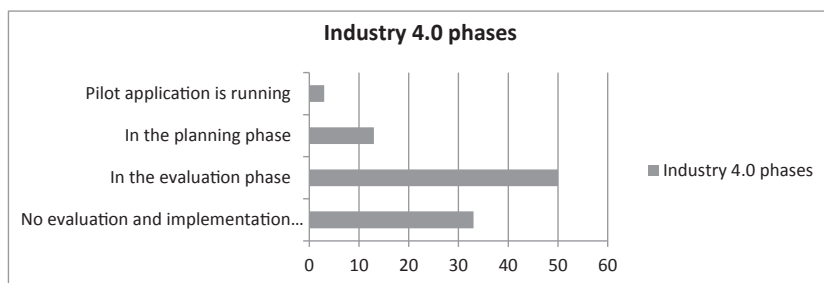
The aim of the study is to determine the view of the firms operating in the manufacturing industry in Konya and their awareness to Industry 4.0. As an application scale of Industry 4.0, the questionnaire developed by Türkoğlu (2018) was used. The prepared form is given to the managers of the firms online. This study aims to determine the industrial 4.0 perceptions of the managers working in manufacturing enterprises operating in the province of Konya, and the extent to which enterprises are ready for Industry 4.0.

For this purpose, the questionnaire prepared based on the scale mentioned earlier applied to manufacturing enterprises' managers. Questionnaire forms applied to 30 managers in total were analyzed by SPSS 21 program.

In the study, 17 % of the firms are in main metal sector, 17 % are in rubber and plastic industry, 13 % are in food sector and 10 % are in clothing sector. The firms are generally medium-scale companies 77 % of them have 51–100 employees).

4 Findings

All companies surveyed have heard the Industry 4.0, IoT and Big Data concepts. Only 10 % of them heard “Made in China 2025” and 7 % heard “Advanced Manufacturing Partnership”. They heard these concepts from social media, journals, newspapers, TV and seminars. It is also observed that 60 % of the surveyed companies have infrastructure required for industry 4.0 applications and 53 % of them have organizational content to support industry 4.0 applications. On the other hand, only 40 % of the firms got supporting processes in transition to the industry 4.0 system. Only 75 % of the companies have a budget for industry 4.0 and 100 % measure the process outputs of the enterprise in real time. When asked in what phase is your firm in practice of Industry 4.0, the distribution of firms is as in the Graph. 1. Most of the firms are trying to evaluate and plan to apply industry 4.0 to their systems.

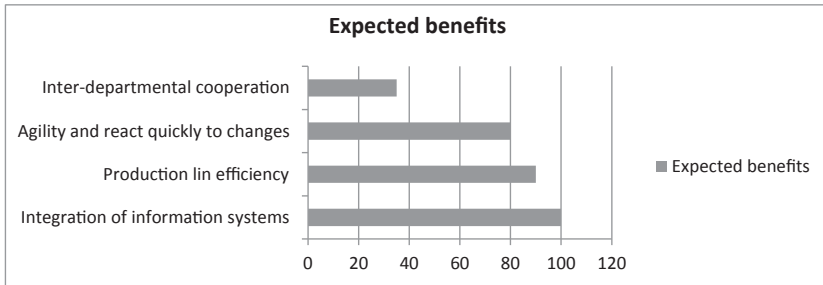


Graph. 1: In what phase are the firms in practice of Industry 4.0? (%)

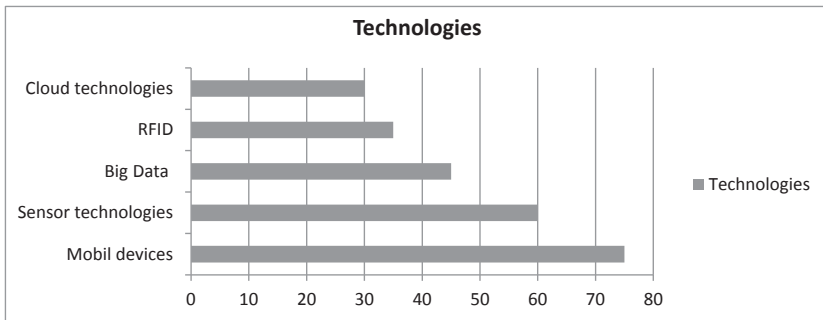
In the integration of Industry 4.0, all companies agree that it is convenient to integrate monitoring and optimization of intra-production processes, remote monitoring of machinery/plant, maintenance of internal machinery and modification of the systems.

When the companies start applying industry 4.0, they point out that they face and expect to face some difficulties like acquiring the necessary new skills and abilities (100 %), controlling increased data volumes, diversity and complexity (90 %) and development of a standard database for all departments (35 %). On the other hand, they expect benefits as shown in Graph. 2. All of them expect integration of information systems.

Only 10 % of the firms have IT department while 65 % of them are getting external support for IT services. In implementing Industry 4.0, they expect



Graph. 2: Expected benefits in the integration of information systems used (%)



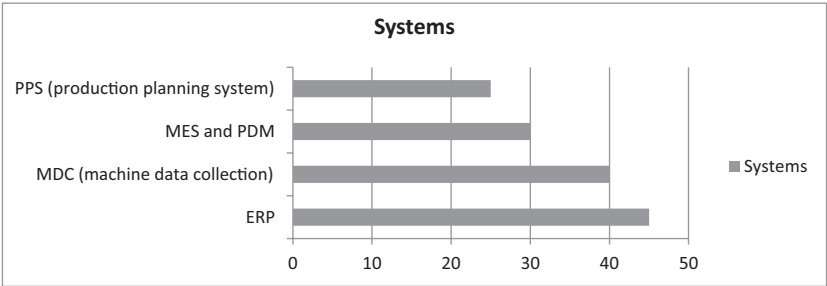
Graph. 3: Which technologies do you apply in your systems? (%)

difficulties like declining manpower requirement, long time for integration, insufficient technological infrastructure and difficulties in training experts.

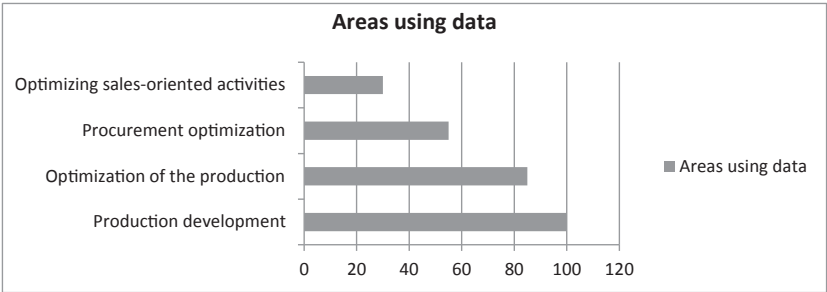
Within the last few years, firms made investments on IT (75 %), production department (55 %), logistics (30 %) and service department (25 %) in order to develop infrastructure for industry 4.0. Firms also think that industry 4.0 could provide added value to business areas like R&D (80 %), production planning (75 %), production and warehousing (60 %), sales department (55 %) and procurement (45 %).

Graph. 3 and Graph. 4 show which technologies and systems are applied by the surveyed firms. They are mostly ready as infrastructure with their sensor, Big Data and RFID technologies, and also with systems like ERP, MDC, MES and PDM.

Most of the firms try to collect data from the manufacturing lines. One the one hand, 55 % of them collect data from only one machine and 15 % of them



Graph. 4: Systems used by enterprises in their own structure (%)



Graph. 5: Areas where they use the data from Industry 4.0 connected applications (%)

collect data from two or three machines. On the other hand, 20 % of the firms do not collect any data. Graph. 5 shows the areas where the firms use collected data.

5 Conclusion

Surveyed companies have potential in terms of infrastructure, organizational content and budget, and they measure the process outputs in real time. They are still in the evaluation and planning phase of Industry 4.0, but they are aware of importance of digitalization, monitoring, optimization and modification of current systems. Big Data and new technological skills seem to be the most alarming difficulties. On the other hand, companies expect that Industry 4.0 integrate their systems and they would gain efficiency, agility and cooperation skills.

Although companies invested much on IT, most of them do not have IT departments. Technological infrastructure and IT investments inside the firm

are the most vital needs for manufacturing companies. Because, they believe these would provide added value to R&D, production planning and running and other activities in the firm.

Manufacturing companies generally use mobile devices, sensor technologies, Big Data, RFID, ERP, MDC, MES and PDM, therefore we can say that they are ready to digitalize all production systems in the near future.

Konya province is one of the cities where manufacturing industry is a pioneer for the economy. Some research on other provinces is needed to get an overview of the country's industry 4.0 awareness. Companies should invest on digitalization step-by-step and train new, skilled workers quickly in order to compete in the industry 4.0 world.

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Vasfi Kahya

The Role of Innovative Human Capital (IHC) for Organizations

1 Introduction

The innovation concept and process are presented as fundamental ingredients of the economic development and growth processes not only on the regional and national levels but also on the organizational level (Teece, 2010). Innovation is also analyzed on the individual basis representing the group of individual competencies that are considered under talent management. In this study, innovation will be used more at the organizational and individual level.

Human capital (HC) represents one of the critical components of the organizations as primarily human capital is seen as an important factor of economic growth (Damanpour, 1991). It is claimed that economies having a higher value of human capital are tended to grow higher (Mincer, 1984). Thus, business organizations are able to grow faster as much as they own the HC amount in terms of quantity and quality (Rauch et al., 2005; Noe et al., 2017). HC may be considered as a useful tool to reach out realizing innovations on the organizational level, and HC capacity of organizations increase with training and employment of more effective methods (Santos-Rodrigues et al., 2010). The mainstream of Intellectual capital takes three basic forms: human capital, customer capital and structural capital; HC includes knowledge, abilities and skills of employees (Edvinsson and Malone, 1997).

2 Innovation and Innovative Human Capital (IHC) Relationship

Innovation is a crucial process for the growth and survival of organizations. Innovation has many definitions and may be summarized as a new commercialized idea in the forms of product or service produced as a result of new dimensions of the process. Innovation has been seen as fundamental for the efficiency and growth of the business organizations by many authors (Miller, 2001; Cohen and Soto 2007; Ahlstrom, 2010) as well as being the leading component of the international competition (Teece, 1992). According to OECD definition, HC means using different and new skills and competencies in order to maintain

a competitive advantage as HC has been considered as an essential dimension of innovation (OECD, 2011). Using HC with innovative competencies is a modern way of gaining competitive advantage (Becker, 1993). In this respect, the concept of IHC has been accepted as a new concept within the literature (McGuirk and Lenihan, 2013; McGuirk et al., 2015). IHC encompasses both tangible and intangible dimensions of the HC that are going to be explained in more detail. IHC may be described as the total valuation of the expertise, skills and competencies in a unit (Ayub et al., 2017). In this context, the innovation side of the HR has been accounted for some specific features including being open to innovation as well as the capacity to develop themselves in terms of competencies, training, skills and abilities. Usually, innovations are not always accepted and digested by the HR, as implementing transformations are always challenges for the business organizations. The HC capacity of the organizations is enhanced by some innovative processes as the HR side starts taking different perspectives on some functions of the organization including R&D, production, organization, marketing and so on.

The concept of HC has been associated with the implicit and explicit knowledge capacity of the organization (Rowley, 2001), as well as linking to the performance dimension of the knowledge and skills from the innovation perspective (Chen and Huang, 2009); and this capacity of the organizations is also called as innovation performance (Cabello-Medina et al., 2011), and value of HC and uniqueness of HC enhance the individual capacity for innovation, and this positively affects the innovative performance (Vinding, 2006). IHC may ensure a competitive advantage to any organization through some generic strategies including low cost, focus, differentiation or any combination for any commercial product and service creation (Porter, 1985). The concept of IHC encompasses some non-imitable features including attitudes, behaviour, competencies, abilities, know-how and personal social environment, and IHC has three dimensions including the value of HC, the uniqueness of HC and proactiveness (Halim et al., 2015). As derived from the literature there have been four factors to be considered for the understanding and measurement of IHC that are formal and business education, job satisfaction and acceptance (readiness) to change (McGuirk and Lenihan, 2013).

3 What Is IHC?

The concepts of innovation and HC are more concerned with innovation capacity of both the organization and the individual, and two capacities also refer to the “absorbent capacity” used in innovation (Roper and Love, 2006; Vinding, 2006).

Innovation capacity has been analyzed in different studies (Love et al., 2011), concluding that some factors such as being open to new ideas, communication and teamwork have positive effects on Innovation (Michie and Sheehan 2003). There have been some local factors including education variables and the diversity of nationality and education variables as well as population density of the region that have an adverse effect on innovation (McGuirk and Jordan 2012). Diversity may serve as a supporting tool for innovation as HC with different features have different perspectives.

It can be thought that different thinking individuals will develop innovation by using different viewpoints. There has been a close linkage between regional innovation and the level of entrepreneurship as the variables are interconnected (Silverberg et., 1988), and entrepreneurs as HC are positively found to be influencing innovation (Romero and Martinez-Roman, 2012).

The IHC literature traditionally started with the development of the HR with some competencies including the number of years in education (McGuirk and Jordan, 2012) as well as the allocated budget for the education and training on the national level (Ederer, 2006). Within the literature on IHC, different dimensions of the HC have been considered starting from the individual to organizational and national levels. Most of the writers claimed that innovation develops HC (McGuirk and Lenihan, 2013). On the other hand, some other writers claimed that HC developments result in innovations (Kato et al., 2015) and also social capital dimension has been influential in this process (Landry et al., 2002). Thus, social capital has a significant role within the IHC acquirement process. In line with the OECD's view as considering the HC as "a strong complement", that is possible through a mutual understanding of the concepts IHC and the social capital.

4 How to Measure IHC?

The measurement of HC has attracted considerable interest in the literature (Vidotto et al., 2017). The most important factor making the measurement of HC is the fact that HC has different dimensions and aspects namely intangible and tangible assets (Soboleva, 2010). According to Becker (1993), HC may be enhanced through some investments such as education and health support for HC. The return on investments may not be evaluated through traditional methods as HC may be in invisible forms. Not only employees of the company but also the founders of the company are considered as HC positively connected to performance (Ganotakis, 2012), and also they enhance the capacity for innovation for their companies (Marvel and Lumpkin, 2007).

4.1 Measurement of Tangible HC

For the tangible part of the HC, there is no agreed method for HC measurement in the literature. However, there are some variables used to measure the HC including the diplomas earned and the total training sessions (Cohen and Soto, 2007) as well as tenure, training and development and private HC (gender, age and family history) (Du and Li, 2009).

The influence of education and training on IHC is quite complicated. In research conducted in Finland, it is found that technical skills contribute more to the innovation capacity of the organizations (Leiponen, 2005) as talents may adopt changes faster supporting innovation. Education and development are considered as the primary sources of HC, and investments made in education and development create a valuable HC as valuable as financial capital (Becker, 1993).

In other words, training and development opportunities of the employees is a tangible HC as with the training and development sessions, the employees' productivity and effectiveness will increase, creating more added value for the company. With the enhanced levels of the competencies and skills, they can contribute more with their more innovative approaches that are a long-term and strategic approach to the HRM (Bratton and Gold, 2017). There is a connection between education levels and innovation capacity.

4.2 Measurement of Intangible (Abstract) HC

The concept of HC has been one of the components of the intellectual capital of the organizations. HC and creativity are partially intangible assets that lead to innovation. Intangible assets are non-physical values that may be grouped as HR tangibles, organizational tangibles and innovation tangibles. Basically, intangible assets may be defined as the difference between book value and market value (Edvinsson and Malone 1997). Some of the intangible assets may be shown on the balance sheet. Some measurement systems of the intangible assets may be used both externally and internally as relating to intangible assets to the tangible and especially financial measures. Measurement of intangible HC may be possible through using concepts of being innovative and open to change. OECD includes culture, relationships and other specific values in intangible assets of the HC and organizational (structural) capital. Stewart (1997) proposed three measurement methods for HC including Tobin's q, market-to-book ratio and calculated the intangible value. Sveiby (1997) claims that the measurement of intangible assets should be based on stability, efficiency, growth and renewal.

5 Conclusion

Competitive advantage for business organizations may be ensured through innovative perspectives of analyzing and differentiating internal and external sources into a valuable product and services (Lengnick-Hall, 1992). IHC has been defined as a set of competencies, skills, abilities and knowledge serving to support new and innovative philosophy, ideas and products. IHC is a fundamental component for all innovative outcomes including high-technologies, new products and new services. The theoretical framework, as well as measurement issues of HC tell us that HC with innovation tangibles has become more crucial for the organizations. There have been many attempts to develop and measure the HC of the organizations, and innovation has been an essential component of such analysis. The level and duration of education have been taking as a means of enhancing the HC capacity of organizations. IHC has been presented as a new strategy to maintain the competitive advantage of the organizations. IHC framework has been expanded to tangible and intangible assets. More abstract level of IHC included many variables not only within HC but also in customer capital and structural capital including motivation, satisfaction, goodwill, work experience and some other variables. IHC has been a useful concept as proposed by Lenihan and McGuirk (2014). Also, the health conditions of the employees are taken as sub-dimensions of HC (Becker, 1993). The new IHC encapsulates four elements: education, training, willingness to change in the workplace and job satisfaction.

In this study, the effects of IHC on organizational level innovations have been reviewed, and it has been concluded that IHC is going to be valuable for all organizations. In this respect, IHC may be considered as a critical concept for the organizations in order to create innovative policies and strategies. This study highlights the significance of HC on innovative performance exploring the conditions of the innovation capacities, and IHC should be seen as a means of ensuring a competitive advantage for the organizations.

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Huseyin Aslan

The Mediating Role of Work Stress on the Effects of Leader-Member Exchange on Turnover Intention

1 Introduction

In the tourism sector rapidly developing in recent years due to its contributions to stocks and employment (Güneysu Atasoy, 2018), leadership plays a crucial and powerful role in institutionalization of organizations (Hırlak & Kara, 2018) and the ability to overcome problems induced by human relations and decrease turnover intention by minimizing stress (Turunç & Avcı, 2015). Individuals' attitudes greatly affect their behaviors (Bakan, Büyükbeşe, Ersahan & Kefe, 2013). Therefore, a high-quality interaction between workers and their leaders contributes to the behaviors of subordinates in a number of positive ways at individual and organizational levels (such as increased work performance, raised job satisfaction levels, reduced stress levels, and a decrease in turnover intention) (Şahin, 2011). However, in such sectors as tourism in which work stress levels are high, work stress may degrade the effects of the interaction between leaders and subordinates (Turunç & Avcı, 2015). Based on this assessment, the main objective of this study is to determine the mediating role of work stress on the effects of leader-member exchange on turnover intention.

2 Theories and Hypotheses

One of the most useful ways to attitudinally investigate leadership processes and their outcomes (Nishii & Mayer, 2009), leader-member exchange is based on the opinion that leaders adopt different attitudes towards their subordinates as opposed to common leadership approaches which suggest that every leader treats their subordinates equally and steadily (Martin, Thomas, Charles, Epitropaki & McNamara, 2005). The LMX theory suggests that leaders do not interact at the same level as their subordinates do and the quality of interaction determines how the leader will treat his/her subordinate (Şahin, 2011). The interaction between leaders and their subordinates results in the formation of two groups, namely the in-group and the out-group. The subordinates with whom the leader develops high-quality and positive relationships based

on mutual trust, respect, and admiration in the cases of high interaction levels are named as the in-group members; while the subordinates with whom the leader develops low-quality and negative relationships, which are only bound by the employment contract and limited to the job description of the subordinate and harbor a lack of trust in the cases of low interaction levels, are named as the out-group members (Brimhall, Lizano & Barak, 2014). Thanks to their high-level interactions with their leader, the in-group members gain more support, information, influence, trust, and interest from their leader; are assigned to crucial and critical missions; and are awarded officially and unofficially more by their leader; on the other hand, due to their low-level interactions with their leader, the out-group members get less support, information, influence, trust, and interest from their leader; are assigned to repetitive, insignificant, and unattractive jobs; and are awarded officially and unofficially much less by their leader (Karcioğlu & Kahya, 2011; Şahin, 2011). The fact that leaders treat their in-group members in a more privileged way than their out-group members due to time pressure and other factors creates a hierarchy-based impact between leaders and their out-group members, and the remoteness and perceived power distance between leaders and their out-group members increase. In addition, more privileged attitudes that leaders adopt toward their in-group members compared to their out-group members lead to differentiation in performances delivered by these groups (Alparslan & Oktar, 2015). In short, high-level and positive interactions between leaders and members increase individuals' achievements at individual and organizational levels as they feel more acknowledged and appreciated, while low-level and negative interactions between leaders and members decrease such individual and organizational achievements (Han & Jekel, 2011). Similarly, conducted studies in the literature state that in the case of high-level interactions between leaders and members, the work performances of subordinates increase (Choy, McCormack & Djurkovic, 2016), job satisfaction levels are raised (Loi, Chan & Lam, 2014), perceived organizational levels increase (Alparslan & Oktar, 2015), and organizational commitment levels are heightened (Aydıntan & Göksel, 2012); whereas in the case of low-level interactions between leaders and members, stress levels of subordinates are maximized (Thomas & Lankau, 2009), and their burnout levels (Huang, Chan, Lam & Nan, 2010) and turnover intention (Nishii & Mayer, 2009) increase. In this context, based on the results obtained from the aforementioned theoretical explanations and research, the following hypotheses have been put forward with regard to the effects of leader-member exchange on work stress and turnover intention.

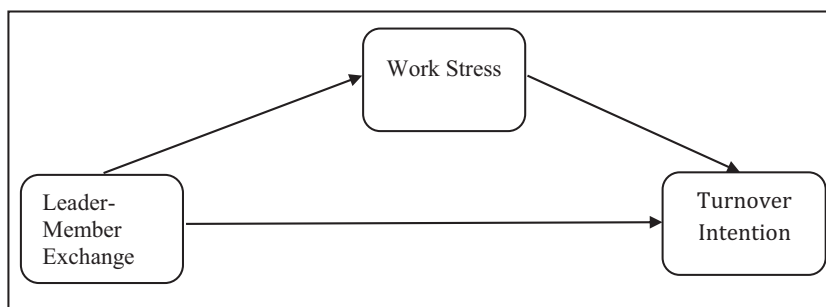


Fig. 1: Research Model

H1: Leader-member exchange has a significantly negative effect on turnover intention.

H2: Leader-member exchange has a significantly negative effect on work stress.

In addition, the fact that the privileged treatment of leaders toward in-group members is seen by out-group members creates stress-induced pressure on out-group members (Thomas & Lankau, 2009), which causes subordinates to quit their jobs (Turunç & Avcı, 2015). In this regard, the following hypotheses have been put forward with regard to the relationships between work stress and turnover intention, and leader-member exchange and turnover intention.

H3: Work stress has a significantly positive effect on turnover intention.

H4: Work stress has a mediating role on the effects of leader-member exchange on turnover intention.

3 Methods

In this research conducted to determine the mediating role of work stress on the effects of leader-member exchange on turnover intention, information on the population and the sample was first obtained. Then, analyses were performed with regard to the model created in light of the data obtained from the population. In this respect, confirmatory factor analysis was performed; afterward, correlations between variables were determined, and tests for goodness of fit were performed for the model through structural equation modeling developed with regard to the current model. While the tests for goodness of fit were being conducted, the results of regression analysis between variables and hypothesis test results were also presented.

Within the scope of the conducted research, the model as demonstrated in Fig. 1 was created to reveal correlations between variables.

3.1 Population and Sample

The population of the research was constituted by approximately 400 people working at three-, four-, and five-star hotels (in the departments of front desk, food and beverage, sales and marketing, and public relations where customer interaction is intense) in Gaziantep, Turkey. The sample of the research was constituted by randomly selected participants from different hotels who accurately represented the population. Data collected from the hotel workers through a questionnaire was used in the research. A total of 230 questionnaires were presented to the participants but 28 of them were unanswered while 17 contained missing answers. The number of analyzed observations was 185. Of the participants in the research, 43 were female while 142 were male. Among the participants, 58 were aged 18–30 years, while 87 were aged 31–40, and 40 were aged 41–50 years. The position of mid-level manager was held by 24 participants, while 56 were junior managers and 105 held other positions at their respective hotels. Fifty-five of the participants worked in front desk, 17 in sales and marketing, 88 in the food and beverage department, 9 in human resources, and 16 in the public relations department. Ten of the participants held less-than-one-year experience of working at hotels, while 61 had 1–3 years of experience, 54 had 4–6 years of experience, 34 had 7–9 years of experience, and 10 had 10 or more years of experience.

3.2 Scales Used in the Research

Leader-Member Exchange Scale: It was developed by Liden and Maslyn (1998) and its Turkish version validation was performed by Başı et al. (2010). The scale is composed of four dimensions, namely affect (3 items), loyalty (3 items), contribution (3 items), and professional respect (3 items), and 12 items in total. In their study, Başı et al. (2010) determined Cronbach's alpha values to be 0.920 for the affect dimension, 0.864 for the loyalty dimension, 0.70 for the contribution dimension, and 0.90 for the professional respect dimension. In the confirmatory factor analysis performed for the scale, it was observed to be composed of four dimensions. The factor loads were calculated to be between 0.69 and 0.89 for the affect dimension, between 0.59 and 0.77 for the loyalty dimension, between 0.69 and 0.86 for the contribution dimension, and between 0.66 and 0.92 for the professional respect dimension. As a result of the reliability analysis performed for the scale, Cronbach's alpha reliability coefficient for the scale was calculated to be 0.79 for the affect dimension, 0.81 for the loyalty dimension, 0.76 for the contribution dimension, and 0.78 for the professional respect dimension.

Perceived Stress Scale: It was developed by Cohen et al. (1983). Comprising of four items and single dimension, the scale was also used in a study conducted by Küçükusta (2007) who determined its reliability coefficient to be $\alpha=0.69$ in the aforementioned study. In the performed confirmatory factor analysis, the scale was observed to be composed of one dimension. Its factor loads were established to be between 0.83 and 0.95. The goodness of fit values of the scale are shown in Tab. 1 along with those of the other scales. As a result of the reliability analysis performed for the scale, Cronbach's alpha reliability coefficient of the scale was calculated to be 0.92.

Intention to Quit Scale: It was developed by Wayne, Shore, and Liden (1997). Composed of three statements, this scale was also used in the study of Küçükusta (2007) who determined its reliability coefficient to be $\alpha=0.76$. In the confirmatory factor analysis performed for the scale, it was observed to be composed of one dimension. The factor loads were calculated to be between 0.92 and 0.95. The goodness of fit values of the scale are shown in Tab. 1 along with those of the other scales. As a result of the reliability analysis performed for the scale, Cronbach's alpha reliability coefficient of the scale was calculated to be 0.90.

Tab. 1: The Goodness of Fit Values of the Scales as a Result of the Confirmatory Factor Analysis

Variable	χ^2	df	CMIN/ DF	GFI	AGFI	CFI	NFI	TLI	RMSEA
Leader-Member Exchange	79.132	36	2.198	0.940	0.902	0.973	0.959	0.964	0.048
Work Stress	7.283	5	1.456	0.964	0.971	0.996	0.984	0.987	0.018
Turnover Intention	3.011	3	1.003	0.967	0.989	0.979	0.986	0.994	0.014

As a result of CFA, it is observed that the scales meet the goodness of fit values and fit well (Yıldız & Çetindaş, 2018).

4 Results

The data obtained from the results of the study were analyzed through SPSS 21 and Amos software. In this regard, in the first stage, mean values and standard deviations of and correlations between the data obtained with regard to the participants' levels of leader-member exchange, work stress, and turnover intention were taken into account.

Mean values, standard deviations, and correlation values obtained as a result of the analysis are given in Tab. 2.

Tab. 2: Descriptive Statistics and Correlation Analysis Results of the Data

Variable	Mean	SD	Leader-Member Exchange	Work Stress	Turnover Intention
Leader-Member Exchange	4.16	.51	(.80)		
Work Stress	3.94	.63	-.197**	(.92)	
Turnover Intention	4.21	.41	-.251**	.789**	(.90)

**0.01 significant at significance level

Upon examining the mean values of the answers given to the questions, it was observed that the hotel workers had mid to high levels of leader-member exchange, work stress, and turnover intention.

In addition, significant relationships exist between all the dependent and independent variables included in the research. Therefore, significant impacts among the variables can be predicted.

In the second stage, a structural model was created in relation to the research model and its goodness of fit values were tested with the Amos software package. Structural equation modeling, the goodness of fit values, and regression weights are given in Fig. 2, Tab. 3, and Tab. 4, respectively.

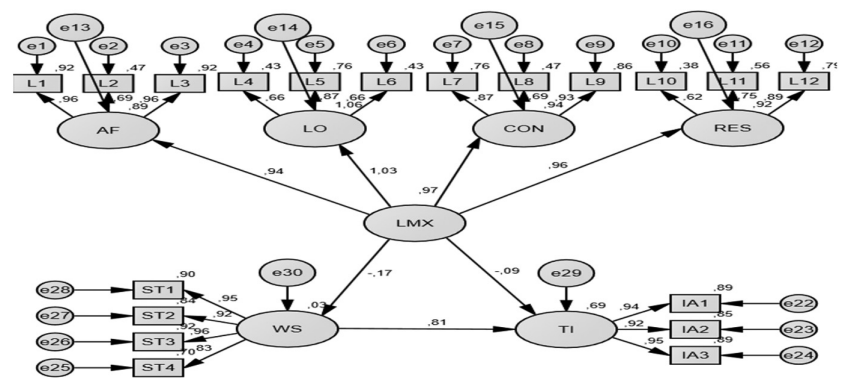


Fig. 2: Structural Equation Modeling

Tab. 3: The Goodness of Fit Values of the Structural Model

Fit Values	χ^2	df	CMIN/DF	GFI	CFI	TLI	RMSEA
Structural Model	143.191	64	2.237	0.942	0.974	0.978	0.048

As can be seen in Tab. 3, structural equation modeling meets the goodness of fit values and the model fits well.

Tab. 4: Regression Weights of the Structural Model

Tested Hypotheses			Estimate	SE	Critical Value	P
TI	<---	LMX	-0.168	0.101	-2.185	0.029
WS	<---	LMX	-0.094	0.075	-1.961	0.038
TI	<---	WS	0.809	0.076	12.691	***

As can be seen in Tab. 4, as a result of the structural equation model, it has been found out that leader-member exchange has a significantly negative effect on work stress and turnover intention while work stress has a significantly positive effect on turnover intention. The hypotheses H1, H2, and H3 have been consequently accepted.

Following this stage, the Sobel test was performed to determine the mediating role of work stress between leader-member exchange and turnover intention.

Tab. 5: Sobel Test Results

Steps	B	SE	β	P
Step 1: Independent Variable: Leader-Member Exchange Dependent Variable: Turnover Intention	-0.366	0.12	-0.233	0.002
Step 2: Independent Variable: Leader-Member Exchange Dependent Variable: Work Stress	-0.221	0.101	-0.167	0.029
Step 3: Independent Variable: Work Stress Dependent Variable: Turnover Intention	0.987	0.076	0.825	***
Step 4: Independent Variable 1: Leader-Member Exchange Independent Variable 2: Work Stress Dependent Variable: Turnover Intention	-0.155 0.965	0.075 0.076	-0.1 0.819	0.039 ***

In Tab. 5, B indicates unstandardized coefficient estimate values while β indicates standardized coefficient estimate values. In Step 1, leader-member exchange has a significantly negative effect on turnover intention. In Step 2, when the mediating variable work stress is included in the analysis as a dependent variable, leader-member exchange has a significantly negative effect on work stress, and in Step 3, work stress has a significantly positive effect on turnover intention. In Step 4, when the independent variable and the mediating variable are both included in the model, the independent variable leader-member exchange has a significantly negative effect on turnover intention, whereas the mediating variable work stress has a significantly positive effect on turnover intention. While the beta coefficient regarding the effects of leader-member exchange on turnover intention was calculated to be -0.233 in Step 1, it dropped to -0.1 in Step 4. This result shows us that work stress partially plays a mediating role on the effects of leader-member exchange on turnover intention.

Z score value should be calculated to test the significance of this mediating role. To calculate Z score value, indirect effects with $\alpha\beta$ were firstly calculated to be -0.218. Afterward, based on the Sobel formula $SE = \sqrt{\alpha^2\sigma^2\beta + \beta^2\sigma^2\alpha}$, Sobel standard error was calculated as 0.101. Following this stage, based on the formula $Z = \text{Indirect effects} / \text{Sobel SE}$, Z score value was calculated to be -2.158 (Sobel, 1982). Z score value being higher than 1.96 ($Z = 2.158$, $P: 0.031$) indicates that the mediating role is significant. As a result of this analysis, the hypothesis H4 has been accepted.

5 Conclusion

The main objective of this study is to reveal the mediating role of work stress on the effects of leader-member exchange on turnover intention. To that end, data collected from 185 people working at hotels in Gaziantep, Turkey, through survey method were analyzed.

As a result of the performed analyses, it has been concluded that leader-member exchange has a significantly negative effect on work stress and turnover intention. This result is in parallel with the conclusions from studies investigating the relationship between leader-member exchange and work stress (Thomas & Lankau, 2009) and turnover intention (Nishii & Mayer, 2009). This conclusion shows that when there is a low-level interaction between leaders and members, subordinates' stress levels and turnover intention increase. Similarly, in the study, it has been observed that work stress has a significantly positive effect on turnover intention. This result is in parallel with the study performed by Turunç and Avcı (2015). Also it has been in the study concluded that work stress plays a mediating role on the effects of leader-member exchange on turnover intention. This result is again in parallel with

the study performed by Turunç and Avcı (2015) who investigated the mediating role of stress on the effects of leader-member exchange on turnover intention in the accommodation sector. This conclusion reveals the effects of stress on turnover intention observed in workers as a result of the privileged treatment of leaders toward certain members in leader-member exchange.

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Digital Marketing: A Conceptual Framework

1 Introduction

In the century we live in, as in every field, marketing activities are changing as well. The rapid development of technology and experiencing new innovations to our lives every day can be considered as reasons for this change. This development and progress create important marketing opportunities that affect the activities of the businesses. In addition to this, due to the development of technology, high-tech products have become an important part of our lives and our behaviours and habits started to become different. The increase in the usage of the Internet, in particular, has made it necessary to move from traditional methods to modern methods in the concept of marketing. This new understanding, which can be expressed as the application of traditional methods in digital environments, has significant advantages compared to traditional marketing.

Using digital marketing channels, businesses are able to reach much larger audiences at lower costs than traditional marketing activities. Thus, they provide easier access to their target audiences and have the opportunity to present their products in digital environments. Due to the change in the marketing of products and shopping behaviours of the consumers owing to the technology, businesses consider digital marketing in budget planning (Ryan, 2016).

Analysts at Jupiter Research have identified seven key points to demonstrate how diffusion and use of technology influence consumer behaviour (Nair, 2016):

- Digital technology helps create new virtual communities and removes geographic boundaries by providing an opportunity to bilateral interaction with tools such as email and instant messaging.
- Thanks to digital technology, consumers can easily compare, access and consume products.
- Digital consumers can use content related to themselves and block insignificant content.
- Consumers can be divided into smaller target groups according to their individual preferences.
- With digital technology, consumers have the opportunity to express their personal thoughts and experiences online.

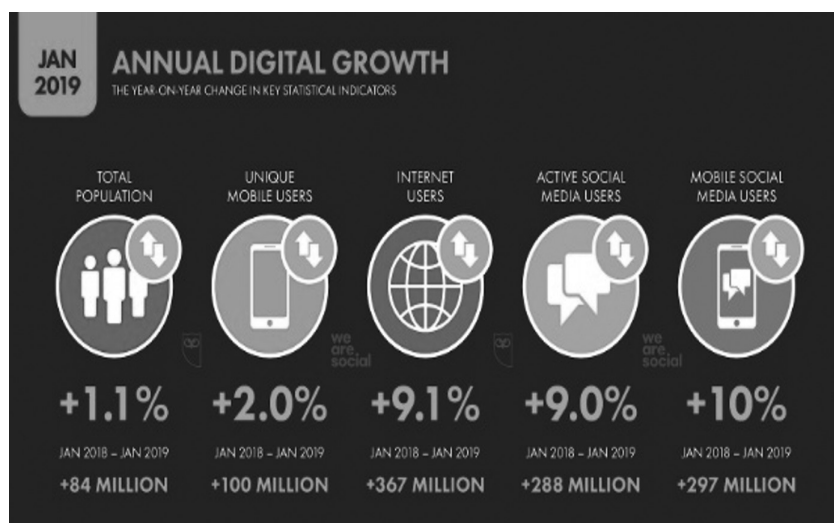


Fig. 1: Annual digital growth. Source: Global Digital Reports (2019)

- Consumers who are directly involved in the creation of products can easily access the products and services they need.

As a result of the increase in the number of Internet users, the importance of digital marketing in businesses has increased (Sawicki, 2016). Technology and Internet, which bring businesses and consumers together in a digital environment, are an integral part of human life while bringing consumers and businesses together in digital environments by enabling marketing actions to be carried out digitally. For this reason, it is important both for consumers and businesses to understand digital marketing conceptually and to examine the environments where it is used.

Today, as of January 2019, the world population is 7.676 billion. And there are 5.11 billion unique mobile users, 4.39 billion Internet users, 3.48 billion social media users and 3.26 billion social media users on mobile devices. Unique mobile users is up 100 million versus January 2018. Internet users is an increase of 366 million in the past year. Active social media users increase is 288 million and growth of people who use social media on mobile devices is 297 million (Global Digital Reports, 2019). These data show how important digitalization is in terms of marketing. Annual digital growth in the world (January 2018 – January 2019) is shown Fig. 1.

2 Definition of Digital Marketing

The Internet has been gradually increasing its presence in human life since the 1990s. It was first introduced as a concept in digital marketing in the 1990s (Kingsnorth, 2017). The first search engine Archie was also introduced within these years. Clickable web banners appeared in 1993 and 1994, respectively, and the first e-advertising campaign was carried out. And in 1998, Google started a revolutionary era in web search. The first mobile advertising campaign was carried out in 2001. Gmail, Facebook and YouTube were launched in 2004 and 2005. In 2006, Amazon's e-commerce revenues reached 10 billion USD. Then, WhatsApp was launched in 2010 (Firat, 2018). Today, the increasing number of various social media channels has also led to the development of digital marketing.

As a concept, digital marketing is carrying out the marketing-related activities in the electronic environment through network tools (Kircova, 2012, p. 24). In other words, it is using the Internet and Internet-related digital technologies to reach marketing goals and support the modern marketing concept (Marangoz, 2014, p.49). Unlike traditional marketing, digital marketing includes the promotion of products and brands in electronic platforms. It uses channels and methods that enable a business or an organization to analyze marketing campaigns in real time and understand what works and what does not (SAS, 2018). Digital marketing offers the possibility of fully integrated application of promoting, positioning, advertising, purchasing, distributing and delivering products and brands by means of one or more electronic media (Nair, 2016).

In general, digital marketing can be defined as the use of digital technologies to meet consumer needs more effectively and to reach potential buyers and targets. Defining it as Internet marketing is one of the most common mistakes in digital marketing. Because the Internet is just one of the numerous ways to reach a customer (Sawicki, 2016; Kingsnorth, 2017). Therefore, it is better to consider digital marketing through an integrated approach that includes the Internet.

All in all, the aim of marketing is to maintain customers and encourage sales in the future. Digital communication tools also make it possible to establish long-term relationships with customers. In this relationship, digital marketing plays an active role in the creation of consumer demand by using the interactive power of the web. Thus, in the digital environment, it can be seen exactly how the marketing campaigns perform, which channel is more beneficial for the business and where the promotion efforts should focus (Stokes, 2013, p. 18).

Nair (2016) considers the increase in the number of digital consumers as one of the main reasons for the emergence and development of digital marketing.

This has created a suitable environment for digital marketing. Another reason is the change of the digital mentality of the businesses. The success of businesses such as Amazon and E-bay in digital media encourages other businesses to use digital channels.

3 Attributes and Basics of Digital Marketing

It is extremely important for businesses to communicate with their customers about their products and brands in real time, to send their messages correctly to their target audience and to receive feedback from them. The importance of this is ever-increasing in our globalizing world. The way to benefit from this situation is through digital marketing.

Digital marketing strategies that are developing every day have some features for businesses and brands. These features include (Pineiro-Otero and Martinez-Rolan, 2016):

- Digital platforms and Internet services provide the opportunity to create brand image thanks to their scope, content and being always up to date.
- It enables business and consumers to become closer by providing consumers with wider and more customized information due to its integration.
- It offers simple and user-friendly platforms for improving user experience in terms of usability and functionality.
- It provides the opportunity to create a positive experience with the brand in a long-term organization-consumer relationship.
- Digital marketing enables visual communication with customers for marketers via various image and video-based tools.
- Customization and segmentation of the advertisements are easier compared to other media tools.
- Digital platforms strengthen and improve the link between communities and the product, brand or business.
- Digital environments can become more effective and more shareable using the viral communication model as a base.
- Data on online platforms can be more easily tracked and measured.

Moreover, through digital marketing, both businesses and consumers have the advantage of acting independently and away from various issues such as place, time, location and physical storage. In addition to the features mentioned above, it can be said that digital marketing is mainly based on four steps; Acquire, Convert, Measure&Optimize and Retain&Grow. These basic steps are briefly described below.

In the *Acquire* phase, there are activities carried out to attract customers to the website or to the platforms where the sales are carried out. E-mail marketing, social media marketing, viral marketing and mobile marketing are the main methods used in this phase (Chaffey & Smith, 2013).

In the *Convert* phase, activities are implemented that will help to achieve the goals after the customers arrive on the website. These activities include practices such as giving persuasive messages in sales texts, usability and accessibility practices, content management, increasing the usefulness of the site, customization according to the customer and separating customer segments (Chaffey & Smith, 2013).

In the *Measure&Optimize* phase, what is right and what is wrong is evaluated and comparisons with competitors are made. In addition, in this phase, the success criteria of the company's website are compared with competitors and various tests such as a/b tests are applied to understand the demands and needs of customers (Smith, 2011).

In the last phase *Retain&Grow*, the practices which satisfy existing customers and make them loyal customers are implemented. Services such as customization, loyalty programs, reference programs and virtual communities are provided in this phase (Chaffey & Smith, 2013).

As can be seen, digital marketing has different characteristics for businesses and brands as well as for consumers. Therefore, the addition of digital marketing to marketing actions will bring strategic benefits. Digital marketing has changed customers' buying behaviour. Consumers have some advantages with it (Yasmin, Tasneem & Fatema, 2015). These advantages are shown in Fig. 2.

4 Tools of Digital Marketing

There are many different areas and tools in which digital marketing is used. In this part of the study, commonly used tools such as e-mail marketing, mobile marketing, search engine optimization (SEO), website marketing, social media marketing and influencer marketing will be summarized.

E-mail marketing: It is an online marketing technique used to gain new customers and send advertising or commercial information to loyal customers. The use of e-mail dates back to 1971. It is one of the main tools used in digital marketing, as it is easy to make changes in its content and usage. The main purpose of e-mail marketing is to manage website traffic, to announce a special campaign, to increase brand popularity and its image, and cost saving (Pineiro-Otero and Martinez-Rolan, 2016). With e-mail marketing, marketing actions can be easily carried out through sales promotion efforts sent to the personal e-mail addresses of the target audience.

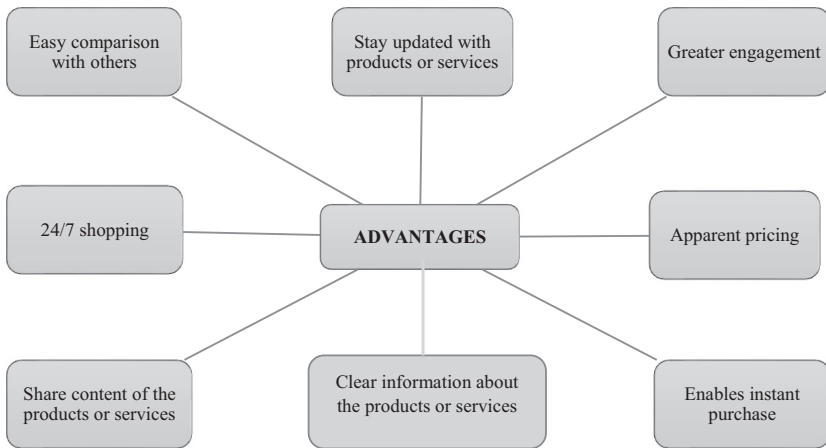


Fig. 2: Advantages of digital marketing for costumers. Source: Yasmin, Tasneem and Fatema (2015)

Mobile marketing: Digital marketing, which could only be used as Internet marketing, content sharing and e-mail promotion activities in the past years can now be utilized in mobile as well with the ever-increasing use of smartphones and tablets. The main reason for the development of mobile marketing, which is one of the most important sub-dimensions of digital marketing, is that mobile tools have the area of usage in almost every place in the world (Telli Yamamoto, 2011: 71). With the help of mobile marketing actions that can be implemented through mobile phones, tablets, etc., businesses can realize marketing efforts related to products, services, brands, ideas and people more quickly and easily. Thanks to mobile marketing, businesses communicate with customers about their products; customers have the opportunity to interact with the products and the business as well.

The purpose of mobile marketing is to provide mobile device users (consumers) in the target audiences of the businesses with a convenient environment for them to access products from any place and place orders, make payments, perform various banking transactions and finish their shopping. In this way, businesses can offer their customers the opportunity to shop at any time and place they want and the freedom to take advantage of mobile application services (Karaca and Gulmez, 2010: 78). In mobile marketing, the response to the product and advertising can be measured more easily and the feedback rate from consumers is quite high.

Website marketing: Website is another way to be present and communicate with customers in the digital world. The websites of organizations or brands are their face to the outside. Customers are informed about brands, products and services by looking at these digital pages.

For an effective and efficient website, brands should mind the following tips (Ryan, 2017):

- How the users will access the website and what they want to find should be determined.
- The website should have descriptive information about the products and should be easy to use.
- It should be analyzed that which search on the search engines will lead the target audience to the website.
- The website should be created according to the aims of the brand goals and the needs of the customers.

Search engine optimization (SEO): SEO is the optimization of the websites of businesses and brands for the natural results of the search engine (Kingsnorth, 2017, p. 90). It is based on the process of statistically presenting consumer information, search records and search volumes essential for the brand compiled by search engines (e.g. Google, Yandex) by scanning thousands of web pages through automatic programs (Ryan, 2016). Through search engines, users can search for the closest results related to the keywords they are looking for (Cevik, 2015, p. 123). In this way, the web pages that businesses consider as suitable for the target customers become easily accessible.

Social media marketing: Social media marketing has taken its place as a digital marketing approach since 2004. Several new organizations such as Facebook and YouTube, which are based on the principles of Web 2.0, have been the precursors of the existing business shifting to this channel. The advertisements on these social media channels have a great impact on reaching consumers (Chaffey & Smith, 2013). Businesses are increasingly using social media tools to communicate with their customers, to give them confidence, to gain potential customers and to increase brand awareness (Mills, 2012). Businesses that can communicate closely with consumers through social media platforms can easily deliver their products, brands and services to their target audiences. The fact that social media channels that cover a very wide range are not included in marketing strategies is a negative situation for the businesses. Because the promotion efforts carried out in these channels are spreading very rapidly to a very wide range.

Influencer marketing: Influencer marketing is bringing the influencers and the brands together in a new and dynamic business model which is the reflection of

word-of-mouth marketing in the digital world, considered to be one of the most effective elements of the marketing world, as influencers have close and trustful relationships with their followers (Kolevitoglu, 2017). In other words, people with a high number of followers in social media channels such as YouTube, Facebook, Twitter and Instagram transfer their experiences about a product or brand to their followers through contents (Unal, 2016). The use of influencers is quite new in terms of digital marketing.

5 Conclusion

The main purpose of this study is to provide a guide for digital marketing researches. In this context, after defining digital marketing, a conceptual framework for the tools that are used by/in digital marketing as a marketing strategy was introduced. Because the most important way for today's businesses to be successful in marketing actions is to use digital marketing. It should be noted that digital marketing is not a fixed phenomenon like traditional marketing and tends to change over time. The changing global digital environment requires a re-examination of all concepts, methods and practices of marketing for businesses. For this reason, businesses should transfer their marketing activities to digital media and tend to shift their actions towards new channels continuously by keeping up with the change. For this purpose, businesses should use tools and resources such as video, video story, chatbots, big data, smart speakers, native ads, virtual reality (VR), augmented reality (AR), wearable technologies (e.g. smart glasses, smart watches) and mobile apps in their digital marketing strategies.

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Hamit Kahraman and Yavuz Selim Duger

The Role of Human Resource Management, Leadership, and Marketing in Employer Brand

1 Introduction

The talented employees, who are distinguished by their different characteristics, are very important in both contributing to the business performance and using the current technologies in the enterprise efficiently. In spite of the increasing supply of labor, finding, attracting, and keeping talented employees is still one of the most important problems for enterprises.

The employer brand is a tool that helps businesses to differentiate themselves from other employers and attract attractive talents with the same values. In addition, it has a strategic importance for enterprises because it is a catalyst in increasing efficiency. Thus, the employer brand creates an organization that provides organizational performance and more and better-quality applications (Chunping & Xi, 2011, p. 2088; Foster, Punjaisri, & Cheng, 2010, p. 403).

The employer brand, which attracts great attention as a special form of managing corporate identities, is the efforts of an enterprise to create a different and desirable employer image both inside and outside (Lievens, Van Hoyer, & Anseel, 2007, p. 48; Ören & Yüksel, 2012, p. 38). At this point, businesses benefit from both marketing and recruitment functions while competing to attract limited number of qualified employees. Persuasion and communication on the basis of both reveal the value of integration of marketing and recruitment. Therefore, it is necessary to benefit from marketing orientation due to evaluation of qualified candidates as a customer by organizations and the competition about attracting them (Cable & Turban, 2001, p. 120). In this study, the place and importance of human resource management, leadership, and marketing together in the employer brand are put forward.

2 Employer Brand Concept and Definition

Although companies' branding efforts are generally addressed to improve their products and corporate brands, they can also be used in the field of human resources management. Implementation of brand principles for human resources management is also called as employer branding (Backhaus & Tikoo, 2004, p. 501).

The employer brand concept was first used by Ambler and Barrow (1996) (Ong, 2011, p. 1089), and the first definition was made by them (Schlager, Bodderas, Maas, & Cachelin, 2011, p. 498). According to them, the employer brand is defined as a package of functional, economic, and psychological benefits provided by the employer and identified with the employer (Ambler & Barrow, 1996, p. 187). The primary objective of this initial approach to the employer brand is to simplify and focus priorities, to increase productivity and to provide a consistent framework for management in recruitment, retention, and loyalty (Schlager et al., 2011, p. 498).

In the literature, various definitions are given in different studies related to the employer brand. Some of these definitions are:

- It is a long-term strategy aimed at managing the awareness and perceptions of current and potential employees and relevant stakeholders for a particular company (Sullivan, 2004).
- It is the sum of efforts of declaring that a company is a desired place to work for its current and future staff (Berthon, Ewing, & Hah, 2005, p. 153).
- It is the level of attractiveness of the company for existing and potential employees (Schlager et al., 2011, p. 498).
- It is an emotional relationship that extends from the employer to the current employees as well as to all stakeholders of the organization and to the community and potential employees (Özgen & Akbayır, 2011, p. 81).

When the definitions are examined, the employer brand reveals an image that includes management and business practices that make itself different and attractive to both its internal and external environment for company as an employer (Backhaus & Tikoo, 2004, p. 502; Sullivan, 2004). In this respect, it includes providing a unique employment experience with the total tangible and intangible awards that a particular organization offers to its employees (Edwards, 2010, p. 7).

In the light of all these definitions and evaluations, employer brand can be defined as the level of awareness and attractiveness created by providing unique experiences thanks to functional, economic, and psychological benefits to attract talented potential employees including stakeholders and to keep existing employees.

The main purpose of the employer brand is to attract the most suitable candidates instead of as many candidates as possible, as the employer's advantage is related to the recruitment and sustainability of the employees who can adapt (Erkanlı, Topuz, & Cop, 2015, p. 88). In addition, there are also people who leave their jobs among the most important target groups within the scope of the

employer brand. Persons who leave work should be able to talk well about the organization, should recommend the organization, and even return to the organization if possible (Öksüz, 2012, p. 20).

On the other hand, employer brand offers an effective commercial link between human resources, internal communication, and marketing. Especially in the service sector, employee loyalty is essential in achieving customer satisfaction and loyalty. Therefore, human resources and internal communication practices are increasingly aligned with the approaches and disciplines that are applied to create and present external value, i.e., marketing and brand management (Barrow & Mosley, 2005, p. 164).

3 Benefits of Employer Brand for Consumers, Employees, and Businesses

In today's business world where globalization is very fast, the fact that international companies can achieve sustainable competitive advantage is becoming more and more dependent on the qualified work force they have. Employer brand emerges to be an important strategy in terms of retaining employees who have superior talent and performance and, more importantly to attract such candidates. Nowadays, corporate brand reputation of the companies alone is not enough in the process of attracting high-quality candidates, and at this point, the employer brand is becoming an important determinant. Employer branding is adopted as an important strategy for companies to differentiate from their competitors and achieve sustainable competitive advantage in the long term.

The employer brand provides significant advantages to consumers, employees, and businesses. Some of these are given below (Chignell, 2015; Barrow and Mosley, 2012: p. 93–106; Baş, 2011: p. 43–47):

- Increases motivation of employees to work effectively and efficiently,
- Employees trust the companies and feel safe as a result of which they are highly committed to the companies,
- The highly qualified, capable, and reliable employees of the companies will reduce the labor force turnover rate, and this will lead to significant savings in the costs of recruitment, training, orientation, retention of employees, and human resources management costs,
- Companies with a high level of employee loyalty invest more in their employees, adding value to the brands of the companies,
- As the company is considered to be a perfect place for the employees, it is much easier to attract other highly qualified candidates and keep existing employees,

- The human resources management process of the company becomes mission and business oriented in long-term,
- The way to be a merit-based reference to other employees is opened,
- The improvement in the quality of the employees increases the satisfaction of the managers,
- Ensures the creation of a brand-oriented organizational culture,
- Thanks to the highly qualified workforce, the competitiveness of the company increases,
- As the brand image of the company in the market will reach a significant level, economic gain is achieved by attracting customers, business partners, or investors.

Since the employer brand is an extremely important strategy for the companies that desire to maintain their existence in the market, the necessary actions should be taken by the managers in terms of the effectiveness of the process. As a result of having a strong and positive organizational culture within the company by using an effective employer brand strategy, it will have a positive impact on external stakeholders. In this way, thanks to having highly talented employees of the company, a competitive advantage will be achieved that cannot be imitated.

4 The Role of Marketing in Employer Brand

Just like the goods and services brands, the employer brand aims to differentiate the enterprises in a distinctive way, to present value and to clearly define the value it offers, to strengthen its competitiveness, and to be one of the limited choices of the target group. Human resources and marketing have an important function in achieving these goals (Köse, 2017, p. 64).

Employer brand is an activity where marketing principles, especially branding science, are applied to human resources activities for current and potential employees (Edwards, 2010, p. 6). Employer brand management requires the human resources function should display performance by working closely with employees in the field of marketing and communication (Barrow & Mosley, 2005, p. 162).

The employer brand is similar to the branding of a product as it is considered to be the adaptation of brand discourse and techniques to employees. For example, product branding focuses on loyal customers, employer branding tries to create employees who are related to the institution; product branding focuses on creating less sensitive customers at the price, employer branding focuses on creating employees who are less sensitive to salaries (Özgen & Akbayır, 2011,

p. 81). While the product brands offer value proposition for customers, the employer brand also presents a value proposition for their current and potential employees as customers (Moroko & Uncles, 2008, p. 164). In addition, the purpose of both is to keep the existing customer and to bring new ones into organization (Ören & Yüksel, 2012, p. 38). As in the corporate brand and product brand, the success of the employer brand is related to its ability to differentiate from competitors (Moroko & Uncles, 2008, p. 164). Even corporate brands and product brands focus on creating brand awareness among their customers, while successful employer brands focus being “known” and “noticable” by influencers such as employees, prospective employees, and recruitment consultants (Moroko & Uncles, 2008, pp. 163–164). In the employer brand, existing and potential employees will have more positive feelings and make positive decisions to the companies that have high brand value, such as the customers’ positive approach to the products of the companies with high brand value. In this respect, brand value is also an effective element on the employer brand (Narcıkara, Gürol, & Üzmez, 2016, p. 49). Advertising and public relations activities are important communication tools for both creating corporate identity and attracting qualified workforce (Özgen & Akbayır, 2011, p. 82). Advertising, which is a critical tool for businesses in their efforts to identify, acquire, and retain qualified employees, is now also used to create employer brands (Berthon et al., 2005, p. 153).

In addition, while corporate brands are identified with the identities of their employees, the behaviors of employees are shaped by brand. As a natural consequence of this, marketing experts and institutions have recently focused on in-house branding tactics to create an employer brand (Özgen & Akbayır, 2011, p. 80). Internal branding enables new employees to understand what the corporate brand means and what values they offer to external stakeholders. Internal branding together with employer branding will create a closer working relationship between human resources and marketing functions (Foster et al., 2010, p. 404).

Internal marketing, which is defined as ‘the task of successfully hiring, training and motivating able employees to serve the customer well’ by Kotler, also represents the elements of good human resources management (Ewing, Pitt, de Bussy, & Berthon, 2002, p. 10). Schlager et al. (2011) also states that a strong employer brand is an effective tool to improve the contribution of employees to service branding. Because satisfied and supported employees positively affect the customer experience and thus help to create a service brand (Schlager et al., 2011, p. 504).

Both the employer brand and product brand serve the corporate brand consistently (Chunping & Xi, 2011, p. 2088). The external marketing of the employer brand is primarily designed to attract the target group and then to support and develop the product or corporate brands. In this respect, the basis of employer branding is that the employer brand is consistent with all other branding efforts of the firm (Backhaus & Tikoo, 2004, pp. 502–503).

5 The Role of Human Resources Management and Leadership in Employer Brand

In today's business world, where talent-based work becomes more important, effective human resources practices are needed in order to achieve goals of the companies. The most important practice is the employer brand carried out by human resources management with marketing. The employer brand is a combination of human resources management and marketing, and this combination is based on customer and employee experience (Ambler and Barrow, 1996: p. 187–200). The employer brand has emerged with a focus on employee experience in human resources management, and has also become a concept that has attracted much attention in the field of human resources management in recent years. This approach has become even more important associated with increasing competition and talent wars, which basically aims to attract the talented candidates by reflecting the strong corporate image (Backhaus and Tikoo, 2004: p. 503–504). Moreover, employer brand has transformed human resources management into a strategic level with participating in the strategic decisions of the company beyond daily operations.

Employer brand is divided into two as internal and external. The internal employer brand aims to create a brand-focused organizational culture, develop employees, and create a positive working environment by focusing on existing employees. Thus, employees are ensured to be brand ambassadors, thanks to the good employees' working experience. External employer brand, on the other hand, is directed to potential talented candidates by utilizing many communication channels such as organizing idea-oriented project competitions, participating in career days of universities, and using various tools via social media. All these attempts reflect the attractive features of company to the potential candidates such as wage, work environment, career opportunities, social rights and deliver the message that the company is the perfect place to work. In this way, the external employer brand ensures increasing the brand awareness, informing the advantages of the company for the employees, building a strong corporate image, as well as attracting talented candidates (Stuss and Herdal,

2017: p. 204–205). As a result, it is clear that human resources management is a complementary and enforcer of both internal and external employer branding process.

Human resources management desires to make the candidates' attracting process more effective by employer brand strategy. In this direction, the aim is not to obtain as many applicants as possible, but to attract the most suitable applicants for the organization. Attracting and retaining the best candidates has become the main task of HRM. In addition, another fundamental task of human resources management is to build a psychological contract between employer and employee by using empowerment, motivation, and talent enhancement activities.

According to Mosley (2007), leadership has an important role in the accuracy and sustainability of the employer brand. Leader links followers with the mission and purpose of the organization, as an element that delivers and reflects organizational culture and values. Leader aligns employee values and organizational values. In this way, employees who have high brand loyalty, as well as high organizational commitment, are created (Pathardikar, Sahu & Maurya, 2013: p. 90). Leader has the most important role in the employer brand process by exhibiting brand ambassador behaviors.

6 Conclusion

Nowadays, one of the biggest risks faced by senior executives is the lack of current and future talents which threatens the profitability, productivity, and most importantly the existence of the companies. The most significant strategy to overcome the shortage of talent is the employer brand. In this respect, employer brand strategies are becoming more critical for businesses.

Human resources and marketing departments have many benefits to each other in the success of the employer brand process. Human resources can use the customer segmentation as talent segmentation for employees. Human resources can use the technological communication channels for employees as marketing department uses for its customers. Marketing department can learn from human resources that marketing is established first internally and then externally. All these achievements carry the organization into the future with its strong corporate culture and corporate image (Bluivgroup, 2013).

Employer brand strategy is seen as an implementation of modern human resources, due to the weight and importance of human resources practices in this process. De Chernatony (2006) emphasizes that the control of the employer brand should be in human resources management, because hiring, training, and

development applications directly affect employer branding activities. Moreover, many researchers also argue that employer brand is the responsibility of human resources management. In fact, the governing of employer brand strategy is so important that it is not left to the responsibility of a single unit. Since the employer brand is related to the future existence of the business, it should be carried out under the control of leadership and the responsibility of human resources and marketing.

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Iclem Er

The Evolution of Social Commerce

1 Introduction

The rapid development of Web 2.0 and social media has paved the way for a transformation in e-commerce. Web 2.0 provides the essential technological basis for using social media where users generate content, share information, and collaborate. In this virtual environment, users also share their experiences and knowledge about products and vendors as consumers, thus providing access to socially shared knowledge and experiences for other consumers who would use them in making more informed purchase decisions. Consequently, digital content created by social media users started to gain economic value (Pitta and Fowler, 2005), and information shared by users/consumers has become a significant source in purchase decision-making (Do-Hyung et al., 2007).

As consumers spend more time on social media, new types of online and offline buyer behaviors are observed. For example, “Digital in 2019 Global Overview” by We Are Social shows that social media continues to grow as a source of information about products and brands, and emerging markets lead the way for using social media at every step of their digital buying process (<https://wearesocial.com/blog/2019/01/digital-2019>).

Meanwhile, firms, which have realized the mutual advantages of social media, began to show more interest in benefiting from the economic value of countless social interactions that consumers participate globally every day. Consequently, e-commerce transforms into a more participatory and social interaction centered form with the aid of Web 2.0 features, and this progress is generally associated with the onset of social commerce. This chapter first explains the basic characteristics of Web 2.0 and social media that enable social commerce, then continues with the evolution of e-commerce to social commerce, and finally defines social commerce and describes its progressive features.

2 Essentials of Web 2.0 and Social Media

Although social media is formed on the concept of Web 2.0, these two notions are used interchangeably. O'Reilly Media introduced the term “Web 2.0” in 2004 to depict the fundamental differences and technological advances of the new Internet, and to describe a second-generation of Internet-based tools and

services (Turban et al., 2016). The second generation of Internet is characterized by user-generated content (UGC), interactive data and information sharing, and online collaboration. Karakas (2009) defines Web 2.0 with five major shifts, which shape the new global environment for business, technology, and innovation as creativity, connectivity, collaboration, convergence, and community.

The key factor behind these shifts is considered to be the UGC, which is also known as consumer-generated media (Tuncer, 2013). UGC defines media content that is produced by end-users and is publicly available, such as reviews, recommendations, ads, electronic word of mouth (eWOM), and entertainment (Turban et al., 2016). These content types are posted or shared on various social media platforms, some of which totally consist of UGC like YouTube, LinkedIn, Twitter, Flickr, and Instagram.

Parallel to the rise in UGC on the Internet, the share of the users or the consumers in product, brand, and vendor conversations is increasing (Hajli and Sims, 2015). Similar to traditional word of mouth, consumers exhibit the propensity to follow social media platforms where they can share information, knowledge, and eWOM. Social media, like traditional media, delivers content for entertainment, leisure, learning, collaboration, marketing, and other purposes. However, social media uses Web 2.0 and its tools; but the concept itself comprises the idea of connected users, the interactions among them, and the digital content that is created by them (Turban et al., 2018).

Social media can be defined as online media content using Web 2.0 tools and platforms for conducting social interactions and conversations, such as sharing opinions, generating content, and providing recommendations; and as an effective means of socialization, communication, and collaboration (Turban et al., 2016). Social media can also be described as software tools that enable individuals to generate content and engage in personal sharing, publishing, conversations, and dissemination of this content in a social platform (Safko, 2012). A more comprehensive definition explains social media as “a group of Internet-based applications that is established on the ideological and technological foundations of Web 2.0 and that allow the generation and exchange of user-generated content” (Kaplan and Haenlein, 2010). As can be seen, all of the above definitions of social media focus on technology, content generation, and social interactions. Social media could be considered as a structure composed of three main elements: (1) application tools for media components such as text, images, audios, and videos; (2) delivery platforms such as social networking sites and services; and (3) social media activities such as conversations, sharing, and recommendations (Turban et al., 2016).

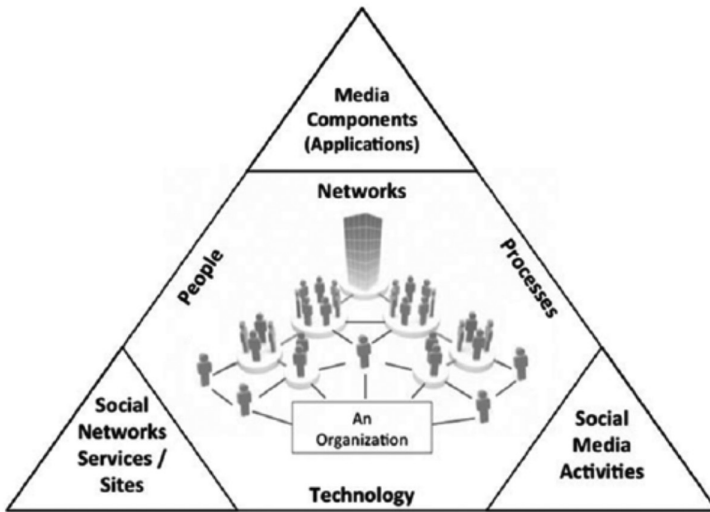


Fig. 1: Three Main Elements of Social Media (Turban et al., 2016: 27)

As We Are Social (2019) reports, worldwide social media user numbers have grown to almost 3.5 billion at the start of 2019, with 288 million new users in the past 12 months pushing the global penetration figure to 45 % (<https://wearesocial.com/blog/2019/01/digital-2019>).

3 Benefits of Social Media for Consumers

UGC of social media does not merely cover personal interactions, socialization, and entertainment (Zhang et al., 2019); but as Pitta and Fowler (2005) state individuals as consumers use social media to share their knowledge and experiences about products, services, and even vendors on the Internet. Social media platforms provide the tools for consumers to interact with each other through information exchange and the content they generate there (Hajli, 2013). Social media offers users various means for searching about products, services, and vendors, accessing information and experiences of other consumers before making a purchase decision (Hajli and Sims, 2015).

Traditionally consumers try to obtain product recommendations from family and friends (Turban et al., 2016) whereas via social media they participate in online communities where trust, interests, knowledge, recommendations, and referrals are shared publicly (Lin et al., 2017). Recommendations and

referrals are considered as major influences in electronic commerce for a long time (Zhang et al., 2019). As the UGC increases in the social media, the users/consumers largely control brand conversations (Turban et al., 2016). Since social media have become readily accessible, more consumers use it as a source of information about companies, brands, products, and services (Zhou et al., 2013). Because social media facilitates individual interactions, it is possible to observe a shift from somehow misleading individual consumption decisions to a more collaborative, sharing based-social shopping decisions as well (Chen and Shen, 2015). “Digital in 2019” Report by We Are Social (2019) illustrates that from using social media, such as Instagram, to search products with customer support on messaging applications, social media has become the center of all digital customer touchpoints (<https://wearesocial.com/blog/2019/01/digital-2019>).

Three basic features of social media allowed the transfer of power from sellers to buyers; (1) accessibility of information anytime anywhere via Web 2.0 tools, which eliminated the information asymmetry between sellers and buyers; (2) elimination of geographic boundaries through Internet which help consumers with similar areas of interest to form online communities and share opinions; and (3) with the help of personal interactions on social media, consumers transformed into active participators rather than passive information takers (Ozata, 2013).

4 Evolution of Social Commerce

The rapid growth of electronic commerce (e-commerce) created a significant effect on the global economic transactions, while the increasing use of social media as an information source produced a shift in consumer shopping behavior (Li and Ku, 2018). We Are Social’s Digital 2019 report shows that the total value of the e-commerce market for consumer goods grew by 14 % in 2018, and the total annual spending is estimated to reach US\$1.78 trillion at the end of 2018, with fashion products representing the largest single category; and 37 % of global population engage in e-commerce activities (<https://wearesocial.com/blog/2019/01/digital-2019>). The increased use of social media together with the technological features of Web 2.0 started a transformation in e-commerce from a product-oriented environment to a more consumer-centered one (Wigand et al., 2008). This e-commerce evolution is commonly associated with the birth of social commerce (Huang and Benyoucef, 2013).

Social commerce has shifted the e-commerce setting from a business-oriented form to one that is shaped around the consumer (Busalim, 2016). In this new environment, consumers could access knowledge and experiences of

other consumers that will allow them to make more informed and accurate online purchasing decisions (Huang and Benyoucef, 2013). When compared to the product-centered e-commerce settings with producer- or vendor-provided information, social commerce emphasizes consumer-centered and socially driven electronic commerce approach where social media helps users to make purchase decision through social connections (Li and Ku, 2018). In this novel social commerce setting, consumers could access UGC about products, services, producers, and even vendors in the form of comments, reviews, tags, likes, complaints, and user profiles, which shifts consumer purchase behavior from a passive information taker to an active engager by seeking and providing eWOM (Henning-Thurau et al., 2004; Busalim, 2016; Li and Ku, 2018). We Are Social's Digital 2019 report suggests that consumers prefer shopping to be a sensory and social experience whether from a physical store or via flipping through Instagram (<https://wearesocial.com/blog/2019/01/digital-2019>).

Meanwhile in this social commerce environment, online businesses could track customers' behavior, apprehend insights about customers' shopping decisions and experiences, and develop effective marketing strategies with the aid of Web 2.0 features (Constantinides and Fountain, 2008). On the business side, firms employ social media and social commerce in order to improve customer relationship, participation, and engagement and deliver greater value (Huang and Benyoucef, 2013).

5 Social Commerce: Definitions and Basic Features

The concept of social commerce emerged in mid-2005 due to the growing commercial use of social media (Zhang and Benyoucef, 2016). Social commerce (SC) involves e-commerce operations provided through social media; in other words, it is mainly a combination of e-commerce, e-marketing, support theories, and social media content (Turban et al., 2016). As shown in Fig. 2, SC is established on an integration of e-commerce and e-marketing with the use of social media content; and this integration is supported by theories from social psychology, consumer behavior, and online collaboration (Turban et al., 2016).

According to Shin (2013), social commerce is a rapidly growing form of e-commerce that includes using social media for supporting online purchasing and sales of products and services with social interactions and user contributions. Social commerce incorporates Web 2.0-based media, such as social media, which facilitates consumers' participation in marketing, comparing, selling, buying, and sharing of products and services in both online and offline marketplaces as well as consumer communities (Zhou et al., 2013). It involves user-generated

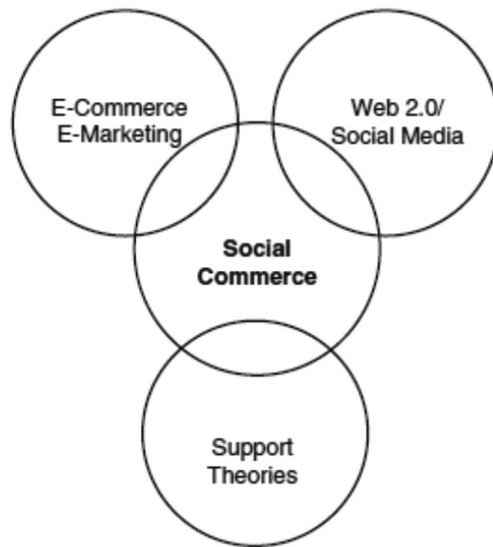


Fig. 2: The Essentials of Social Commerce (Turban et al., 2016)

online product and vendor information such as recommendations, reviews, ratings, posts, photos, and even complaints. Huang and Benyoucef (2013) provided a list of social commerce definitions that are presented in Tab. 1.

After a comprehensive literature review, Huang and Benyoucef (2013) defined social commerce as a new form of social media–facilitated commerce that allows consumers to participate actively in the marketing and selling of products in online marketplaces. The authors identified to different types of social commerce: (1) social commerce that is observed in online retailers such as Amazon, which is intrinsically a part of e-commerce but provides some tools for consumers to generate content; (2) the second type of social commerce evolves in social media but integrates e-commerce features (Huang and Benyoucef, 2013). The first type of social commerce restrains social interaction among consumers since consumers cannot directly interact with each other such as sending private messages. The second type of social commerce, on the other hand, provides devices for direct consumer interaction, value creation through collaborative activities such as information and experience sharing (Li, 2017). The basic differences between e-commerce and social commerce are presented in Tab. 2.

Other features of social commerce that distinguish it from e-commerce can be summarized as follows (Yildiz, 2019):

Tab. 1: Various Definitions of Social Commerce*

Definition	Author(s)
"Social commerce can be defined as word-of-mouth applied to e-commerce"	<i>Dennison et al., 2009</i>
"Social commerce refers to a more social, creative and collaborative approach used in online marketplaces"	<i>Parise and Guinan, 2008</i>
"Social commerce is a concept that applies social media applications to shape business, and transforming a market for goods and services into a socially centered and user-driven marketplace"	<i>Wigand et al., 2008</i>
"Social commerce is a significant trend in online marketplaces where businesses leverage social media or Web 2.0 as a direct marketing tool to support customers' decision making processes and buying behavior"	<i>Constantinides and Fountain, 2008</i>
"Social commerce is an online mediated application combining Web 2.0 technologies with interactive platforms, such as social networking sites and content communities in a commercial environment"	<i>Lee et al., 2008</i>
"Social commerce involves utilizing web-based social communities by e-commerce companies, focusing on the impact of social influence which shapes the interaction among consumers"	<i>Kim and Srivastava, 2007</i>
"Social commerce includes the psychology of social shopping, where people are influenced by relevant information from people within a networked community when they shop online"	<i>Marsden, 2009</i>

* Adapted from Huang and Benyoucef (2013)

- Social commerce utilizes social intelligence by harnessing consumer participation and user-generated content;
- The dynamic force behind social commerce is user-generated viral marketing where consumers' favorable or unfavorable eWOM drives purchasing decisions;
- Social commerce allows especially firms to focus on behavioral targeting and micromarket segments.

As stated in We Are Social's Digital 2019 report, social commerce is maturing with native functionality and a supportive environment, where major social media platforms enhanced their "native" social commerce capabilities, and the environment optimizes the path from social platform to purchase. In addition,

Tab. 2: The Basic Differences Between e-Commerce and Social Commerce*

Features	E-commerce	Social commerce
<i>Consumer interaction</i>	Consumers interact with e-commerce platforms individually and independently	Consumers participate in online communities that facilitate social interaction and consumer conversations
<i>Consumer control</i>	Consumers have limited or no control	Consumers are empowered to have control
<i>System interaction</i>	Provides one-way browsing for shopping	Provides a more interactive, social, and collaborative shopping experience
<i>Business objectives</i>	Focuses on maximizing shopping efficiency	Focuses on social interactions
<i>Media design</i>	Product-centered	User/consumer-centered

* Adapted from Li and Ku, 2018.

influencers in social media have an expanding impact on consumers' purchase decisions, and brand collaborations are estimated to increase (<https://wearesocial.com/blog/2019/01/digital-2019>).

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Onur Cakir and Ece Dogantan

Local Tourism Entrepreneurship and Entrepreneurial Climate: The Case of İğneada, Kırklareli

1 Introduction

Rural areas are one of the rare sites that can offer a lifestyle that is intertwined with nature that many people have longed for, or that will allow them to experience a short-term experience in the context of tourism (Çakır and Ergüven, 2016). However, in order to develop rural tourism in a certain region, it is necessary to create supply elements to meet this demand. The task of producing and introducing touristic goods and services in rural destinations falls to local entrepreneurs. However, it would not be a realistic approach to expect local people to become entrepreneurs in the tourism sector, which is a sector they have no experience with. Therefore, it is imperative to prepare a suitable climate in order to encourage locals to become tourism entrepreneurs.

The fact that the West Marmara region, which includes the research area İğneada region, is among the regions with the lowest intention of entrepreneurship (Karadeniz, 2014) also supports the fact that it is not realistic to expect the local people to intend to invest in tourism in their own self. Therefore, the establishment of a supportive and encouraging entrepreneurial climate in increasing the local people's intention to become an entrepreneur in the field of tourism is even more important for the İğneada region. Considering the regional differences in entrepreneurial intentions, it is thought that a positive entrepreneurship climate in Kırklareli-İğneada region will contribute to the emergence of local tourism entrepreneurs. From this point of view, Kırklareli-İğneada town, which is one of the rural tourism destinations in West Marmara, was chosen as the research area, and the impact of entrepreneurship climate on local people's entrepreneurial intentions was evaluated. Thus, suggestions have been made on how to create an entrepreneurial climate that will increase entrepreneurial intentions of the local people to invest in tourism sector.

2 Rural Tourism Supply and Local Entrepreneurship

Agriculture-based development projects prepared for the development of rural areas do not fully respond to the expectations of people living in rural areas. It is

argued in the literature that better results can be achieved by creating a balanced supply in sectors such as tourism, industry, and agriculture (Tozak, 2017). In recent years, rural tourism has been seen as a rural development tool, particularly in terms of economic aspects and in terms of conservation and management of local culture and resources (Kline, 2007). When the development plans of the countries are examined, it can be seen that the plans and policies related to the support of rural tourism development in rural areas are included. As of the Seventh Five-Year Development Plan, it is noteworthy that the focus of tourism policies have been shifted from increasing the bed capacity to the development of environment-friendly tourism types such as rural tourism, nature tourism, etc. In this respect, the Ninth Development Plan (2007–2013) states that tourism will be developed in the less developed regions with tourism potential so that the economic and social development of these regions could be realized.

Although entrepreneurship is not a solution to all rural problems, it is considered as an important opportunity to increase the efficiency and income of rural areas (Çakır and Doğanatan, 2018). Research shows that tourism entrepreneurship is lucrative and advantageous for rural areas in terms of generating economic solutions for society, mobilizing local talents, creating a local identity, encouraging regional development, and supporting the local purchasing philosophy (Kline and Milburn, 2010). Within this framework, one of the most important tools for the development of rural tourism destinations is the creation of job opportunities to stimulate the economy in the region and the encouragement of entrepreneurship. As a matter of fact, in order to meet the demand created by the desire to experience the lifestyle of the locals in rural destinations, it is necessary to develop and market the tourism supply elements. The task of producing and supplying the goods and services required to meet this demand falls to the local entrepreneurs residing in rural areas (Çakır and Ergüven, 2016).

3 Entrepreneurial Climate and Entrepreneurial Intention

Entrepreneurial climate is defined as the sum of the necessary environmental conditions for increasing the number of entrepreneurs that can have an impact on the emergence of new establishments and/or on existing enterprises (Lordkipanidze, 2002; Başar, 2016). Goetz and Freshwater (2001) state that although there are several studies aiming to measure entrepreneurial climate, there is no consensus yet on the definition of entrepreneurial climate. Başar (2016) likewise points out that the factors affecting entrepreneurial climate may differ in different economies. In the literature, it is observed that the factors of entrepreneurial climate are local government's support, local people's attitudes,

community size, proximity to big cities, physical infrastructure quality, high speed internet, quality of life, financial resources, business services, business networks, education opportunities for small businesses, availability of building area, technology usage, R&D, etc. (MacKenzie, 1992; Kuratko, Hornsby & Naffziger 1999; Dabson, 2001; Henderson, 2002; Başar, 2016). Kline (2007), by organizing meetings with locals, identified the needs of rural entrepreneurs, which are education, financial capital, legal support, infrastructure, and cooperation with service providers.

In a different study examining the determinants of the entrepreneurial climate, Chatman, Altman, and Johnson (2008) found that fair treatment by the local community, local support, business networks, and high internet speed have positive effects on the perception of entrepreneurial climate. Kline (2007) emphasized that although the studies on the entrepreneurial climate and especially the measurement of the entrepreneurial climate factors are a relatively new, some common factors can be found such as entrepreneur-friendly physical, financial, and commercial infrastructure. In addition to these factors, he states that technical and educational support, and network relations and human capacity are of vital importance for entrepreneurial activities, and the factors related to local culture cannot be ignored. From this point of view, Kline (2007) evaluated the effect of entrepreneurship climate on tourism development in rural areas. As a result, he revealed a seven-factor structure of entrepreneurial climate consisting of education and assistance, institutional support, quality of life, community/local culture, economic development level and infrastructure, natural tourism resources, and commercial support factors. When the researches in the literature are evaluated, it is observed that there are many researches that reveal the dimensions of entrepreneurial climate, and these researches are concentrated on the economic, social, legal, cultural, and technological axis of the entrepreneurial climate in general. In this research, the factors defined by Kline (2007) was utilized to examine the entrepreneurial climate since they present a comprehensive structure including dimensions revealed in other studies.

4 The Impact of Entrepreneurial Climate Perception on the Local's Entrepreneurial Intentions in Rural Tourism

In the study, which was designed with quantitative research design, a questionnaire form was utilized as data collection tool. In the first part of the three-part questionnaire, questions about the demographic characteristics of the participants were included, while the second part consisted of items measuring

entrepreneurial intentions (Brice, 2002), and the last part of the questionnaire consisted of the scale items developed by Kline (2007) to measure entrepreneurial intention. Via convenience sampling method, 202 people were included in the sample out of 1966 people residing in İğneada, and when the demographic characteristics of the participants are examined, it is seen that the majority of them were male (60.4 %); only 14.85 % were currently entrepreneurs with an existing investment. Majority of the participants were university graduates (34.17 %) and high school graduates (31.16 %). The monthly income of the participants varies between 1000 TL and 10,000 TL, and their average monthly income is 2,631.95 TL. The age of the participants ranged between 17 and 64 years, and the mean age was 33.05.

The perception of the entrepreneurial climate of the local people was examined within the framework of entrepreneurial climate factors (Tab. 1). When the average values of the factors were examined, it was found that the local population had the highest mean perceptions about the tourism potential of the region ($\bar{x}=3,76$); followed by local institutional support ($\bar{x}=3,69$), entrepreneurial infrastructure ($\bar{x}=3,61$), local culture ($\bar{x}=3,50$), and education and assistance ($\bar{x}=3,01$).

Tab. 1: Factors of Entrepreneurial Climate

Factors of Entrepreneurial Climate	Mean
Infrastructure	3.61
Institutional Support	3.69
Tourism Potential	3.76
Local People's Attitude	3.01
Local Culture	3.50
Education	3.21
General Entrepreneurial Climate Perception	3.47

As a result of the descriptive analysis, the average of entrepreneurial intention of local people was found to be 3.31. This shows that the local population is neutral to entrepreneurship in the tourism sector and points to the necessity of a climate that encourages investment and entrepreneurship. When the effect of entrepreneurship climate on entrepreneurship intention is examined via structural equation modeling, it was found that the entrepreneurial climate perceived by local people has a positive and statistically significant effect on entrepreneurial intention ($\beta = 0.35$; $p < 0.01$).

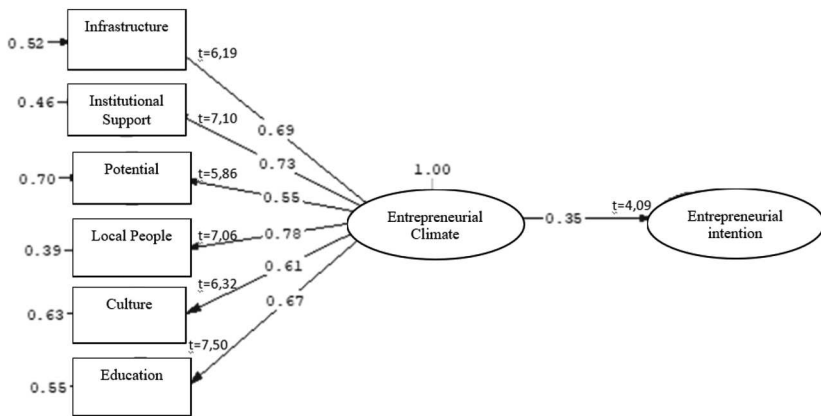


Fig. 1: The Effect of Entrepreneurial Climate on Entrepreneurial Intentions

5 Conclusion and Recommendations

The aim of this study is to determine the effect of the entrepreneurial climate on the locals' (residing in the Kırklareli-İğneada region, which is planned to be developed as an eco-tourism destination in the 2023 Tourism Strategies) intentions to invest in tourism sector. In this context, the components of the entrepreneurial climate have been determined, and the factors necessary for the formation of an entrepreneurial climate that will promote and encourage tourism entrepreneurship in the region are listed as entrepreneurial infrastructure, local governments and institutional support, tourism potential, support of local people, local culture, and education and assistance.

In developing rural tourism destinations, it is a great risk for the local people to leave their livelihoods and invest in the tourism sector. Furthermore, due to the negative attitude of the local people to tourism and tourists, it may be possible to boycott tourism enterprises in the region, and local entrepreneurs might be faced with exclusion from their social environment. This situation may pose various social problems for local entrepreneurs and also could make it difficult for their enterprises to operate in the region outside of active tourism seasons. Therefore, the local entrepreneurs seek the acceptance and support of the community in which they live in, before they decide to invest in the tourism sector as a local entrepreneur (Çakır and Doğan, 2018). For this reason, the first thing that should be done in promoting and encouraging local tourism

entrepreneurship in a rural region is to ensure that the local people would develop positive perceptions, attitudes, and behaviors towards tourism.

In the study, it has been determined that the local governments and institutions (municipalities, foundations, universities, etc.) are the second most important factor that positively effects the perceptions of entrepreneurial climate of the local people in the rural destinations. It is important that local administrations in the region, such as nongovernmental organizations, foundations, and associations etc. support tourism and show interest in the needs of people with entrepreneurial intentions. Local governments should participate in tourism congresses and fairs where local entrepreneurs cannot participate due to the cost, organize various activities that can attract people to the region such as festivals and concerts, and provide incentives and assistance to tourism entrepreneurs in order to create a positive climate for entrepreneurship activities of local people. In addition, associations, foundations, and communities could contribute to the establishment of a positive entrepreneurial climate by organizing activities that can create tourist attraction towards rural regions such as mountain hiking, nature photography, orienteering, and bird watching, which can create tourist attraction in the region; can educate local people; and raise awareness on tourism development and entrepreneurship.

The third important factor shaping the perception of entrepreneurial climate was found to be the entrepreneurial infrastructure of the region. People cannot be expected to invest in tourism in regions where there is no suitable transportation and communication network, and entrepreneurs do not have areas and shops where they can establish their business and have an inadequate infrastructure that cannot sustain the development of tourism. Therefore, it is necessary to establish a strong and adequate infrastructure in creating a positive entrepreneurial climate for the tourism sector in a rural destination.

One of the factors influencing the local people's entrepreneurial climate in the research is education and assistance. It is a known fact that highly educated people tend to migrate from rural areas such as İğneada, which is the study area of the research. It is almost impossible for the local people to learn about tourism, how the tourism sector in the region can create entrepreneurship opportunities, and how to become entrepreneurs without any training and assistance. For this reason, in creating a positive entrepreneurial climate, it is essential to provide appropriate training, consultancy services, and assistance for local entrepreneurs.

Aside from the entrepreneurial factors mentioned above, in order to promote local tourism entrepreneurship, it is important to have a local culture open to ethnic differences and change. Also, the rural areas should have built or natural

tourist attractions that creates potential for further tourism development in the region, so that the local people could see a future to invest in tourism.

When the entrepreneurial intentions of the local people in the region were evaluated, it was revealed that the local people are not very willing to invest in the tourism sector. On the other hand, it was concluded that the perceived entrepreneurial climate had a significant effect in the establishment of a local tourism supply structure that would enable the majority of the profit to be maintained in the region by improving the entrepreneurial intentions of the local people. For this reason, it is predicted that the entrepreneurship climate in the region can be improved with the development of the abovementioned factors and the local people's intention to invest in tourism sector and to be an entrepreneur can be increased significantly.

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